BoneCraft Manual

BoneCraft 9.2.5. BoneCraft is a rigging and avatar toolkit for Bforartists.

The navigation and tool paths in this edition cover BoneCraft 9.2.5, including the Starter Rigs catalog, Setup Rigging, Skin, Avatar Tools and Facial. See Getting Started for the interface map and visibility settings.

The Crescent Rose tutorial includes eight FBX preparation steps, 30 Mechanisms steps and 52 annotated screenshots checked against the 9.1.2 interface.

Chapters

Video supplement: IK/FK beginner walkthrough and practice file.

Use the chapter links below. PDF pagination changes when the manual is rebuilt.

1. Getting Started — what BoneCraft is, installing it, and where everything is.
2. Guides 1–7 — first avatar, VRoid, MMD, bone names, humanoid mapping, hair, clothing.
3. Guides 8–13 — lip sync, performance, poses, merging rigs, upload problems, CATS.
4. The CATS Pipeline — the six phases, the ledger, and why the order matters.
5. Rig Tools — rig UI, animation, weight and rig-control tools.
6. Rig Builder — building a new rig, an illustrated Simple Human → mannequin → Auto-Rig → IK/FK workshop, and Bone Merge.
7. Mechanisms — joints and links for props and machines: hinges, slides, gears, sequences.
8. VRChat Tools — VRChat, VRM and MMD tools, the I/O Hub and the Task Board.
9. CATS Tools — every CATS tool, one by one.
10. Quick Fixes and Glossary — common problems and the words this manual uses.
11. What's New — changes by version.
12. Tutorial: Crescent Rose with Mechanisms — a worked example: rig a transforming scythe-rifle step by step, with screenshots.

Getting Started

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What is BoneCraft?

BoneCraft is an add-on for Bforartists that helps you rig characters and props, and prepare 3D avatars for VRChat, VRoid / VRM and MMD. It adds panels and buttons that automate the most tedious steps: building a rig, painting weights, fixing bone names, setting up lip sync and eye tracking, and exporting.

What BoneCraft cannot do: - It cannot model or sculpt your avatar's body shape - It cannot create textures or materials from scratch - It cannot upload to VRChat directly (you still need the VRChat Creator Companion / SDK)

BoneCraft and BoneForge

BoneCraft runs only in Bforartists (the Blender fork). For standard Blender there is a separate add-on, BoneForge, with a smaller set of tools. Install one or the other, never both. In standard Blender, BoneCraft does not load: it shows one notice with a button to get Bforartists.

BoneForge 9.0.0 and later target standard Blender. BoneCraft is the separate Bforartists add-on. Both have Auto-Rig Wizard, Quick Rig, Mannequin and the shared CATS/Material Atlas tools. BoneCraft also has Mechanisms, the Control Picker and its production control-rig, joint-detection and profile-export tools. Similar placement does not add those tools to BoneForge.

Installing BoneCraft

What you need: - Bforartists (BoneCraft is tested on Bforartists 5.2) - The BoneCraft `.zip` file

Steps:

1. Open Bforartists.
2. Open **Edit** → **Preferences** → **Add-ons**.
3. Click **Install from Disk…** and pick the BoneCraft `.zip` file.
4. Tick the checkbox next to BoneCraft to enable it.
5. Expand BoneCraft's entry to see its settings: **Language**, **Tabs**, **Shelf Width**, **Show Helpers** and the **Tools** list.

You should see: the BoneCraft and Rig Builder tabs on the left Tool Shelf (press **T**) and the CATS tab on the right sidebar (press **N**).

Where things are

Tab names select a workspace group. Setup Rigging, Skin, Review, Avatar Tools and Facial are expandable sections inside those tabs, not separate tabs. The paths below use the English UI labels; translated labels

depend on the Language preference.

Tab	Default location	Sections
Rig Builder	Left Tool Shelf (T)	Auto-Rig Wizard, Quick Rig - premade armatures, Mannequin
BoneCraft	Left Tool Shelf (T)	Overview, Setup Rigging, Skin, Mechanisms, Review
CATS	Right sidebar (N)	CATS, Avatar Tools, VRChat Tools, Import / Export

Tool	Open this path
Bone Merge	BoneCraft → Setup Rigging → Bone Merge
Delta Mush / Proximity Wrap	BoneCraft → Skin → Delta Mush / Proximity Wrap
Viseme Library / Lip Sync	CATS → Avatar Tools → Viseme Library → Lip Sync
VRChat Viseme Mapping / Face Tracking	CATS → VRChat Tools → Facial
Humanoid / Hair	CATS → Avatar Tools → Set Up Humanoid Bones / Add Hair Physics
VRChat export	CATS → Import / Export → Unity / VRChat
VRM / MMD / I/O Hub	CATS → Import / Export → VRM / VRoid, MMD / PMX, Game Engine Export
Performance / Clothing / Naming	CATS → VRChat Tools → Check Performance / Attach Clothing / VRChat Naming Extras

Select an armature for rig and viseme-library controls; select a mesh in Object Mode for Delta Mush and Proximity Wrap. Lip Sync requires an armature in Pose Mode. Expand the section triangle to show its buttons. Options folds hold optional settings; tools that run immediately can be adjusted in Adjust Last Operation before running another operation.

These are three different facial workflows: CATS viseme generation creates mouth shape keys, Viseme Library / Lip Sync stores poses and keys animation, and Facial maps existing shape keys for VRChat or configures face tracking. UVToolkit-derived controls are in CATS → Material Atlas → Override Inherited Settings → UV Packing; this is the shared atlas workflow, not the full standalone UVToolkit add-on.

Move a tab: open **Edit** → **Preferences** → **Add-ons** → BoneCraft → **Tabs**. Each tab has a show switch and two buttons, Left and Right. A change applies at once. While a tab is hidden, its Left and Right buttons turn grey, but they still work.

Properties editor: Rig Builder, Mechanisms, CATS, Avatar Tools, VRChat Tools and Import / Export are also shown in the Properties editor's Scene tab while **Scene Properties** is on in Preferences → Tabs. This switch starts on. Click its **Show** button to hide or show all of these sections; blue means they are shown. It does not change the viewport tabs or the Armature data tab. With an armature selected, BoneCraft's armature tools also appear in the Armature data tab, independently of these visibility switches. Rig Builder, Mechanisms and the CATS sections use the Scene tab instead. To remove a tool everywhere, switch its package off in the **Tools** list, or disable the add-on.

Shelf Width: Widen Narrow Shelves starts on. It widens shelves holding BoneCraft tabs once if they are below the Left or Right minimum: 240 and 260 pixels by default, at 1× UI scale. A shelf you drag narrower afterwards stays at your chosen width. Turn the switch off to manage widths yourself; its width fields become inactive.

Menus and built-in Tool Shelf tabs: you can also reach tools from the 3D Viewport's header menus:

Location	Tools
Add → Armature	Quick Rig, Auto-Rig Wizard, Mannequin for Active Rig
Pose → BoneCraft (Pose Mode)	Pose tools, the picker, bone collections, rig setup and checks
Weights → BoneCraft (Weight Paint mode)	Flood fill, mirror and normalize weights
Object → Mechanisms (Object Mode)	Add parts, joints and links; test motion
Tool Shelf → Pose or Weights → BoneCraft	Buttons for common pose or weight operations

The Pose and Weights shelf buttons show text when the shelf is wide and become an icon grid when it is narrow. Hover an icon to read its tooltip. These access points use the same tools as BoneCraft's own tabs; their availability follows the enabled tool packages.

Viewport tools: Place Rig Markers places the Auto-Rig Wizard's next marker when you click the mesh; Build Mechanism starts a guided joint or link when you click a part. See Rig Builder and Mechanisms for their setup and controls.

Run first, adjust afterwards: tools such as Quick Rig, Fit Dropped Rig, Weight Mirror, Weight Transfer, Flood to Custom, Ribbon, Spline IK and Delta Mush run when clicked. Immediately afterwards, use the Adjust Last Operation panel in the viewport to change their options. Adjust before running another operation. Other panels put optional settings in a fold below the action buttons; expand it when you need those settings. A panel that needs a different selection may stay visible and explain what to select.

Show Helpers: BoneCraft explains its sections with short "what / when" lines and "How this works" notes. Untick **Show Helpers** in the add-on's preferences to hide them. In the Avatar Tools section, **Compact Mode** is a shortcut for the same switch.

Language: the **Language** setting in the add-on's preferences changes BoneCraft's own panels to English, 日本語, 한국어, 中文, Français, Português or Español.

Fix Model first, every time: in the CATS tab, always start with **Fix Model** before any other CATS tool. The CATS pipeline uses a ledger to track which steps you have done, and each tool warns you if you run it out of order. See The CATS Pipeline.

Quick panel: press **Ctrl+Shift+R** in the 3D Viewport to open BoneCraft's rig panel as a pop-up under the cursor.

Where Should I Start?

Choose the description that best fits you:

> **"You are a person who has a brand new 3D model file (FBX, OBJ, or Blender file) and wants to rig it for VRChat from scratch."** → Go to Guide 1: Get Your First Avatar Into VRChat

"You are a person who made your avatar in VRoid Studio and exported a VRM file." → Go to Guide 2: Bring a VRoid Avatar into VRChat

"You are a person who has an MMD model (PMX file) and wants to use it in VRChat." → Go to Guide 3: Bring an MMD Avatar into VRChat

"You are a person who already has a rigged avatar but it won't upload because the bones have wrong names." → Go to Guide 4: Fix Your Avatar's Bone Names

"You are a person who has a rigged avatar and wants to get it fully VRChat-ready — lip sync, eye tracking, clean mesh, all at once." → Go to Guide 13: Get Your Avatar VRChat-Ready with CATS

"You are a person who wants to fix a specific problem." → Go to the Quick-Fix Index

Next: Guides 1–7

Guides 1–7

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Guide 1: Get Your First Avatar Into VRChat

For a practice mesh before using your own avatar, follow the illustrated mannequin workshop: Simple Human, a rigless mannequin, Auto-Rig, then IK/FK.

> **Time:** About 45–60 minutes for a complete first run **Result:** A rigged, VRChat-ready avatar exported for the VRChat SDK

Before you begin — check these: - [ ] Your 3D model is imported into Bforartists (File > Import, choose your format) - [ ] The model is in a T-pose (arms out to the sides, body upright) — or close to it - [ ] The model has no obvious broken geometry (no random floating triangles) - [ ] You can see your model in the 3D viewport as a solid grey shape

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Step 1 — Open the Rig Builder

Open the **Rig Builder** tab on the left Tool Shelf (press **T** in the 3D Viewport). The same Rig Builder is in the **Properties** editor's **Scene** tab while Preferences → Tabs → Scene Properties → Show is on. It has three parts: the **Auto-Rig Wizard**, Quick Rig (ready-made armatures) and **Mannequin**.

Expand **Auto-Rig Wizard** and click **Start Auto-Rig**.

You can also start the wizard from the 3D Viewport's Add → Armature → Auto-Rig Wizard menu, then continue in Rig Builder.

Step 2 — Select Your Mesh

The wizard lists the meshes in the scene. Click your avatar's mesh in that list, then click **Next**.

What you should be seeing: A line "Selected:" with your mesh's name, then the rig type step.

Step 3 — Choose the Rig Type

Click **Body + Face** for an avatar whose face you want to animate (blinks, expressions, lip sync), or **Body Only** for a body without facial controls. Under Generation Options, **Kinematics** sets IK/FK, IK only or FK only. Click **Next**.

Step 4 — Place Body Markers

This is the most important step. Markers are dots that show BoneCraft where each major joint is: the pelvis, the spine, the neck and head, the shoulders, arms, hands, legs and feet. BoneCraft works out the bones from them.

The quick way: in the Smart Detection box, click **Detect Markers**. BoneCraft traces the limbs in your mesh and places every marker, each with a confidence. Click **Accept High** to keep the sure ones, and **Reset Low** to clear the doubtful ones so you can place them yourself. **Guess Markers from Mesh** is a simpler guess from the mesh's proportions.

With the viewport tool: for placing several markers, activate Place Rig Markers in the viewport toolbar in Object Mode. The tool header names the next marker and has axis locks and a mirror switch. Click the mesh to place and confirm that marker; the tool moves on to the next unconfirmed marker. When the step is done, click **Next** in the wizard.

One marker at a time: each marker has a row of icon buttons. Hover a button to read its tooltip.

1. Click **Place Marker** (the arrow icon) on the marker's row, then click the right spot on your avatar.
2. Click **Confirm Marker** (the check mark) when it sits correctly. **Reset Marker** (the circular arrow) clears a placed marker so you can place it again.

Symmetry: Toggle Symmetry (the first icon on a left or right marker's row) places the other side's marker whenever you place this one. **Mirror L to R** and **Mirror R to L** copy every confirmed marker to the other side.

When every marker is placed, click **Confirm All Placed**, then **Next**.

Step 5 — Place Finger Markers

Under Finger Detail, choose how many fingers to build: "5 — Full Human Hand" for a standard hand, or fewer (down to "0 — No Fingers"). Click **Guess Finger Markers**. BoneCraft finds each finger from the shape of the hand and traces it from knuckle to fingertip. Adjust any marker that is off, then click **Next**.

Step 6 — Place Face Markers (Body + Face only)

Click **Guess Face Markers**, then adjust the eyes, lids, brows, mouth corners and jaw as needed. Eye markers stay where you put them. Click **Next**.

Step 7 — Review and Generate

The review step lists any warnings and the generation options. Click **Generate Rig**.

What you should be seeing: A skeleton (orange bones) inside your avatar, with IK/FK limbs, finger bones and, for a Body + Face rig, a face rig, and your avatar's skin attached to it so it deforms when the bones move. The last step offers **Build Control Rig** for the review options Generate Rig does not apply, and **Close Wizard**; Rig Builder describes both.

> **Skin weights** = numbers that decide which bones move which parts of your avatar's mesh. BoneCraft sets them automatically; you can refine them with the Weight Tools (see Rig Tools).

Step 8 — Map to VRChat's Humanoid Bones

Open the **CATS** tab on the right sidebar (press **N**), expand **Avatar Tools** and then **Set Up Humanoid Bones**. Click **Auto-Map All**: BoneCraft connects your bones to the body slots VRChat uses to move an avatar with you. Click **Normalize Names** to give the mapped bones the names VRChat expects, then **Validate**.

What you should be seeing: The required slots (Hips, Spine, Head, LeftUpperArm and so on) each show a bone. Guide 5 explains the slots.

Step 9 — Export for VRChat

In the **CATS** tab, expand **Import / Export** → **Unity / VRChat**. Set the **Folder** and the **Avatar** name, then click **Export to VRChat (Unity)**. The duplicate export panel in Avatar Tools was removed in 9.2.3.

What you should be seeing: The exported avatar in your chosen folder, ready for the VRChat Creator Companion and the VRChat SDK in Unity.

What This Unlocks: You now have a rigged VRChat avatar. From here you can add hair physics (Guide 6), attach clothing (Guide 7), set up lip sync (Guide 8) and check performance (Guide 9). For a one-stop cleanup and VRChat setup with the CATS tools, see Guide 13.

Guide 2: Bring a VRoid Avatar into VRChat

> **Time:** About 15–25 minutes **Result:** Your VRoid avatar ready for VRChat export

Before you begin — check these: - [ ] You have exported your avatar from VRoid Studio as a `.vrm` file, or downloaded it from VRoid Hub as a `.zip` - [ ] BoneCraft is installed and enabled

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Step 1 — Install the VRM Add-on

BoneCraft uses the VRM add-on to read `.vrm` files. In the **CATS** tab, expand **Import / Export** → **VRM / VRoid**. If it says the add-on is not installed, click **Auto-Install Latest from GitHub**. Without an internet connection, download the add-on yourself and use **Install from .zip on Disk...**; **Open Extensions Page** opens its download page.

Step 2 — Import Your Avatar

In the same **VRM / VRoid** section, click **Import VRM...** and pick your `.vrm` file. For a VRoid download, click **Import VRoid (.vroid / .zip)...** and pick the `.zip`.

If you pick a VRoid Studio project file (`.vroid`), BoneCraft shows a "VRoid Studio Project" card instead: a project is not an avatar yet. Export it to VRM in VRoid Studio, then import that file.

What you should be seeing: Your VRoid character in the viewport with its skeleton in place.

Step 3 — Map to VRChat's Humanoid Bones

In the **CATS** tab → **Avatar Tools** → **Set Up Humanoid Bones**, click **Auto-Map All**. VRoid skeletons follow a standard layout, so this usually fills every slot. If a slot is wrong, **Fix Humanoid Map** in the **VRM / VRoid** section (under Lint) repairs the usual VRoid name mix-ups.

Step 4 — Fix Bone Names

Click **Normalize Names** in **Set Up Humanoid Bones**. VRoid has its own bone names; this gives the mapped bones the names VRChat expects.

Step 5 — Set Up Visemes (Lip Sync)

VRoid avatars already have the mouth shapes for lip sync. In the **VRM / VRoid** section, click **Map VRoid Visemes**. BoneCraft matches VRoid's shape keys to VRChat's lip-sync shapes.

Step 6 — Hair Physics (Optional)

VRoid hair uses SpringBones. **Convert SpringBones → PhysBones** in the **VRM / VRoid** section turns them into VRChat PhysBones. Guide 6 covers hair physics in more detail.

Step 7 — Export

Export as in Guide 1, Step 9.

What This Unlocks: Your VRoid avatar is ready for the VRChat Creator Companion. You can also check performance (Guide 9) before uploading.

Guide 3: Bring an MMD Avatar into VRChat

> **Time:** About 20–30 minutes **Result:** Your MMD `.pmx` model ready for VRChat

Before you begin — check these: - [ ] You have a `.pmx` or `.pmd` MMD model file - [ ] BoneCraft is installed

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Step 1 — Install mmd_tools

BoneCraft uses the mmd_tools add-on to read MMD files. In the **CATS** tab, expand **Import / Export** → **MMD / PMX**. If it says mmd_tools is not installed, click **Auto-Install Latest from GitHub** (or **Install from .zip on Disk...** with a downloaded copy).

Step 2 — Import Your PMX File

In the **MMD / PMX** section, click **Import PMX / PMD...** and pick your model.

What you should be seeing: Your MMD character in the viewport with Japanese bone names.

Step 3 — Fix Bone Names

VRChat cannot read MMD's Japanese bone names. In the **MMD / PMX** section, click **Convert Bone Names -> Unity**. For names in another language, use the **CATS** tab → **CATS** → **Translate Bone Names**: pick the **Language** and run it.

Step 4 — Clean Up the Model

MMD models often have extra geometry and many separate parts. In the **CATS** tab → **CATS**, run **Fix Model** first, then **Join Meshes** to combine the body parts into one mesh (recommended for VRChat). Guide 13 walks through the whole CATS pipeline in order.

Step 5 — Map to VRChat's Humanoid Bones

In the **CATS** tab → **Avatar Tools** → **Set Up Humanoid Bones**, click **Auto-Map All**. MMD skeletons are close to VRChat's, so most slots fill automatically. For an empty slot, click its button and choose the bone.

Step 6 — Physics and Export

Convert Physics -> PhysBones in the **MMD / PMX** section turns MMD's rigid-body hair and skirt physics into VRChat PhysBones. Then export as in Guide 1, Step 9.

What This Unlocks: Your MMD avatar is ready for VRChat. You can add hair physics (Guide 6) and set up lip sync (Guide 8) before uploading.

Guide 4: Fix Your Avatar's Bone Names

> **Time:** About 5–15 minutes **Result:** The bones VRChat needs renamed to VRChat-compatible names

Before you begin — check these: - [ ] Your avatar is open with its skeleton (armature) visible

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Step 1 — Detect the Current Naming Style

In the **CATS** tab, expand **VRChat Tools**, find its Naming group and click **Detect**. BoneCraft shows the naming style it found (for example Mixamo, VRoid or Unity) and a score; **Force Rescan** checks again after you change names.

For bone names in Japanese, Chinese, Korean, Portuguese, Spanish or French, use the **CATS** tab → **CATS** → **Translate Bone Names** instead: pick the **Language** and run it. It converts the names to English in one step.

Step 2 — Rename to VRChat Names

Map the bones first (Guide 5, Step 2), then click **Normalize Names** in **Set Up Humanoid Bones**: every mapped bone gets the name VRChat expects.

What you should be seeing: Bone names such as `Hips`, `Spine`, `Chest` and `LeftUpperArm`.

Step 3 — Manual Fixes (If Needed)

The Naming group in **VRChat Tools** also has batch tools for the bones that are left:

- **Find and Replace** — replace one piece of text with another in every bone name
- **Add Prefix** and **Add Suffix** — add text to the start or end of every bone name
- **Remove Prefix** and **Remove Suffix** — strip it again
- **Regex Replace** — the same, with a regular expression

What This Unlocks: Correct bone names let the humanoid mapping (Guide 5) work automatically. Without them, VRChat cannot tell how your character should move.

Guide 5: Map Your Avatar to VRChat's Body System

> **Time:** About 10–15 minutes **Result:** Your avatar's skeleton connected to VRChat's humanoid movement system

Before you begin — check these: - [ ] Your avatar's bones are named sensibly (see Guide 4 if not) - [ ] The skeleton is selected - [ ] You are in **Object Mode** (check the dropdown in the top-left of the viewport)

Step 1 — Open Set Up Humanoid Bones

In the **CATS** tab, expand **Avatar Tools** → **Set Up Humanoid Bones**.

Step 2 — Auto-Map

Click **Auto-Map All**. BoneCraft scans your skeleton and fills the body slots.

> A humanoid slot is like a labelled hook VRChat uses: "Hips go here, Head goes here, Left Hand goes here." BoneCraft matches your bones to these hooks. Humanoid mapping now reads back correctly when you reopen a file.

What you should be seeing: The required slots (15: Hips, Spine, Head, both upper arms, lower arms, hands, upper legs, lower legs and feet) and the optional ones (Chest, UpperChest, Neck and others), each with a bone, and the share of required slots that are filled.

Step 3 — Check the Mapping

Click **Validate**. BoneCraft checks that every required slot has a bone and that the hierarchy makes sense.

Step 4 — Fill Empty Slots by Hand

A slot without a bone shows "(click to map)". Click it and choose the right bone. Then click **Validate** again, and **Normalize Names** so the mapped bones get VRChat's names.

Step 5 — Set Up Eye Tracking (Optional)

If your avatar has eye bones, use the **CATS** tab → **CATS** → **Eye Tracking**: click **Auto-Detect** to find the eye bones (or pick the **Left Eye** and **Right Eye** bones), set the limits, and click **Create Eye Tracking**. Guide 13, Phase 3 explains the settings.

What This Unlocks: With the humanoid mapping done, your avatar moves properly in VRChat: IK (inverse kinematics, the system that makes your in-game hands follow your real controllers) works, and your avatar follows your real-world movements.

Guide 6: Add Hair Physics

> **Time:** About 15–25 minutes **Result:** Your avatar's hair (and other soft parts) swinging naturally in VRChat

Before you begin — check these: - [ ] Your avatar is open with the skeleton selected - [ ] Set Up Humanoid Bones is done (Guide 5)

> PhysBone = VRChat's component that makes bones bounce and swing as if they had weight. BoneCraft sets them up on chains of bones: hair, ears, tails, skirts and accessories.

Step 1 — Open Add Hair Physics

In the **CATS** tab, expand **Avatar Tools** → **Add Hair Physics**. It has four parts: Hair Chain Detection, Hair Chain Generation, PhysBone Configuration and Hair Physics Colliders.

Step 2 — Find the Hair Chains

If your hair already has bones, click **Detect Hair Physics Chains**. BoneCraft finds the chains of bones that look like hair (bones linked end to end from a root) and lists each one with its root bone and length.

If your hair has no bones yet, select the hair meshes and use Hair Chain Generation: choose the **Density** (Low, Medium or High: how many chains), then click **Generate**. **Replace** builds them again with a new density.

Step 3 — Choose a PhysBone Preset

Select a bone of a chain. Under PhysBone Configuration, click the preset that fits: **Hair**, **Skirt** (skirts and capes) or **Tail** (tails and ropes).

What you should be seeing: The chain's parameters: Pull, Spring, Stiffness, Gravity and Max Angle.

Step 4 — Fine-Tune (Optional)

The panel shows the chain's values; it does not edit them. More Stiffness makes a chain bend less, more Gravity pulls it down, and a smaller Max Angle limits how far it swings. For different values, click another preset, or change the PhysBone component in Unity after you import the avatar.

Step 5 — Add Colliders (Recommended)

Colliders are invisible shapes that stop hair from passing through the head and body. Under Hair Physics Colliders, click **Place Default Colliders** to add standard colliders on the head, shoulders and chest. **Add Collider** and **Remove Collider** add or remove one at a time.

What This Unlocks: Your avatar's hair now moves naturally in VRChat when you turn your head, jump or dance. The same steps work for tails, ears, skirts and dangling accessories.

Guide 7: Attach Clothing That Moves With Your Body

> **Time:** About 20–30 minutes **Result:** A clothing item attached to your avatar's skeleton, moving with your body

Before you begin — check these: - [ ] Your base avatar has a finished skeleton, and its body is weight painted - [ ] The clothing item is in the same file (with its own armature if it came with one) - [ ] The clothing roughly fits around your avatar's body

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Step 1 — Open Attach Clothing

In the **CATS** tab, expand **VRChat Tools** and find its Clothing group. It has three parts: Bone Matching, Collision Detection and Clothing Merge.

Step 2 — Match Bones

If the clothing came with its own armature, select it and click **Match Bones**. BoneCraft pairs each clothing bone with the equivalent bone of your avatar (the clothing's left arm with your avatar's left arm, and so on).

What you should be seeing: A list of bone pairs: clothing bone → avatar bone. For clothing without an armature, skip this step.

Step 3 — Merge the Clothing

Select the clothing — its armature and meshes, or only its meshes if it has no armature — then Shift-click your avatar's armature last, so it is the active object. Click **Merge Clothing**. With a clothing armature, BoneCraft moves the clothing onto your skeleton and merges the bones; without one, it transfers your body's weights to the clothing meshes.

> Weights (weight painting) = numbers that tell each point of a mesh how much each bone moves it. If your left sleeve has weight on the left arm bone, moving that bone pulls the sleeve with it.

What you should be seeing: The clothing under your avatar's skeleton. Moving a bone moves both the body and the clothing.

Step 4 — Check for Clipping

Click **Detect Collisions** to find places where the clothing pokes through the body. Fix them by adjusting the weights there (see the Weight Tools in Rig Tools).

What This Unlocks: Clothing that moves naturally with your body. Repeat for each outfit piece.

Next: Guides 8–13

Guides 8–13

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Guide 8: Set Up Lip Sync

> **Time:** About 10–20 minutes **Result:** Your avatar's mouth moving correctly when you speak in VRChat

Before you begin — check these: - [ ] Your avatar's head mesh has mouth shape keys (also called blend shapes or morph targets — the different mouth positions for speaking) - [ ] You know roughly what the shape keys are named (look in the Shape Keys list of the mesh's Object Data properties)

> Viseme = a mouth shape for a sound. `aa` is the "ahh" sound, `o` the "ohh" sound, and so on. VRChat uses 15 visemes to drive your avatar's lip sync.
>
> Shape key (blend shape) = a saved version of your mesh in another position. Your mouth-open shape key is your mesh with the mouth open.

Step 1 — Open VRChat Viseme Mapping

In the **CATS** → **VRChat Tools** → **Facial** → VRChat Viseme Mapping.

Step 2 — Auto-Map Visemes

Click **Auto-Map All Visemes**. BoneCraft reads your mesh's shape keys and matches them to VRChat's 15 viseme slots by name.

What you should be seeing: The viseme list, with most slots showing a mapped shape key.

Step 3 — Fill in Missing Slots

A slot without a shape key shows "(unmapped)". Click a slot's button to open a search list of your mesh's shape keys, and pick the closest one (type to filter the list). Pick "(unmapped)" to clear a slot. Common matches:

VRChat viseme	What it sounds like	Shape key to look for
`aa`	"ahh"	mouth_open, A, aa
`o`	"ohh"	mouth_o, OH, oh
`CH`	"ch" / "sh"	mouth_ch, CH
`PP`	lips together (P, B, M)	mouth_m, MM, lips_together
`SS`	"sss" / "zzz"	mouth_s, SS

No viseme shape keys at all? If the mouth has at least three base shapes (A, O and CH), the CATS tab's **Visemes** section creates all 15 from them. Without any mouth shapes, make those three first. See Guide 13, Phase 2.

Step 4 — Check a Mouth Shape

To see a mapped mouth shape, set its shape key's value to 1.0 in the mesh's Shape Keys list (Properties → Object Data), then set it back to 0.

Step 5 — Face Tracking (Optional)

Below the viseme list, **Face Tracking (SteamVR)** prepares expression shapes for face-tracking hardware. It is optional: most users do not need it.

What This Unlocks: Your avatar's mouth moves when you talk in VRChat. Expressions build on the shape keys you just mapped.

Guide 9: Improve Your Avatar's Performance

> **Time:** About 15–25 minutes **Result:** A better VRChat performance rank and faster loading for other players

Before you begin — check these: - [ ] Your avatar is complete (rigged, clothed, lip sync set up) - [ ] You know your target rank (Good or Excellent for most users)

> Performance rank = VRChat's rating of how demanding your avatar is. Very Poor avatars may be hidden by other players. Good and Excellent avatars load fast and are always shown.

Step 1 — Check the Current Rank

In the **CATS** tab, expand **VRChat Tools** and find its Performance group. Click **Calculate Performance Rank**. BoneCraft shows the overall rank and a breakdown — polygons, skinned meshes, material slots, bones, PhysBones and contacts — with how far each is from the next tier.

Step 2 — Clean Up

The VRChat Optimizer in the same section lists safe fixes. **What-If Mode** is on at first: each fix reports what it would change without changing anything. Untick it, because the fixes only report while it is on, and run the fixes you want:

1. **Remove Zero-Weight Bones** — deletes bones that move no part of the mesh
2. **Merge Same-Material Meshes** — joins meshes that share a material
3. **Remove Unused Shape Keys** — deletes shape keys nothing uses
4. **Remove Unused Vertex Groups** — removes empty weight groups

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Step 3 — Reduce the Polygon Count (If Needed)

If the polygon count is too high, use VRChat Decimation in the same section with the mesh selected: set the target ratio (0.5 halves the polygons, 0.8 removes 20%), click **Preview** to see the result, and **Apply** to keep it. Shape keys are protected while decimating.

> Start at 0.8 and work downward: small reductions rarely change how the avatar looks but can improve performance a lot.

Step 4 — Merge Materials (Optional)

If your avatar uses many materials, combine them into one texture sheet (an atlas). In the **CATS** tab → **CATS** → **Material Atlas**:

1. Click **Analyze** to see your materials and textures. Choose which ones to include.
2. Click **Smart Combine Materials** (or **Bake Atlas** for a plain bake). BoneCraft bakes the atlas and points each mesh at it; a backup of the originals is kept.
3. Click **Accept — Delete Backup** to keep the result, or **Revert — Restore Originals** to go back.

CATS Tools: Material Atlas Combiner explains every setting.

Step 5 — Check the Rank Again

Click **Calculate Performance Rank** again to see the improvement.

What This Unlocks: A better rank means more players see your avatar. A Good or Excellent avatar loads quickly, uses less GPU, and is shown to other players by default.

Guide 10: Save and Reuse Poses

> **Time:** About 5–10 minutes **Result:** A saved library of poses you can apply with one click

Before you begin — check these: - [ ] Your avatar has a finished skeleton - [ ] You are in **Pose Mode** (select the armature and press Ctrl+Tab, or use the mode dropdown in the top-left of the viewport)

Step 1 — Open the Pose Library

In the **BoneCraft** tab on the left Tool Shelf, expand **Review** → **Pose Library**.

Step 2 — Pose Your Avatar

Move the bones into the position you want to keep: rotate arms, tilt the head — any combination of bone positions is a pose.

Step 3 — Save the Pose

Click **Save** and give the pose a name (for example "Peace Sign" or "Thinking"). **Category** files it under a group such as "Greetings"; the **Filter** at the top shows one category at a time. A thumbnail of the viewport is captured with the pose.

Step 4 — Apply a Saved Pose

Select a pose and click **Apply**.

- **Blend** — applies the pose at partial strength, to mix two poses
- **Mirror** — applies the pose flipped left to right

Rename and **Delete** manage the list.

Step 5 — Export and Import Poses

Export saves your pose library to a `.bfpose` file you can keep with the project or share. **Import** loads poses from a `.bfpose` file.

What This Unlocks: A personal pose library you can reuse across projects, blend for animation keys, or share with other BoneCraft users.

Guide 11: Merge Two Rigs Together

> **Time:** About 25–40 minutes depending on complexity **Result:** Two skeletons combined into one, with all weights kept

Before you begin — check these: - [ ] The base avatar and the second rig (a clothing or accessory rig) are in the same file - [ ] Both are rigged and weighted

> Rig merge = absorbing one skeleton into another, so you end up with one combined skeleton. Useful when a body rig and a hair or outfit rig must become one.

Step 1 — Open Bone Merge

In the **BoneCraft** → **Setup Rigging** tab, expand the **Bone Merge** section. It works in three stages; each opens when the one before it is done.

Step 2 — Stage 1: Scope

Pick the **Base** armature (the body skeleton that stays) and the armature that merges into it (the clothing or hair rig, which is added to the base and then hidden). Click **Analyze**.

BoneCraft compares the two skeletons and lists: - ✓ Matched — bones that exist in both and line up - + Source only — bones of the merging rig with no match in the base - − Target only — base bones the merging rig does not have

When you have read the list, click **Acknowledge Scope — Proceed to Rename**.

Step 3 — Stage 2: Rename

For each source-only bone, decide what it is:

- Give it the name of the base bone it stands for, and click **Apply Rename**
- Click **Mark as Unique** if it is a new bone with no equivalent (it is added to the base as it is)

Normalize Both Armatures renames the standard bones (spine, arms, legs and so on) that BoneCraft recognises in both rigs. **Auto-Propose Renames** suggests names for the rest from each bone's position; **Apply Batch Pattern** renames many bones with one pattern.

When every source-only bone is resolved, click **Verify Rename — Proceed to Merge**.

Step 4 — Stage 3: Merge

Click **Run Merge Preview** first: it reports what the merge will do without changing anything.

When the report looks right, click **Merge Armatures**. BoneCraft backs up both armatures, then combines them.

What you should be seeing: One armature containing all bones of both skeletons, with every mesh weighted to it.

What This Unlocks: A single rig that is easier to export and edit. Needed for any avatar with separate clothing or accessory rigs.

Guide 12: Fix Upload Problems

> **Time:** About 5–15 minutes depending on the issue

Before you begin: Find your problem in the list below and go to that section.

"Upload failed — bones not recognized"

Your bones have names VRChat does not understand. → Guide 4: Fix Your Avatar's Bone Names

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"Avatar is in T-pose / not moving with me in VRChat"

The humanoid mapping is missing or wrong. → Guide 5: Map Your Avatar to VRChat's Body System

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"Mesh is deforming weirdly / skin stretching wrong"

The bone weights need adjusting. → See Weight Transfer and Weight Mirror in Rig Tools, or run **Smart Automatic Weights** in the CATS tab.

"Hair is clipping through the head"

Hair colliders are missing or too small. → Guide 6: Add Hair Physics, Step 5.

"Avatar is Very Poor performance and being blocked by other players"

→ Guide 9: Improve Your Avatar's Performance

"Lip sync not working"

The visemes are not mapped correctly. → Guide 8: Set Up Lip Sync

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"Rig validator shows errors"

In the **BoneCraft** tab, expand **Review** → **Rig Validator** and click **Run Validation**. Each result names the problem and the bone; follow its suggestion, or check the Quick-Fix Index. **Export Report** saves the results to a file.

"VRM import not working"

The VRM add-on may not be installed. → Guide 2, Step 1.

"MMD import not working"

The mmd_tools add-on may not be installed. → Guide 3, Step 1.

Guide 13: Get Your Avatar VRChat-Ready with CATS

> **Time:** About 20–35 minutes for a full pipeline run **Result:** A clean, optimized avatar with lip sync, eye tracking, and correct transforms — ready for VRChat upload

Before you begin — read this first:

The CATS tools follow a pipeline: each phase must be done in order. The CATS tab's Pipeline Status shows a checkmark for each phase you have done (Fix Model, Material Atlas, Visemes, Eye Tracking, Pose To Shape, Apply Transforms), and the Operation Ledger lists every CATS operation you ran. A tool whose earlier phase is not done still runs, but shows a warning such as "Run Fix Model first for best results".

Always start with Fix Model. Every time. Without exception.

Read The CATS Pipeline before you continue, if you have not already.

Before you begin — check these: - [ ] Your avatar is imported and visible in the 3D viewport - [ ] The CATS tab is open on the right sidebar (press N if the sidebar is hidden) - [ ] Your avatar is selected (click it in the viewport or the outliner)

Phase 1 — Fix Model

Fix Model is the mandatory foundation for everything else in CATS. Run it first, even if your model looks fine: it removes hidden problems that would otherwise silently break the tools that come after.

What Fix Model does (each part can be switched off in its options): - Applies modifiers - Removes duplicate vertices and loose geometry (triangles floating away from the body) - Recalculates normals (the directions that decide which side of the mesh faces outward) - Removes empty vertex groups and zero-weight bones - Removes constraints and rigid bodies VRChat cannot use

Steps: 1. In the **CATS** tab, expand **CATS** → **Fix Model** 2. Make sure your avatar's mesh is selected 3. Click **Fix Model** and wait — for large meshes this takes a few seconds 4. Check that Pipeline Status shows a checkmark (✓) for Fix Model

What you should be seeing: Your avatar looks the same or slightly cleaner. Pipeline Status checks Fix Model, and the Operation Ledger lists the run.

> **If your model disappears or looks inside-out after Fix Model:** the mesh's normals were inverted. Select the mesh, enter Edit Mode (Tab), select all (A), then use Mesh > Normals > Flip. Then run Fix Model again.

Phase 2 — Viseme Generation

Run after: Fix Model ✓

The Visemes section creates all 15 VRChat lip-sync mouth shapes from three base shapes you already have. If your avatar already has complete viseme shape keys, you can skip this phase — but check them in VRChat Viseme Mapping (Guide 8).

VRChat needs 15 mouth shapes (visemes) to drive your avatar's lips when you speak. Most avatars have only a few basic mouth shapes. CATS combines your base shapes to make the rest.

The three base shapes: - **A Shape** — wide open mouth (the "ahh" shape) - **O Shape** — rounded mouth (the "ohh" shape) - **CH Shape** — narrow open mouth with teeth showing (the "ch" / "sh" shape)

Steps: 1. In the **CATS** tab, expand **CATS** → **Visemes** 2. Click **Auto-Detect** to fill the three shapes, or pick them yourself (for keys named like `mouth_open`, `vrc.v_0` or `mouth_wide`, pick the closest match) 3. Click **Generate Visemes** 4. CATS creates 15 shape keys with VRChat's names (`vrc.v_aa`, `vrc.v_0`, `vrc.v_CH` and so on) 5. Check that Pipeline Status shows Visemes ✓

What you should be seeing: 15 new shape keys in the mesh's Shape Keys list (Properties → Object Data → Shape Keys).

> **If your avatar has no mouth shape keys at all:** create at least A, O and CH yourself (sculpt or edit them as shape keys) before CATS can generate the rest. See the Glossary entry for Shape Key.

Phase 3 — Eye Tracking

Run after: Fix Model ✓

Eye Tracking sets up your avatar's eye bones for VRChat's built-in eye tracking, so the eyes move naturally and look at other players.

What it does: - Finds the left and right eye bones - Names them as VRChat requires (`LeftEye` and `RightEye`) - Adds the rotation constraints VRChat needs to move the eyes - Limits eye rotation to a natural range (up, down and to the side)

Without Fix Model first, the eye detection can lock onto leftover duplicate vertices instead of the real eyes and place the constraints in the wrong spot — avatars arrive in VRChat with both eyes stuck looking down.

Steps: 1. In the **CATS** tab, expand **CATS** → **Eye Tracking** 2. Click **Auto-Detect** — CATS looks for bones whose names or positions match eye bones 3. Check the **Left Eye** and **Right Eye** fields; pick the bones yourself if they are wrong 4. Set the **Up Limit**, **Down Limit** and **Side Limit** if needed 5. Click **Create Eye Tracking** 6. Check that Pipeline Status shows Eye Tracking ✓

What you should be seeing: Both eye bones renamed, with rotation constraints in the Bone Constraints panel.

> **If CATS cannot find eye bones:** your skeleton may have none. Some formats (especially older MMD models) blink with shape keys instead of bones. Skip this phase: VRChat falls back to shape-key eye animation when there are no eye bones.

Phase 4 — Pose to Shape Key

Run after: Fix Model ✓

Pose to Shape Key turns your avatar's current pose into a shape key. Use it to capture custom expressions or rest poses for VRChat's expressions menu.

Without Fix Model first, the vertex order can still contain gaps from removed duplicates, so the shape key captures distorted geometry — shape keys that make the mesh explode outward when triggered in VRChat.

Steps: 1. Pose your avatar in Pose Mode (select the armature, press Ctrl+Tab) 2. Move the bones into the expression or position you want to capture 3. Return to Object Mode (Ctrl+Tab again) 4. In the **CATS** tab, expand **CATS** → **Pose & Shape Keys** 5. Click **Pose to Shape Key** and name the new shape key 6. Check that Pipeline Status shows Pose To Shape ✓

What you should be seeing: A new shape key in the mesh's list. Set its value to 1.0 to check that it shows the pose.

> **Shape Key to Basis:** the companion tool does the opposite — it bakes a shape key into the mesh's neutral rest shape. Use it to lock in a corrected rest pose. Run it after Fix Model too.

Phase 5 — Apply Transforms

Run after: every earlier phase you need

The last phase. Apply Transforms freezes all position, rotation and scale into the mesh and skeleton, so everything reads as clean values (position 0,0,0 / rotation 0°,0°,0° / scale 1.0). The VRChat SDK needs clean transforms: un-applied scale in particular makes avatars appear at the wrong size or with physics that misbehave.

Applying transforms before the earlier phases bakes their problems into the mesh for good — avatars that look right in Bforartists but spawn at a fraction of their height in VRChat.

The tools in Transforms & FBT: - **Apply All Transforms** — applies position, rotation and scale to the mesh and the armature together - **Fix FBT** — a transform correction for Full Body Tracking setups (moves the root bone to floor level) - **Remove FBT** — removes that correction again

Steps: 1. Confirm that the phases your avatar needs are checked in Pipeline Status (Material Atlas and Pose To Shape are optional, and Visemes stays unchecked when you mapped existing visemes instead) 2. In the **CATS** tab, expand **CATS** → **Transforms & FBT** 3. Click **Apply All Transforms** 4. Check that Pipeline Status shows Apply Transforms ✓ 5. Export your avatar (VRChat → Export to VRChat, Guide 1 Step 9)

What you should be seeing: Apply Transforms checked, after the phases you needed. Your avatar is ready for the VRChat Creator Companion.

What This Unlocks: A fully processed avatar with clean geometry, all 15 visemes, eye tracking (when it has eye bones), any custom expressions you made, and clean transforms — everything a VRChat upload needs to work the first time.

Next: The CATS Pipeline

The CATS Pipeline

← Contents

Read this before using any CATS tool for the first time.

The CATS tools are not a menu of independent options — they are a pipeline. Each step builds on the previous one. Running them out of order produces broken results that stay invisible until you are already in VRChat, and by then they cannot be fixed without starting over. You may skip a phase your avatar does not need, but the phases you run must keep this order.

The Six Phases, In Order

The CATS tab's Pipeline Status lists the phases and checks each one off when it is done:

1. **Fix Model** — Clean the mesh: remove duplicates and broken geometry, fix normals. Every other step depends on this one.
2. **Material Atlas** — Combine the avatar's materials into one texture sheet (optional, for performance).
3. **Visemes** — Generate the 15 VRChat lip-sync mouth shapes from your three base shapes (A, O, CH).
4. **Eye Tracking** — Set up VRChat's eye movement with your avatar's eye bones.
5. **Pose to Shape Key** — Capture custom poses or expressions as shape keys.
6. **Apply All Transforms** — Freeze all position, rotation and scale so your avatar has clean transforms for upload.

Pipeline Status and the Operation Ledger

Pipeline Status shows one line per phase. When a phase finishes, its line shows a checkmark. The status is kept in the scene, so it stays with your file.

The Operation Ledger lists every CATS operation you ran, newest last, with its time and outcome: Changed (it changed the model), Clean (nothing needed changing) or Failed. **Copy** puts the ledger on the clipboard (useful when you ask for help); **Clear** empties it.

The order is advice, not a lock: a tool whose earlier phase is not done still runs, but shows a warning such as "Run Fix Model first for best results". Take the warning seriously.

Why This Order

Each phase changes the mesh or skeleton in ways the next one depends on:

- **Fix Model must be first** because it changes the vertex count, removes duplicates and corrects normals. Any tool that reads vertex positions (viseme generation, eye bone detection, pose capture) produces wrong results on a mesh that still has duplicate or broken vertices.
- **Visemes before Eye Tracking** because both change shape key and bone data. In this order, the shape keys exist before the constraints are written.
- **Eye Tracking before Pose to Shape** because Pose to Shape captures the whole mesh, including bone-driven deformation. With the eye constraints in place first, the neutral eye position in the saved shape is right.
- **Apply Transforms last** because it freezes everything for good. Any unfixed geometry, unmapped shape keys or wrong constraints are locked in. After applying transforms, the only ways back to an earlier phase are undo (Ctrl+Z) or re-importing from the source file.

What Breaks If You Skip a Step

Skip Fix Model: Every later tool works on a mesh that may hold duplicate vertices at the same position. A generated viseme can then move the visible vertex but not its hidden duplicate, so in VRChat the two pull apart and tear the mouth at every lip-sync shape. Eye tracking may attach its constraints to the duplicate eye mesh instead of the visible one, locking the eyes in a fixed stare. Apply Transforms then bakes all of these errors in.

Skip Visemes: Without generated or mapped visemes your avatar has no lip sync. If your avatar already has viseme shape keys, map them in VRChat Viseme Mapping (Guide 8) instead of generating new ones; Pipeline Status then shows Visemes unchecked, which is fine.

Skip Eye Tracking: Your avatar has no eye movement in VRChat and stares straight ahead. That is acceptable when it has no eye bones.

Skip Pose to Shape: Optional: skip it if you have no custom expressions to save.

Next: Rig Tools

Rig Tools

← Contents

This chapter covers BoneCraft's rigging, animation and weight tools in detail. Use it when you want to understand a feature more deeply, or when the guides do not cover your exact situation.

Each entry says what the tool does, when you would use it, its main controls, and where to find it. Each entry's "Where to find it" line names the mode needed to use it: for example, rig controls in Pose Mode, Weight Mirror and Weight Transfer in Weight Paint mode, and Delta Mush in Object Mode. Some sections stay visible with a message about what to select when their tools cannot run.

In 9.1, many tools run at once and put their options in the Adjust Last Operation panel. Change those options immediately after the click, before running another operation. The Pose → BoneCraft and Weights → BoneCraft menus, and the BoneCraft button panels in Bforartists' Pose and Weights Tool Shelf tabs, offer shortcuts to common operations. Getting Started lists these access points.

Rig UI Tools

These tools help you manage the visual side of working with armatures — which bones are visible, how they are organized, and quick access.

Bone Collections

What it does: Shows your skeleton's bone collections as rows. Click a collection's name to select its bones; the eye icon shows or hides it.

> A bone collection is a named group of bones — for example "IK Controls", "Face Bones" and "Deform Bones". Hiding a collection hides those bones in the viewport, so you can focus on one part of the rig.

Key controls: - Each row: the accent color, **Toggle Collection Visibility** (eye icon), the collection's name (selects its bones), **Solo Collection** and **Rename Collection Display**, which sets a custom display name for the row. Hover an icon-only button to identify it by its tooltip - **Show All Collections** and **Hide All Collections** (open and closed eye icons above the list) — everything at once - Sections — group several collections under a collapsible header

Where to find it: BoneCraft tab → Review → Collections

Visibility Bookmarks

What it does: Saves which bone collections are visible, so you can switch between saved views at once.

Example use: You set up a view with only the face bones for expression work and save it as "Face Only". Then you show everything for weight painting and save it as "Full Rig". Now one click switches between them.

Key controls: each slot is a row. - The bookmark's name — click it to apply the saved view - **Save Bookmark** — saves the current visibility in the slot; hover its icon to see the tooltip - The color swatch — color-codes the bookmark - **Clear Bookmark** and **Remove Bookmark** (close icons) — empty the slot, or remove an added one - **Add Bookmark** — adds a slot

Default bookmarks: FK Arms, IK Body, Face Only, Full Rig

Where to find it: BoneCraft tab → Review → Collections → Visibility Bookmarks

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Hotkey Quick Panel

What it does: Opens BoneCraft's rig panel (collections and bookmarks) as a pop-up wherever your cursor is.

How to use: Press **Ctrl+Shift+R** in the 3D viewport. Click outside the pop-up to close it.

Where to find it: the **Ctrl+Shift+R** shortcut in the 3D viewport (it has no panel of its own)

Animation Tools

Pose Library

What it does: Stores named poses with thumbnails that you can apply to your rig with one click.

Key controls: - **Save** — stores the current bone positions as a named pose, with a thumbnail of the viewport - **Apply** — snaps the bones to the saved pose - **Blend** — applies the pose at partial strength, mixed with the current position - **Mirror** — applies the pose flipped left to right - **Delete**, **Rename** and **Category** — manage the list; **Filter** shows one category - **Export** and **Import** — save to or load from a `.bfpose` file

Blend applies immediately. Set the blend strength in Adjust Last Operation after clicking it.

Where to find it: BoneCraft tab → Review → Pose Library

Control Picker

What it does: A clickable map of your rig's controls. Click a shape to select its bone; the picker highlights what is selected.

Key controls: - **Generate Picker** — builds the layout from your rig - **Edit Layout** — move, resize and relabel the controls (**Move Picker Control**, **Resize Picker Control** and **Relabel Picker Control** appear as icons on each control's row) and **Snap** them to a grid; **Rebuild Layout** builds it again - **Mirror Selection** — selects the mirrored controls - **Pop-up Picker** — opens the picker in a floating window, which also has **Add Selection Set**

Where to find it: BoneCraft tab → Review → Control Picker

Control Layer (IK/FK)

For an illustrated hands-on comparison, follow Test weights and use IK/FK.

What it does: Animator controls for the limbs of a rig made by **Build Control Rig**: switch a limb between IK and FK without the limb jumping, and bake one into the other over a frame range.

Key controls: - **IK** and **FK** — switch the limb and snap the new controls to where the limb is, so the elbow or knee does not pop - **Bake IK** and **Bake FK** — convert the animation over a frame range to IK or to FK - **Pole Follow**, **Pin** and **Lock-Free**, and **Global Rig Scale**, which keeps protected locks

Where to find it: BoneCraft tab → Review → Control Layer (with a rig made by **Build Control Rig** selected; see Rig Builder)

Rigify Tools

What it does: Detects a Rigify control rig and connects BoneCraft's collection panel, bookmarks and property sliders to Rigify's standard controls.

> Rigify is Bforartists' built-in system for generating animation-ready rigs. If you built your rig with Rigify, this connects BoneCraft's UI to Rigify's IK/FK controls.

Key controls: - **Auto-Enhance** — sets up BoneCraft's panels for the selected Rigify rig - **Re-run Enhancement** — rebuilds them from scratch - **Enable Rigify** is in Quick Rig; expand its Troubleshooting fold for **Generate Control Rig (active metarig)** (see Rig Builder) - Rigify Properties — the rig's IK/FK, stretch, parent-space and head/neck follow sliders

Where to find it: BoneCraft tab → Setup Rigging → Rigify Tools (and the Rigify Properties section)

Retarget

What it does: Moves animation from one skeleton to another — for example a motion-capture clip onto your avatar. Bones are matched by name and by direction, so limbs keep their orientation; collarbones are matched loosely so shoulders do not over-rotate.

Key controls: - **Import Clip** and **Browse Library** — load a clip, or set a library folder; click a clip's name to select it - **Auto-Match Bones** (the check mark on a clip's row) — pairs source and target bones (the panel's Bone Mapping list shows the pairs) - **Preview** and **Stop Preview** — try the result - **Apply** — bakes the clip onto your rig

Where to find it: BoneCraft tab → Setup Rigging → Retarget

Corrective Shape Keys

What it does: Shape keys that switch on when a bone reaches an angle, to fix pinching or collapsing at extreme poses.

Example use: Your character's elbow collapses when fully bent. You sculpt a corrected elbow and link it to the arm bone so it applies automatically as the elbow bends.

Key controls: - New Corrective — choose the **Shape Key**, the **Driver Bone**, the rotation axis, the angle at which the shape is fully on, and how gradually it fades in, then **Create Corrective** - The list of correctives: click a corrective's shape key name to select it; **Edit** and **Delete** under the list act on the selected one

Edit starts from the corrective's current values. Change its angle and falloff in Adjust Last Operation after clicking it.

Where to find it: BoneCraft tab → Skin → Correctives (the mesh needs a basis shape key)

Graph Tools

What it does: Animation refinement tools for keyframes and curves.

- **Interactive Breakdown** (Breakdowner; shortcut **Ctrl+Shift+B** in the 3D viewport) — drag left and right to blend the current pose between the nearest keyframes
- **Apply Delta** (Delta Mover; shortcut **Ctrl+Shift+D** in the 3D viewport) — nudge selected bones by an exact offset in screen space or world space
- **Capture** and **Swap** (Buffer Curves) — keep a copy of the curves and swap between it and your edits
- **Smart Bake** — bakes constraint-driven animation to keys, dropping redundant keys (**Tol** sets how much)
- **Euler Filter** — fixes rotation flips caused by gimbal lock
- Tangent Handles — set handle types on selected keys: **Auto**, **Vector**, **Aligned**, **Free** or **Auto Clamped**

Where to find it: BoneCraft tab → Review → Graph Tools

Weight Tools

These tools control how your mesh deforms when bones move. Weights tell each part of your mesh which bones to follow, and how much.

Automatic Bone Weight Recalculation

What it does: Recalculates the weights of a mesh from the existing rig, without changing the mesh. On dense meshes it works through a lighter proxy mesh, so it does not fail silently.

Key controls: choose the target mesh and armature, then Body or Clothing mode. Body calculates weights from the existing rig; Clothing transfers them from a correctly weighted source mesh. Set the method, blend and smoothing, the maximum number of bone influences per vertex, a minimum weight, and whether to keep a backup, then run **Automatic Bone Weight Recalculation**. These settings are above the button. The CATS shortcut, **Smart Automatic Weights**, opens a dialog for the target mesh, armature and mode, plus the source mesh in Clothing mode.

Where to find it: BoneCraft tab → Skin → Weight Tools

Weight Mirror

What it does: Copies weights from one side of your avatar to the other. Essential for symmetrical characters.

Key controls: - **Mirror All Weights** — mirrors every bone group to its opposite side - **Mirror Active Weight** — mirrors only the active bone group - Symmetry axis and mirror direction (both ways, left to right, or right to left) - **Topology Mirror** — pairs vertices by mesh topology instead of position - **Search Distance** — how far apart two vertices may be and still count as a pair (raise it for meshes that are not perfectly symmetrical) - **Select Unmatched** — selects vertices without a partner; **Normalize Weights** makes the weights sum to 1.0 afterwards

Click a mirror button first, then set the axis, direction, topology mode and search distance in Adjust Last Operation. **Mirror Active Weight** is also in Weights → BoneCraft and the Weights Tool Shelf tab.

Where to find it: BoneCraft tab, in Weight Paint mode → Weight Mirror

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Weight Transfer

What it does: Copies weights from a source mesh to the active mesh. Used when attaching clothing to a body, or moving weights from a high-resolution mesh to a lower one.

Key controls: - Pick the source mesh before transferring - After the click, set the transfer method and bone filter in Adjust Last Operation - **Normalize** in Adjust Last Operation — keeps the weights summing to 1.0 - **Transfer Weights** — runs it; the result shows mean and maximum distance, and **Select Problem Vertices** selects vertices that matched badly

Where to find it: BoneCraft tab, in Weight Paint mode → Weight Transfer (also under Skin → Weight Tools)

Weight Table

What it does: Shows the exact weights of the selected vertex for each bone and lets you set a precise value.

How to use: Select one vertex in Weight Paint mode; the table lists its bones and weights. Click the hand icon on a bone's row, then set **Weight** in Adjust Last Operation. Add Influence uses the bone you choose; set its new weight in Adjust Last Operation too. With several vertices selected, **Normalize All Vertices** normalizes them all.

Where to find it: BoneCraft tab, in Weight Paint mode → Weight Table

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Flood Fill

What it does: Sets the active group's weight on the selected vertices in one click: **Flood to 0.0**, **Flood to 1.0**, or **Flood to Custom**. After Flood to Custom, set **Value** and **Normalize** in Adjust Last Operation. The Weights menu and shelf call the fixed-value buttons Flood to Zero and Flood to One.

Where to find it: BoneCraft tab → Skin → Weight Tools; also Weights → BoneCraft and Tool Shelf → Weights → BoneCraft

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Deform vs Control

What it does: Sorts the rig's bones into deform bones (they move the mesh) and control bones (the animator moves them), tags deform bones, renames them to a deform naming, and exports a deform-only skeleton.

Where to find it: BoneCraft tab, in Pose Mode → Deform vs Control

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Delta Mush

What it does: A smoothing deformation that reduces pinching at joints. The mesh keeps its shape at rest but deforms more cleanly in motion.

How to use: Click **Add Delta Mush** to add the modifier and bind it to the rest shape. Immediately afterwards, set **Iterations**, **Smooth Strength**, **Pin Borders** and **Only Smooth Deform** in Adjust Last Operation. Once the modifier exists, the panel shows **Rebind** and **Remove**, with a Settings fold containing the modifier's iterations, smoothing strength and border pinning controls.

Where to find it: BoneCraft → Skin → Delta Mush (Object Mode; Mesh)

Proximity Wrap

What it does: Makes one mesh follow the surface of another, like a second skin. Useful for clothing that hugs the body.

How to use: select the target mesh, choose a different **Driver Mesh**, then click **Bind Proximity Wrap**. Keep the driver's armature in its rest pose when binding. The Options fold below the button lists **Falloff Type** and **Max Distance**; in 9.1.2 these values are stored with the setup but do not change its Surface Deform binding. Once bound, the panel shows **Rebind** and **Unbind**. Unbind disables the modifier in the viewport and render; it does not remove it.

Where to find it: BoneCraft → Skin → Proximity Wrap (Object Mode; Mesh)

Rig Controls

These tools appear in the BoneCraft tab in Pose Mode.

Space Switch

What it does: Changes which space a bone follows during animation. A hand holding a prop can switch from following the body to staying still in the world with one click.

Key controls: - **Add Space** and **Remove Space** (X icon under Manage Spaces) — the spaces a bone can follow (World, Origin or another bone) - One button per space, named after it; the current space is marked ●. Click another space's button to switch the bone to it

Where to find it: BoneCraft tab, in Pose Mode → Space Switch

Spline IK

What it does: Makes a chain of bones follow a curve. Used for tails, tentacles, ropes, long hair and spines that need smooth, sweeping motion.

How to use: Make the starting bone active, then click **Generate Spline IK**. Set **Chain Length**, **Curve Resolution**, **Stretch** and **Volume Preservation** in Adjust Last Operation. **Remove Spline IK** removes the setup.

Where to find it: BoneCraft tab, in Pose Mode → Spline IK

Chain Dynamics

What it does: Physics-like secondary motion on a bone chain: the bones lag behind when the parent moves and bounce back when it stops. Used for hair strands, tails and accessories.

How to use: select the bone chain and click **Add Chain Dynamics**. Set **Stiffness**, **Damping**, **Mass** and **Gravity Scale** in Adjust Last Operation. The panel also lists the parameters in a Settings fold below the action. Once dynamics exist, it shows **Bake** and **Remove**. Collision uses a simplified sphere shape, so concave surfaces may not collide correctly.

Where to find it: BoneCraft tab, in Pose Mode → Chain Dynamics

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Ribbon

What it does: Builds a ribbon of Bendy Bones — bones that curve and twist smoothly — for lips, eyebrows, belts and other soft curved areas.

How to use: Select the bone chain and click **Generate Ribbon**, then set **Control Bones** and **Ribbon Width** in Adjust Last Operation. **Remove Ribbon** removes the setup.

Where to find it: BoneCraft tab, in Pose Mode → Ribbon

Viseme Library

What it does: Creates and manages sets of visemes (mouth shapes) linked to shape keys, and generates lip-sync keys from text. For VRChat viseme mapping use VRChat Viseme Mapping; to generate visemes from scratch use the CATS Visemes section.

Key controls: - **New Set** and **Delete** — create or remove a viseme set - In each set, one button per viseme (with a speaker icon) previews it - Lip Sync — text to viseme keys, with frames per phoneme and smoothing

Where to find it: CATS → Avatar Tools → Viseme Library (Armature); Lip Sync (Pose Mode)

Set Driven Key

What it does: Links one value to another without Python expressions. Pose the driver (for example an eyebrow bone), set the driven value (for example a "Brow Raised" shape key), and record the point;

BoneCraft builds the driver curve.

Key controls: - **Driver Bone** and **Driver Attribute** — what drives - **Target Bone** and **Target Attribute** — what is driven - **Set Value** and **Record Point** — record a point on the curve - **Create SDK** — builds the driver

Where to find it: BoneCraft tab, in Pose Mode → Set Driven Key

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Rig Validator

What it does: Checks your rig against a set of rules and reports problems — naming, missing bones, broken constraints, hierarchy, weights, drivers that need Python, and the Mechanisms rules. Bone names with hyphens (such as `DEF-upper_arm.L`) pass.

Key controls: - **Run Validation** — runs every check and lists the results - **Export Report** — saves the results to a file

The Task Board (BoneCraft tab → Overview → Task Board) shows a validator task when a check fails.

Where to find it: BoneCraft tab → Review → Rig Validator

Rig Notes

What it does: Written notes attached to the rig's bones — what you set up, reminders, notes for collaborators.

Key controls: Add Note, and **Remove** on each note. Rig Readme shows the rig's notes as a read-only page.

Where to find it: BoneCraft tab → Overview → Rig Notes, and Review → Rig Readme

Next: Rig Builder

Rig Builder

← Contents

For a narrated demonstration of both rig types, see the IK/FK beginner walkthrough and practice file.

Building a New Rig

The Rig Builder makes a new rig for a character. It is in its own **Rig Builder** tab on the left Tool Shelf (press **T**), and in the **Properties** editor's **Scene** tab while Preferences → Tabs → Scene Properties → Show is on. In a narrow Tool Shelf it switches to a compact layout with short labels. The Auto-Rig Wizard comes first, followed by Quick Rig and Mannequin. You can also start the wizard or add a Quick Rig from the viewport's Add → Armature menu.

Auto-Rig Wizard

What it does: A step-by-step guide that places markers on your mesh and builds a complete rig with weights from them. The main way to rig a character from scratch.

Guide 1 walks through it step by step.

Steps: Select Mesh → Rig Type → Body Markers → Finger Markers → Face Markers → Review → Generate

Key controls: - **Start Auto-Rig** opens the wizard. **Back** and **Cancel** are at the top; **Next** is at the bottom of the mesh, rig-type and marker steps. Review uses **Generate Rig** instead of Next; the final step uses **Close Wizard** - The mesh list (click your mesh), then the rig type: **Body + Face**, **Body Only** or **Face Only** (for an existing armature) - **Detect Markers** (in Smart Detection) — traces the limbs in the mesh and places every body marker with a confidence; **Accept High** keeps the sure ones and **Reset Low** clears the doubtful ones - **Guess Markers from Mesh** and **Guess Precision** (body), **Guess Finger Markers** (finger markers are placed from the shape of the hand) and **Guess Face Markers** - Finger Detail — how many fingers to build, from "0 — No Fingers" to "5 — Full Human Hand" - Each marker's row: **Place Marker** (arrow icon), **Confirm Marker** (check mark) and **Reset Marker** (circular arrow); **Toggle Symmetry** (first icon) on left and right markers - **Mirror L to R** and **Mirror R to L** — copy confirmed markers across the centre line - **Confirm All Placed** — confirms every placed marker of the step - **Generate Rig** — builds the rig from the confirmed markers; then, on the last step, **Build Control Rig** (see below) and **Close Wizard**

What Generate Rig builds: - A body rig with the spine and neck segments you chose, and IK/FK, IK-only or FK-only arms and legs (the **Kinematics** option) - Finger bones for the finger count you chose - For **Body + Face**: a face rig, with control shapes when **Generate Control Shapes** is on - Skin weights at the chosen quality. Dense meshes are weighted through a lighter proxy mesh, so a 678,000-vertex character takes about a minute instead of failing silently.

Build Control Rig: the review step also shows **Twist Bones**, **Skirt Bones**, **Face Controls** and **Joint Correctives**. Generate Rig does not apply them. They belong to the engine control rig (twist bones, skirt bones that follow the legs, eye aim and blink controls, and joint corrective shape keys). After Generate Rig, the last step offers **Build Control Rig**, which builds that rig on the generated armature with the options

you chose. Click it before **Close Wizard**, because closing resets the wizard's options. The Control Layer panel (Rig Tools) appears only on a rig built this way.

Where to find it: Rig Builder → Auto-Rig Wizard

Place several markers with the viewport tool: start the wizard and advance to a body, finger or face marker step. In Object Mode, activate Place Rig Markers in the viewport toolbar. The tool header names the next unconfirmed marker and shows axis locks and a mirror switch. Click the wizard's mesh to place and confirm that marker. Symmetry follows the wizard's settings; the per-marker Place buttons still work. When the tool says the step is done, press **Next** in the wizard. Clicking elsewhere does not place a marker.

Quick Rig

What it does: Adds a ready-made armature in one click: BoneCraft's Basic Humanoid (19 bones, no Rigify needed) or one of Rigify's metarigs. Faster than the wizard, less tailored.

Key controls: - The Basic Humanoid button, and the Rigify metarigs (**Enable Rigify** first, if Rigify is off) - Click a rig button to add it immediately, then adjust fitting to the mesh, automatic weights, **Add IK/FK Controllers** and **Initial Pose** in Adjust Last Operation - **T-Pose** and **A-Pose** — switch the selected rig's rest pose - Expand Troubleshooting for **Diagnose Quick Rig (print to console)**. With a non-metarig armature active it also shows **Inspect Active Rig Controls** - With a Rigify metarig active, Troubleshooting shows **Generate Control Rig (active metarig)** instead of Inspect Active Rig Controls; it generates the Rigify control rig

The BoneCraft Rigs starter catalog: all nine bundled starters share a **Starter Rigs** catalog in the native bottom asset shelf. Open the shelf's catalog menu at the left, choose **All** libraries, and enable **Starter Rigs** once to add its tab beside the existing catalog tabs. Select that tab to browse the starters. **Add → Armature → Starter Rig Catalog** opens the dependency-aware popup for fitting and adding a starter.

Basic Humanoid needs no Rigify. The other thumbnails are labelled **Rigify**. If Rigify is inactive, choosing or dropping one offers **Enable Rigify & Add Rig** and Cancel. Accepting enables the installed Rigify and continues with that starter; cancelling adds nothing. You can also enable Rigify in Quick Rig; the dependency-aware starter popup offers Enable Rigify in its asset context menu when needed. The top-left viewport starter controls were removed in 9.2.5. A missing dependency or a preset absent from your installed version produces an explanation without adding a substitute rig.

Click a thumbnail to add a starter, or drag it into the viewport. With a mesh selected, the starter is fitted to that mesh; automatic weighting and control generation start off. Adjust them afterwards in **Adjust Last Operation**. With no mesh selected, select the new starter and pick the **Mesh** in Quick Rig, then click **Fit Dropped Rig**. Fitting scales and positions the skeleton; inspect and edit joint placement before generating controls or weighting. Dragging outside the viewport, pressing Esc, or right-clicking cancels without adding anything.

Where to find it: Rig Builder → Quick Rig; Add → Armature → Quick Rig also adds the premade rigs

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Mannequin

What it does: Builds a preview mesh around an existing rig so you can inspect its deformation without a finished character mesh. Select the armature, or a mesh parented to it, first. If you have no rig yet, add one

with Quick Rig. The mannequin is a preview only: replace it with your own mesh before final use.

Key controls: - **Quick Mannequin** — builds a preview with fixed defaults, replacing the previous mannequin for this rig - **Add Mannequin** — builds a preview using the scene settings; it replaces an existing mannequin by default. Replace Existing is in Adjust Last Operation after the click - **Regenerate** — rebuilds the existing preview using its saved settings - **Remove** — deletes the mannequin mesh and keeps the armature - Expand Options below the buttons for **Mode**, **Body Type**, density and **Radius Scale**; Advanced inside Options holds the region scales and output settings. To use changed scene settings, click **Add Mannequin** again

Where to find it: Rig Builder → Mannequin

Illustrated mannequin workshop

Follow these three parts in order: A — Simple Human to mannequin, B — rigless mannequin through Auto-Rig, and C — test weights and use IK/FK. Every numbered step has a real interface screenshot; red numbers match the caption. Screenshots were captured with the **BoneCraft 9.1.2 release** in `Bforartists 5.2.0`. Blender's panel arrangement and shortcuts can differ; the named commands are the reference.

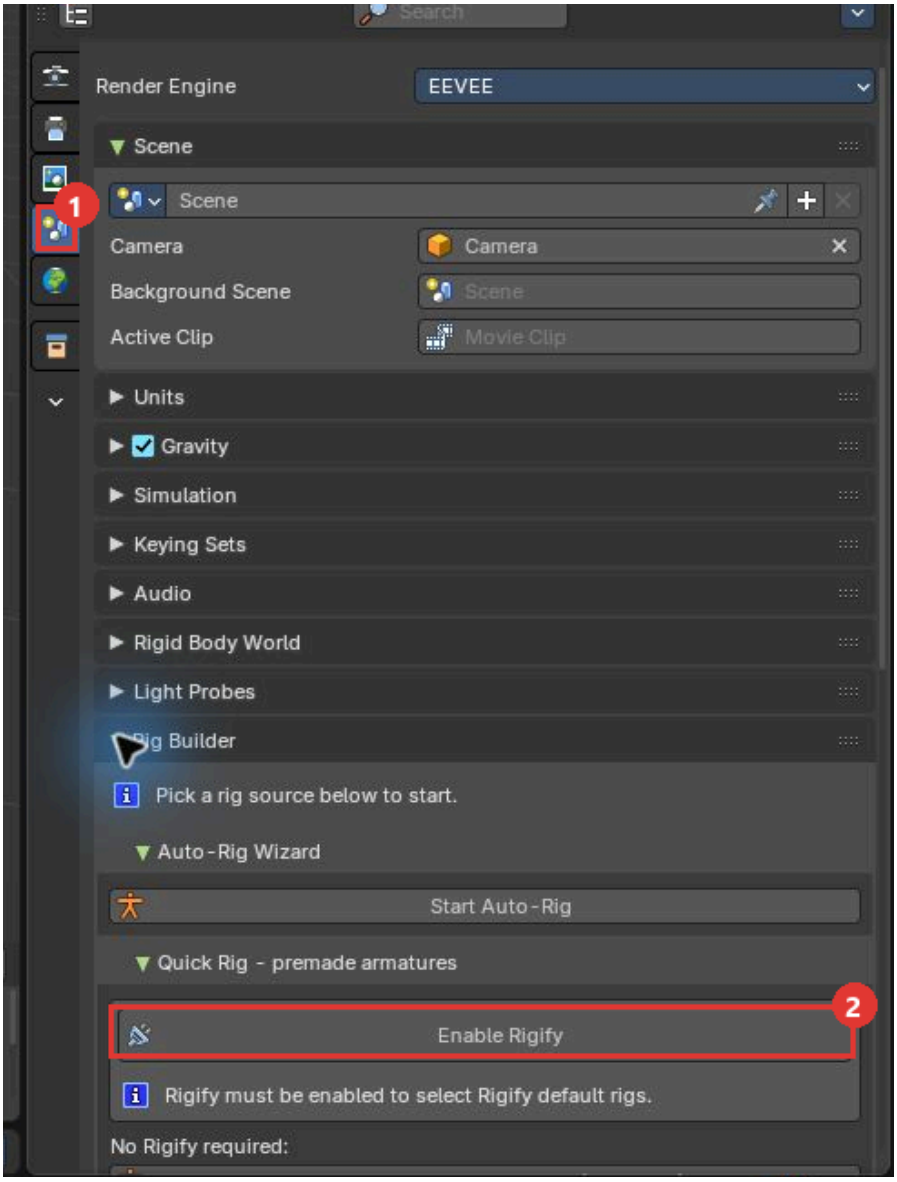
This exercise produces a simple capsule preview for learning rigging. Its head, hands and torso are placeholders, and it is not a finished character model. Begin in a new file; keep the two saved checkpoints so you can repeat the conversion without losing the original metarig.

The repository includes both checkpoints in `examples/mannequin/`: `SimpleHuman_Mannequin.blend` is the neutral Part A result, and `Mannequin_AutoRig.blend` is the completed, posed Part C example with corrected weights.

Part A: Simple Human to mannequin

A1. Enable Rigify

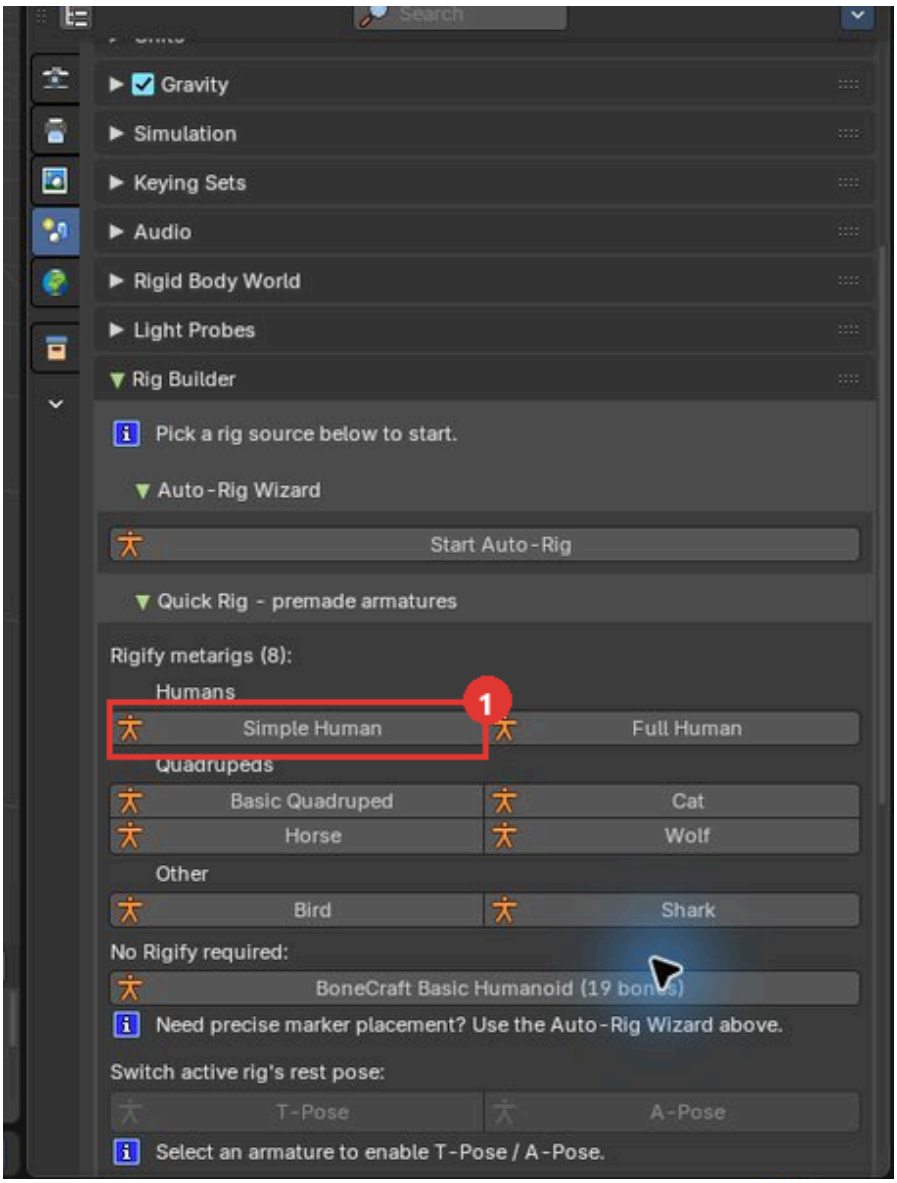
Start in a new scene. Open **Scene Properties** → **Rig Builder** → **Quick Rig**. If the Rigify presets are unavailable, click **Enable Rigify**. This tutorial uses the Rigify `Simple Human` metarig; **BoneCraft Basic Humanoid (19 bones)** is a different preset.



1 — Scene Properties. 2 — Enable Rigify.

A2. Add Simple Human

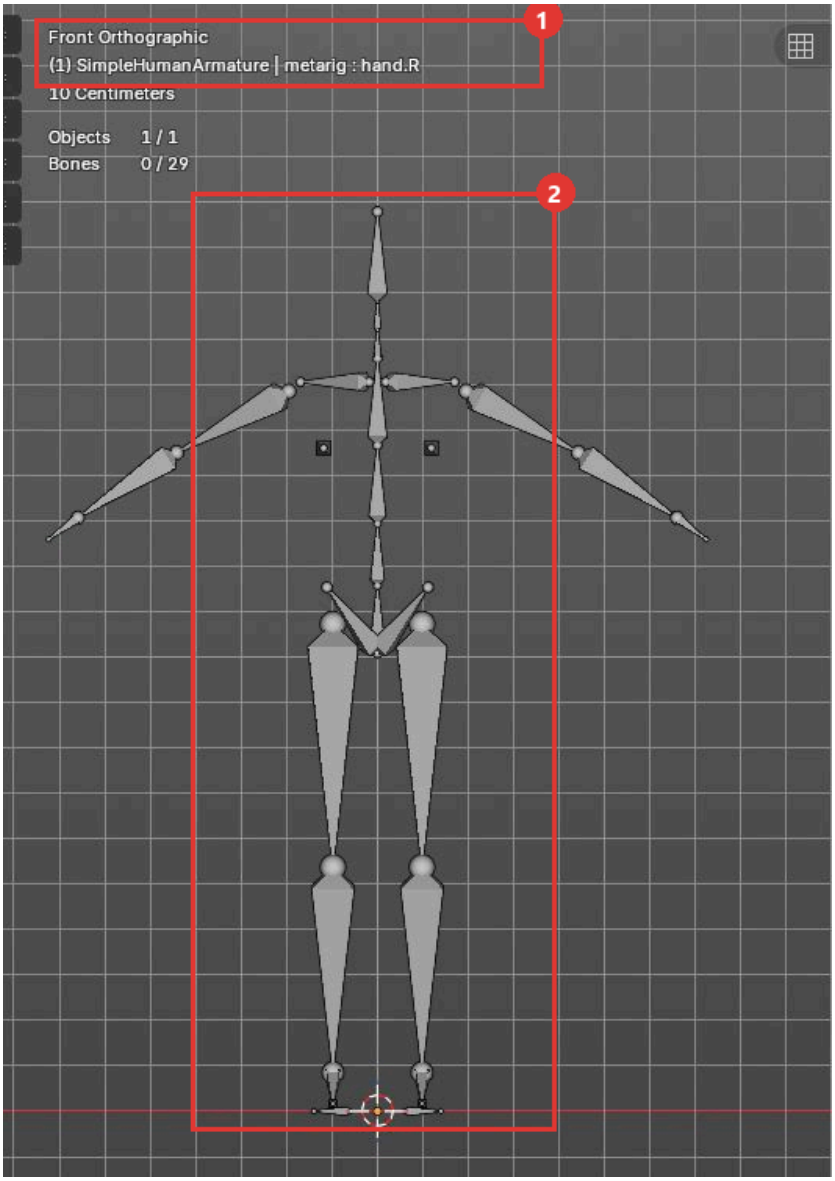
Click `Simple Human` under Quick Rig. The result is an editable metarig, not yet a mesh. It has 29 bones in this release.



1 — Simple Human preset.

A3. Inspect the metarig

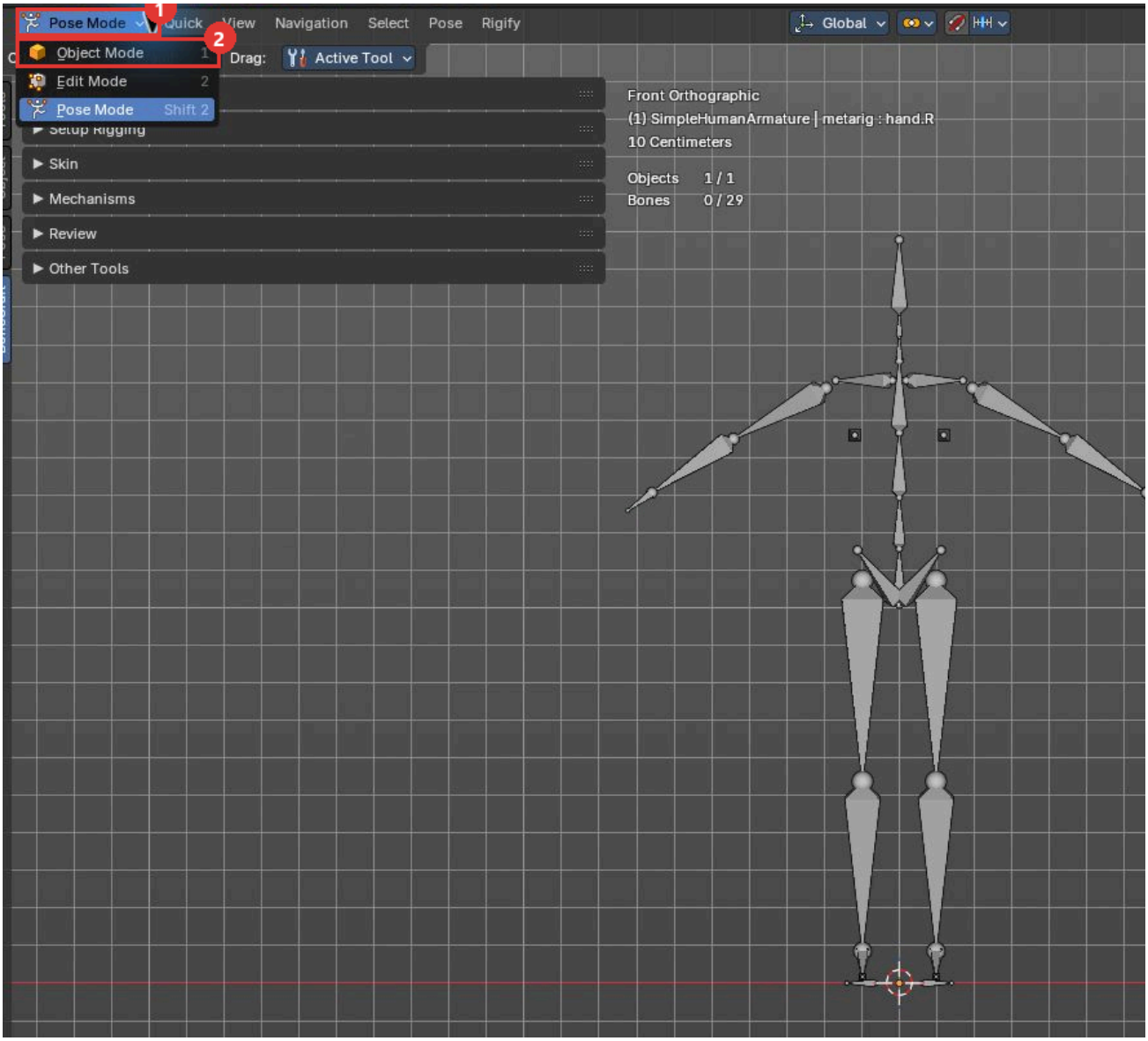
Move the pointer into the viewport and press `Numpad 1` for Front Orthographic. Check that the whole skeleton is visible. Use the mouse wheel to frame it. The new metarig may start in Pose Mode.



1 — Front view. 2 — Simple Human skeleton.

A4. Return to Object Mode

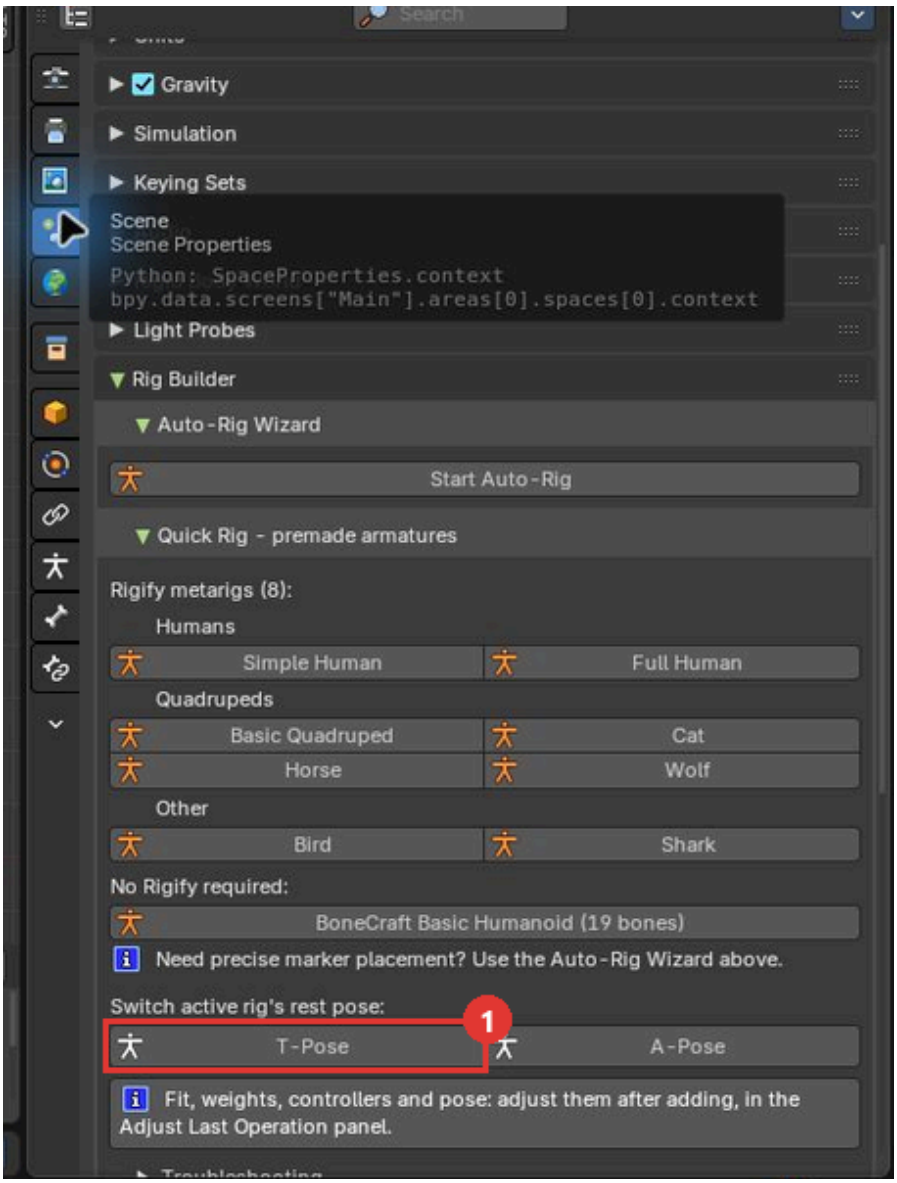
Use the viewport mode dropdown and choose **Object Mode**. Keep `SimpleHumanArmature` selected. Choose Object Mode before changing the rest pose or generating the mannequin.



1 — Mode dropdown. 2 — Object Mode.

A5. Use a T-pose

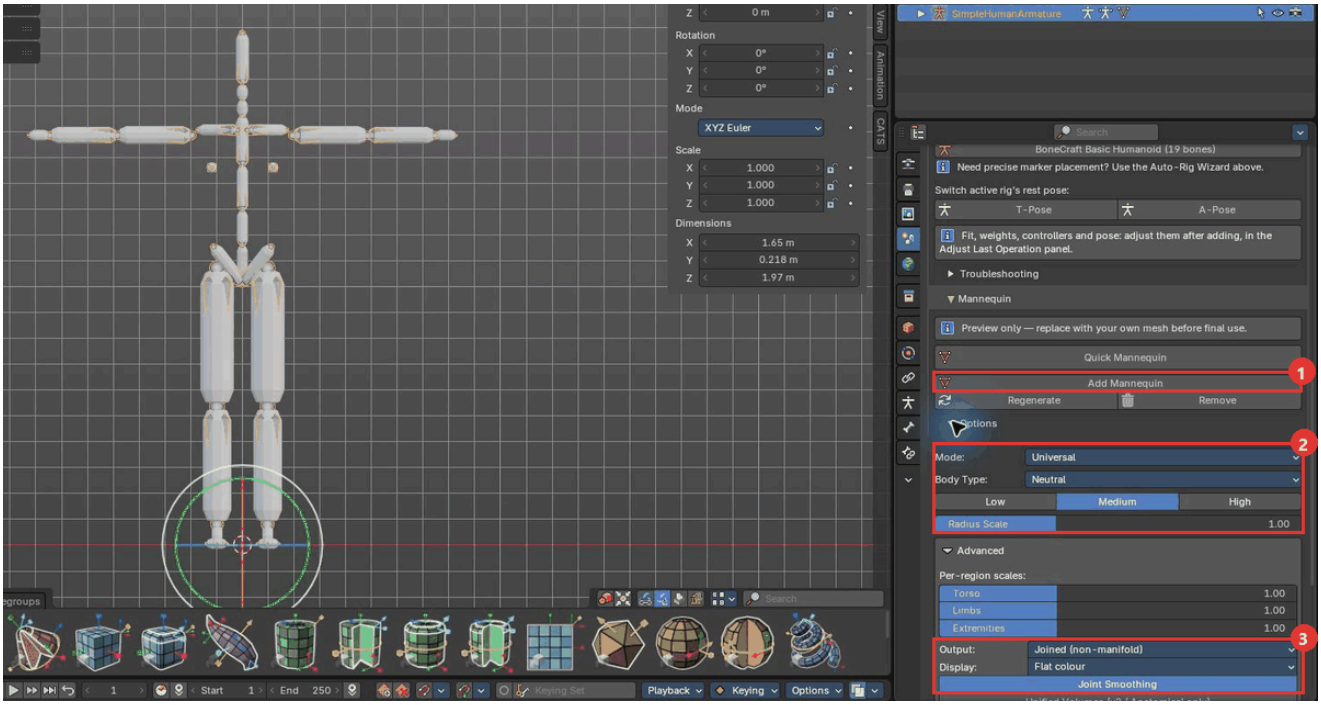
In Quick Rig, click **T-Pose**. The arms extend horizontally. For a custom body, fit the metarig in Edit Mode before generating its preview; keep left and right joints symmetric for this exercise.



1 — T-Pose command.

A6. Generate the mannequin

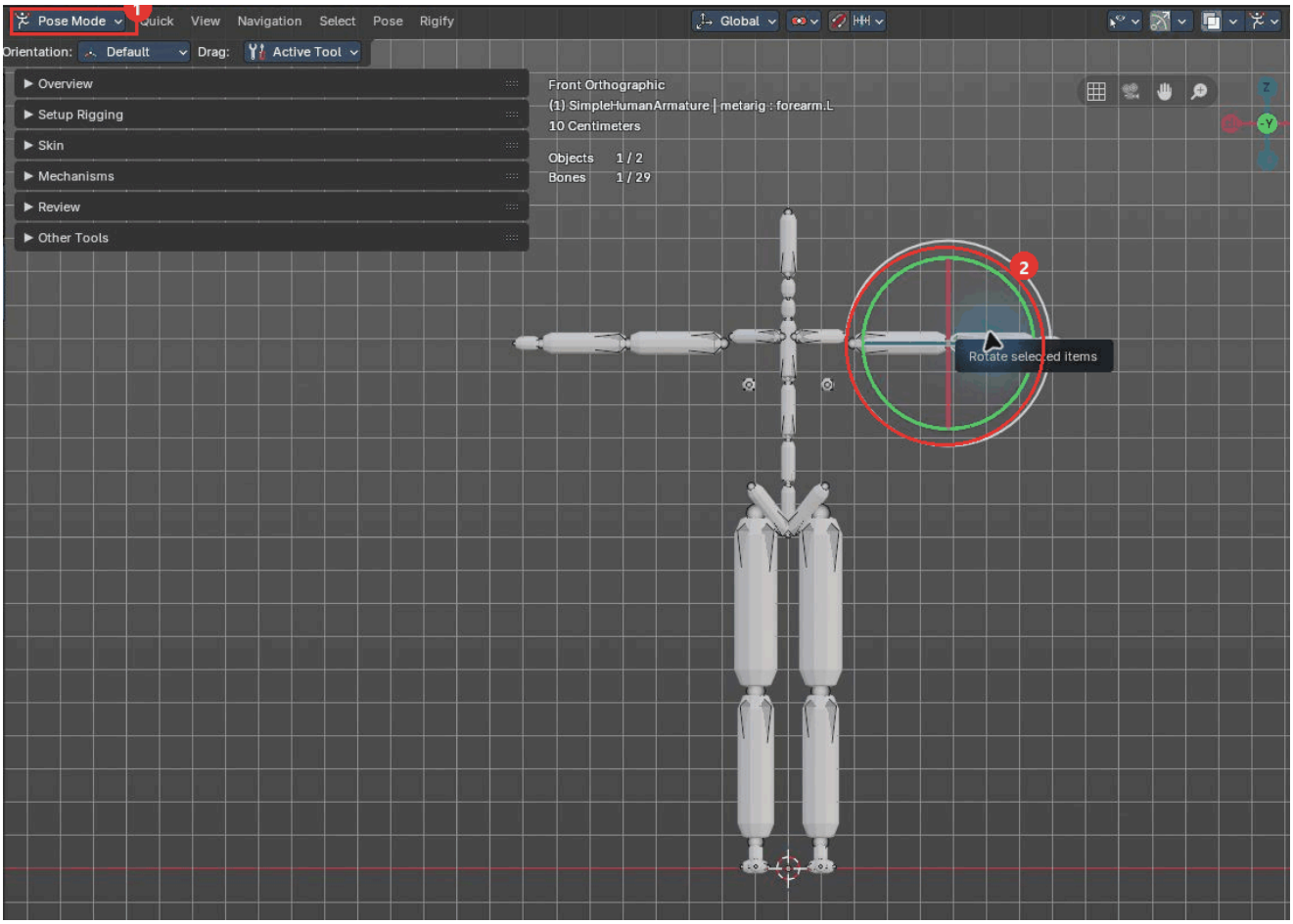
Expand **Mannequin** → **Options**. Set `Mode: Universal`, `Body Type: Neutral`, **Medium**, and `Radius Scale: 1.00`. Under Advanced, keep region scales at 1.00, choose `Joined (non-manifold)`, **Flat colour**, and enable **Joint Smoothing**. Leave Unified Volume off. Click **Add Mannequin** after setting these options. The screenshot shows the resulting capsule mesh. Use Add for current options; Regenerate reuses the saved mannequin settings.



1 — Add Mannequin. 2 — Universal and body settings. 3 — Joined output and smoothing.

A7. Try a bone pose

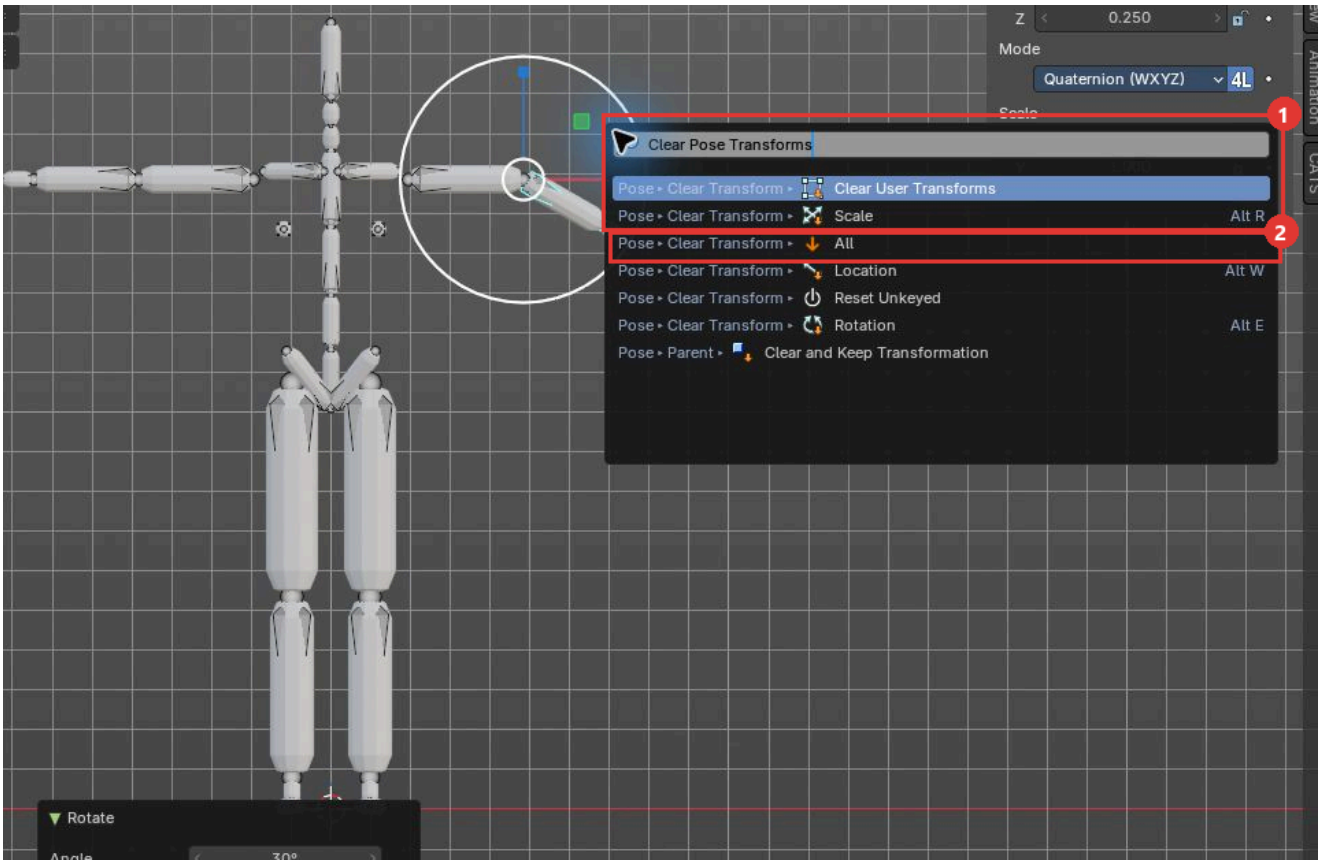
Select the armature, switch to **Pose Mode**, and click the left forearm bone (the character's left is on your right in front view). Rotate it slightly with the Rotate tool; the attached mannequin follows. This is a weighted preview on the metarig. It does not yet have the animator controls used in Part C.



1 — Pose Mode. 2 — Selected forearm and its transform gizmo.

A8. Restore and save the preview

With the pointer in the viewport, select all pose bones. Press **F3**, search `Clear Pose Transforms`, and choose **All**. Confirm the mannequin returns to its T-pose. Use **File** → **Save As** and name the file `SimpleHuman_Mannequin.blend`. Keep this checkpoint before continuing.

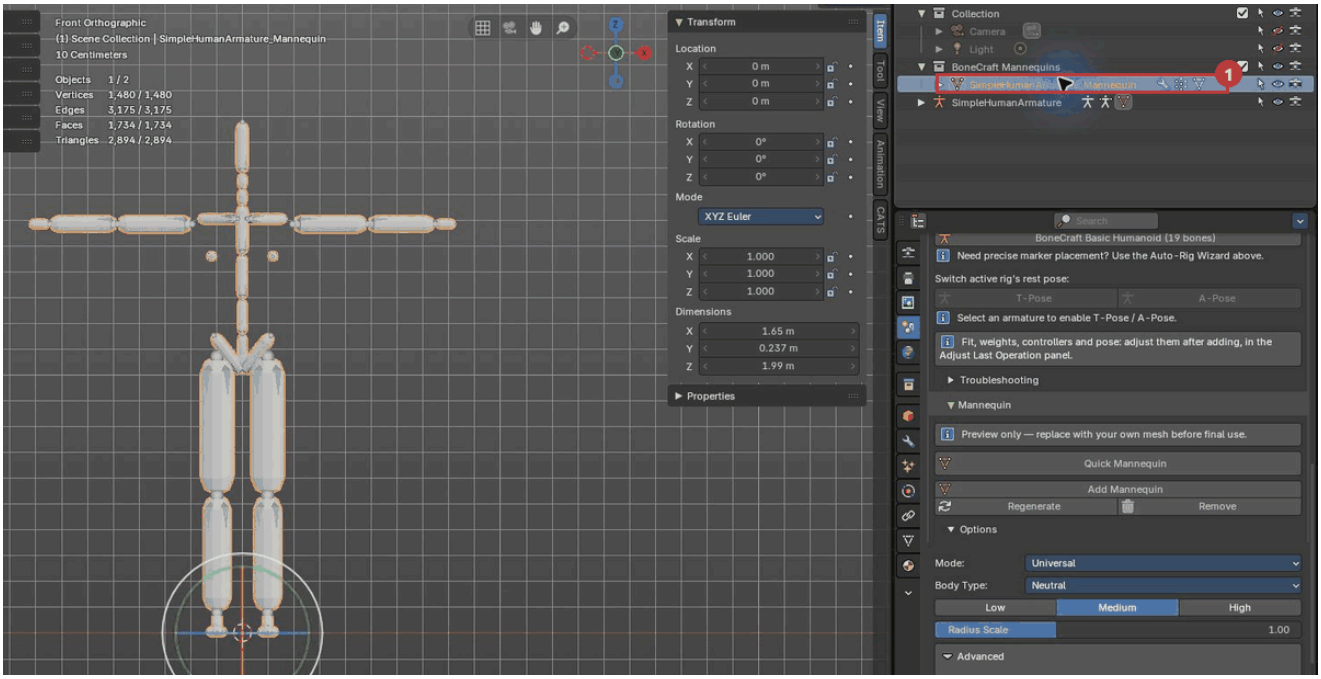


1 — Clear Pose Transforms search. 2 — All resets the selected pose bones.

Part B: Rigless mannequin through Auto-Rig

B1. Select the mannequin mesh

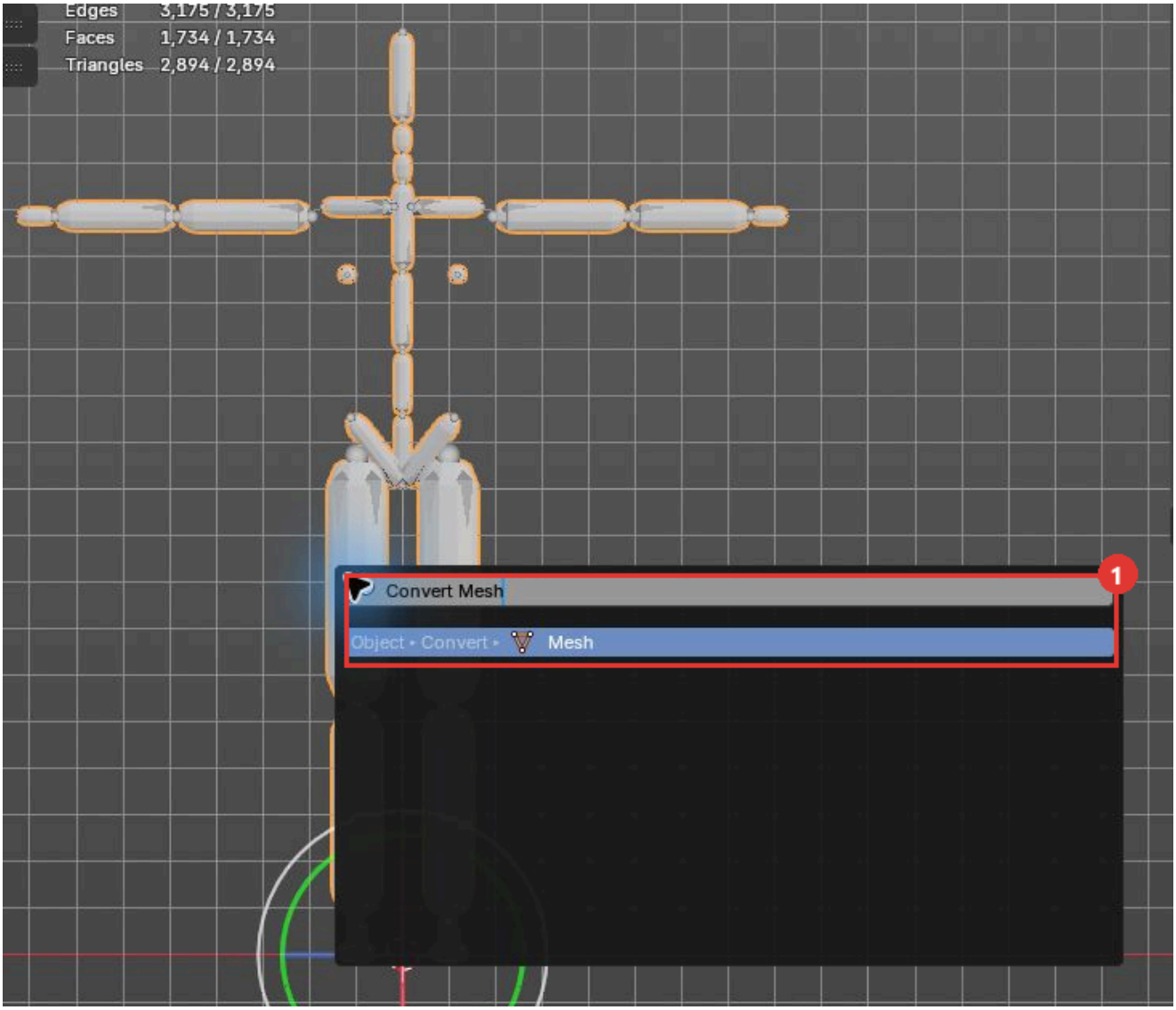
Continue from the neutral checkpoint in Part A. Switch to **Object Mode** and select `SimpleHumanArmature_Mannequin` in the Outliner. We will deliberately remove its rig before running the wizard. Save the result under a different filename to preserve Part A.



1 — Select the mesh, not the armature.

B2. Bake the preview into a mesh

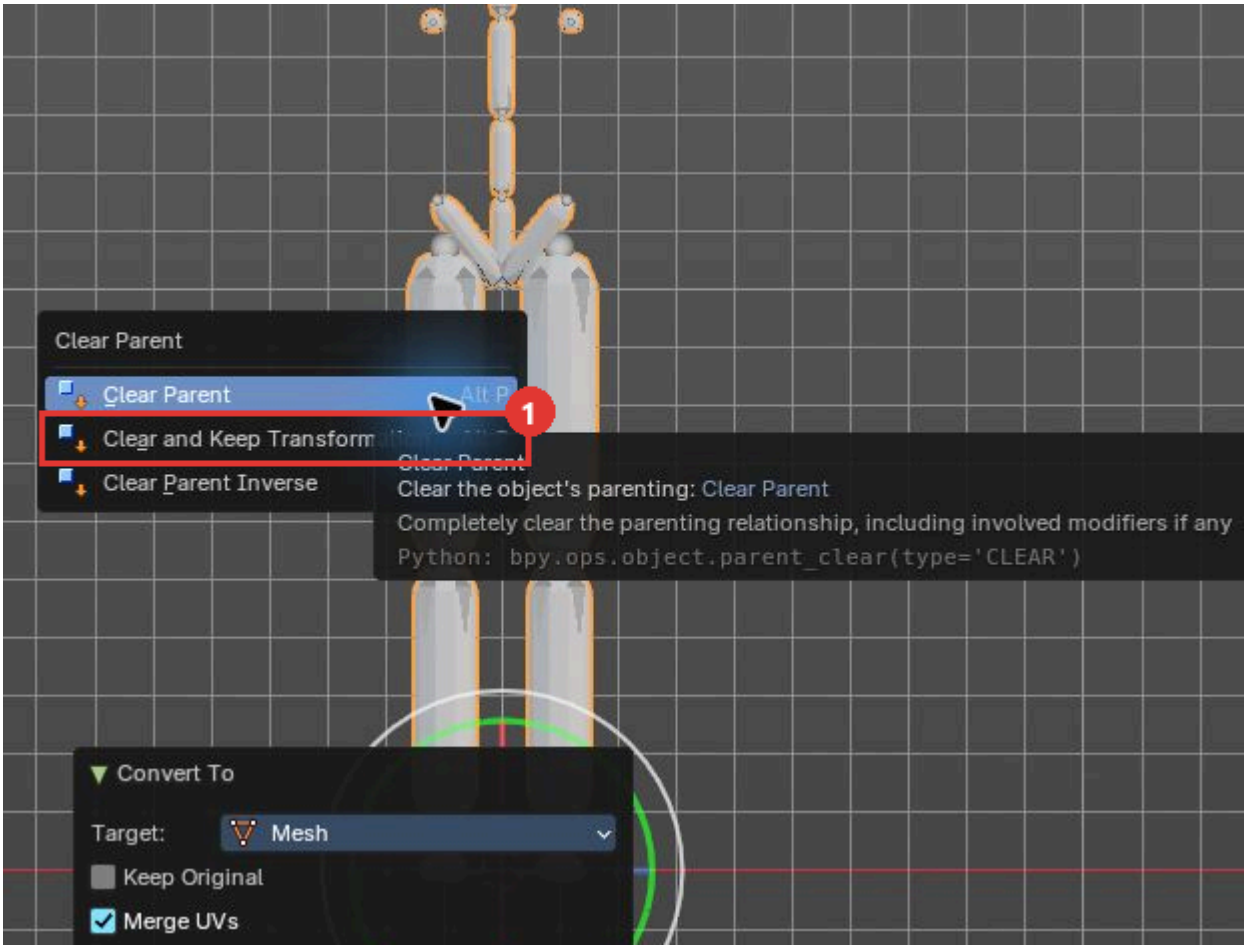
With only the mannequin selected, press **F3**, search `Convert Mesh`, and execute it. This keeps the visible neutral geometry and removes the Armature modifier. Do this with the mannequin in its restored T-pose.



1 — Convert Mesh command.

B3. Clear the parent

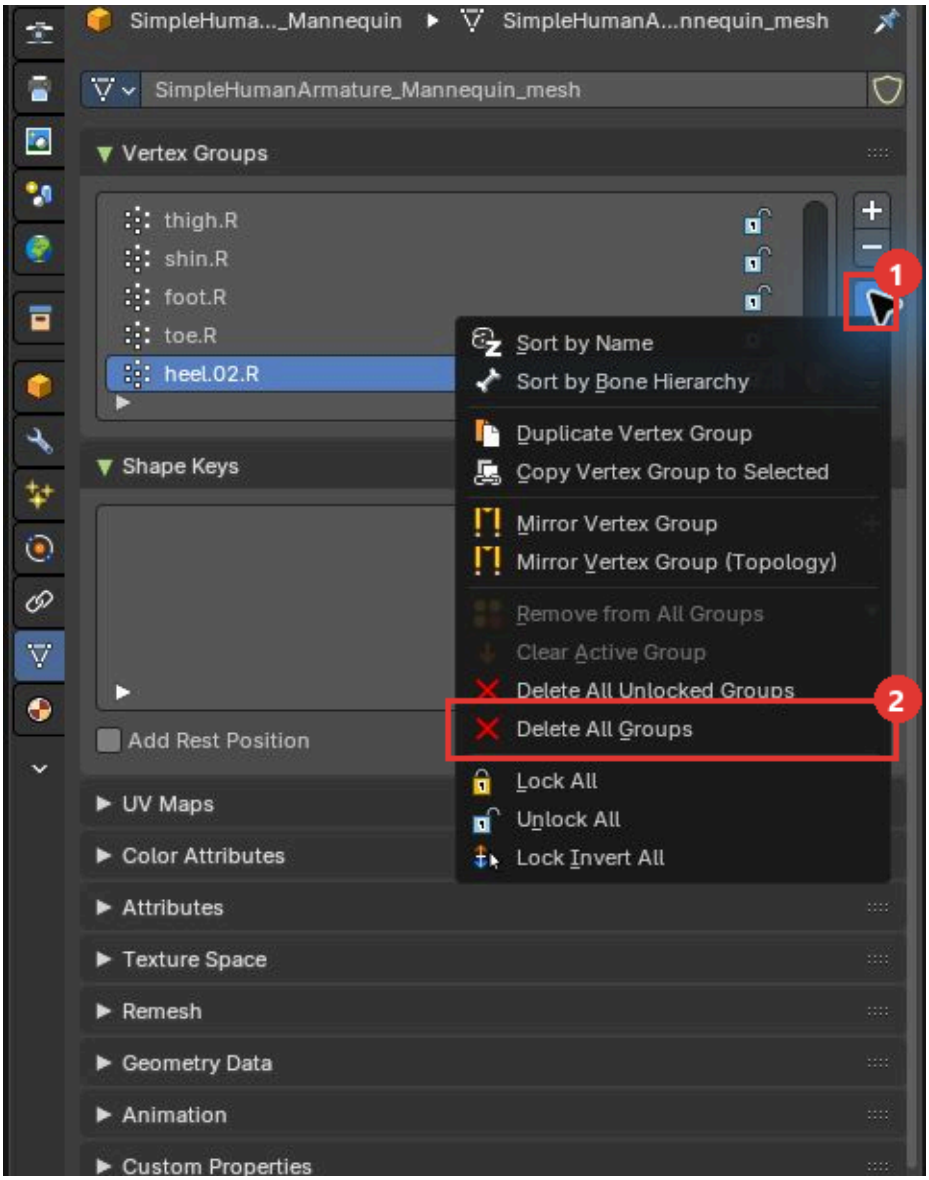
Press **Alt+P** and choose `Clear and Keep Transformation`. This detaches the mesh from the old metarig without moving it.



1 — Clear and Keep Transformation.

B4. Remove the old weights

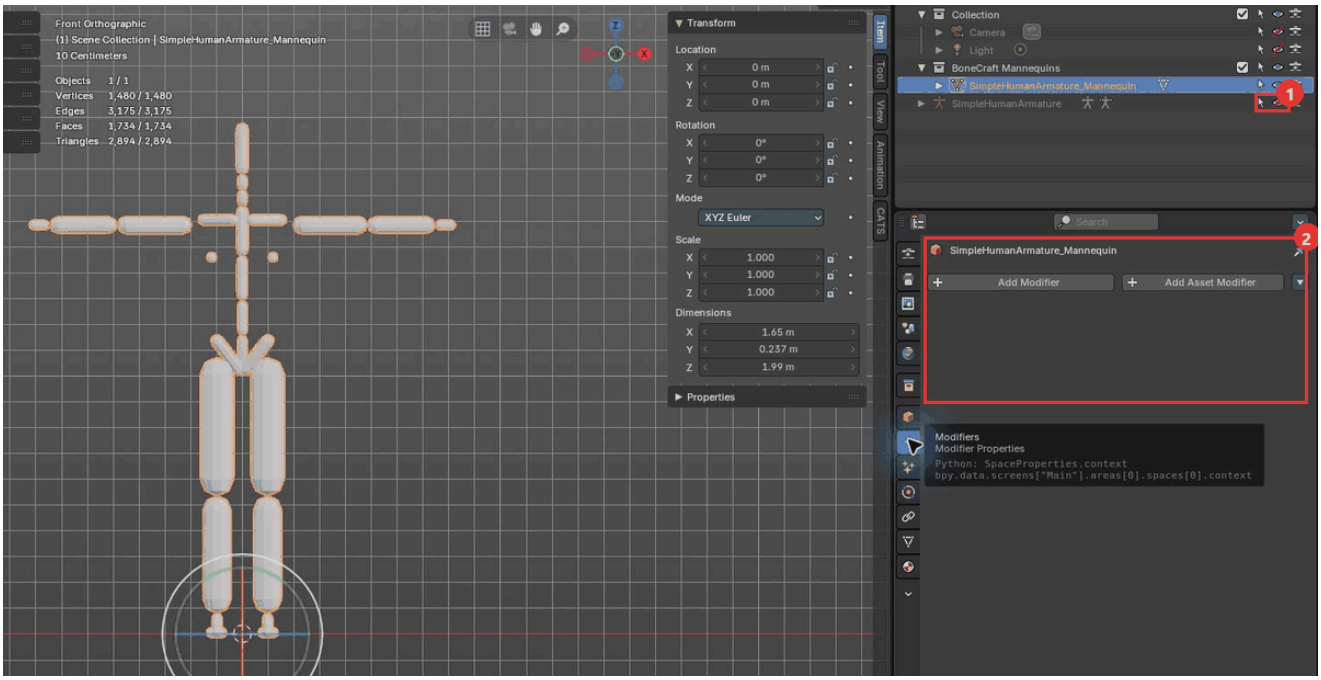
Open **Object Data Properties** → **Vertex Groups**. In the list's dropdown menu choose `Delete All Groups`. These groups belong to the old metarig; the wizard will create new ones. This action is for this disposable training mesh.



1 — Vertex Groups menu. 2 — Delete All Groups.

B5. Verify the rigless checkpoint

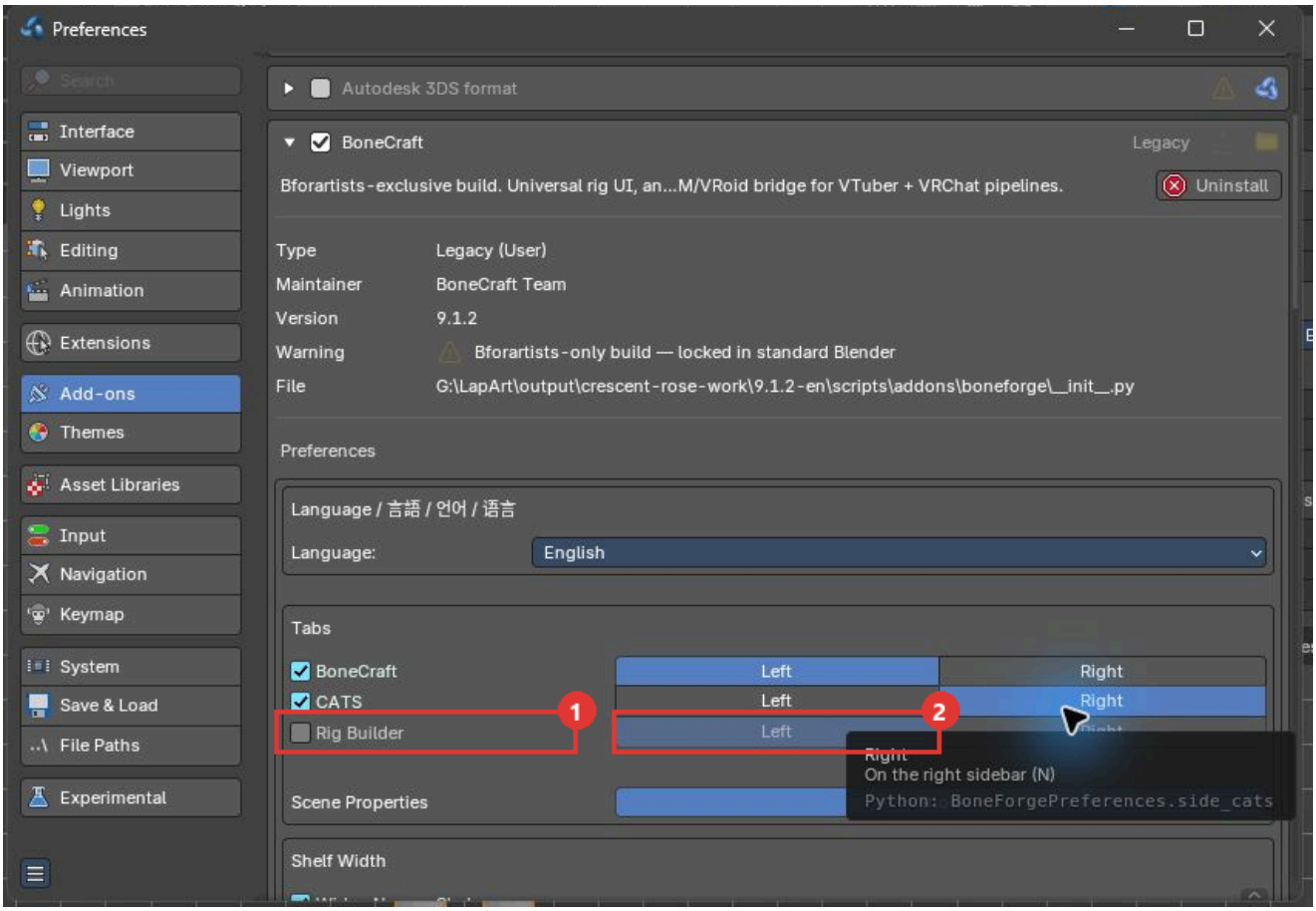
Hide the old `SimpleHumanArmature` using its Outliner eye icon. Check the mannequin remains visible, its Modifiers panel has no Armature modifier, and its Vertex Groups list is empty. This is the rigless mannequin input. BoneCraft 9.1.2 generates mannequins from an armature; detaching this preview is the route to a rigless training mesh.



1 — Hide the old metarig. 2 — Empty modifier stack.

B6. Enable the viewport Rig Builder tab

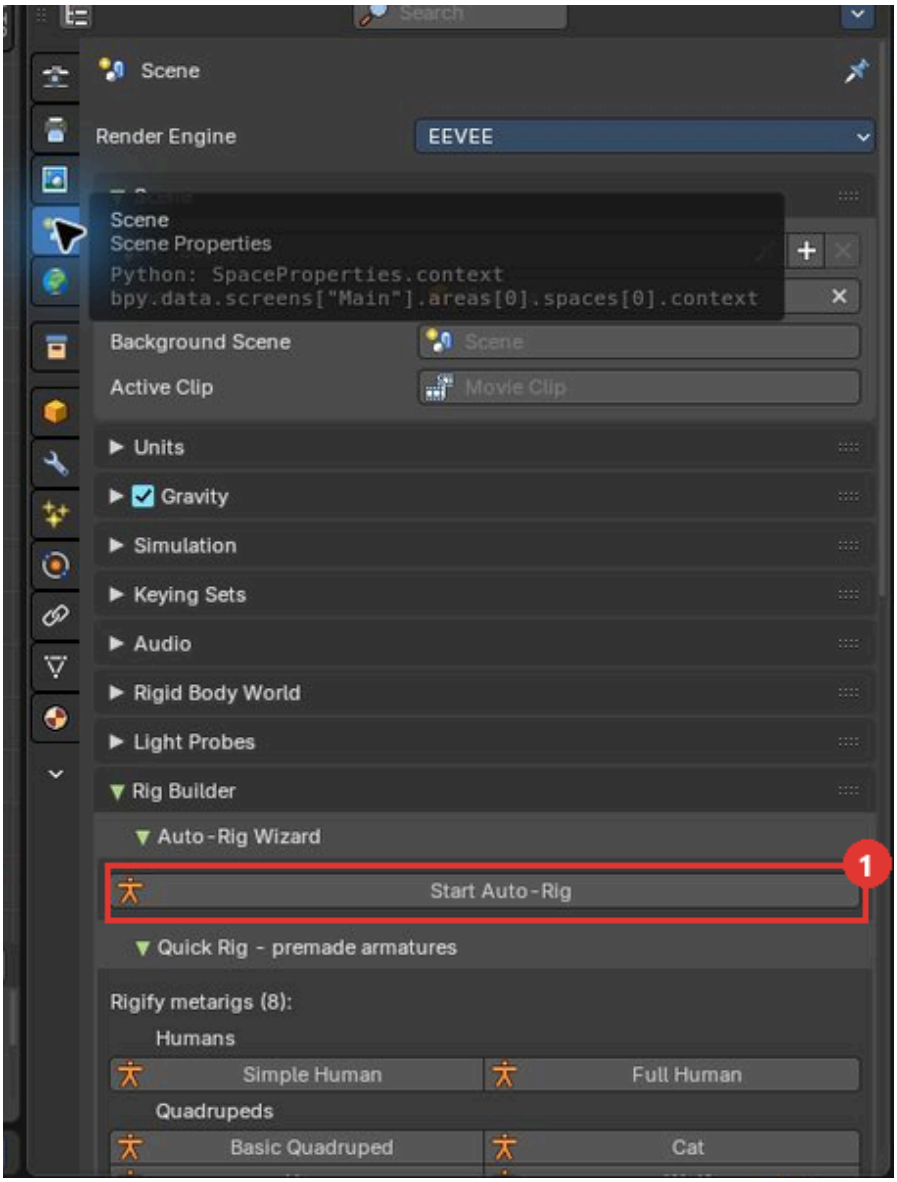
Open **Edit** → **Preferences** → **Add-ons** → **BoneCraft** → **Tabs**. Enable **Rig Builder** and choose **Left**. Close Preferences, place the pointer over the viewport, and press **T** if the left Tool Shelf is hidden. Use this viewport tab for marker placement: in the tested 9.1.2 host, starting Place Marker from Scene Properties did not accept viewport clicks.



1 — Enable Rig Builder. 2 — Left Tool Shelf placement.

B7. Start Auto-Rig

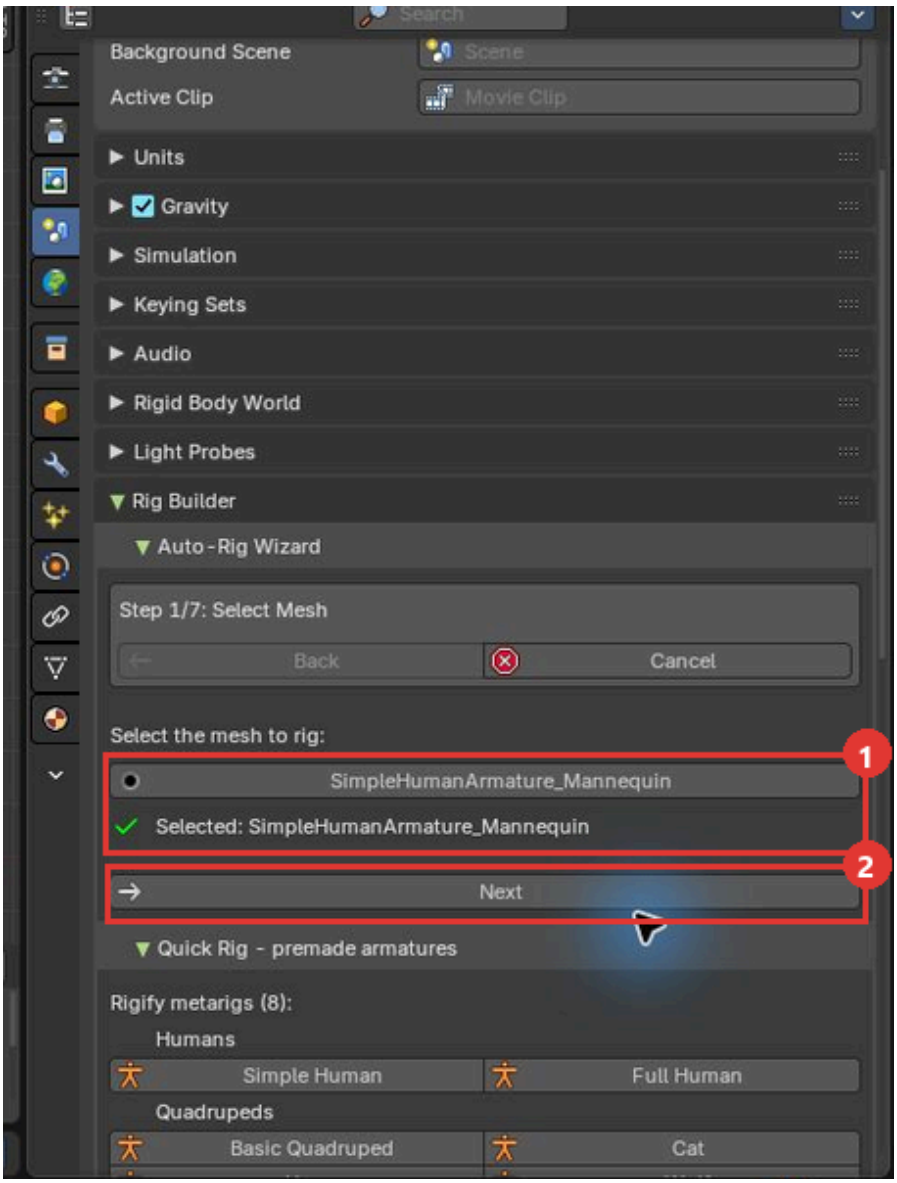
With the rigless mannequin selected, click **Start Auto-Rig**. The screenshot shows the equivalent Scene Properties button; the same command is at the top of the viewport Rig Builder tab enabled in the previous step.



1 — Start Auto-Rig.

B8. Choose the input mesh

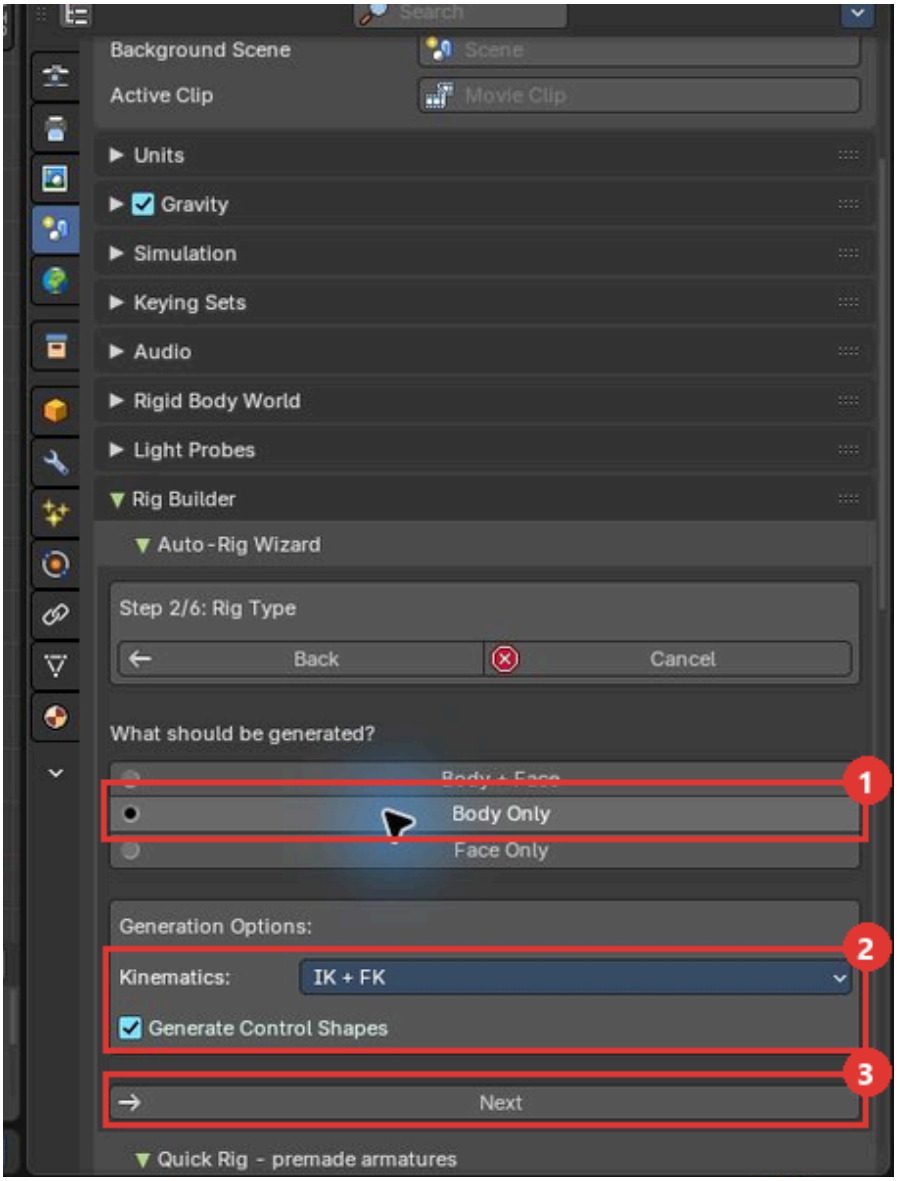
In `Select Mesh`, choose the mannequin and verify the selected-mesh check, then click **Next**. Do not choose a widget or another preview object.



1 — Mannequin input. 2 — Next.

B9. Choose a body rig with both methods

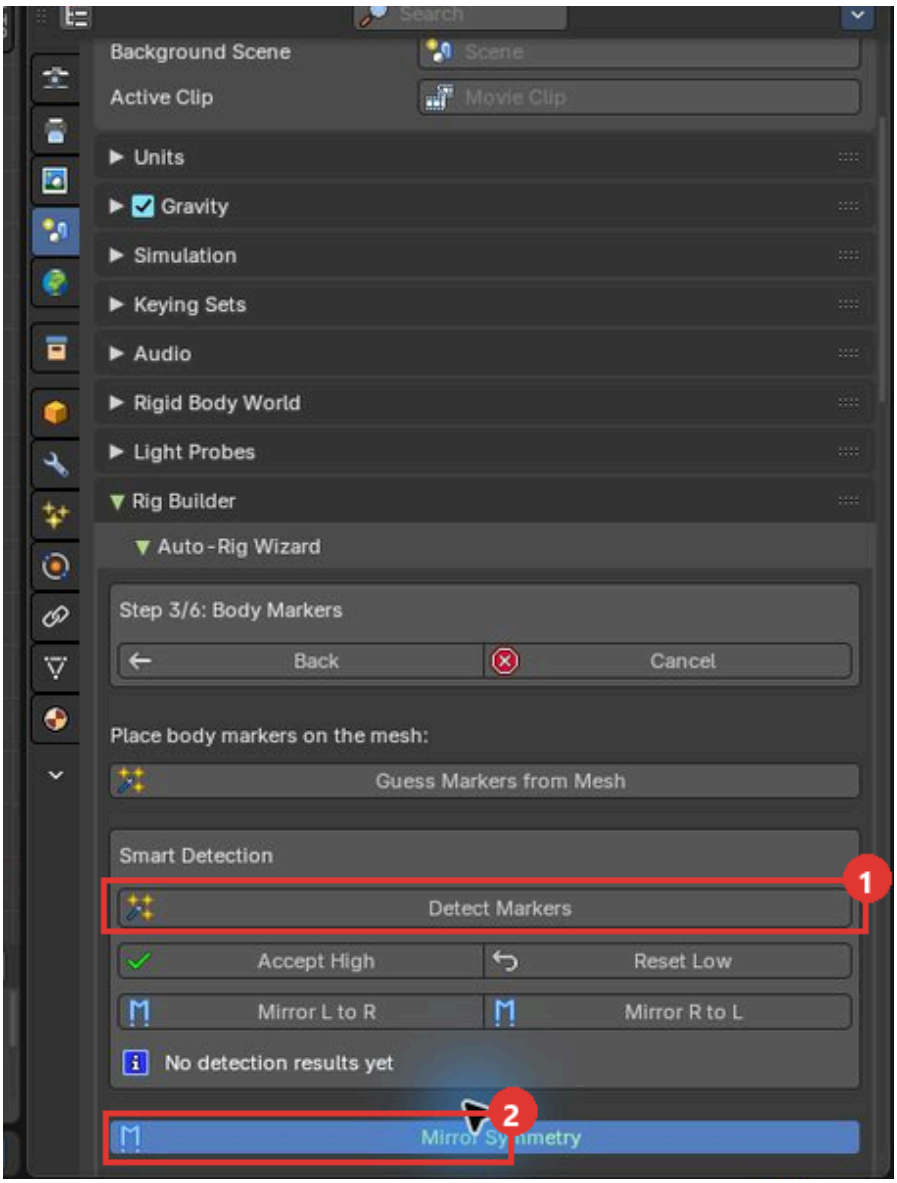
Select **Body Only** and `Kinematics: IK + FK`. Keep **Generate Control Shapes** enabled. Click **Next**. This capsule mannequin has no facial topology, so Body + Face is unnecessary.



1 — Body Only. 2 — IK + FK. 3 — Next.

B10. Detect initial body markers

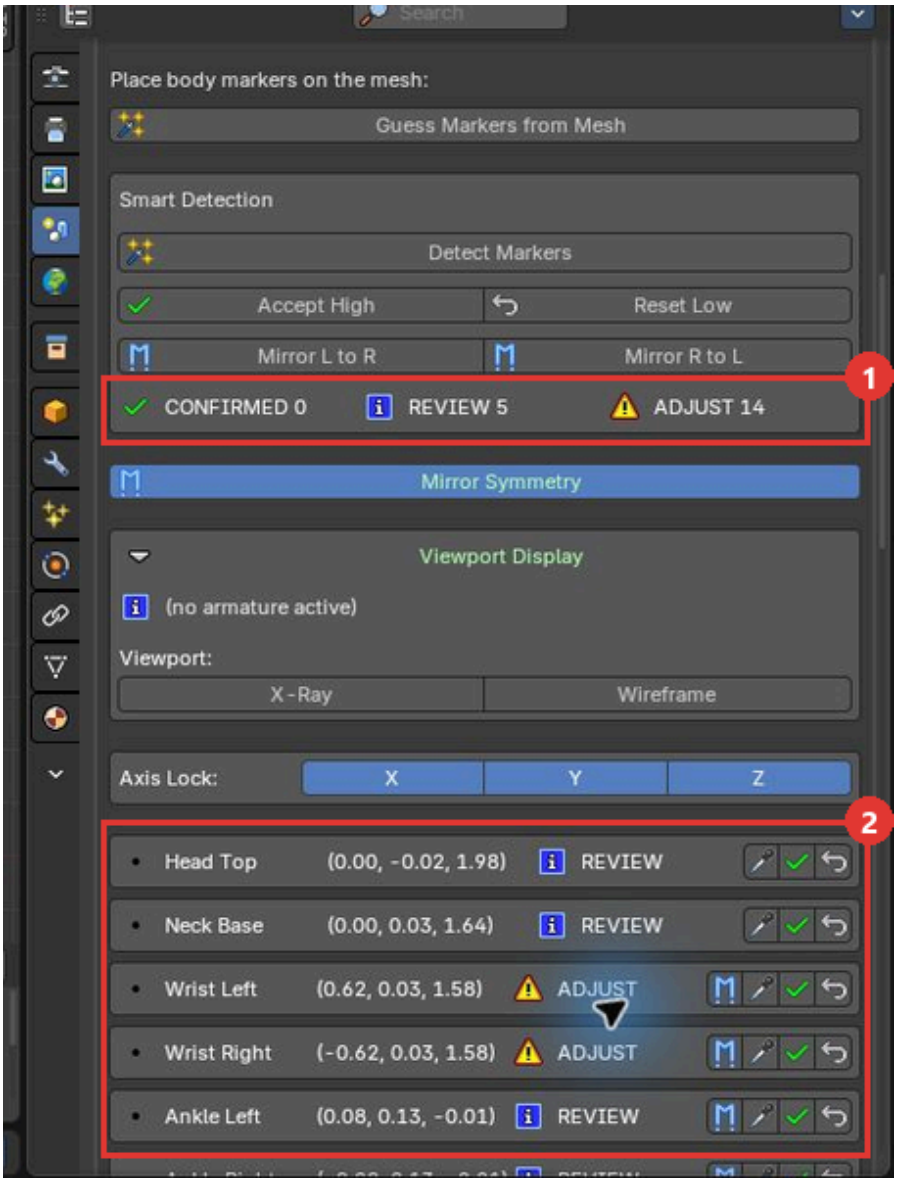
In Body Markers, click **Detect Markers** under Smart Detection. Treat its guesses as a starting point. Keep **Mirror Symmetry** enabled for this symmetric mannequin. Do not accept every guess without checking it.



1 — Detect Markers. 2 — Mirror Symmetry.

B11. Review the detected positions

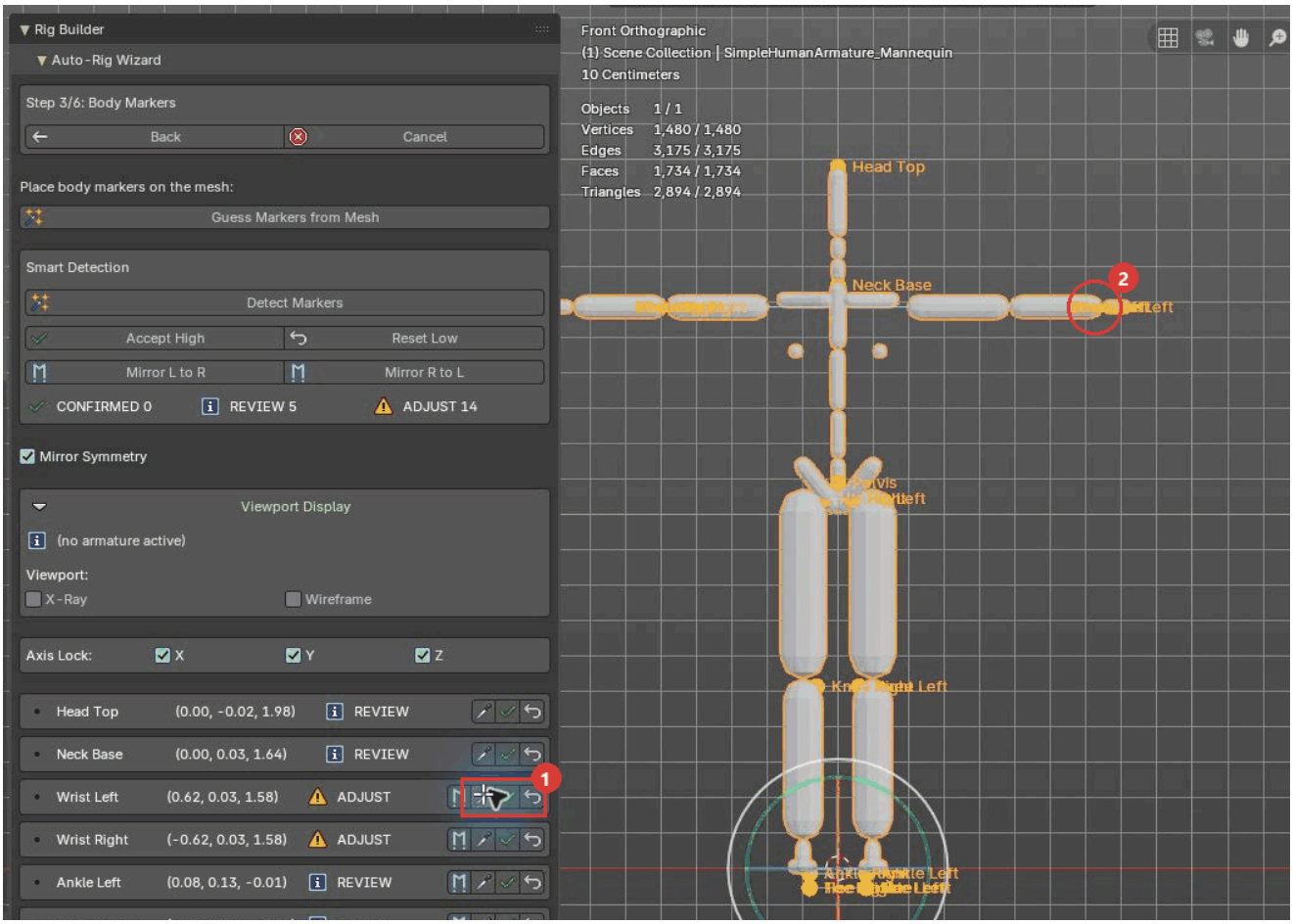
Check head top, neck base, pelvis, wrists and ankles against the visible mesh. This segmented preview produces several low-confidence guesses. Review and correct both the main landmarks and the optional precision landmarks before generating. The confidence labels describe the original detection; a green check at the start of a row indicates that the current placement is confirmed.



1 — Detection confidence summary. 2 — Body marker rows.

B12. Place a wrist on the mesh

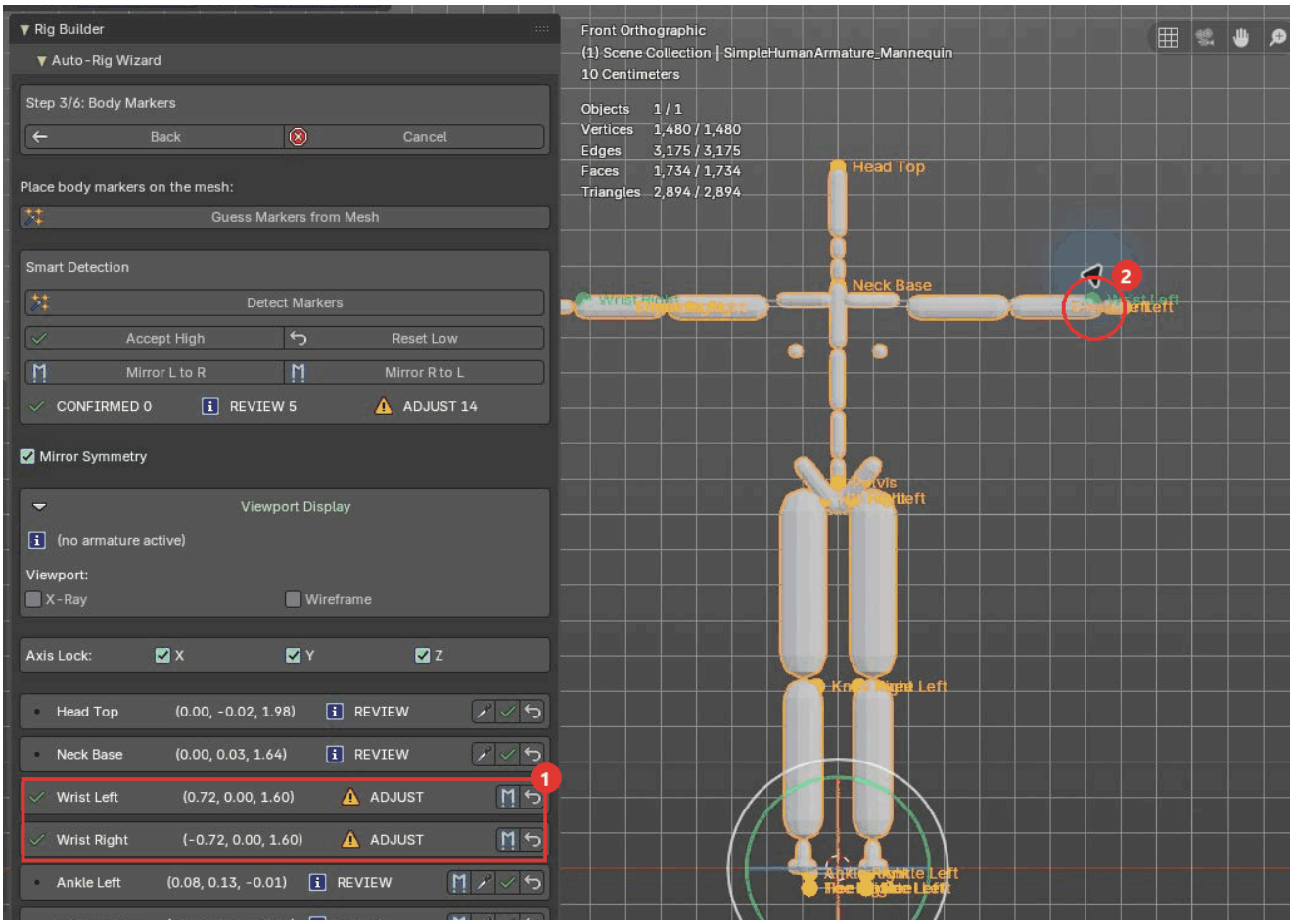
In the viewport **Rig Builder** tab, click the **Place Marker** eyedropper on Wrist Left, then click the wrist surface in the viewport. For an already confirmed marker, reset that row first to expose its placement button. In front view, the character's left wrist is on the screen's right.



1 — Wrist Left placement control. 2 — Click the wrist surface.

B13. Check the mirrored wrist and ankles

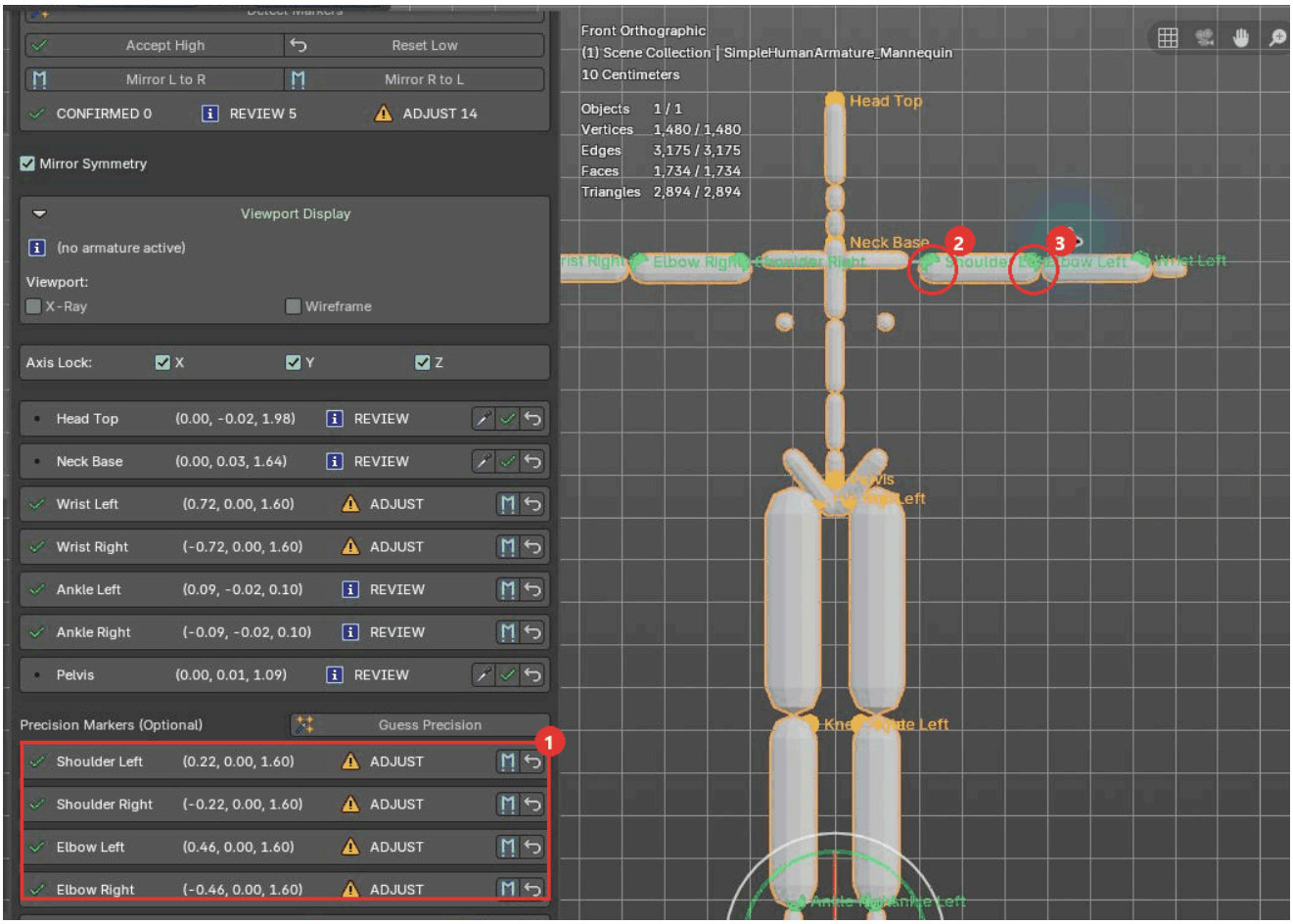
Verify both wrist rows now show green checks and opposite X coordinates. Repeat placement for `Ankle Left`; symmetry supplies Ankle Right. Place each ankle at its joint, not at the toe. Check symmetry against the mesh rather than copying the example coordinates onto a differently sized character.



1 — Confirmed wrist pair. 2 — Mirrored wrist on the mesh.

B14. Refine shoulders and elbows

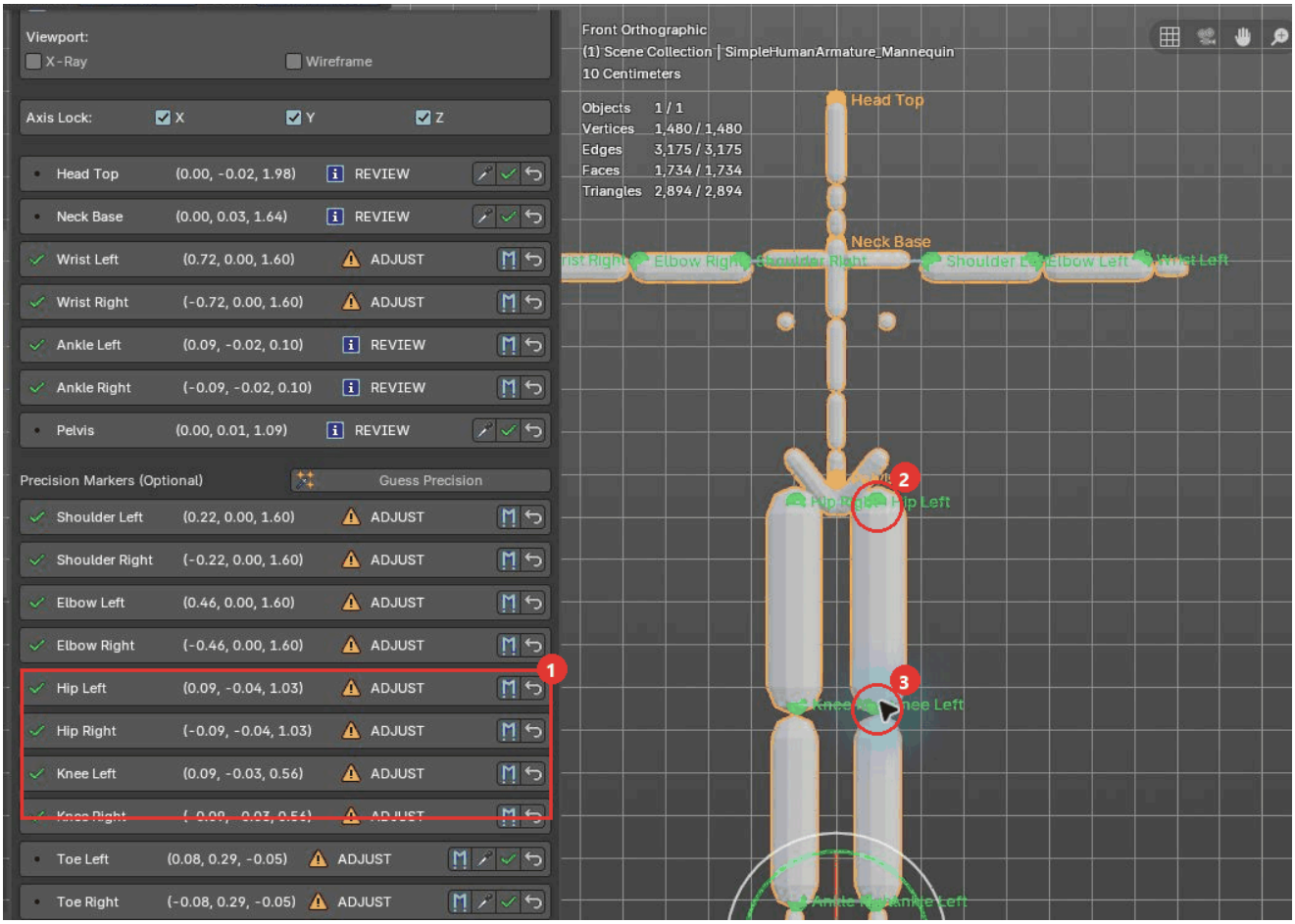
Scroll to Precision Markers. Reset and place `Shoulder Left` at the upper-arm pivot and `Elbow Left` at the elbow gap. Confirm their mirrored partners. Keep the wrist at the hand end of the forearm. These points determine the arm chain, so a shoulder or elbow guessed on the wrong capsule needs correction.



1 — Shoulder and elbow marker rows. 2 — Shoulder. 3 — Elbow.

B15. Refine hips and knees

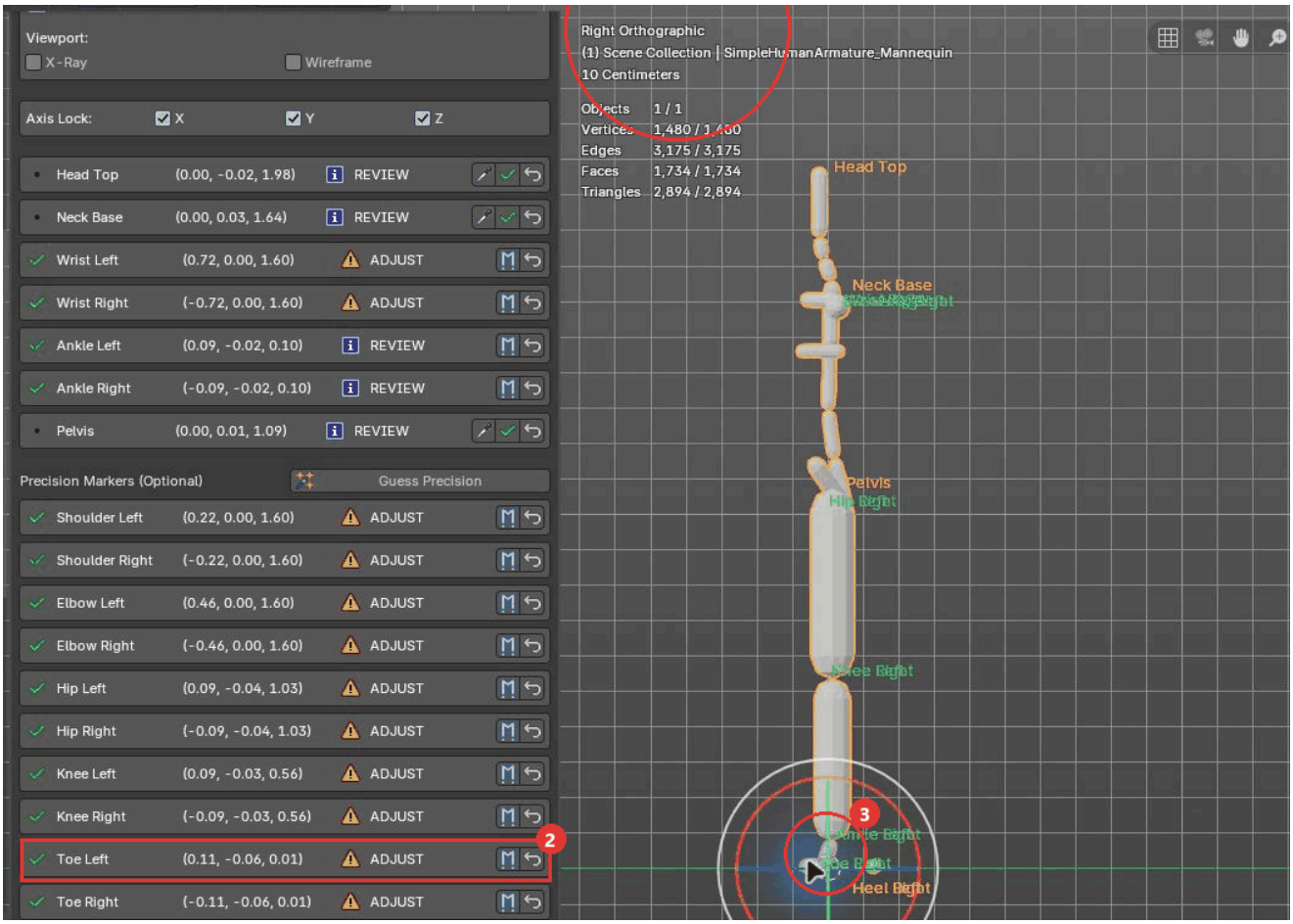
Place `Hip Left` at the top of the thigh and `Knee Left` at the thigh/shin gap. Check the opposite side supplied by symmetry. Keep the pelvis on the centre line, between the hips.



1 — Hip and knee rows. 2 — Hip surface. 3 — Knee gap.

B16. Check depth and toes from the side

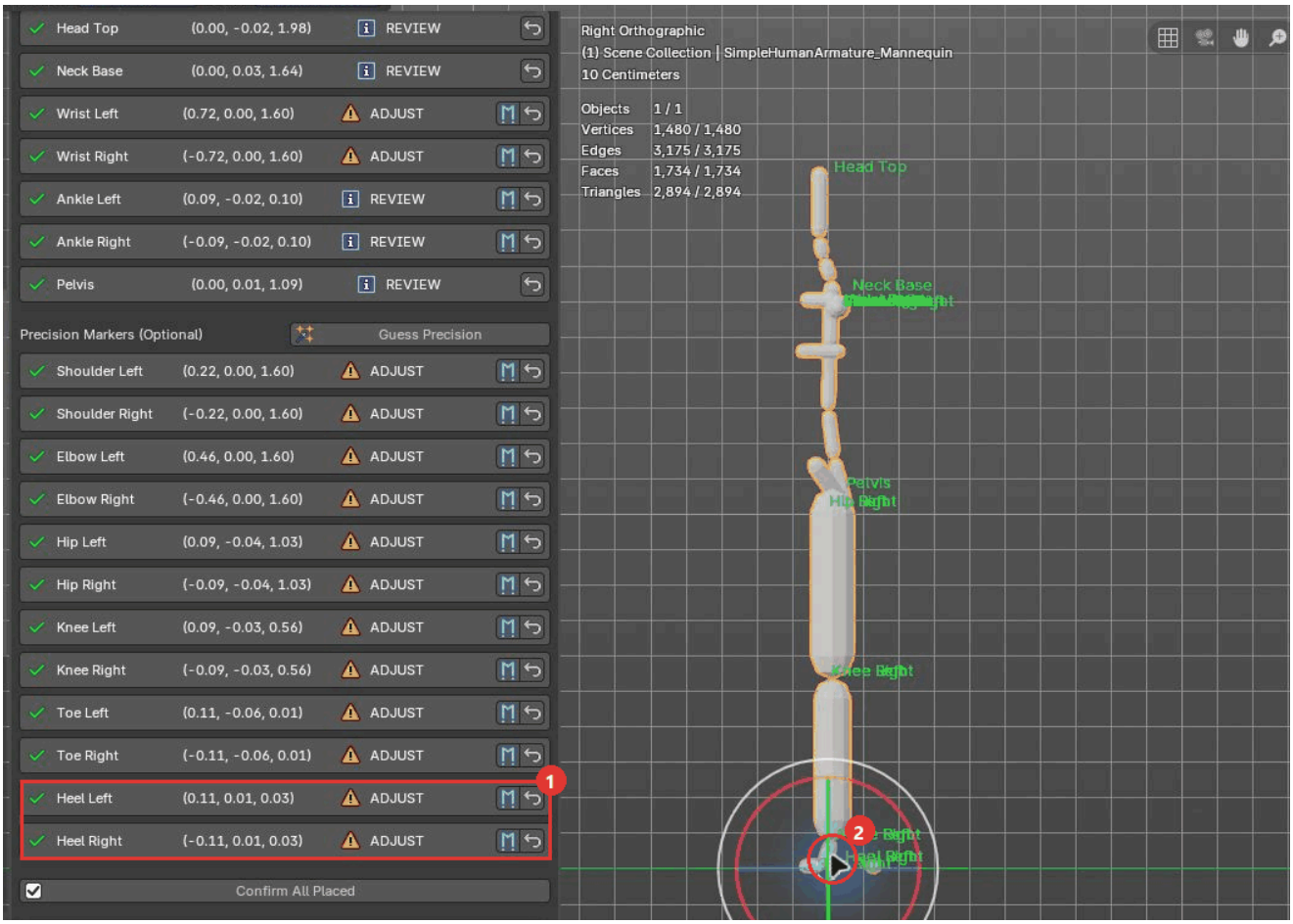
Press `Numpad 3` with the pointer over the viewport. Check the landmarks in Right Orthographic as well as from the front. Place `Toe Left` near the front of the foot; verify its mirrored partner. A marker can appear correct from the front yet be too far forward or behind the body.



1 — Right Orthographic. 2 — Toe marker row. 3 — Foot surface.

B17. Place the heels

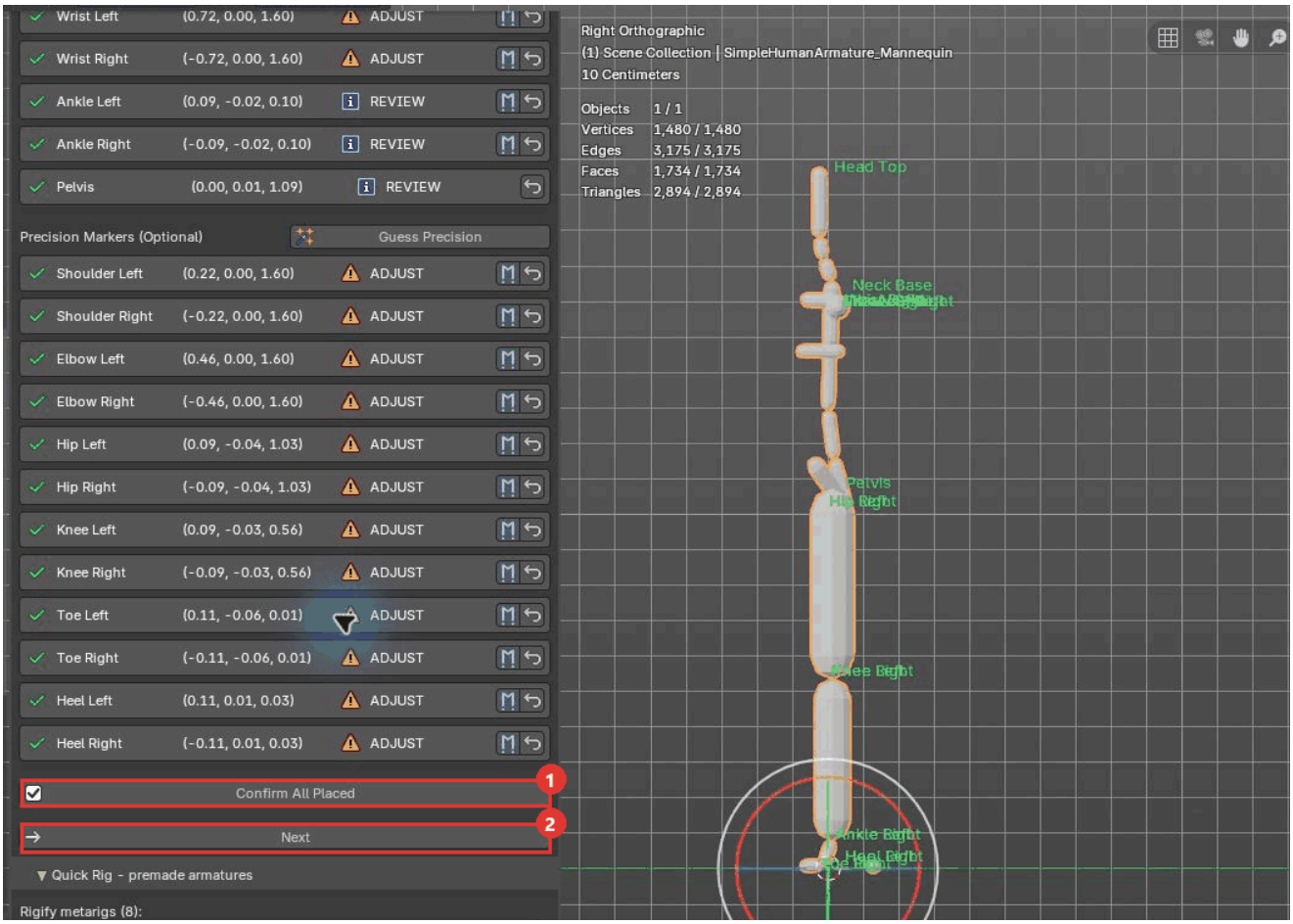
Reset a wrongly guessed heel, then place `Heel Left` near the back of the foot in side view. Verify Heel Right. Although these markers are optional, leaving both at zero produced a proximity warning in the tested release. Placing them on the feet cleared that warning.



1 — Confirmed heel pair. 2 — Heel location.

B18. Confirm and continue

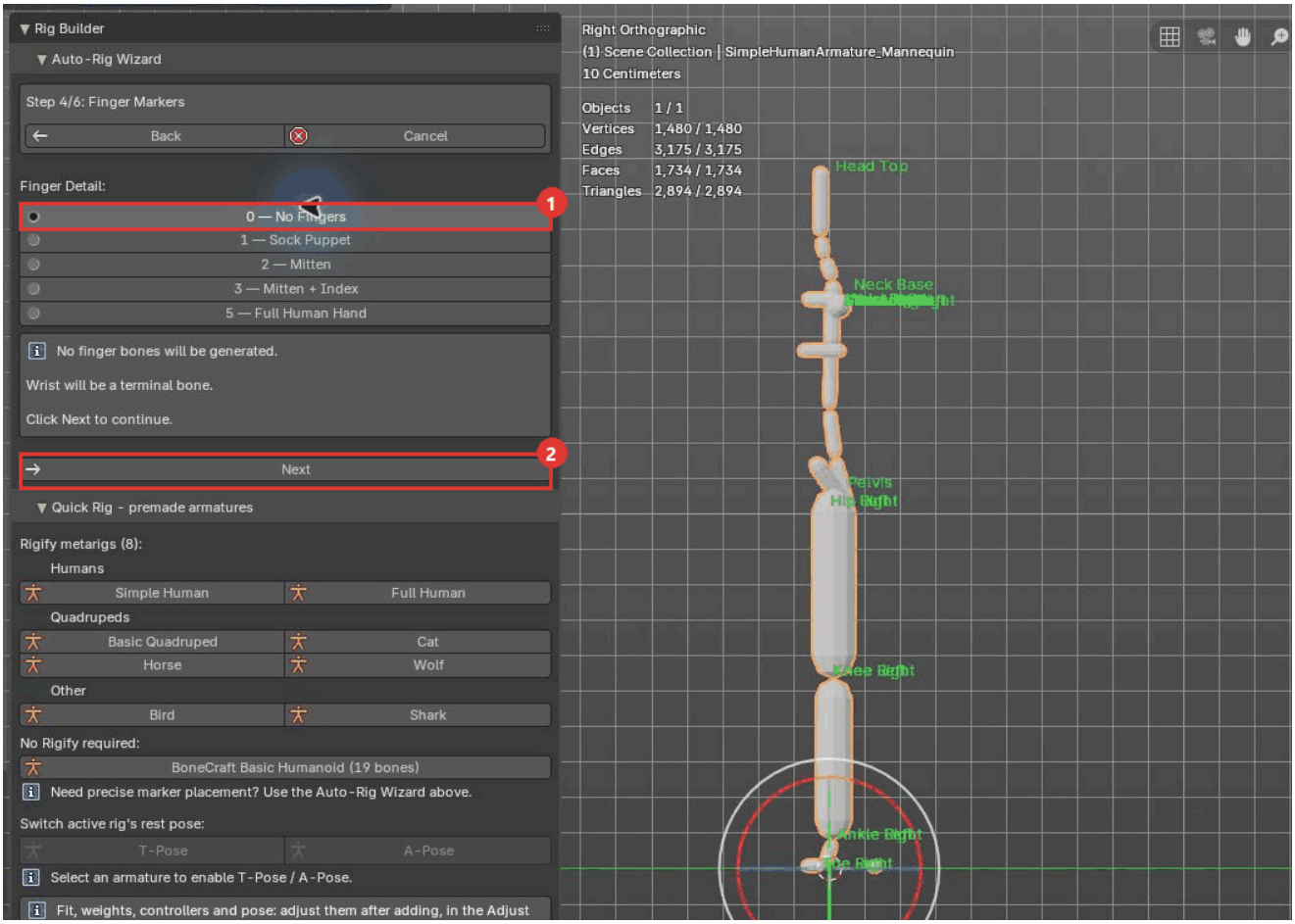
Review head top, neck base and pelvis as well as every paired landmark. Click **Confirm All Placed**, then **Next**. The green checks beside the names are the confirmation state; the original REVIEW/ADJUST confidence labels may remain visible in 9.1.2.



1 — Confirm All Placed. 2 — Next.

B19. Skip fingers for this mesh

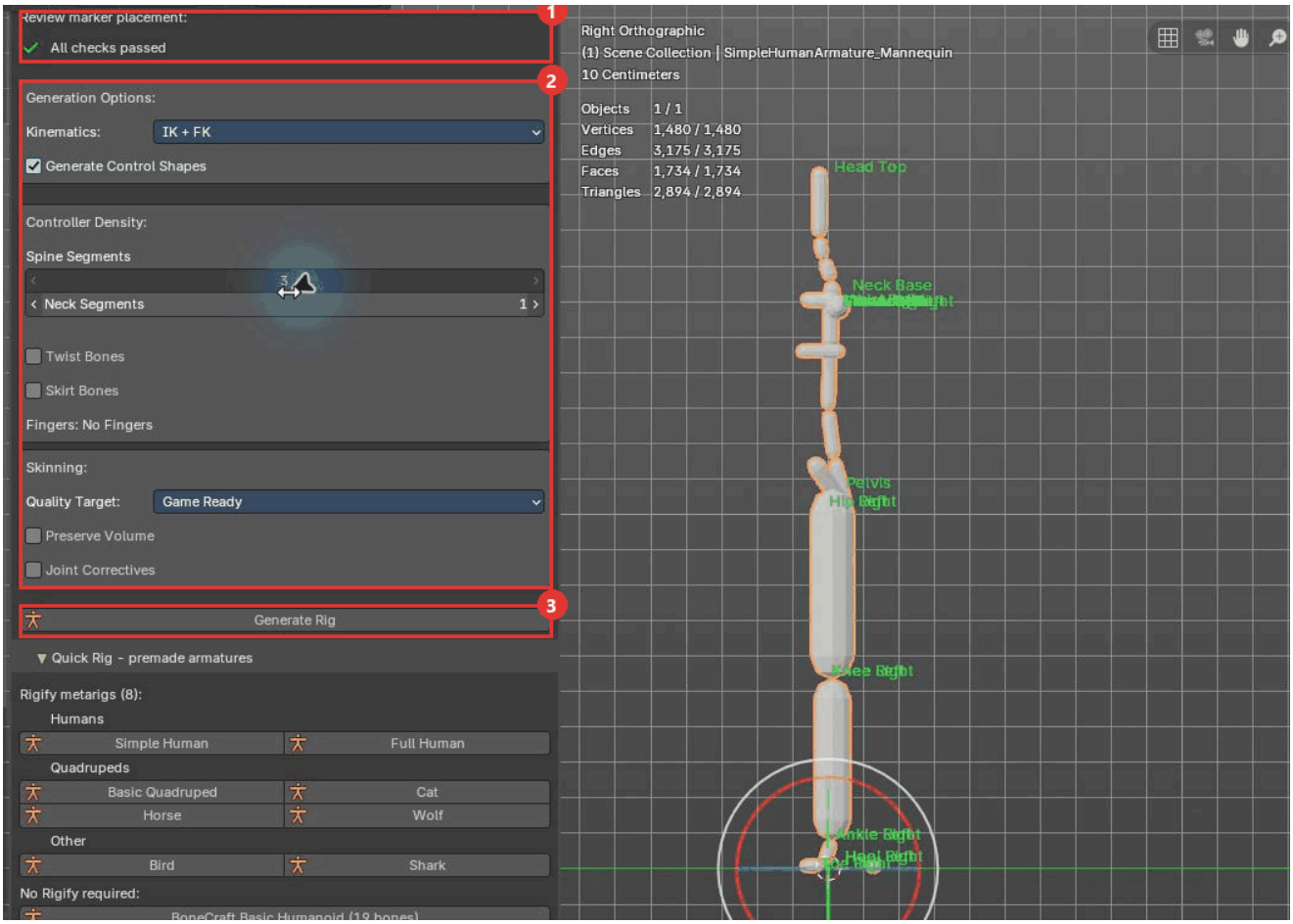
Choose `0 — No Fingers` and click **Next**. This mannequin has hand capsules but no finger geometry. On a real hand mesh, choose the appropriate count and place its finger markers instead.



1 — No Fingers. 2 — Next.

B20. Review and generate

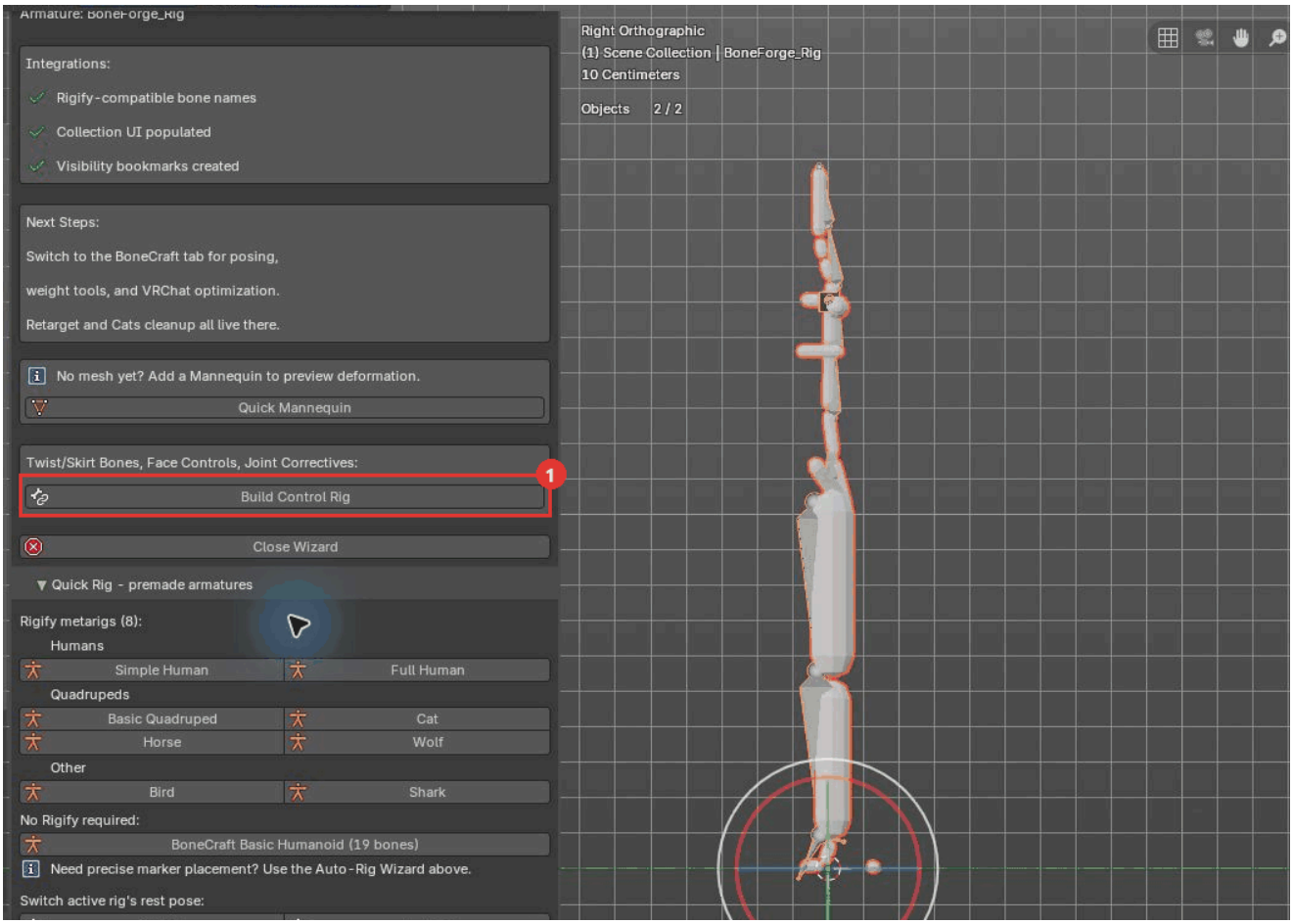
Require **All checks passed** before proceeding. Use **IK + FK**, control shapes on, `Spine Segments: 3`, `Neck Segments: 1`, `No Fingers`, and **Game Ready** for this exercise. Turn Twist Bones, Skirt Bones, Preserve Volume and Joint Correctives off. Click **Generate Rig**. These settings describe this simple training mannequin, not a universal export prescription.



1 — Marker checks. 2 — Generation settings. 3 — Generate Rig.

B21. Build the animator controls

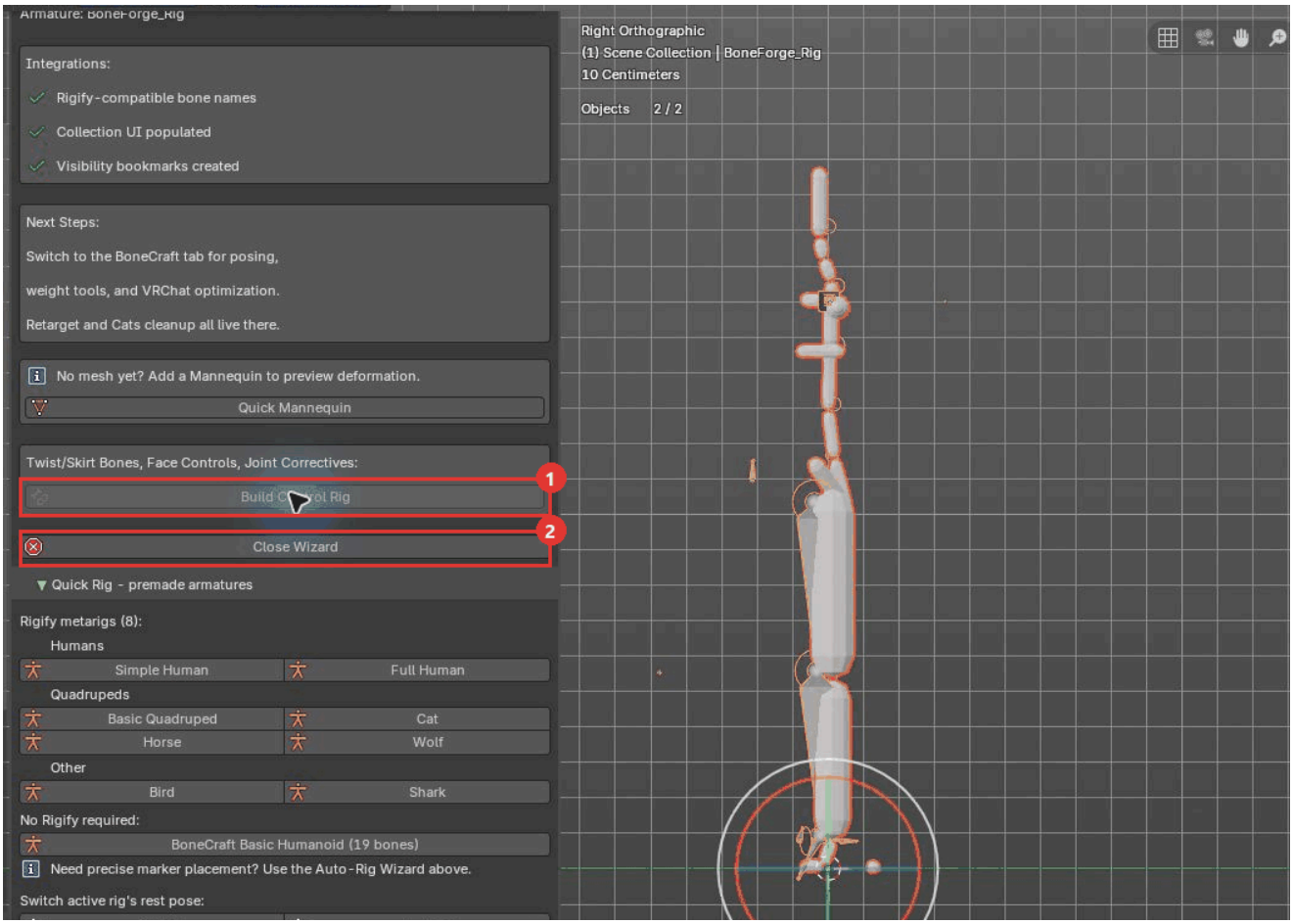
After generation, click **Build Control Rig**. Generate Rig supplies the body rig and weights; Build Control Rig adds the control layer used below. Do not click Quick Mannequin again—the input mesh is already present, even if the completion panel displays its generic mannequin offer.



1 — Build Control Rig.

B22. Finish the wizard

Wait for the control-rig completion message, then click **Close Wizard**. The new armature appears in the Outliner. Keep the old Simple Human metarig hidden. The new rig must deform the mesh; merely seeing bones in the viewport is not sufficient validation.



1 — Build button is now disabled. 2 — Close Wizard.

Part C: Test weights and use IK/FK

The generated armature has this object name in the tested release (use it in the Outliner and the weight tool):

```
BoneForge_Rig
```

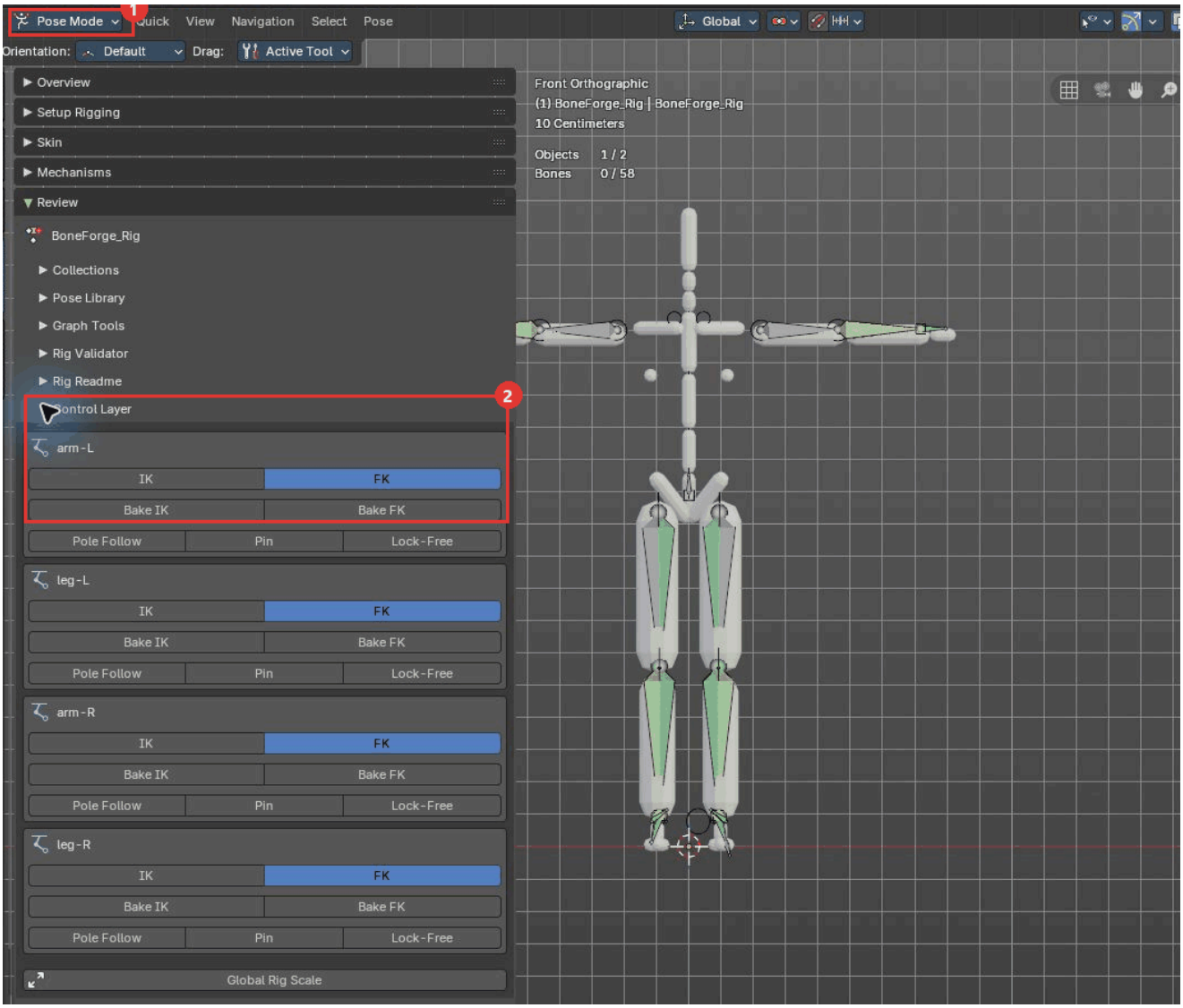
`FK (Forward Kinematics)` follows rotations down a hierarchy: rotate the upper arm, then forearm, then hand. It suits broad swings and freely moving limbs. `IK (Inverse Kinematics)` starts at an endpoint: move a wrist or ankle target and the solver bends the chain to reach it. It suits planted feet and hands that must meet a surface.

Task	Control to use	What to watch
Swing an arm	FK shoulder and elbow controls	Child controls follow their parent
Place a hand	IK wrist target	Shoulder and elbow solve toward the target
Plant a foot	IK ankle target for that leg	Move the body while the reachable target stays placed
Aim an elbow or knee bend	IK pole control	The pole sets the bend direction; it is not the wrist/ankle target
Change method	The limb's IK/FK buttons	Check the snapped pose before keying animation

The exercise below compares one arm in both modes. Each limb has its own switch. `Bake IK / Bake FK` are animation conversion tools; a single-pose switch does not replace baking or keying an animation. See Control Layer for the other controls.

C1. Open the control layer

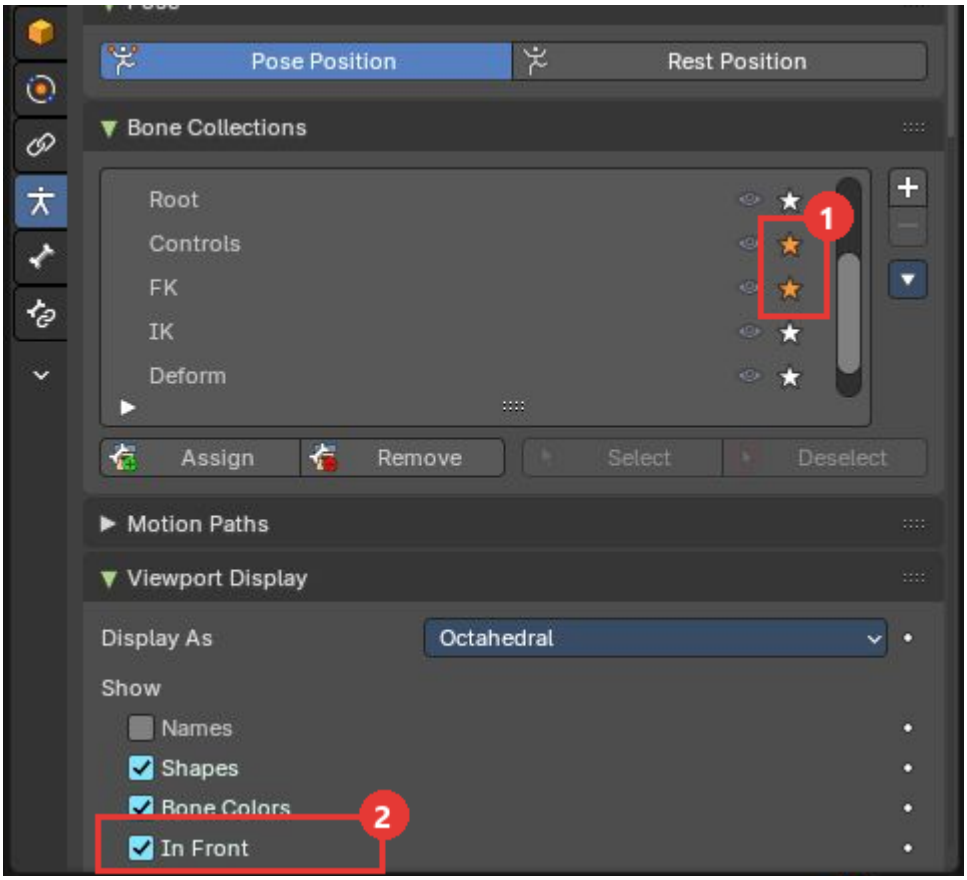
Select **generated armature**, choose **Pose Mode**, then open **BoneCraft → Review → Control Layer**. There is a separate IK/FK switch for each arm and leg. Start with **arm-L → FK**. Left and right refer to the character, so arm-L appears on your right in front view.



1 — Pose Mode. 2 — Control Layer and arm-L switch.

C2. Show the animator controls

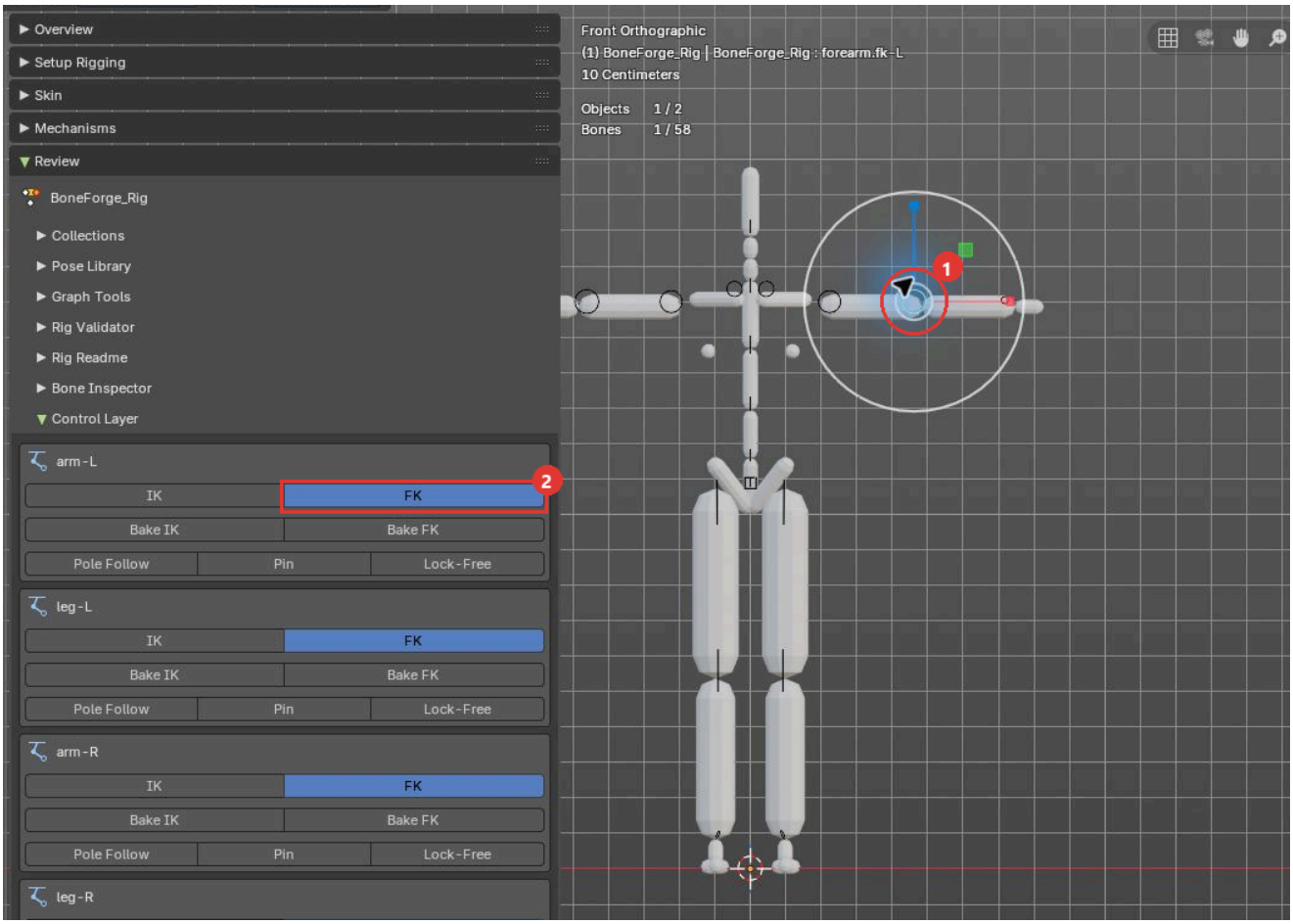
Open the armature's **Object Data Properties** → **Bone Collections**. Use the Solo stars for `Controls` and **FK** to isolate their control shapes. Enable **In Front** under Viewport Display when controls are obscured by the mesh. Later include **IK** when using targets. This native collection list also works if the BoneCraft Collections panel fails to draw in this host.



1 — Controls and FK Solo stars. 2 — In Front.

C3. Select the forearm FK control

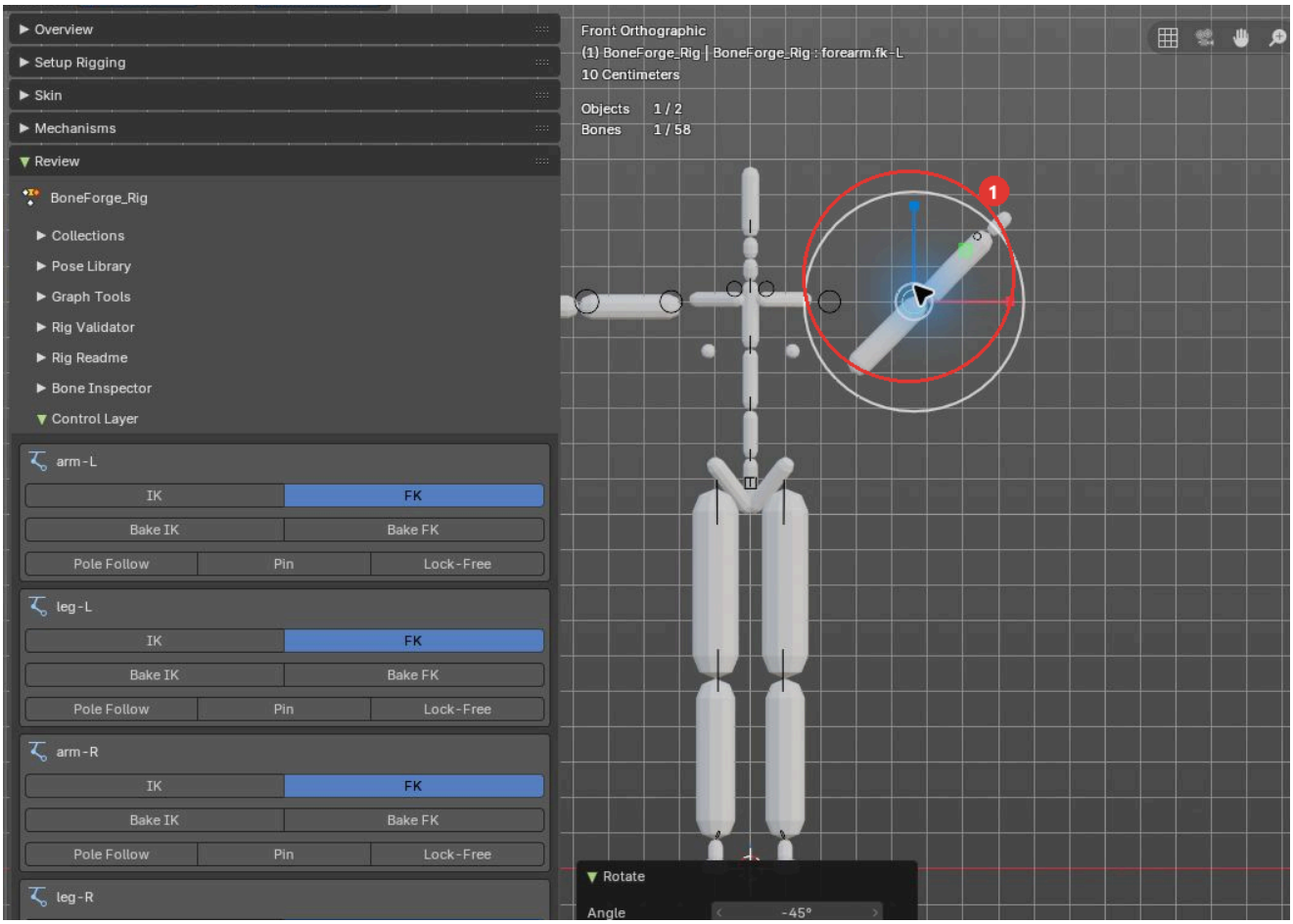
In front view, click the circular control at the character's left elbow. Check the viewport selection name is **forearm.fk-L**. An FK forearm control rotates around the elbow; selecting the shoulder control would rotate the upper arm and its children instead.



1 — Selected forearm.fk-L control. 2 — FK is active for arm-L.

C4. Rotate and inspect the deformation

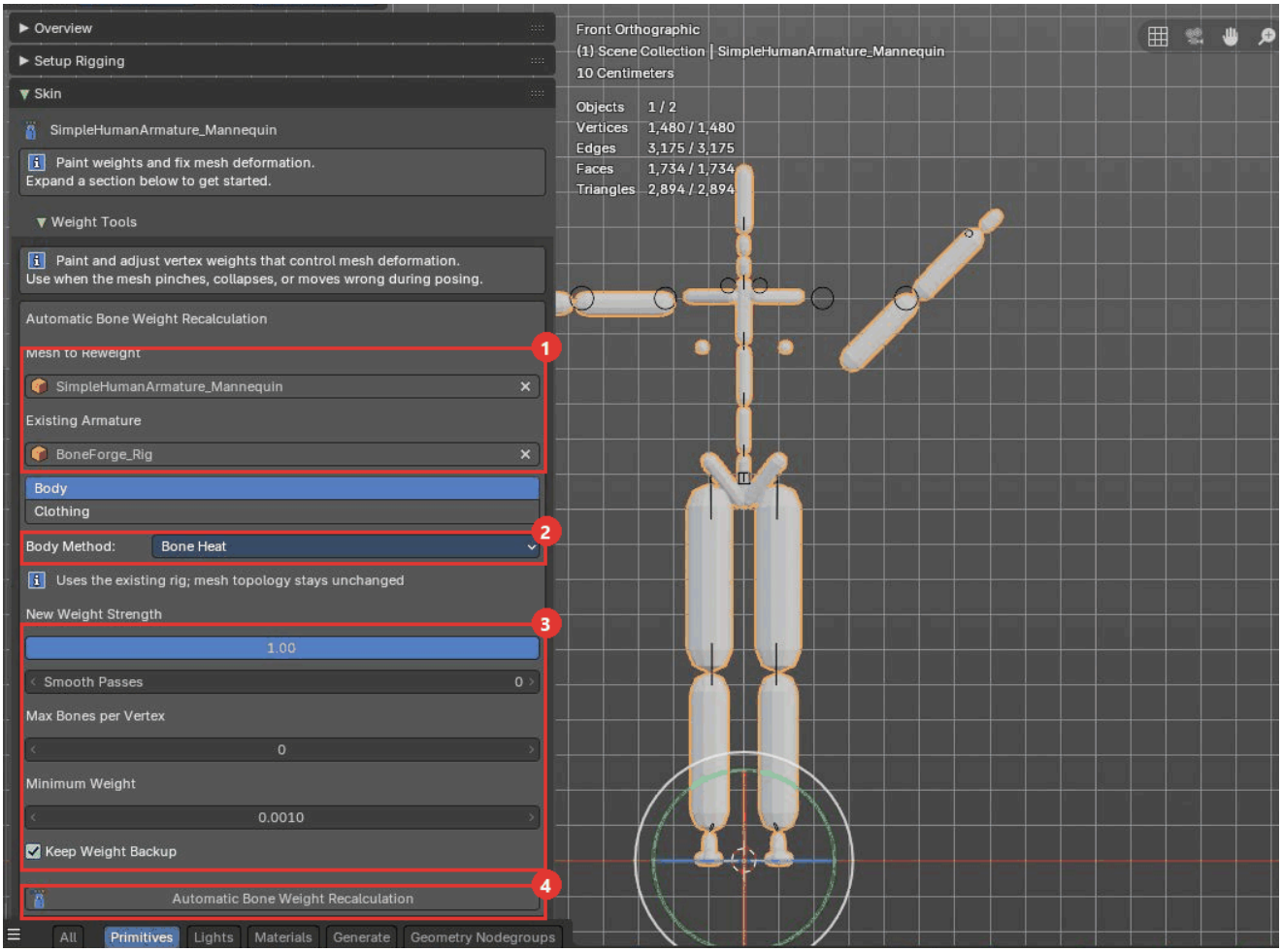
Rotate the selected FK control about global **Y**, approximately –45° for this front-view example. Use the Rotate tool; in the shown Bforartists keymap, **Ctrl+E**, **Y**, type **-45**, then **Enter** performs that rotation. Blender's default rotate shortcut is **R**. The screenshot deliberately shows the initial weight problem: the upper-arm capsule tilts with the forearm. Do not accept this as a finished rig.



1 — Incorrect upper-arm movement during the FK test.

C5. Recalculate the mannequin weights

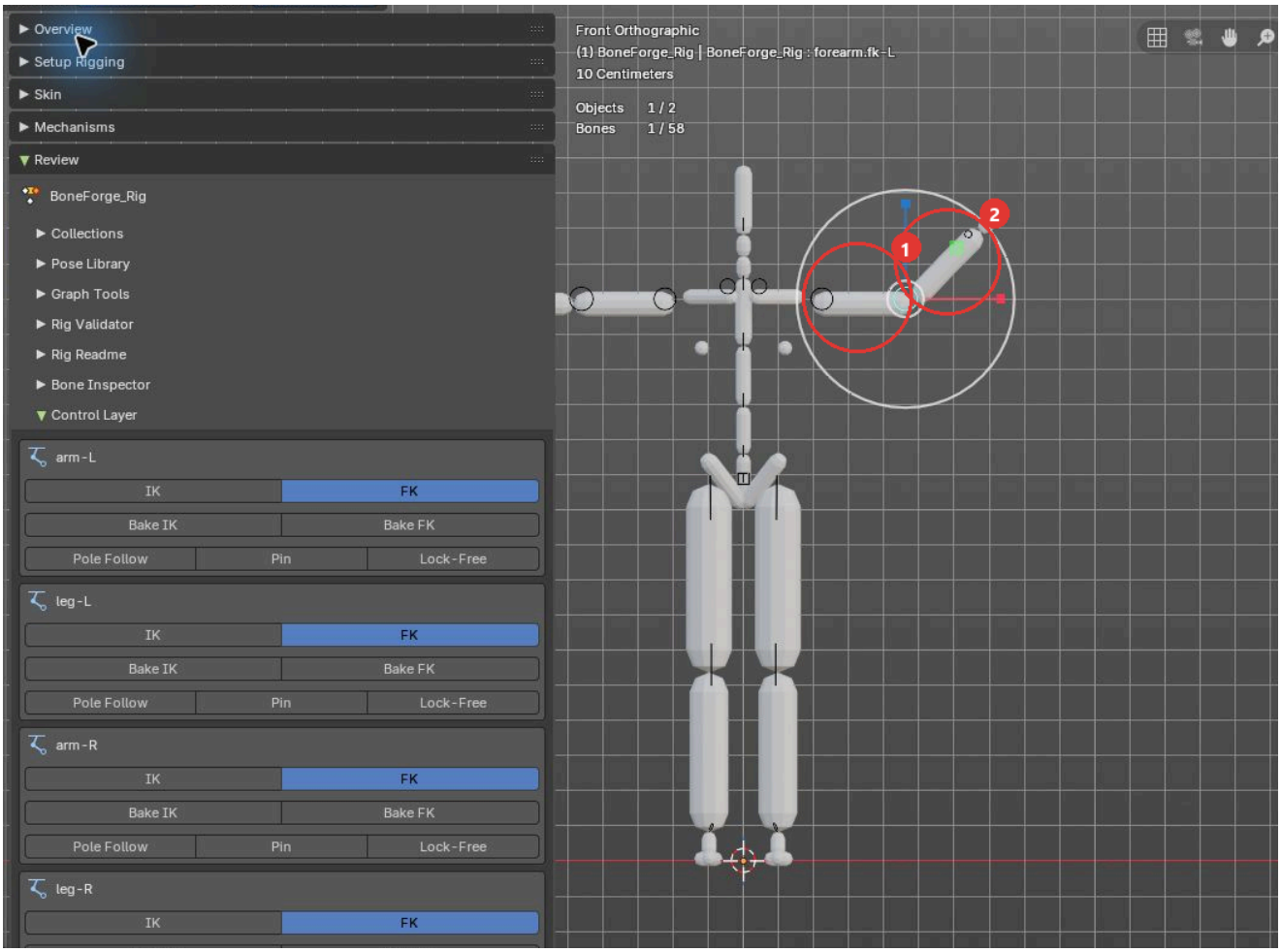
If the pose test above fails, switch to **Object Mode**, select the mesh, and open **BoneCraft** → **Skin** → **Weight Tools** → **Automatic Bone Weight Recalculation**. Set Mesh to Reweight to the mannequin and Existing Armature to **generated armature**. Choose **Body**, `Body Method: Bone Heat`, `New Weight Strength: 1.00`, `Smooth Passes: 0`, `Max Bones per Vertex: 0`, and `Minimum Weight: 0.001`. Keep **Keep Weight Backup** on, then run **Automatic Bone Weight Recalculation**. This repaired the segmented preview in the tested release.



1 — Mesh and armature. 2 — Bone Heat. 3 — Replacement settings and backup. 4 — Recalculate.

C6. Verify the repaired FK bend

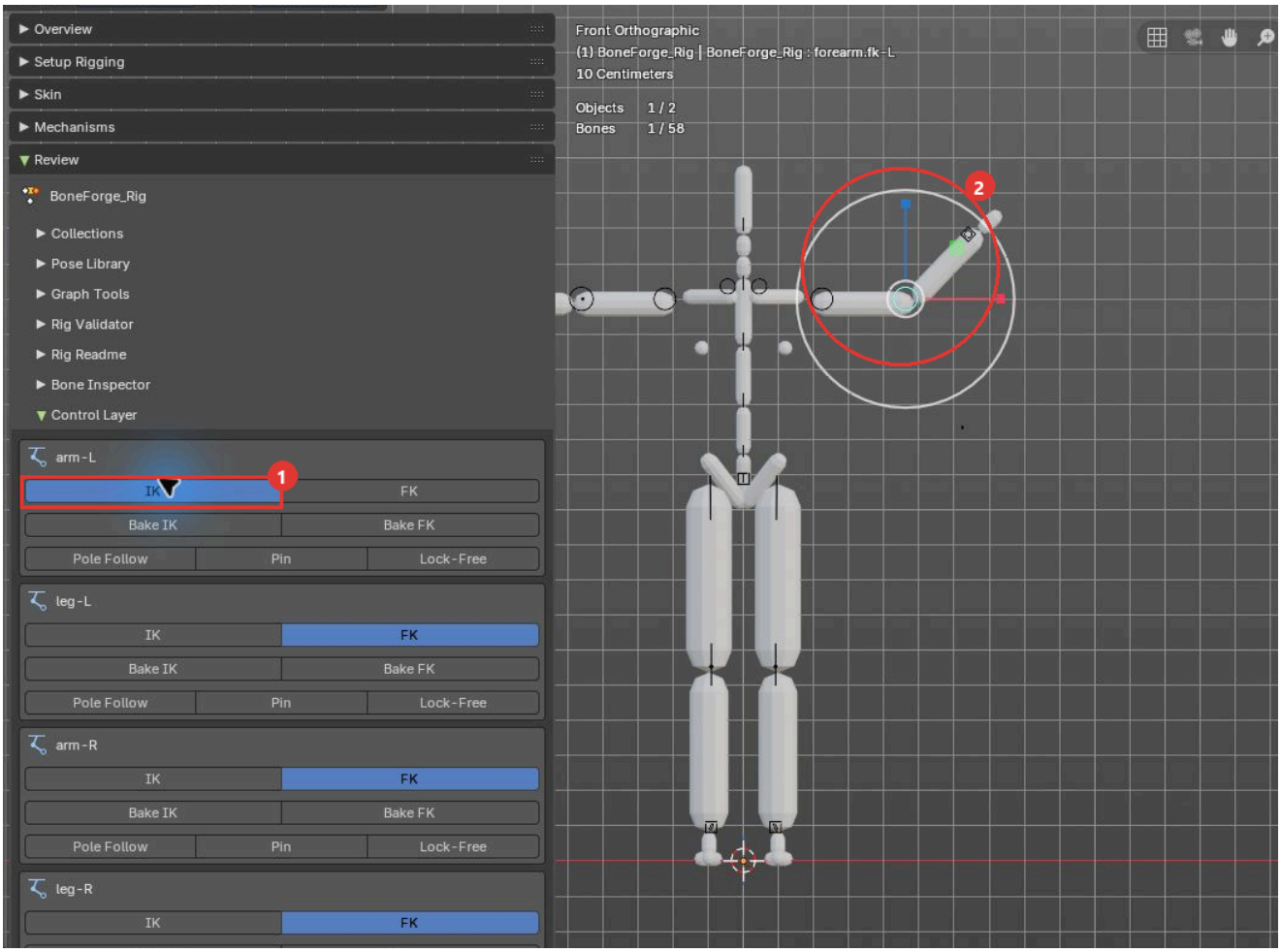
Select the generated armature again and return to **Pose Mode**. The upper arm should now stay horizontal from shoulder to elbow while the forearm rises from the elbow. The hidden `WEIGHTS_BACKUP_...` object preserves the old weights. Test other limbs too before using your own character in animation. A generated skeleton alone is not proof of correct skinning.



1 — Upper arm stays in place. 2 — Forearm bends at the elbow.

C7. Switch the same arm to IK

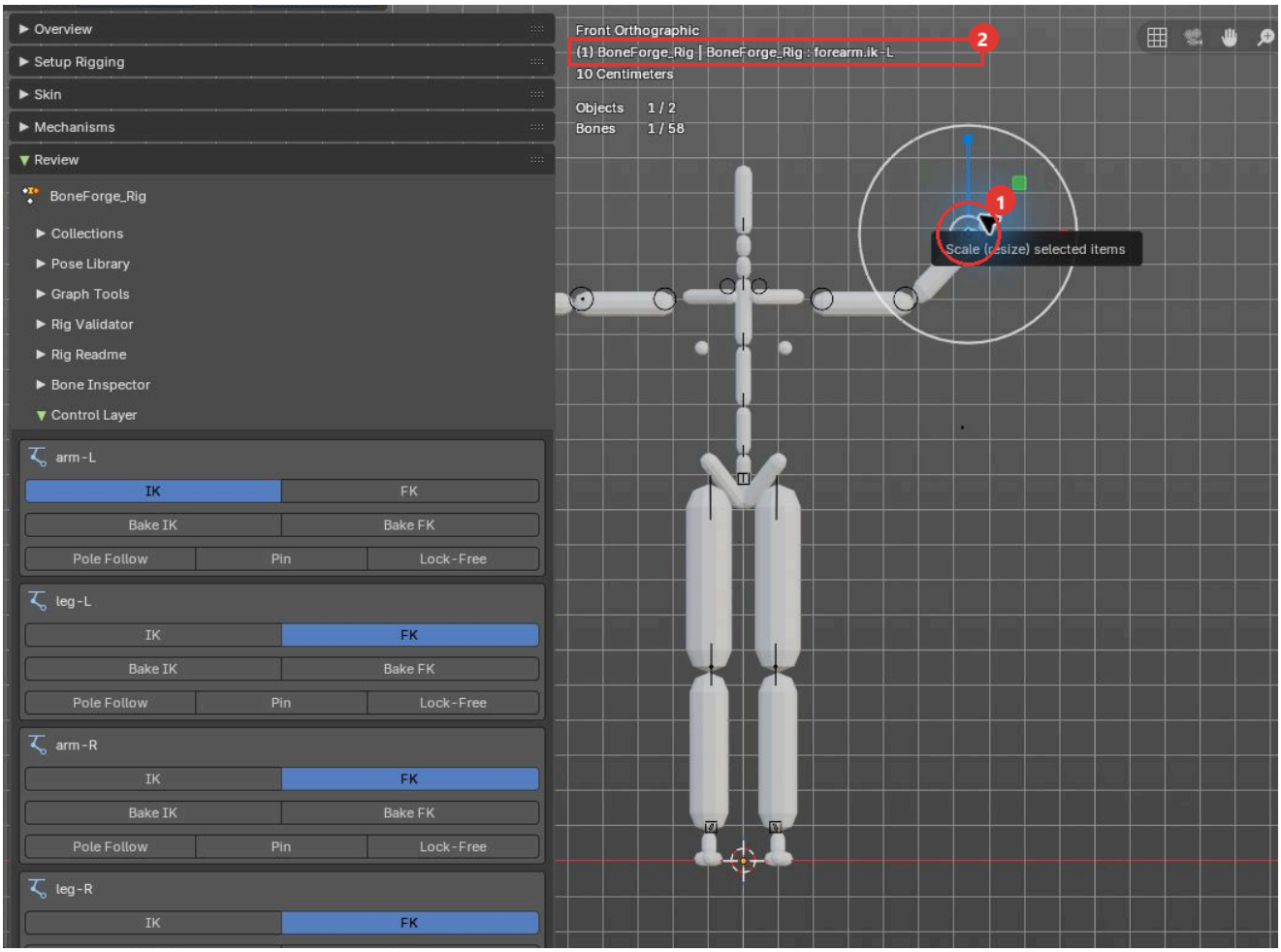
Click **arm-L** → **IK**. The switch snaps the target to the current pose, so the arm should remain in its raised pose. Add **IK** to the visible Solo collections if its targets are hidden. Switching changes which controls drive the chain; it does not mean a new animation has been recorded.



1 — arm-L IK switch. 2 — Pose retained when switching.

C8. Select the wrist IK target

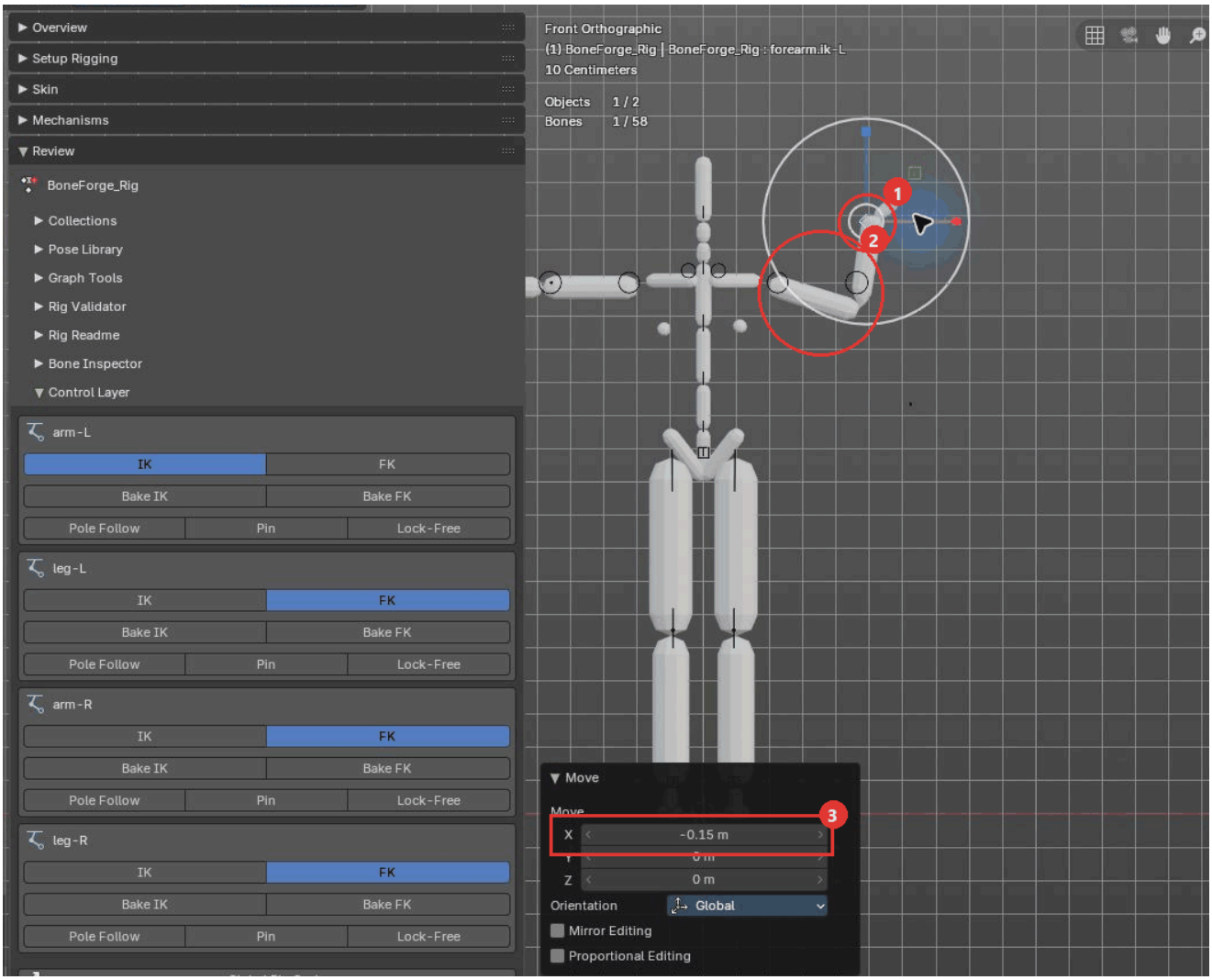
Click the control at the wrist and check the selection name **forearm.ik-L**. Despite its forearm name, this is the endpoint target for the arm chain. Move this target to place the hand; the solver changes shoulder and elbow angles to reach it.



1 — Selected wrist target. 2 — Selection name.

C9. Move the target and compare

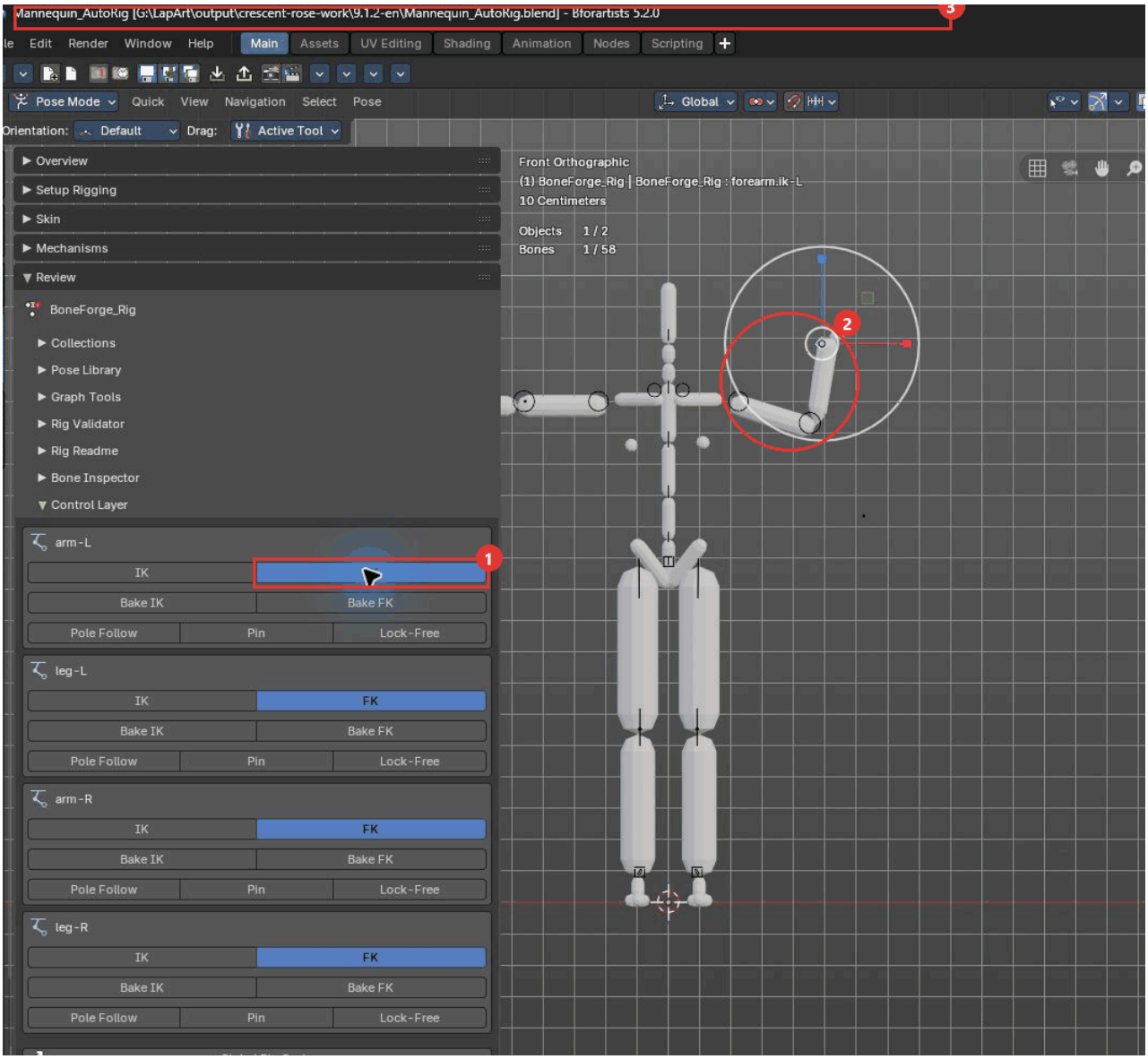
Move the wrist target slightly inward along global **X**. In this example, `-0.15 m` creates a visibly different bend. In the shown keymap use **Ctrl+W**, **X**, type `-0.15`, then **Enter**, or drag the Move tool's X arrow. Blender's default move shortcut is **G**. Compare the whole arm with the FK example: the endpoint leads and both arm segments follow. Keep the target within the arm's reach.



1 — Wrist endpoint. 2 — Chain responds. 3 — Example X movement.

C10. Switch back and save the example

Click **arm-L** → **FK** again. The FK controls snap to the current IK pose. Verify the mesh does not jump, then use **File** → **Save As** to save `Mannequin_AutoRig.blend`. The screenshot shows the saved example with FK active. For a neutral training copy, keep all four limbs in FK, select all visible pose controls and use **F3** → **Clear Pose Transforms** → **All** as in Part A; save that under a separate name.



1 — FK active again. 2 — Pose preserved. 3 — Saved file name.

Completion check: you have preserved the original metarig preview, detached a rigless mannequin, generated a new body rig and its controls, corrected the preview weights, and compared an FK rotation with an IK target move. Keep the saved rig as a learning example; replace the preview with suitable character geometry for production.

Animation Retargeting

Moving animation from one skeleton to another is covered in Rig Tools: Retarget.

Bone Merge

What it does: Combines two armatures into one — for example a body rig and a clothing or hair rig — keeping every weight.

Guide 11 walks through it step by step.

The three stages: 1. Scope — pick the two armatures and review how they differ 2. Rename — resolve name conflicts and mark new bones 3. Merge — preview, then merge

Key controls: - **Base** — the armature that stays; the other armature merges into it and is then hidden - **Analyze** — compares the two skeletons; **Acknowledge Scope — Proceed to Rename** opens stage 2 - **Normalize Both Armatures** — renames the standard bones of both rigs to one naming standard (Mixamo Prefixed, Mixamo Stripped or custom) - **Auto-Propose Renames** — suggests names for the merging rig's own bones - **Apply Rename** — renames one entry (one undo step); **Apply Batch Pattern** renames many with a pattern (with `{bone}`, `{side}` and `{index}`) - **Mark as Unique** — keeps a bone as a new bone instead of merging it - **Verify Rename — Proceed to Merge** — opens stage 3 - **Run Merge Preview** — reports what the merge would do, without changing anything - **Merge Armatures** — backs up both armatures and merges them

Where to find it: BoneCraft → Setup Rigging → Bone Merge

Next: Mechanisms

Mechanisms

← Contents

What Mechanisms does

Mechanisms turns hard-surface parts into a working mechanical rig: hinges, slides, telescopes, pistons, gears, followers and whole transformations played by one slider. It builds ordinary Blender bones and constraints, so you can see and change everything it makes, and the rig works in Blender without scripts. Engines receive the `DEF-` bones. Export of baked mechanism motion to engines comes later (see Limits of this version).

BoneCraft ships the tools, not a weapon. You bring the parts.

A complete worked example, with a screenshot for every step, is Tutorial: Crescent Rose with Mechanisms.

Words used in the panels and in this page:

Word	Meaning
Part	A piece that moves as one solid block: an object, or a loose piece of one mesh. A part can ride on another part.
Joint	How a part may move: Hinge (turns), Slide (moves), Telescope (extends nested segments).
Link	What moves a part for you: Piston, Gear Link, Follow.
Handle	The bone you drag in the viewport. Each joint has one. Its value is also a slider in the panel.
Sequence	A transformation played by one slider, made of steps.
Form	A saved set of handle and Sequence values, such as "Stowed" or "Rifle".

Where it is

- **BoneCraft tab** → **Mechanisms** in the 3D Viewport. The BoneCraft tab is on the left Tool Shelf (press **T**) unless you moved it to the right sidebar (**N**) in Preferences. It sits between Skin and Review. It shows when a mesh is active (to start a rig) or when an armature with mechanisms is active; otherwise it explains what to select.
- **Properties editor** → **Scene tab** → **Mechanisms**, right after Rig Builder. It is shown while Preferences → Tabs → Scene Properties → Show is on. With nothing usable selected it shows one line: "Select a mesh to start, or an armature with mechanisms." Switching the BoneCraft viewport tab off does not remove it. The Scene Properties switch hides the Scene copy only. The `mechanisms` tool switch (**Preferences** → **Add-ons** → **BoneCraft** → **Tools**) removes the tool from every location. Scene-tab sections start collapsed, so click **Mechanisms** to expand it.
- **Object menu** → **Mechanisms** gives shortcuts to add parts, joints and links, and to test motion.
- **Viewport toolbar** → **Build Mechanism** starts a guided joint or link from a click on a part, in Object Mode or Pose Mode. Add the parts first, then select the mechanism rig or one of its parts and choose the type in the tool header.

Inside Mechanisms there are four subsections: **1. Parts**, **2. Joints & Links**, **3. Sequences** and **4. Controls**. They appear once the rig has parts. Before that you see only **Add Parts from Selection**.

Every button is one undo step. If a build fails, everything it changed is put back and the panel shows the reason.

Parts

1. Select the part objects (or one mesh made of loose pieces) in Object Mode.
2. Click **Add Parts from Selection**. - Several objects: one part per object. Each object is parented to its part's bone and stays where it is. - One mesh with loose pieces: one part per piece. Each piece gets a vertex group for its bone and the mesh gets an Armature modifier. - The first time, BoneCraft makes the rig (an armature) for you.
3. Select a part in the list to edit it.

For each part:

- **Rides on:** choose the part this one moves with. A child part follows its parent part. "the rig" means it rides on the rig's root.
- **Size** sets the bone length, which sets the handle size.
- **Edit Pivot** puts a marker on the part. Drag it to where the part should turn or slide from, then click **Apply Pivot**. The part's bones move there and the object stays in place.
- **Apply Transforms** appears when an object part has unapplied rotation or scale. Click it. Unapplied transforms tilt the axes.
- The **X** button removes the part. It refuses while a joint, link or other part uses it, and names them.
- **Merge for VRChat** (two or more object parts) joins the object parts into one mesh with one vertex group per former object and an Armature modifier. Do this last. The joined parts become loose parts of that one mesh. It removes every vertex group named after a bone of this armature (other than the part's own DEF- bone), so merge only parts that are rigged to the mechanism rig alone.

You can rename a part's object: parts are tagged on their objects, so the part finds its object again. A copy of the object (Shift+D) does not take over the part. If a part's object is deleted, the part shows "Part '' is missing" with **Reassign** and **Delete**. Other mechanisms keep working.

- **Reassign** uses the mesh you have selected. It refuses a non-mesh, and it refuses an object that another object part already uses. Loose parts of one mesh may share that mesh.
- **Delete** removes the part (see Editing below for when it is refused).

Joints and links, one example each

Add a joint with **Add Joint**, a link with **Add Link**. Each choice starts a guided build. They work from the viewport tab and from the Properties editor's Scene tab; from the Scene tab the guide uses the largest 3D Viewport on the screen. Alternatively, activate Build Mechanism in the viewport toolbar, choose the joint or link type in the tool header, and click its first part. That click starts the same guide and supplies its first pick. Continue with the picks and axis choices requested in the viewport; the Add Joint and Add Link menus remain available.

Short rules for every guided build:

- Click parts in the viewport. Clicked parts get an outline and a number.

- Where a step needs an axis, press **X**, **Y** or **Z**, or **A** to align to the selection. **Align to Selection** is for angled hinges. It needs the part's mesh to still be in Edit Mode with one edge or face selected when you press **A**, and the guide ignores **Tab**. So enter Edit Mode on the part's mesh and select one edge or face first. Stay in Edit Mode, start **Add Joint** → **Hinge**, click the part, press **A**. If nothing usable is selected the guide tells you and you can press another key.
- The last step previews the mechanism moving in a loop. **Enter** keeps it. **Esc** removes it.
- The header and status bar show a full sentence for each step. With Show Helpers off they show a short label.

Each row of the Joints & Links list has four icon buttons: **Test** (play triangle), **Edit** (pencil, shows the settings), **Detach** (broken chain) and **Delete** (X).

After the build, the selected item in the list shows only the settings its type uses.

Hinge

Blade Hinge, 0°…90°:

1. Add Parts from Selection with a Blade part.
2. **Add Joint** → **Hinge**. Click the Blade. Press **X**. Press **Enter**.
3. You get a half-circle handle, `CTRL-Blade`, and a *Blade Hinge* in the list. Set **Min** 0 and **Max** 90 (° for a Hinge).
4. Try it: drag the half-circle handle. It stops at 90°. **Set Min** and **Set Max** copy the handle's current pose into the limits.

A limit that is under a full turn must stay within −180°…180°. Limits such as 170°…190° are refused with a message. A range of a full turn or more (for example −180°…180°) spins freely with no limit.

Slide

Barrel Slide, 0…0.3:

1. **Add Joint** → **Slide**. Click the Barrel part. Press **Y**. Press **Enter**.
2. You get an arrow handle. Set **Min** 0 and **Max** 0.3 (scene units).
3. Try it: drag the arrow. The barrel slides and stops at 0.3.

Telescope

Mast with three segments:

1. Make the base and three segment parts.
2. **Add Joint** → **Telescope**. Click the base. Click each segment from the base outwards. Press **Enter**. Press the axis key. Press **Enter** to keep it.
3. You get one arrow handle on the base. **Length** is the total travel; **Shares (empty = even)** splits it between the segments. To try custom shares, set **Length** to 1 first, then type `0.5, 0.25, 0.25`. The shares must add up to the Length, so the tip travels the full length. A typo (for example `0.5, abc`) is refused with the same plain message: the shares must be numbers, one per segment, that add up to the Length.
4. Try it: drag the one arrow. All three segments extend by their shares. **Set Max** sets Length from the handle's pose.

Piston

Lift: cylinder on the frame, rod on the arm.

1. Give the arm a Hinge first, and set the arm to ride on the frame.
2. **Add Link** → **Piston**. Click the cylinder part (on the frame). Click the rod part (on the arm).
3. You get a Piston with no handle. Each part points at the other's attach point. **Stretch** (optional) stretches the rod, for springs and cables.
4. Try it: move the arm's handle. The piston follows. If nothing moves the Piston yet, the build works but has no preview and **Test** says so.

Gear Link

Gear 20:40:

1. Both parts need a joint first. The driving part needs a Hinge. For Gear and Belt the driven part needs a Hinge; for Rack it needs a Slide.
2. **Add Link** → **Gear Link**. Click the driving part. Click the driven part.
3. In its settings set **Source Teeth** 20 and **Target Teeth** 40 (or type a **Ratio**). **Kind** is Gear, Belt or Rack. **Reverse** flips the direction.
4. Try it: turn the source. With **Kind** Gear the target turns −0.5× as far (+0.5× with Belt). A Gear reverses direction, a Belt keeps it, and a Rack turns rotation into a slide (distance per radian = **Ratio** × **Pinion Radius**).

How BoneCraft builds it depends on the source Hinge's range:

- Under a full turn: a Transform constraint. It follows everything that moves the source, Sequences included.
- A full turn or more: a driver that needs no Python. The target spins forever with the source. It follows handles and gears only. The Rig Validator warns when a Sequence or Follow would be lost.

Follow

Blade 0–90° lifts Cover 0–40°:

1. Both parts need a Hinge or a Slide. A source Hinge needs limits under a full turn; a free-spinning source is refused.
2. **Add Link** → **Follow**. Click the leading part (Blade). Click the following part (Cover).
3. The ranges start as the full range of each joint. Set **Source Range** 0…90 and **Target Range** 0…40. Past either end the target holds.
4. Try it: move the Blade handle. The Cover follows.
5. Tick **Curve** to reshape the mapping in the Graph Editor. BoneCraft never overwrites your curve (see Editing below).

Sequence

Record steps to fold a weapon:

1. In **3. Sequences**, click **Add Sequence**. It makes a slider handle `CTRL-SEQ-<name>`.
2. Click **Move Slider to End**. The slider jumps to its end (1), and the button becomes Record Step followed by the next step number.

3. Pose the handles of the joints for the first step. Click Record Step 1. Each handle that moved becomes a step with its target value, and the handles go back to zero.
4. Repeat steps 2 and 3 for more steps. Each new step takes the next quarter of the slider by default.
5. Set **Target**, the ease (Linear, Ease In, Ease Out, Ease In Out, Snap), **Start** and **End** for each step. Outside its slice a step holds its end value. Several Sequences can stack.
6. Click **Test**: the slider plays 0 → 1 → 0.

Next to the step-recording button and **Test**, the selected Sequence has **Detach** (broken chain) and **Delete** (X). A held Sequence (for example a step whose curve you reshaped) shows its message and a **Restore** button. Restore regenerates the steps from the record. Keep mine is not offered for a Sequence.

Test

Test (the play button in the list, and in 3. Sequences) sweeps one mechanism through its full range and back. It writes no keys and puts the pose back when it ends. Press **Esc** to stop it early; the pose is restored then too. For a Gear Link or Follow it sweeps the source. For a Piston it sweeps the joint that moves it. Watch for parts that clash; BoneCraft does not detect collisions.

Forms

In **4. Controls**:

- Every joint and Sequence has one slider. It is the handle's own channel, so dragging the handle and using the slider are the same thing.
- **Save Form** records every handle and Sequence slider value under a name. Rename it in the list.
- **Go to Form** sets every control to the saved values.
- **Go and Key** does the same and also inserts keys on the current frame.
- The **X** next to the list removes a form.

What BoneCraft builds

For each part:

Bone	Job	Visible
`CTRL-<part>`	The handle. Its value is the panel slider. Only Hinge, Slide and Telescope make one.	Yes
`MCH-<part>`	Adds the handle and every link, then applies the limits.	Hidden
`DEF-<part>`	Copies `MCH-<part>`. The mesh is bound to it. Engines receive this bone.	Hidden

The bones sit in the collections `Mechanism Controls`, `Mechanism MCH` and `Mechanism DEF`. A part with no joint has only `MCH-` and `DEF-` bones. A Telescope's handle is `CTRL-<telescope name>`. A Sequence's is `CTRL-SEQ-<name>`. A Piston adds `MCH-<part>-aim` and `MCH-<part>-target` bones for each of its two parts.

The constraints on `MCH-<part>` always come in the same order: Copy Transforms from the handle, then each link's contribution, then the joint's Limit Rotation or Limit Location. `DEF-<part>` has one Copy Transforms from `MCH-<part>`.

Everything BoneCraft builds is tagged with its record: constraint names start with `BC·<id>·<role>`, and bones carry the properties `bc_mech_id` and `bc_mech_role`. BoneCraft finds its bones by tag, not by name, so you can rename any bone. A rebuild finds it again and makes no duplicate.

With Show Helpers on, each joint and link has a **How this works** note. It lists the constraints by name, says what each does, and has a **Select** button (an arrow icon) to inspect the bone.

Your own constraints, drivers and bones are never touched. A constraint you add to an MCH bone stays below BoneCraft's after a rebuild.

Editing and hand edits

- **Value edits** (limits, ratio, ranges, timing, ease, handle look) update the existing constraints in place. There is no rebuild.
- **Structural edits** (a Gear Link's **Kind**, a Follow's **Curve**, a Piston's **Stretch**) replace that mechanism's constraints and rebuild only that mechanism.
- **Changed by hand.** If you edit a BoneCraft constraint (for example a Limit), the mechanism's message ends with "(changed by hand)". BoneCraft never overwrites it silently. Choose:
- **Keep mine** copies your values into the record. It refuses what it cannot read back, and says to use Restore or Detach.
- **Restore** throws your edit away and rebuilds from the record.
- A held mechanism also holds every mechanism that uses its bones, until you settle it.
- A reshaped curve is the same. If you reshape a Follow's curve (**Curve** ticked) in the Graph Editor and then change a value that would replace it, such as its Target Range, the Follow's message ends with "(changed by hand)". **Keep mine** keeps your curve. **Restore** regenerates it.
- A held Sequence shows its message and **Restore** in **3. Sequences**, with **Detach** and **Delete** next to it. Keep mine is not offered for Sequences.
- **Detach** (broken-chain button) deletes the record and the tags but keeps the bones and constraints as your own hand-made rig. Detach and Delete both refuse a record that another record uses, and name the users.
- **Delete** (X button) removes the record and everything BoneCraft built for it.
- A link loop is refused before anything is built, and the loop is named, for example "Blade Hinge → Cover Follow → Blade Hinge would loop."

The Rig Validator and the Task Board report link loops, missing parts, DEF bones moved from outside their mechanism, drivers that need Python, and Gear Links that lose a Sequence or Follow. The validator's naming check accepts hyphens (`CTRL-`, `MCH-`, `DEF-`), and its drivers check warns only on expressions that need Python, so a clean Mechanisms rig shows no validation errors — only the one warning about the `root` bone described in Limits of this version.

Show Helpers

Show Helpers is one preference for all of BoneCraft: **Preferences** → **Add-ons** → **BoneCraft** → **Helpers**. Off hides the teaching text everywhere: the what / when lines, the **How this works** notes and the full guided-step sentences (those become short labels). The steps themselves, the lists, Test, warnings and Blender's hover tooltips always stay.

The VRChat **Compact Mode** button is its shortcut: Compact Mode on means Show Helpers off.

Limits of this version

- Click Turn, Track and Linkage mechanisms come later.
- Engine export of baked mechanism motion comes later, after the desktop hand check. Until then the `DEF-` bones are the bones to export.
- BoneCraft does not detect collisions. Use Test and look.
- A free-spinning Gear Link follows handles and gears only.
- The rig's `root` bone is in no bone collection, so the Rig Validator's collections check warns about it and the Task Board shows "Rig validation warnings (1)".

Next: VRChat Tools

VRChat Tools

← Contents

VRChat Tools

The VRChat tools are in the **CATS** tab on the right sidebar (press **N**), in the **Avatar Tools** section. Its subsections cover humanoid bones, hair physics and Viseme Library / Lip Sync. VRChat export is under CATS → Import / Export → Unity / VRChat. The separate **VRChat Tools** section below it groups Check Performance, Attach Clothing, Facial and VRChat Naming Extras. The same section is in the **Properties** editor's **Scene** tab while Preferences → Tabs → Scene Properties → Show is on. This switch does not hide the CATS viewport tab.

Set Up Humanoid Bones

Maps your skeleton's bones to VRChat's humanoid slots and checks the mapping.

See Guide 5: Map Your Avatar to VRChat's Body System.

Add Hair Physics

Sets up PhysBone chains on hair, ears, tails and accessories, with colliders.

See Guide 6: Add Hair Physics.

Attach Clothing

Where to find it: CATS tab → VRChat Tools → Clothing

Merges clothing onto your avatar's skeleton and transfers its weights.

See Guide 7: Attach Clothing That Moves With Your Body.

VRChat Naming Extras

Where to find it: CATS tab → VRChat Tools → Naming

What it does: Detects your skeleton's naming style and renames bones in bulk.

See Guide 4: Fix Your Avatar's Bone Names.

Batch tools: Find and Replace, **Add Prefix**, **Remove Prefix**, **Add Suffix**, **Remove Suffix** and **Regex Replace**

VRChat Viseme Mapping

Where to find it: CATS → VRChat Tools → Facial. Face Tracking is in the same section.

What it does: Maps your mesh's shape keys to VRChat's 15 lip-sync visemes.

See Guide 8: Set Up Lip Sync. To create visemes your avatar does not have yet, see CATS Tools: Visemes.

VRChat's 15 viseme slots: `sil`, `PP`, `FF`, `TH`, `DD`, `kk`, `CH`, `SS`, `nn`, `RR`, `aa`, `E`, `I`, `O`, `U`

Check Performance

Where to find it: CATS tab → VRChat Tools → Performance

What it does: Measures your avatar's VRChat performance rank and offers fixes.

See Guide 9: Improve Your Avatar's Performance.

Performance tiers (best to worst): Excellent → Good → Medium → Poor → Very Poor

Tools: - **Calculate Performance Rank** — the current rank, with a breakdown by polygons, skinned meshes, material slots, bones, PhysBones and contacts - **Remove Zero-Weight Bones**, **Merge Same-Material Meshes**, **Remove Unused Shape Keys** and **Remove Unused Vertex Groups** — the optimizer's fixes (**What-If Mode** shows what they would change) - VRChat Decimation — reduces the polygon count by a ratio (**Preview** first, then **Apply**), keeping shape keys safe

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Export to VRChat

What it does: Exports your finished avatar as an FBX for the VRChat SDK in Unity. Shape keys (blend shapes) are kept unless **Remove Shape Keys** is on.

Settings: - **Folder** — where the export goes - **Avatar** — the exported file name - **Sidecar** — also writes a `.bfvrc` settings file for BoneCraft's Unity importer - **Merge Meshes** — joins the meshes of the export copy into one - **Separate Clothing** — keeps clothing as separate meshes - **Remove Shape Keys** — applies the shape keys to the export copy's basis - **Embed Textures** — packs the textures into the FBX, so Unity finds them - **Helper Meshes** — includes hidden or helper meshes; leave it off for normal avatars

Location: CATS → Import / Export → Unity / VRChat. Use **Export to VRChat (Unity)** to write the FBX, or **Check Performance Rank** before exporting.

Cleanup

Mesh cleanup before export lives in the CATS tools: **Fix Model**, **Join Meshes** and **Apply All Transforms**. See CATS Tools.

VRM Bridge

The VRM tools are in the **CATS** tab → **Import / Export** → **VRM / VRoid**. They need the VRM add-on; the section installs it (**Auto-Install Latest from GitHub**, **Install from .zip on Disk…**, **Open Extensions Page**).

VRM and VRoid Import

What it does: Imports `.vrm` avatars (VRoid Studio, Virtual Cast and other VRM avatars) with their materials, skeleton and shape keys.

- **Import VRM...** — a `.vrm` file
- **Import VRoid (.vroid / .zip)...** — a VRoid Hub download (`.zip`). A VRoid Studio project (`.vroid`) shows a card that explains how to export it to VRM first.

VRM Export

What it does: Exports your rigged character to VRM, or to a target app's format.

Settings: - **Folder** and **File** — where it goes and its name; the extension follows the target - **Target** — VRM 1.0, VRM 0.x, VRChat FBX, VSeeFace, Warudo or Resonite - **Scope** — the active armature, or every armature with VRM data - Skip lint — export without the target check; use only when you understand the reported risk

Click **Export...**.

VRM Lint

What it does: Checks the armature against the chosen target before export: humanoid mapping, VRM metadata, the target's visemes, and the target app's expectations.

Lint Now runs the check without changing anything. Errors block export unless you skip the lint; warnings explain what the target app may still import but run worse.

If the lint says humanoid bones are missing although they exist, click **Fix Humanoid Map**: BoneCraft detects the humanoid slots again and saves the mapping, without renaming any bone.

Also here: Convert SpringBones → PhysBones (VRoid hair physics to VRChat) and **Map VRoid Visemes**.

MMD Bridge

The MMD tools are in the **CATS** tab → **Import / Export** → **MMD / PMX**. They need the mmd_tools add-on; the section installs it (**Auto-Install Latest from GitHub**, **Install from .zip on Disk...**).

MMD Import

- **Import PMX / PMD...** — MMD models
- **Import VMD Motion...** — MMD animations
- **Import VPD Pose...** — MMD poses

MMD Export

Settings: the **Folder**, then for each format a file name and a scope (the active model, or every MMD model in the scene): **PMX File** and **PMX Scope**, **VMD File** and **VMD Scope**, **VPD File** and **VPD Scope**.

Buttons: Export PMX..., **Export VMD Motion...** and **Export VPD Pose...**

Also here: Convert Bone Names -> Unity and **Convert Physics -> PhysBones**.

Game Engine Export

What it does: Exports for game engines with a validated profile: Unreal (humanoid or universal), Unity humanoid, Godot humanoid, and generic FBX or GLB. Every fix the profile needs is shown and applied only when you click it, never silently.

Settings: the **Profile**, the **Animation** to include, **Root Motion**, **Interpolation**, and the **T-Pose** used for export (by default a T-pose for Unity humanoid and VRChat).

Buttons: - **Check for Export** — checks the rig against the profile and lists findings - **Apply Transforms (Fix)** — one of the explicit fixes the check can offer - **Export (Profile)** — exports with the profile

Where to find it: CATS tab → **Import / Export** → **Game Engine Export**

Task Board

What it does: A summary of the current rig: a health score, quick wins, and a list of tasks BoneCraft found (missing humanoid bones, unresolved names, missing visemes, rig validator warnings and so on), grouped by category. **Quick Wins** shows up to three recommended tasks; **Remaining Tasks** contains only the others. **Open Health Check** runs validation and opens its results; **Refresh Task Board** checks again.

Where to find it: BoneCraft tab → **Overview** → **Task Board**

Bone Inspector

What it does: Shows the selected bone in detail — its name, parent, constraints, custom properties and drivers — and lets you edit simple properties without entering Edit Mode.

Where to find it: BoneCraft tab → **Review** → **Bone Inspector**

Next: CATS Tools

CATS Tools

← Contents

The CATS tools are in the **CATS** tab on the right sidebar (press **N**), in the **CATS** section, and in the **Properties** editor's **Scene** tab while Preferences → Tabs → Scene Properties → Show is on. The CATS viewport switch and the Scene Properties switch control their own copies independently. They follow the pipeline described in The CATS Pipeline. Pick the armature they work on at the top of the section.

In 9.1, several panels put optional settings in a fold below their action buttons. Inputs needed before running a tool still come first: Fix Model's cleanup switches, Translate Bone Names' source language, and the mouth shapes or eye bones. Bone tools that offer Adjust Last Operation run immediately and let you tune their options afterwards. Hover a button or option to read its tooltip.

Fix Model

Pipeline phase: 1 (always first)

What it does: A one-click cleanup before any other CATS operation. It removes hidden problems that would otherwise silently break every later step.

Options (each can be switched off): - Apply modifiers - Remove doubles — merge duplicate vertices at the same position - Recalculate normals — fix inside-out surfaces - Remove loose geometry — faces not attached to the body - Remove empty vertex groups - Remove zero-weight bones — bones that move no vertex - Remove constraints and rigid bodies VRChat cannot use

Click **Fix Model** to run them on the selected mesh in one pass.

Translate Bone Names

What it does: Translates your skeleton's bone names to English VRChat names. It works offline, from a built-in dictionary of known bone names.

Supported languages: Japanese (日本語, the most common, from MMD and many community models), Chinese (中文), Korean (한국어), Portuguese (Português), Spanish (Español) and French (Français).

Key controls: the **Language** of the source names, and the translate button (**Translate Bone Names**).

This is about the language of the model's bone names, not your interface language: for a Japanese MMD model, use it whatever language Bforartists is set to.

Join Meshes

What it does: Combines your avatar's mesh objects into one mesh. VRChat performs best with one mesh per avatar. When the meshes have different shape keys, CATS pads the missing ones with neutral shapes,

so the joined mesh has one consistent set.

Key controls: the list of meshes (refresh it with the button next to it), then **Join Meshes**.

When to use: after all clothing and accessories are attached and weighted, and after Fix Model. Joining before Fix Model spreads duplicate-vertex problems from one object to another; the joined mesh then keeps phantom vertices from every original object, and generated visemes tear it apart at the seams.

Material Atlas Combiner

Pipeline phase: 2 (optional)

What it does: Bakes several materials into one texture sheet (an atlas). Fewer materials give a better VRChat performance rank.

Key controls: - **Analyze** — lists your materials and textures, with diagnostics, duplicate groups and size overrides. Choose which materials and which textures to include: untick any you want to keep as they are. - **Smart Combine Materials** — the recommended path: bakes the atlas and points each mesh at it; the atlas UV becomes UV0 for export - **Bake Atlas** — a plain bake; **Add Group** and **Remove Group** bake groups of materials separately - Override Inherited Settings — atlas size, UV method and seed, packing preset, bake padding, island rotation step, and the bake passes (Albedo by default; Normal, Emission and Roughness can be baked as separate outputs) - **Preserve Backup Duplicate** — keeps a copy of the originals - **Accept — Delete Backup** — keeps the result and deletes the backup - **Revert — Restore Originals** — goes back to the original materials - **Copy Debug Report** — copies diagnostics to the clipboard for a bug report

VRM / MToon materials keep their transparency: the alpha mode is respected and checked against the real texture alpha, emission does not blow out, and problem tiles are flagged in magenta after baking.

Lineage credits: this workflow continues the CATS integration lineage, credits the original Cats Blender Plugin, credits Grim-es/material-combiner-addon for the Material Combiner lineage, and credits UV Toolkit by Alexander Belyakov plus the oRazeD/UVToolkit archival repository for UV workflow inspiration. BoneCraft implements its own atlas UV helpers instead of including UVToolkit source.

Visemes

Pipeline phase: 3

What it does: Creates the 15 VRChat visemes from three base mouth shapes.

Key controls: A Shape, **O Shape** and **CH Shape** (pick them, or **Auto-Detect**), then **Generate Visemes**. Expand Options below the buttons to choose whether to overwrite existing visemes before generating. See Guide 13, Phase 2.

Eye Tracking

Pipeline phase: 4 (run after Fix Model)

Without Fix Model first, eye detection may latch onto leftover duplicate geometry instead of the real eyes and put the constraints in empty space — avatars arrive in VRChat staring at the floor.

What it does: Finds your avatar's eye bones, names them as VRChat requires (`LeftEye` and `RightEye`), and creates the rotation constraints that move the eyes naturally in VRChat.

Key controls: - **Auto-Detect** — finds bones that match eye bone names and positions - **Left Eye** and **Right Eye** — the eye bones, if you pick them yourself - **Up Limit**, **Down Limit** and **Side Limit** — how far the eyes may turn - **Create Eye Tracking** — renames the bones and creates the constraints

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Pose & Shape Keys

Pipeline phase: 5 (optional)

Pose to Shape Key

Turns the current pose into a new shape key — for custom expressions or a corrected rest shape. Pose the armature in Pose Mode, return to Object Mode, then click **Pose to Shape Key** and name the shape key.

Shape Key to Basis

The reverse: bakes a shape key into the mesh's neutral rest shape. Use it to lock in a corrected rest pose for good. Run it after Fix Model too.

Transforms & FBT

Pipeline phase: 6 (last)

Apply All Transforms

Applies position, rotation and scale to the mesh and the armature together, so everything reads as clean values. The VRChat SDK needs this: un-applied scale makes avatars the wrong size in VRChat.

Fix FBT

A transform correction for Full Body Tracking setups: moves the root bone to floor level.

Remove FBT

Removes the Fix FBT correction again.

Bone Tools

- **Create Root Bone** — adds a root bone (named by **Root Name**) at the bottom of the hierarchy
- **Merge Bones** — merges bones shorter than the **Merge Threshold** into their parents
- **Duplicate Bones** — duplicates the selected bones

Armature Tools

What it does: Merges one armature into another, for example an outfit's armature into the body's.

Key controls: - **Base / Root** — the armature that stays (the result keeps its name) - **Incoming** — the armature that merges into it; bones with the same name fold into the base - Naming Prep — makes the two rigs' bone names match before merging. It works on any rig, not just Mixamo imports: choose Mixamo Prefixed and click **Normalize Merge Names**, then choose Mixamo Stripped and normalize again; both rigs end with the same names. - Options to parent the bones automatically and to join the meshes afterwards - **Merge Incoming Into Base** — merges

For a staged merge with previews, use Bone Merge (see Rig Builder).

Mesh Tools

- **Separate by Materials** — one mesh per material
- **Separate by Loose Parts** — one mesh per connected piece
- **Separate by Shape Keys** — splits off the parts that shape keys move

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Smart Automatic Weights

Smart Automatic Weights opens a dialog: check the target mesh and armature, then choose Body to recalculate from the existing rig or Clothing to transfer from a weighted source mesh. The dialog uses smart defaults and keeps a hidden backup. **Advanced Recalculation** expands a location hint for the full controls in BoneCraft → Skin → Weight Tools; it does not open those controls here. See Rig Tools.

Pipeline Status and Operation Ledger

Pipeline Status shows which of the six phases are done. The Operation Ledger lists every CATS operation with its time and outcome (Changed, Clean or Failed); **Copy** puts it on the clipboard and **Clear** empties it. See The CATS Pipeline.

Next: Quick Fixes and Glossary

Quick Fixes and Glossary

← Contents

Quick-Fix Index

Use this section when something has gone wrong and you need the answer fast.

Problem	Where to look
Upload fails — bones not recognized	Guide 4 + VRChat Naming Extras
Avatar in T-pose / doesn't track movement	Guide 5 + Set Up Humanoid Bones
Mesh deforming weirdly / skin stretching	Weight Transfer + Weight Mirror, or recalculate with Automatic Bone Weight Recalculation
One side of the body has different weights	Weight Mirror
Hair clips through the head	Guide 6, Step 5 — add colliders
Hair physics not moving	Guide 6 — check chain detection and the PhysBone preset
Clothing clips through the body	Guide 7 — run **Detect Collisions**
Lip sync not working	Guide 8 + VRChat Viseme Mapping
Avatar is Very Poor performance	Guide 9 + Check Performance
VRM import fails	Guide 2, Step 1 — install the VRM add-on
MMD import fails	Guide 3, Step 1 — install mmd_tools
Rig validator shows errors	Rig Validator — run validation and follow the messages
Bones vanished / can't see any bones	Bone Collections → **Show All Collections**
Can't find BoneCraft's panels	BoneCraft and Rig Builder are on the left Tool Shelf (press T), CATS on the right sidebar (press N) — see Where things are
I want a complete Mechanisms example	Tutorial: Crescent Rose — 8 preparation steps and 30 Mechanisms steps with screenshots, and the finished file in `examples/crescent-rose/`
A tab is on the wrong side	Preferences → Add-ons → BoneCraft → Tabs → Left or Right
BoneCraft's sections are missing from Properties → Scene	Preferences → Add-ons → BoneCraft → Tabs → Scene Properties → Show; blue means on. Also check the tool packages in Tools
I want to hide BoneCraft in Properties → Scene	Turn off Scene Properties → Show in Preferences → Tabs. It leaves the viewport tabs and Armature data tab alone
A tool runs before I can choose its options	In 9.1, adjust its options immediately afterwards in the viewport's Adjust Last Operation panel, before running another operation

Problem	Where to look
Place Rig Markers does not place a marker	Start the Auto-Rig Wizard, go to a marker step, use Object Mode and click the wizard's selected mesh — see Rig Builder
Build Mechanism does not know which parts to use	Add Parts from Selection first, select the mechanism rig or one of its parts, choose the type in the tool header, then click the first part — see Mechanisms
Export blocked: humanoid bones reported missing although they exist	Run **Auto-Map All**, then **Fix Humanoid Map** in the VRM / VRoid section's lint. If a required bone really does not exist, the rig needs it first: see Guide 5
Exported FBX shows giant helper shapes or tubes	Keep **Helper Meshes** off unless you need control or custom-shape meshes
Unity or Unreal imports grey materials	Export with **Embed Textures** on, then use Unity's material import/extract options or Unreal's FBX material import
Weights are all on the wrong bones	Re-run the wizard's Generate Rig, use Weight Transfer, or **Smart Automatic Weights** in the CATS tab
Two rigs need to be one	Guide 11: Merge Two Rigs Together
Corrective shape not switching on	Check the bone axis and angle in Corrective Shape Keys
Animation looks wrong on a different rig	Retarget — check the bone mapping
A CATS tool warns "Run Fix Model first"	Run Fix Model — The CATS Pipeline
Mouth tears apart or splits when speaking	Fix Model was skipped — re-run from Phase 1 — Guide 13, Phase 1
Avatar's eyes stuck looking down in VRChat	Eye tracking ran before Fix Model — re-run from Fix Model — CATS: Eye Tracking
Shape key makes the mesh explode when triggered	Pose to Shape Key ran before Fix Model — re-run from Fix Model — CATS: Pose & Shape Keys
Avatar spawns at the wrong size in VRChat	Transforms were applied before the pipeline was done — re-import from the source file and restart from Fix Model
Bone names are Japanese / Chinese / Korean	CATS: Translate Bone Names
Avatar has no lip sync and no mouth shape keys	CATS: Visemes — generate 15 visemes from 3 base shapes
A mechanism shows "changed by hand"	Mechanisms — use **Keep mine** or **Restore**

Glossary

Adjust Last Operation: The viewport panel for changing the options of the operation you just ran. Many BoneCraft 9.1 tools run first and take their options here; adjust them before running another operation.

Armature: Bforartists' word for a skeleton: a set of bones that can be posed and animated.

Blend shape: See Shape key.

Bone: One segment of a skeleton. Bones form a hierarchy (parent → child) in which child bones follow their parents.

Bone collection: A named group of bones. You can show or hide a whole collection at once.

CATS: The CATS tools: model preparation for VRChat with a guided pipeline (cleanup, visemes, eye tracking, transforms). They are in the CATS tab.

CATS pipeline: The six-phase order of the CATS tools: Fix Model → Material Atlas → Visemes → Eye Tracking → Pose to Shape → Apply Transforms. Tools warn when an earlier phase they depend on is not done.

Corrective shape key: A shape key that switches on when a bone reaches an angle, to fix the mesh at extreme poses.

Deform bone: A bone that moves the mesh. Not every bone deforms; some are only controls.

Eye tracking: The CATS tool that sets up the eye bones (`LeftEye`, `RightEye`) and the rotation constraints VRChat uses to move the eyes.

FBT (Full Body Tracking): A VRChat feature that tracks your whole body with extra hardware (such as Vive Trackers), including hips and feet. Fix FBT adjusts the root bone for it.

FBX: A file format that carries 3D models, skeletons and animation between programs. The standard format for VRChat.

Fix Model: The CATS cleanup: duplicate vertices, loose geometry and bad normals. Always the first CATS step.

FK (Forward Kinematics): Posing by rotating each bone of a chain in turn: shoulder, then elbow, then wrist. Natural for broad body poses.

IK (Inverse Kinematics): Posing by placing the end of a chain (a hand or foot); the bones in between follow. Natural for precise hand and foot placement.

Humanoid: VRChat's avatar system that maps bones to standard body slots, so every avatar uses the same movement controls.

Mechanism: A rig for a prop or machine part: a joint (Hinge, Slide, Telescope) or a link (Piston, Gear Link, Follow), built from ordinary bones and constraints. See Mechanisms.

Mesh: The 3D surface that makes up your avatar's visible body.

MMD (MikuMikuDance): Free 3D animation software popular in Japan and the anime community. Uses `.pmx` model files and `.vmd` animation files.

Morph target: See Shape key.

Operation Ledger: The CATS list of every operation you ran, with its time and outcome.

PhysBone: VRChat's component that makes bones swing, bounce and collide. Used for hair, tails and dangling accessories.

Pipeline Status: The CATS list of the six phases, with a checkmark for each phase that is done.

PMX: MikuMikuDance's main 3D model format.

Pose Mode: The mode for posing and animating bones. Select the armature and press Ctrl+Tab, or use the mode dropdown in the top-left of the viewport.

Rig / rigging: Building a skeleton inside a 3D model and connecting the mesh to it so it can be posed and animated.

Shape key: A saved version of a mesh in a deformed position. Shape keys blend together at different strengths; they drive facial expressions, lip sync and body morphs.

SDK (Software Development Kit): For VRChat, the VRChat Creator Companion and its Unity tools for uploading avatars.

Spline IK: A chain of bones that follows a curve. Used for tails, tentacles, long hair and spines.

T-pose: A reference pose: standing upright with the arms out to the sides.

Tool Shelf: The panel on the left side of the 3D Viewport (press T), where the BoneCraft and Rig Builder tabs are by default.

Vertex: A single point in 3D space. Meshes are made of many vertices joined by edges and faces.

Vertex group: A named set of vertices; it defines which vertices a bone moves.

Viseme: A mouth shape for a speech sound. VRChat uses 15 visemes for lip sync.

Weight / weight painting: Values from 0.0 to 1.0 that say how strongly each bone moves each vertex. Weight painting is the visual way to adjust them.

Wizard: The Rig Builder's Auto-Rig Wizard: places markers on your mesh and builds the rig from them.

Zero-weight bone: A bone that moves no vertex. It costs performance without changing how the avatar looks; Fix Model and the performance optimizer remove them.

Next: What's New

What's New

← Contents

Newest first.

9.2.5 — Starter Rigs catalog tab

Starter Rigs now uses one catalog tab in the native bottom asset shelf. The top-left viewport controls have been removed. Enable **Starter Rigs** once in the shelf's catalog menu to show its tab beside the other catalogs. Native asset import and **Add** → **Armature** → **Starter Rig Catalog** remain available.

9.2.4 — Task Board and native tools

Open Health Check now runs rig validation and opens its results. Select an armature or its bound mesh. **Quick Wins** shows up to three recommendations; **Remaining Tasks** excludes those cards. **Refresh Task Board** checks again.

Place Rig Markers and Build Mechanism remain available as viewport tools. Their registration no longer breaks native Grab/Move or tool cycling in Object and Pose modes.

9.2.3 — VRChat export location

Removes the duplicate Export to VRChat panel from Avatar Tools. Export remains under CATS → Import / Export → Unity / VRChat.

9.2.2 — Current interface

Tab names select a workspace group. Setup Rigging, Skin, Review, Avatar Tools and Facial are expandable sections inside those tabs, not separate tabs. The paths below use the English UI labels; translated labels depend on the Language preference.

- Bone Merge → BoneCraft → Setup Rigging
- Delta Mush / Proximity Wrap → BoneCraft → Skin
- Viseme Library / Lip Sync → CATS → Avatar Tools
- VRChat Viseme Mapping / Face Tracking → CATS → VRChat Tools → Facial

9.2.0 (October 2026)

- **Complete starter catalog:** all nine bundled skeletons/metarigs appear as thumbnails from the start: Basic Humanoid, Simple Human, Full Human, Basic Quadruped, Cat, Horse, Wolf, Bird and Shark. The catalog selector groups Humans, Quadrupeds and Other Animals.
- **Rigify when needed:** Basic Humanoid works independently. Rigify starters stay visible while Rigify is off; selecting one offers **Enable Rigify & Add Rig** or Cancel. Cancellation, activation failure or an unavailable preset adds no rig.

- **Optional shelf:** the viewport header's **Starter Rigs** switch returns to the regular asset shelf. The choice persists in preferences. **Add → Armature → Starter Rig Catalog** opens the chooser as a popup.
- **Click or drag:** click to add, or drop inside the viewport. Mesh fitting starts on; weighting and control generation start off and can be changed in Adjust Last Operation. Native click, drop, cancellation and undo/redo were checked in Bforartists 5.2.0.

9.1.2 (October 2026)

- **Scene Properties switch:** Preferences > Tabs has a new Scene Properties row. Its Show button (blue while on) shows or hides all of BoneCraft's sections in Properties > Scene tab (Rig Builder, Mechanisms, CATS, Avatar Tools, VRChat Tools, Import / Export) without disabling the add-on. It starts on, and it leaves the viewport tabs and the Armature tab alone. See Where things are.

9.1.1 (October 2026)

- **Clearer icons:** each icon now means one thing everywhere. About 60 buttons got a better-fitting Bforartists icon (for example Apply All Transforms, Decimate, Normalize Names, Translate, Merge, Separate, Blend Pose and Lip Sync), and a button shows the same icon in every panel.
- **Calmer panels:** headings and fixed text no longer carry topic icons. Hints that report a state show the info, warning or check icon.
- **Readable icon grid:** every button in the Pose and Weights Tool Shelf tabs has its own icon, so a narrow shelf stays readable.

9.1.0 (October 2026)

- **Tools where Bforartists puts its own:** BoneCraft entries in the Add → Armature, Pose, Weights and Object menus, and a BoneCraft panel of buttons in Bforartists' Pose and Weights Tool Shelf tabs.
- **Act first, tune after:** Quick Rig, Fit Dropped Rig, the weight tools, Ribbon, Spline IK, Delta Mush and the CATS bone tools run at once with the values you last used; you tune them afterwards in the Adjust Last Operation panel. Fewer "are you sure" pop-ups.
- **Actions first, options folded:** several panels show their buttons first and optional settings in a fold below. Required inputs remain before the action, as do Fix Model's cleanup switches. Settings look like settings (checkboxes, never buttons), and panel headers have no icons.
- **Gesture tools:** Place Rig Markers (click the mesh to place the Auto-Rig Wizard's next marker) and Build Mechanism (click a part to start a joint or link) are in the viewport tool bar.
- **Avatar sections:** Avatar Tools holds humanoid mapping, hair, visemes and export. The separate VRChat Tools section groups Performance, Clothing and Naming utilities. The guides use these current panel names.
- **Tooltips:** every operator and every option explains itself.
- **Show, don't hide:** sections stay visible and say what to select when the rig is gone, the Auto-Rig Wizard comes first in Rig Builder, and the shelves start wide enough to read.
- **Fixes:** Generate Spline IK works again; Add Influence uses the chosen bone; viseme previews use their set.

9.0.0 (October 2026)

- **Mechanisms:** a new section of the BoneCraft tab (and of the Properties Scene tab) that rigs props and machines: Hinge, Slide, Telescope, Piston, Gear Link, Follow and Sequence, built with guided steps and checked with Test. BoneCraft never overwrites a hand edit: you choose **Keep mine** to keep it or **Restore** to rebuild. See Mechanisms.
- **Example file:** `examples/crescent-rose/` holds Crescent Rose rigged with Mechanisms (scythe, rifle and compact forms), and the manual's chapter 12 builds it step by step.
- **Tab sides:** each viewport tab can sit on the left Tool Shelf or the right sidebar (Preferences → Tabs → Left / Right). BoneCraft and Rig Builder start on the left, CATS on the right.
- **Rig Builder:** always in the Properties Scene tab, plus its own viewport tab with a compact layout for a narrow Tool Shelf. A Basic Humanoid rig ships in the BoneCraft Rigs asset library; Fit Dropped Rig fits a rig dropped from the asset shelf.
- **CATS tab:** Import / Export moved into the CATS tab. CATS, VRChat and Import / Export are also always in the Scene tab.
- **Show Helpers:** one preference shows or hides BoneCraft's teaching text; Compact Mode is its shortcut.
- **Rig Validator:** bone names with hyphens pass; drivers warn only for expressions that need Python; Mechanisms rigs are checked too. The Task Board's validator task reports again.
- **Name:** the add-on is called BoneCraft.

8.9.x (September–October 2026)

- **8.9.6:** the whole dress piece follows the skirt bones (no hip clipping); jaw-open corner correctives with sticky lips.
- **8.9.5:** IK/FK elbow switch without a pop; symmetric shoulder detection; eye markers no longer snap; the Control Picker shows the selection; rounder mouth corners; a dress-over-thigh correction; relaxed retarget collarbones; T-pose export.
- **8.9.4:** rounder knees (correctives replace Preserve Volume); contact keys for deep bends; a shoulder seam blend.
- **8.9.3:** two-segment skirt bones that follow, and are pushed by, the legs.
- **8.9.2:** blink with shape keys (no lid folds); an oval mouth opening.
- **8.9.1:** much faster skinning on dense meshes (678,000 vertices: 232 s to 57 s); bone heat through a lighter proxy mesh instead of failing silently.
- **8.9.0:** joint corrective shape keys; face controls (eye aim, blink, lip follow); clothing that bends with the joints.
- Also in 8.9: automatic bone weight recalculation, and a clearer atlas selection.

8.7–8.8 (September 2026)

- **8.8.0:** the retarget panel works on Blender 5 hosts; finger markers placed from the hand's shape; IK stretch to the skin; a Preserve Volume option; a Delta Mush bind fix.
- **8.7.0:** an integrated control rig; limb-traced joint detection; direction-matched retargeting; twist bones.

8.5–8.6 (July 2026)

- **8.6.1–8.6.3:** Material Atlas: tile diagnostics after baking; MToon transparency checked against the real texture alpha and kept during atlasing; better partitioning of mixed materials.

- **8.5.5–8.5.10:** VRM / MToon materials: the alpha mode is respected, alpha works on Blender 5 hosts, no emission blow-out, magenta diagnostics and safe atlas names.
- **8.5.4:** humanoid mapping reads back correctly; Naming Prep guidance.
- **8.5.2–8.5.3:** import a VRoid download (.zip); a VRoid project hand-off card.
- **8.5.1:** FBX export that keeps blend shapes.

Earlier versions

8.5.0: Smart Combine now makes `atlas_uv` the export UV0 by default after baking. The pre-atlas source UV map is removed from the generated atlas mesh unless **Keep Source UV Maps** is enabled in Advanced settings. The CATS / Material Combiner / UVToolkit-derived controls are now shared with the open Blender build; B4Artists exclusivity remains on the production rigging, control, retarget/export, and B4Artists-only release systems.

8.5.0 also updates the export and validation workflow. VRChat, VRM, MMD, and Unreal exports now expose folder and file-name controls in the BoneCraft panels instead of forcing you to rely only on Blender's file browser defaults. VRChat/Unity and Unreal FBX exports default to **Embed Textures** for easier material import, while VRChat export keeps helper/control-shape meshes out of the FBX unless **Helper Meshes** is enabled. The VRM panel now includes **Lint Now** plus **Fix Humanoid Map**, which can repair stale humanoid alias mapping before export without renaming your bones.

8.4.5: Smart Combine now forces source textures to sample the preserved pre-atlas UV during baking, validates the generated atlas mesh before hiding originals, and resets all atlas faces to the final atlas material slot to prevent gray or scrambled robe/clothing chunks.

8.4.4: The Material Atlas Combiner adds quality-parity inspection controls before baking: shader/material diagnostics, texture role labels, duplicate/shared source markers, per-material size overrides, packing presets, optional Normal/Emission/Roughness output bakes, and preflight blockers for Metallic/ORM channel packing until that path is verified.

8.4.3: The Material Atlas Combiner includes selectable UV packing methods inside the existing Material Atlas workflow. It includes Smart Pack, Grid Pack, Seeded Variation, Oriented Pack, Fit 0-1 Bounds, Advanced Variation, and Rotation Step, with UV method details recorded in the debug report and bake ledger.

8.4.2: The Material Atlas Combiner now targets the generated atlas UV during bake while copied source texture nodes explicitly read the pre-atlas UV. This fixes broken robe/clothing atlas results caused by source textures and the baked atlas sampling different UV maps. The selectable material and texture rows from 8.4.1 remain available.

7.1.3 (preference label cleanup): Two add-on preference toggles were renamed so they now match the sidebar tab they control. "VRChat Avatar Tools" is now labelled **CATS** (matches the CATS sidebar tab). "Task Board & Sidebar" is now labelled **Rig Builder** (matches the Rig Builder sidebar tab). No tools were removed — only the on/off labels in Edit > Preferences > Add-ons > BoneCraft changed.

7.1.1: BoneCraft now includes the CATS tools — a complete suite of model preparation tools specifically designed for getting VRChat avatars clean, optimized, and fully configured. CATS lives in its own sidebar tab and uses a pipeline system that guides you through the right order of operations every time.

Next: Tutorial: Crescent Rose with Mechanisms

Tutorial: Crescent Rose with Mechanisms

← Contents

What you will build

Crescent Rose is the scythe that is also a rifle. In this tutorial you rig a model of it with Mechanisms, so that one slider turns the scythe into the rifle and a second slider folds the rifle into its compact form. On the way you use every joint and link Mechanisms has today except Gear Link: eight Hinges, three Slides, a Telescope, a Piston and a Follow, then two Sequences and three Forms.

The forms are teaching poses for this fan-made model. Mechanisms does not detect collisions or make a physically engineered replica: inspect clearances when adapting the rig.

The result is the file `examples/crescent-rose/CrescentRose_Mechanisms.blend`. Open it any time to compare.

Before you start

- The files: `examples/crescent-rose/CrescentRose_parts.blend` is the model already split into 15 part objects; start the tutorial from it. `CrescentRose_Mechanisms.blend` is the finished rig.
- Time: allow a separate session for FBX preparation and another for the Mechanisms lesson; these are substantial exercises, not timed speed runs. Save checkpoints between stages.
- Words: read "What Mechanisms does" in Mechanisms first: it defines part, joint, link, handle, Sequence and Form.
- Language: the screenshots show Bforartists in English with BoneCraft's panels in English. BoneCraft's own panels follow the **Language** setting in its preferences; the rest of Bforartists follows Bforartists' interface language. If your Bforartists is in a language BoneCraft does not translate, its panels stay English and this edition still applies.
- Starting route: use P1-P8 below for the original FBX, or skip to Step 1 with the prepared file. Both routes use the same 15 parts.
- Version: instructions and genuine interface captures were checked with BoneCraft `9.1.2` in Bforartists `5.2.0`, using its default keymap.

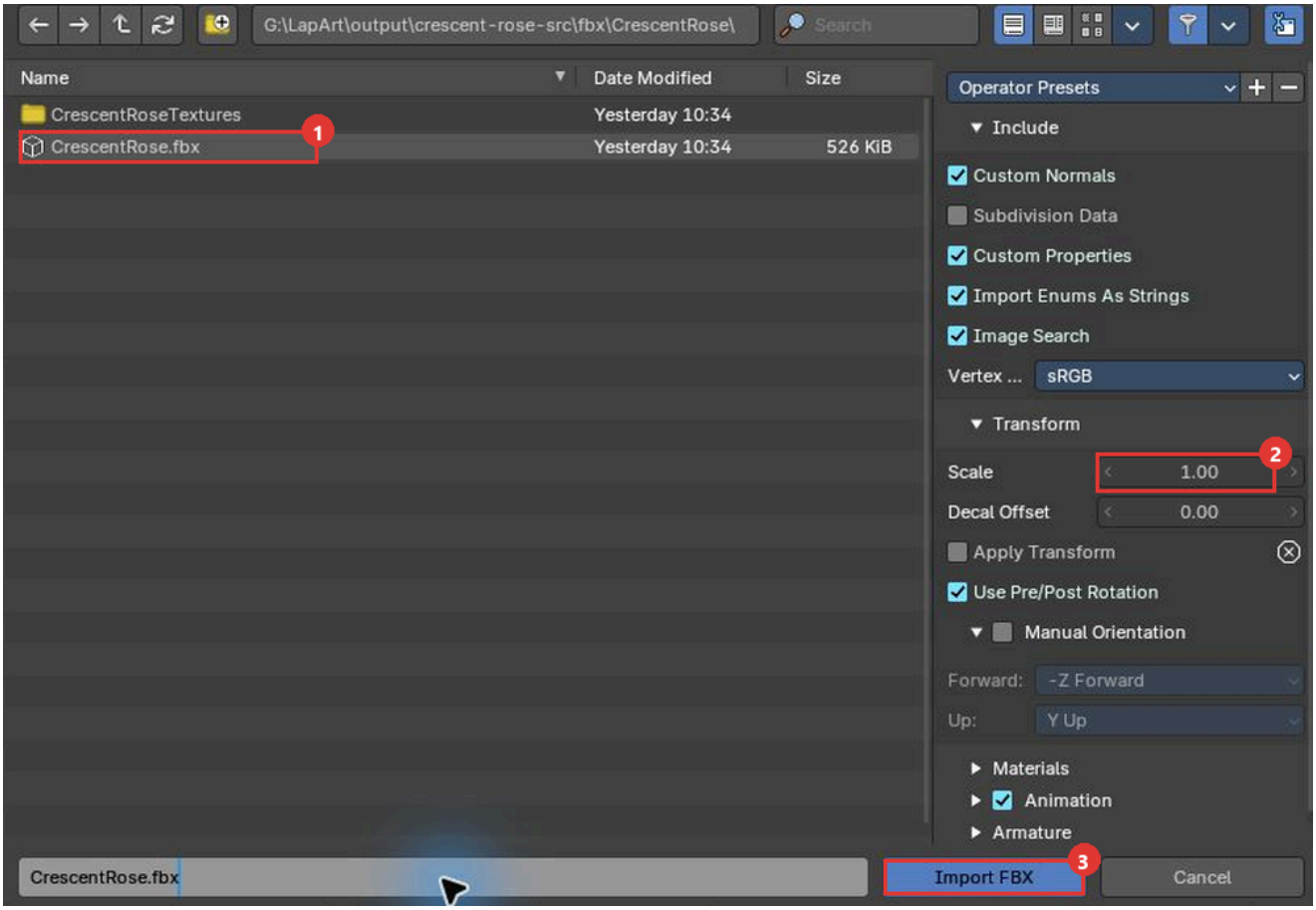
Red numbered marks identify what to click, type into, or inspect. Follow each image's numbered action key in order (1, then 2, then 3). Numbers restart in each image. A boxed result is a check; a circled part or button is the interaction target.

Prepare the original FBX (P1-P8)

Skip this section if you are using `CrescentRose_parts.blend`. Work in a new file and keep the source FBX and its `CrescentRoseTextures` folder together. These steps were replayed in Bforartists 5.2.0 with the default keymap. Use the visible mode menu: Tab can toggle overlays in this keymap rather than enter Edit Mode.

P1 - Import the FBX

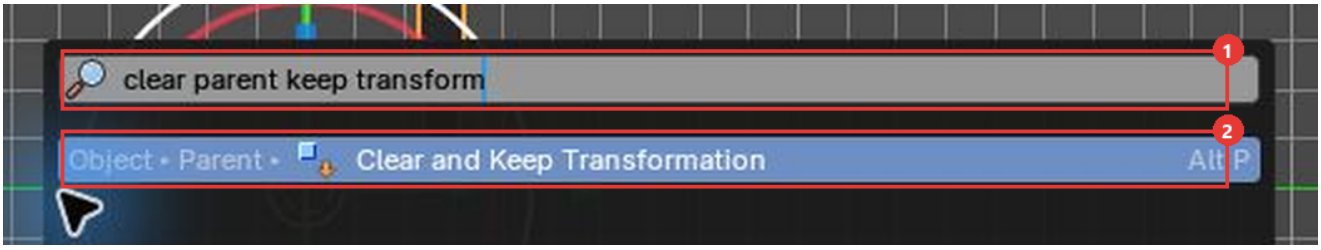
Start with an empty scene. Choose **File > Import > FBX (.fbx) (Legacy)**, select `CrescentRose.fbx`, and leave import Scale at `1.00`. Use the Legacy importer for this recipe. The Outliner contains a root and three material meshes: `Black01`, `Blades01`, and `Red01`. Press Numpad 3 for Right view and Home to frame them.



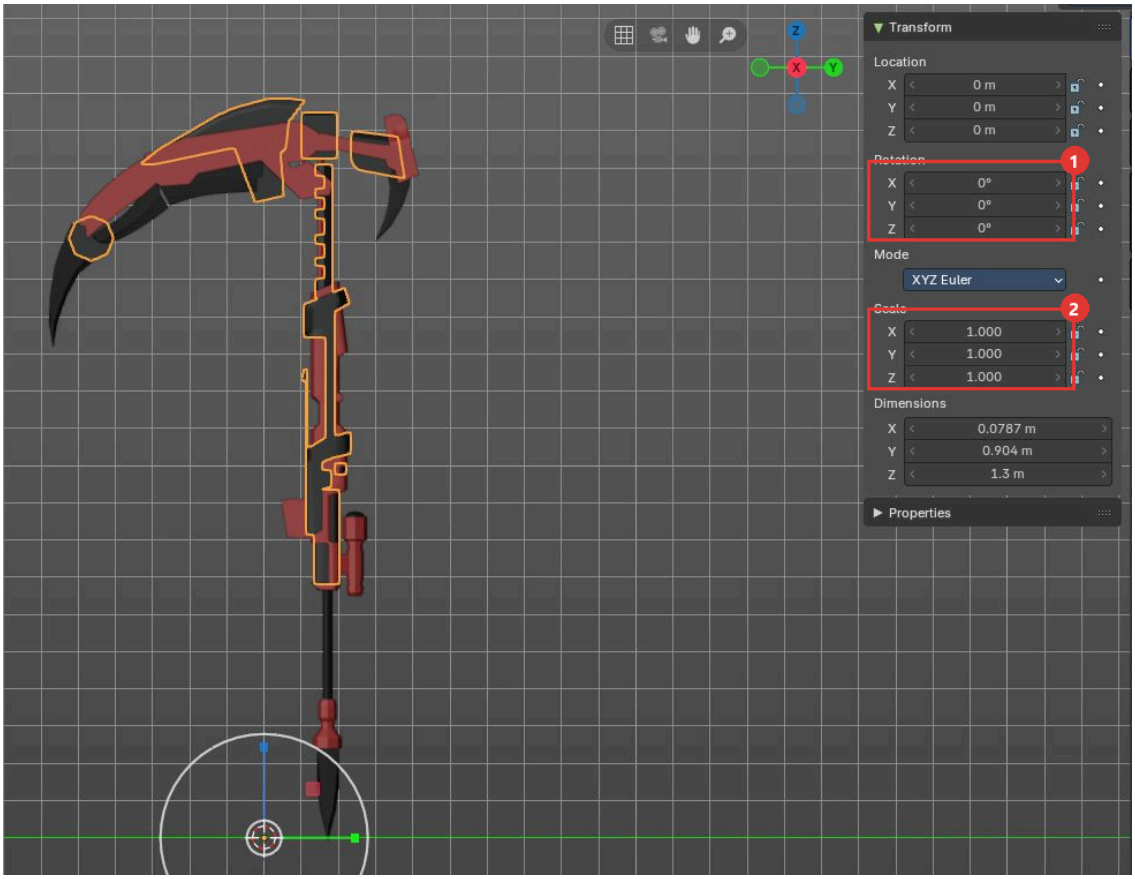
1: Select CrescentRose.fbx. 2: Keep Scale at 1.00. 3: Click Import FBX; use the Legacy importer.

P2 - Preserve placement and set the size

In Object Mode select the three meshes. Use **Object > Parent > Clear and Keep Transformation**. Remove the now-unused root from your working scene. With all three meshes selected, use **Object > Apply > Rotation & Scale**. The imported weapon is approximately `0.32 m` tall. Scale all three together by `6.25` (in this Bforartists keymap, Ctrl+R starts Scale; type 6.25 and confirm), then apply Rotation & Scale again. The complete weapon is now approximately `1.997 m` tall. Do this once; the prepared shortcut is already scaled.



1: Press F3 and search Clear Parent Keep Transform. 2: Choose Clear and Keep Transformation. Repeat for all three meshes before removing the root.

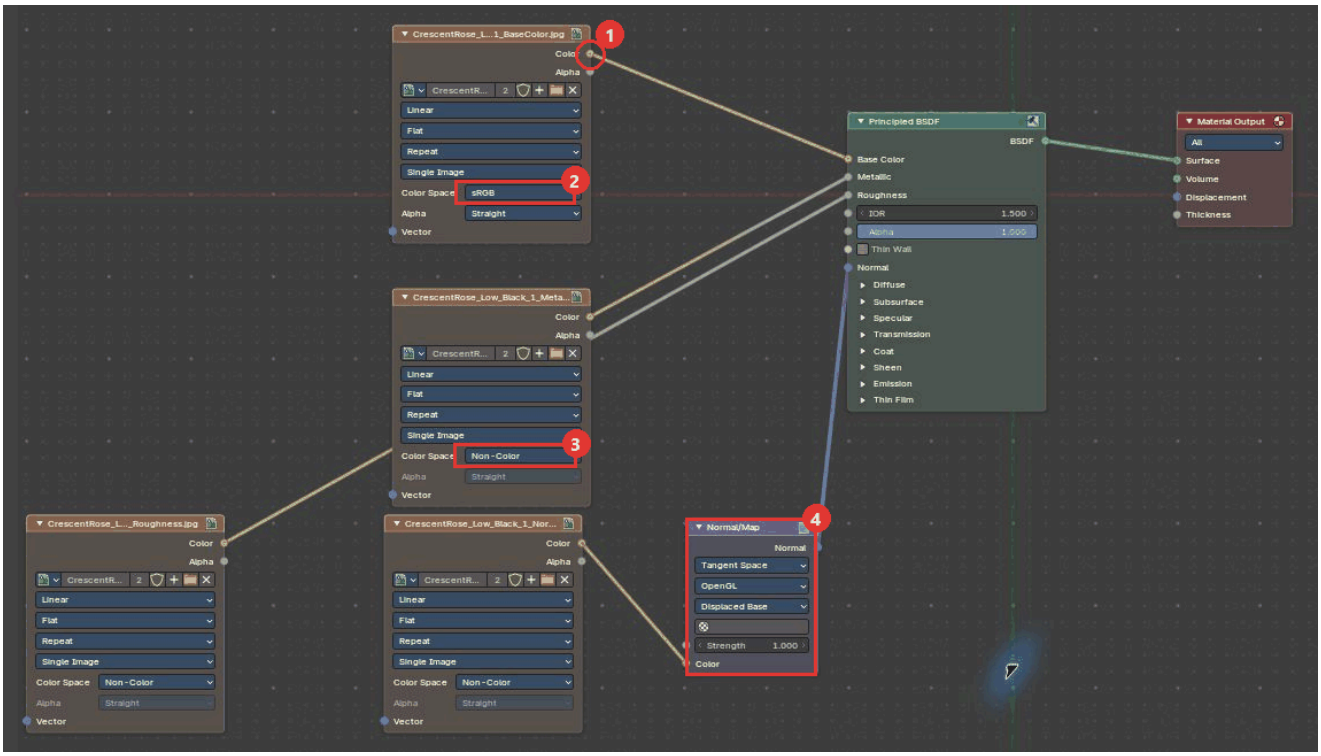


1: After scaling the three meshes together by 6.25, apply Rotation & Scale; Rotation must read 0. 2: Scale must read 1 on all axes.

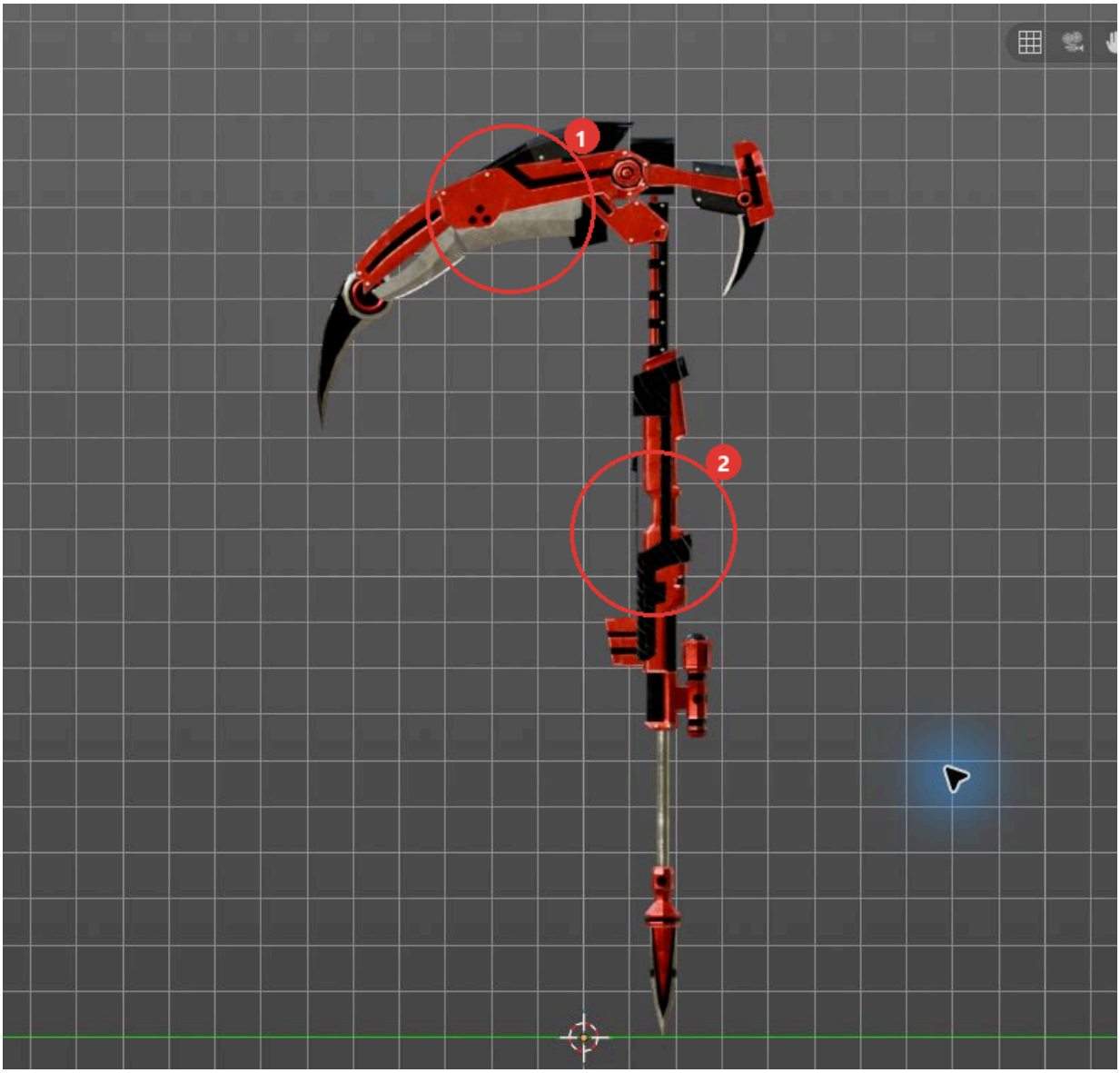
An individual object's Dimensions field measures only that material mesh. Check the combined bottom-to-top extent when measuring the weapon. All three objects must show rotation zero and scale 1 before separation.

P3 - Connect the materials

Switch to the Shading workspace and Material Preview. For each of the three materials, load its matching `BaseColor`, `Metallic`, `Roughness`, and `Normal` images from `CrescentRoseTextures`. Connect Image Texture Color to the matching Principled BSDF input. BaseColor uses **sRGB**; Metallic, Roughness, and Normal use `Non-Color`. For Normal, insert a **Normal Map** node, keep `Tangent Space`, connect image Color to its Color, and connect its Normal output to Principled BSDF Normal. Keep the existing UVs.



1: Connect BaseColor Color to BSDF Base Color. 2: Keep BaseColor on sRGB. 3: Set all data maps to Non-Color. 4: Connect Normal image > Normal Map > BSDF Normal, using Tangent Space.

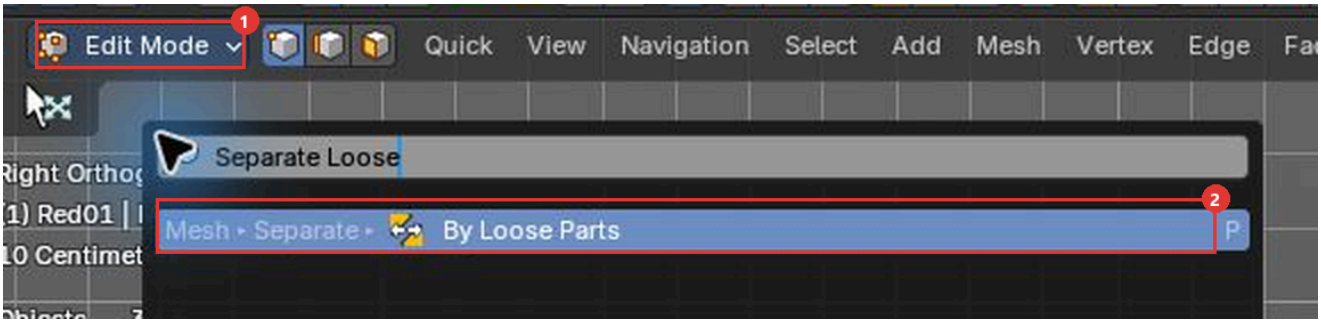


1: Check the blade material. 2: Check the red/black body materials in Material Preview; no pink missing textures.

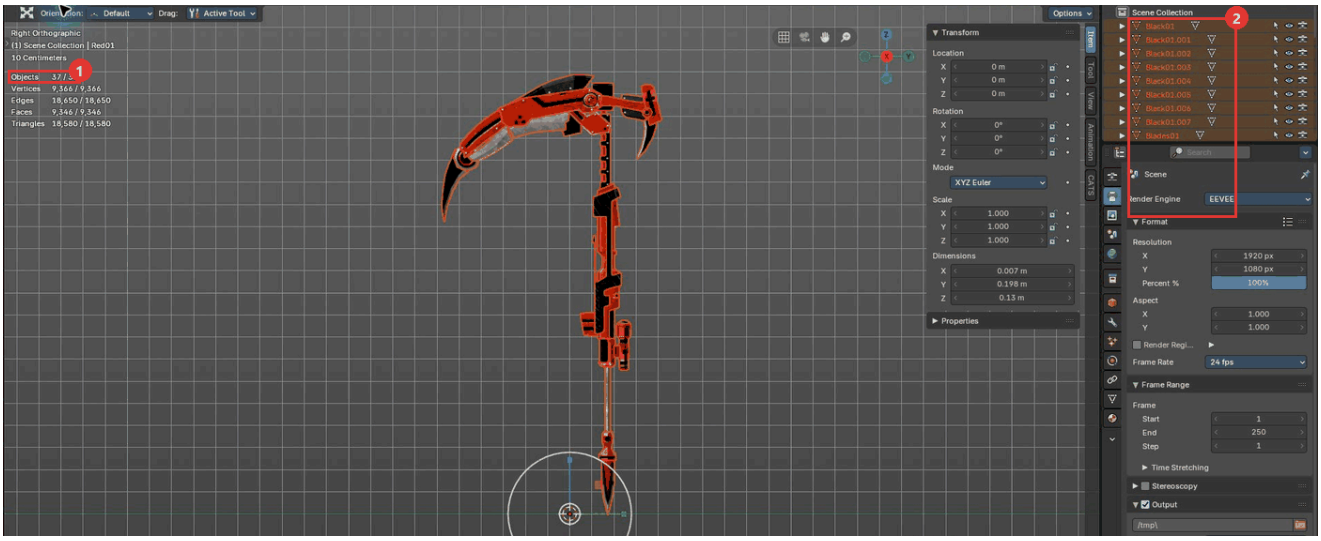
Repeat the same connections for the blade and red materials. Check each image's color space after choosing it; the setting belongs to the image. The weapon should be red, black and metallic, with no pink missing textures.

P4 - Separate loose pieces

Select all three material meshes, choose **Edit Mode** from the viewport mode menu, select all geometry, then use **Mesh > Separate > By Loose Parts**. Return to Object Mode. There are **37** mesh objects at this point. Their generated numeric suffixes can vary, so identify them by position and shape.



1: Switch to Edit Mode and select all geometry. 2: Press F3, search Separate Loose, and choose By Loose Parts.



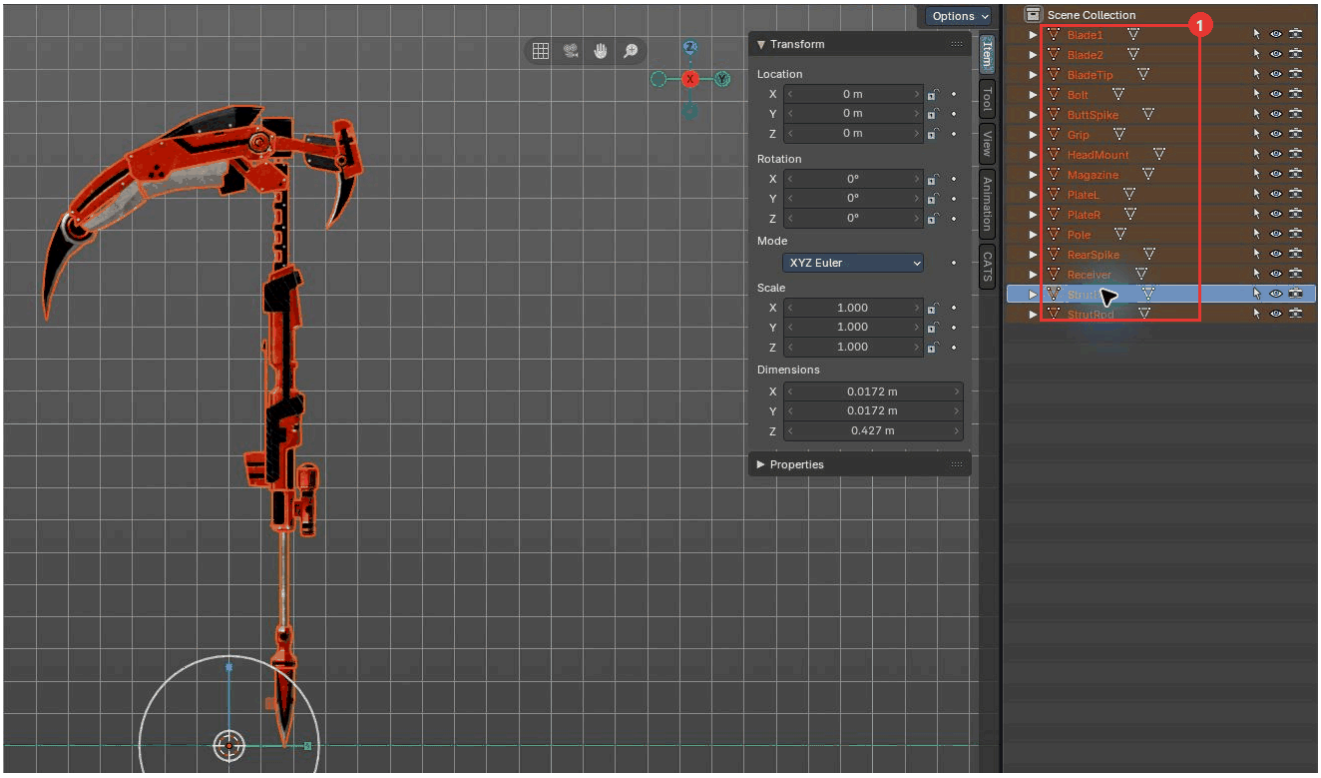
1: Check the object total is 37 after separation. 2: Use the Outliner to select pieces by shape and location; suffixes can vary.

P5 - Join the 15 moving groups

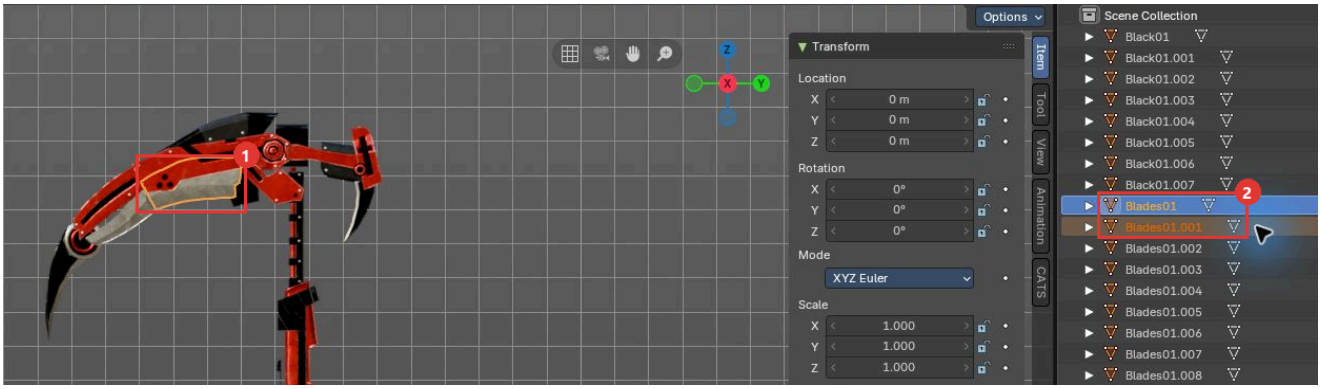
Select the pieces that must move together and use **Object > Join** (Ctrl+J). Rename the resulting object in the Outliner. For paired pieces on opposite sides, orbit the view to select both; check the selection before joining. Work from the blade tip inward, then the receiver and pole. The table lists every source island; a material name describes its appearance, not its final object name.

Final object	Pieces to combine	Mesh vertices after joining
`BladeTip`	Black tip-joint knob and metal tip spike	252
`Blade2`	Two metal second-segment plates and two red covers	1000
`Blade1`	Two metal first-segment plates nearest the head	512
`HeadMount`	Black rear block, top cap and core; two red inner plates and two thin top covers	1395
`PlateL`	One red outer head plate on negative X	676
`PlateR`	One red outer head plate on positive X	676
`RearSpike`	Two metal spike blades, two red pivot caps and two red mount plates	1194
`Receiver`	Black body and small knob, metal ring above the bolt, long red housing	2609
`Grip`	Red grip block below the receiver	194
`Magazine`	Black tube beneath the body	96
`Bolt`	Red bolt cylinder and its two knob pieces	362
`Pole`	Metal lower shaft and red collar	148
`ButtSpike`	Metal bottom spike	110
`StrutBase`	Thin red rod between receiver and head	34
`StrutRod`	Black notched sleeve around that rod	108

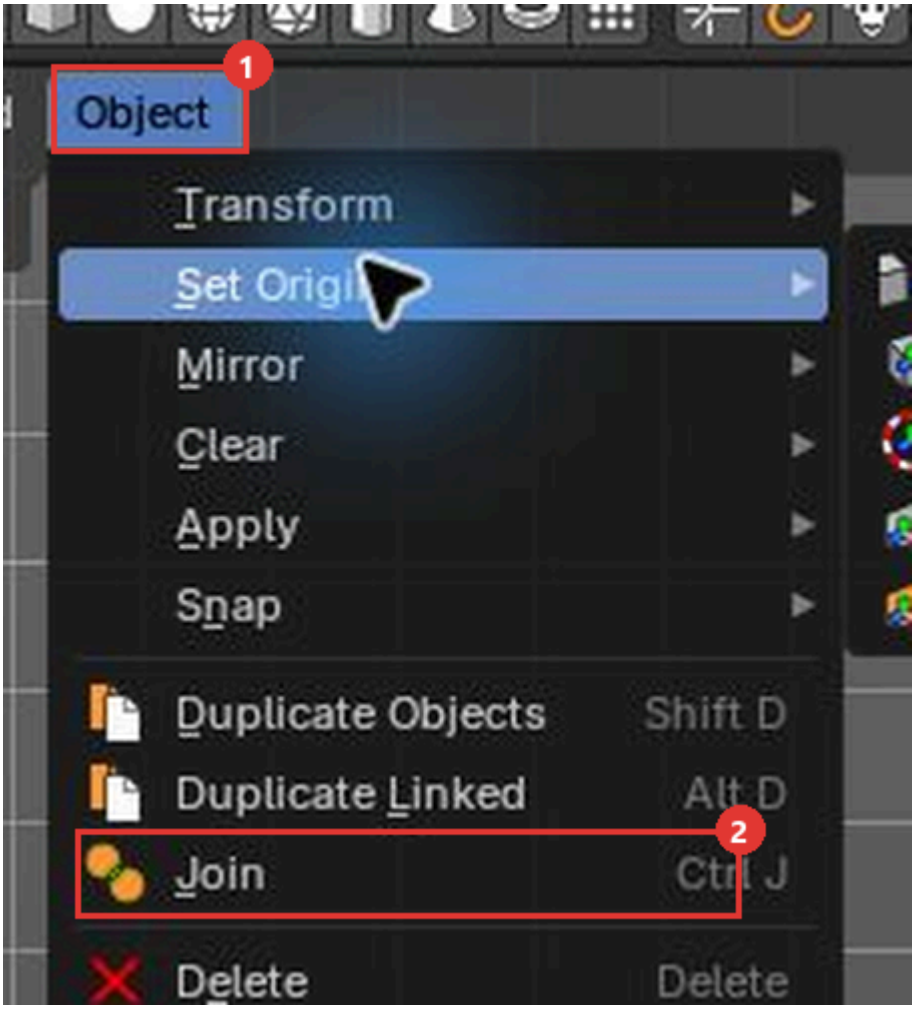
There must be **15 meshes and 9,366 vertices** in total. Do not merge the outer plates into Blade1, or the two strut pieces together. The strut already has two separate pieces; no bisect is needed. Step 2 repeats the finished part names and their later parenting.



1: Rename every joined group in the Outliner to match the 15 names in the table.



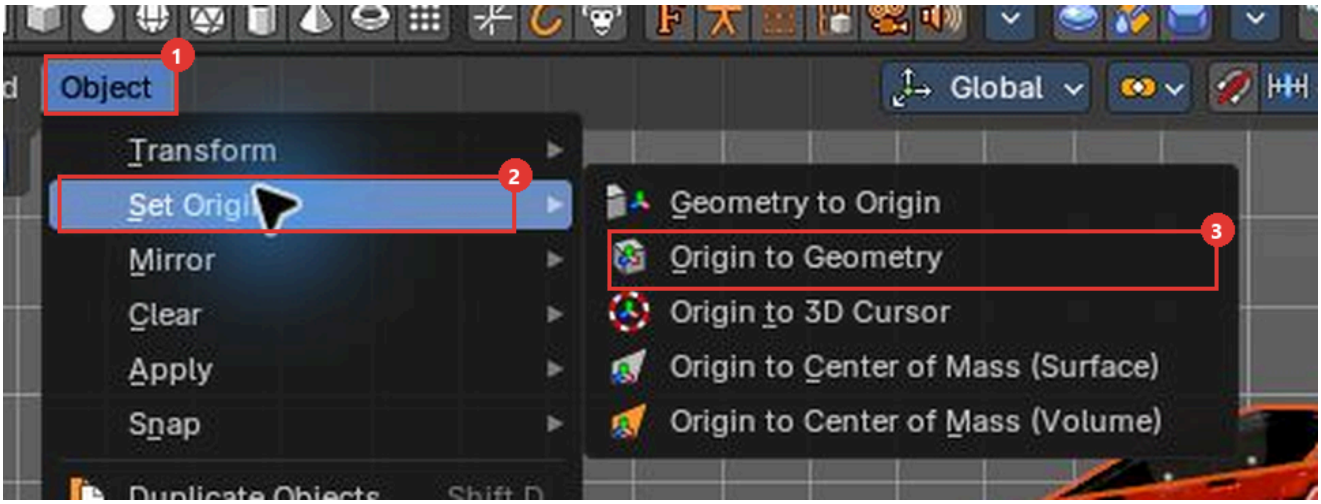
1: Select the two metal Blade1 plates on opposite sides. 2: Verify both objects are selected, then use Join and rename the result Blade1.



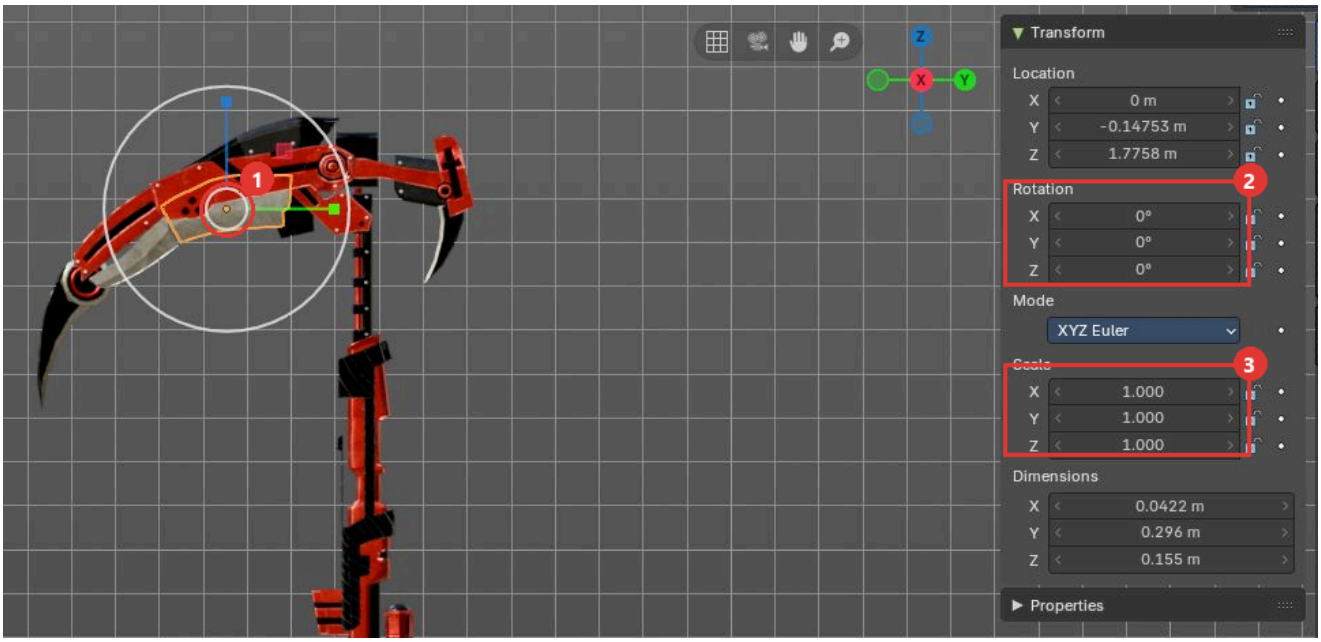
1: Open Object after selecting only the pieces of one group. 2: Choose Join (Ctrl+J). This menu crop identifies the command; use the two-piece selection shown above.

P6 - Centre the origins

In Object Mode select all 15 meshes. Choose **Object > Set Origin > Origin to Geometry** and set the operation's Center option to `Bounds Center`. Check rotation zero and scale 1. The objects still have no parents and there is no armature. These are convenient preparation origins; Step 6 later moves the mechanism pivots to their actual joints.



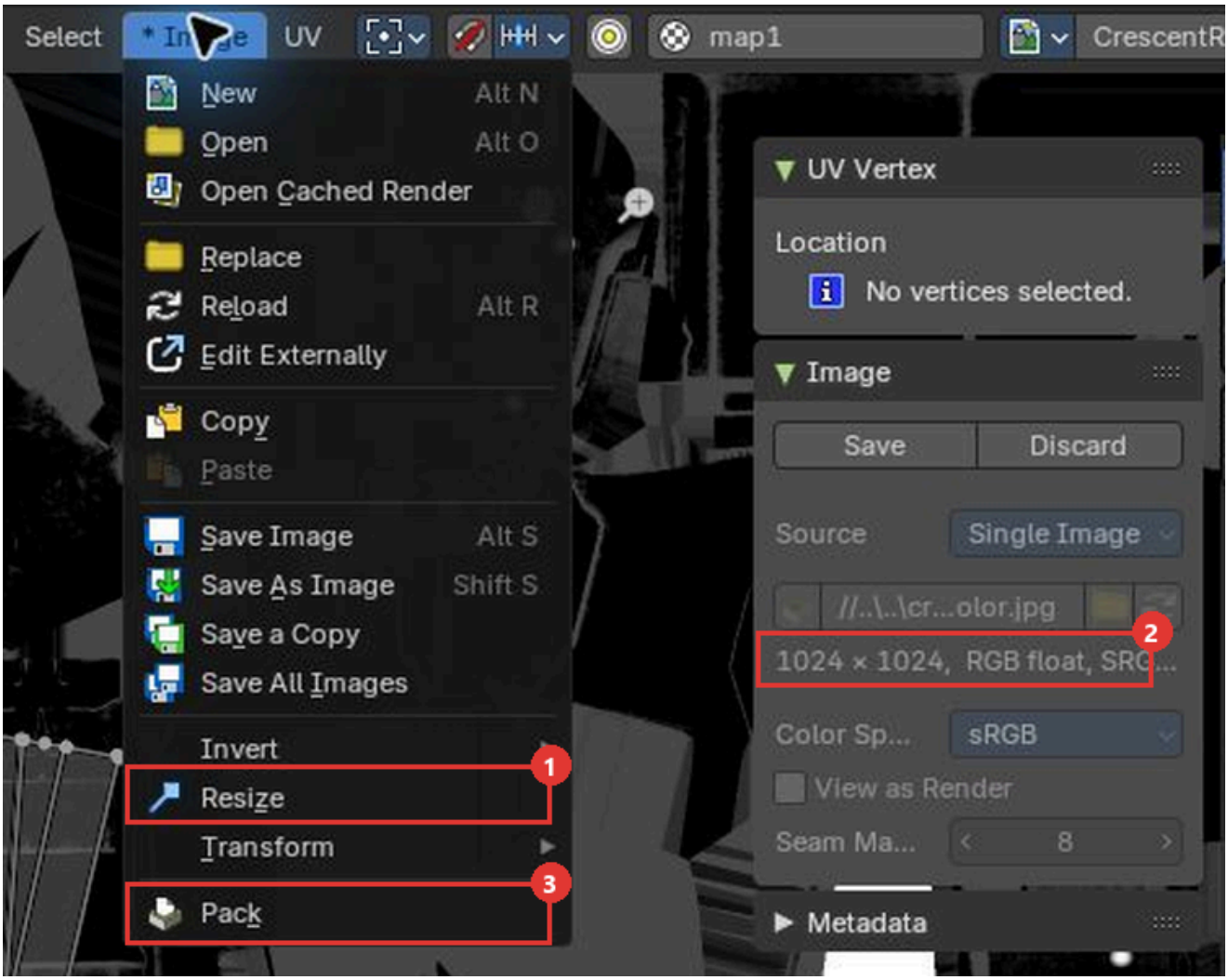
1: Open Object. 2: Hover Set Origin. 3: Choose Origin to Geometry; set Bounds Center in the operation options.



1: The orange origin is now centred on the mesh. 2: Confirm Rotation 0. 3: Confirm Scale 1. Mechanism pivots are moved later.

P7 - Resize and pack the textures

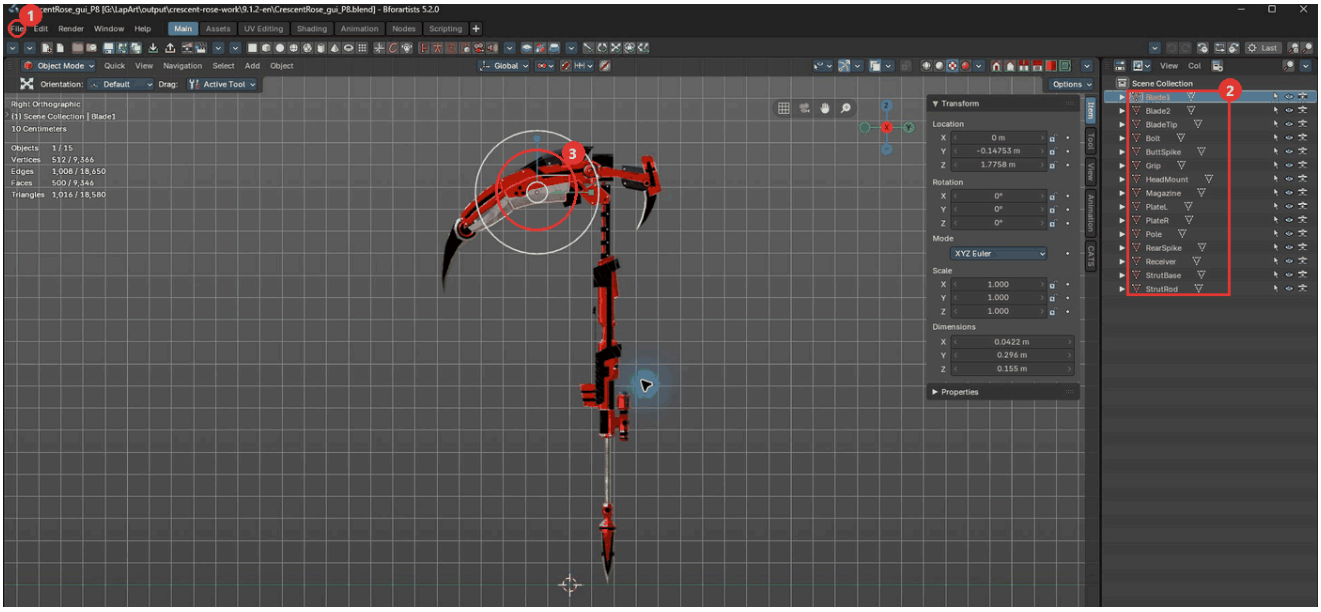
In the Image Editor select each of the **12 used maps** (four per material). Choose **Image > Resize**, set width and height to **1024**, confirm, then choose **Image > Pack**. Repeat for every used map. Packing the resized image preserves those pixels inside the blend; do not overwrite the source JPEGs. Check Material Preview afterward.



1: Choose Image > Resize and enter 1024 for width and height. 2: Check the resulting 1024 x 1024 dimensions. 3: Choose Pack. Repeat for all 12 used maps.

P8 - Save and reopen the prepared file

Save your working copy as `CrescentRose_parts.blend`. Reopen it and check the named 15 meshes, their transforms, and the red/black materials. All 12 maps should remain available without relying on the original texture directory. Use Right view and Home again, then continue with Step 1.

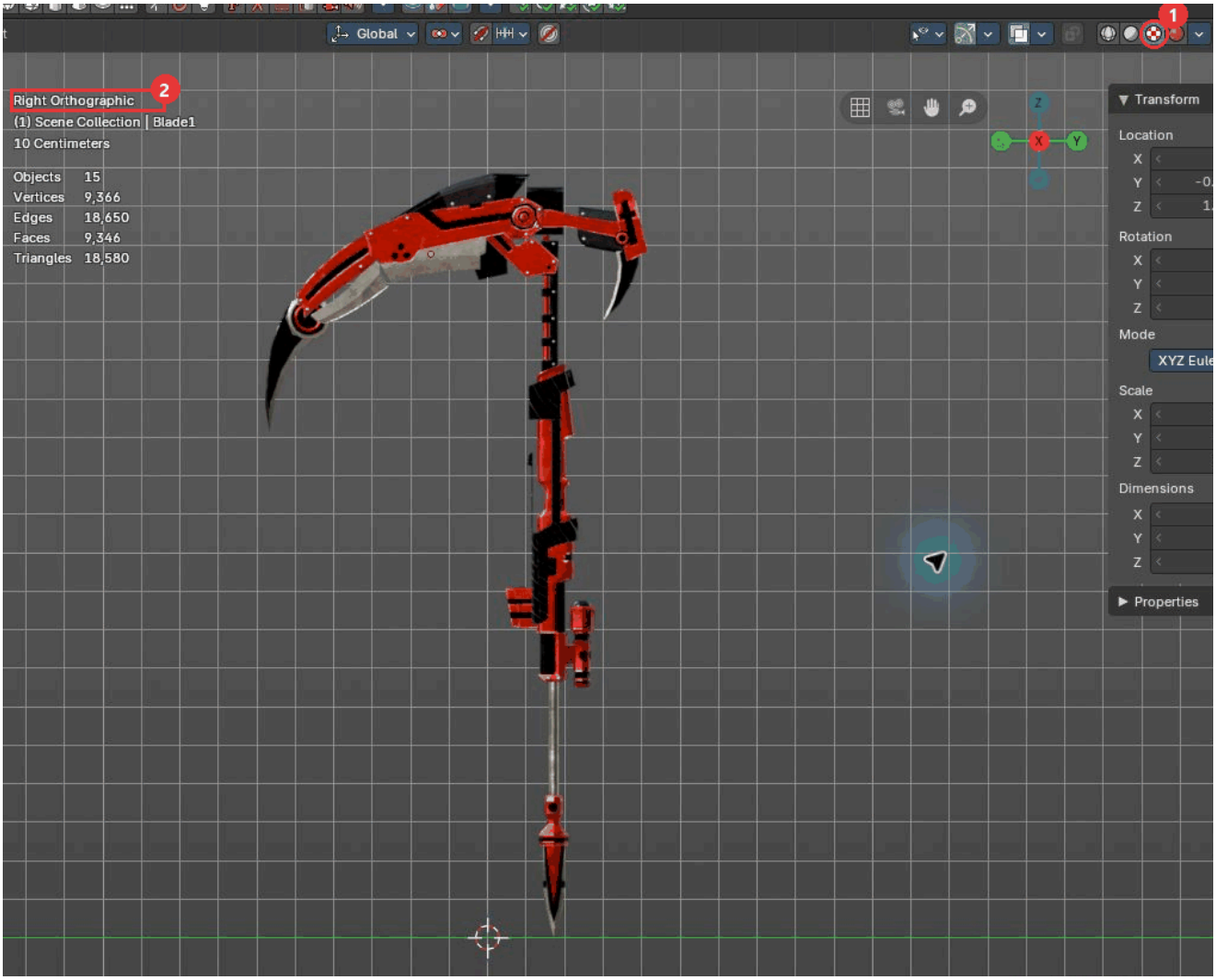


1: Open File for Save As, then Open to reload your working copy. 2: Confirm the 15 named meshes survived. 3: Check the materials; the title bar identifies the reopened file.

A. Open the file

Step 1 — Open the parts file

Open `examples/crescent-rose/CrescentRose_parts.blend`. Switch the viewport to Material Preview shading, and press Numpad 3 for the Right view: the blade points left, the butt spike down. Press Home to frame the weapon.



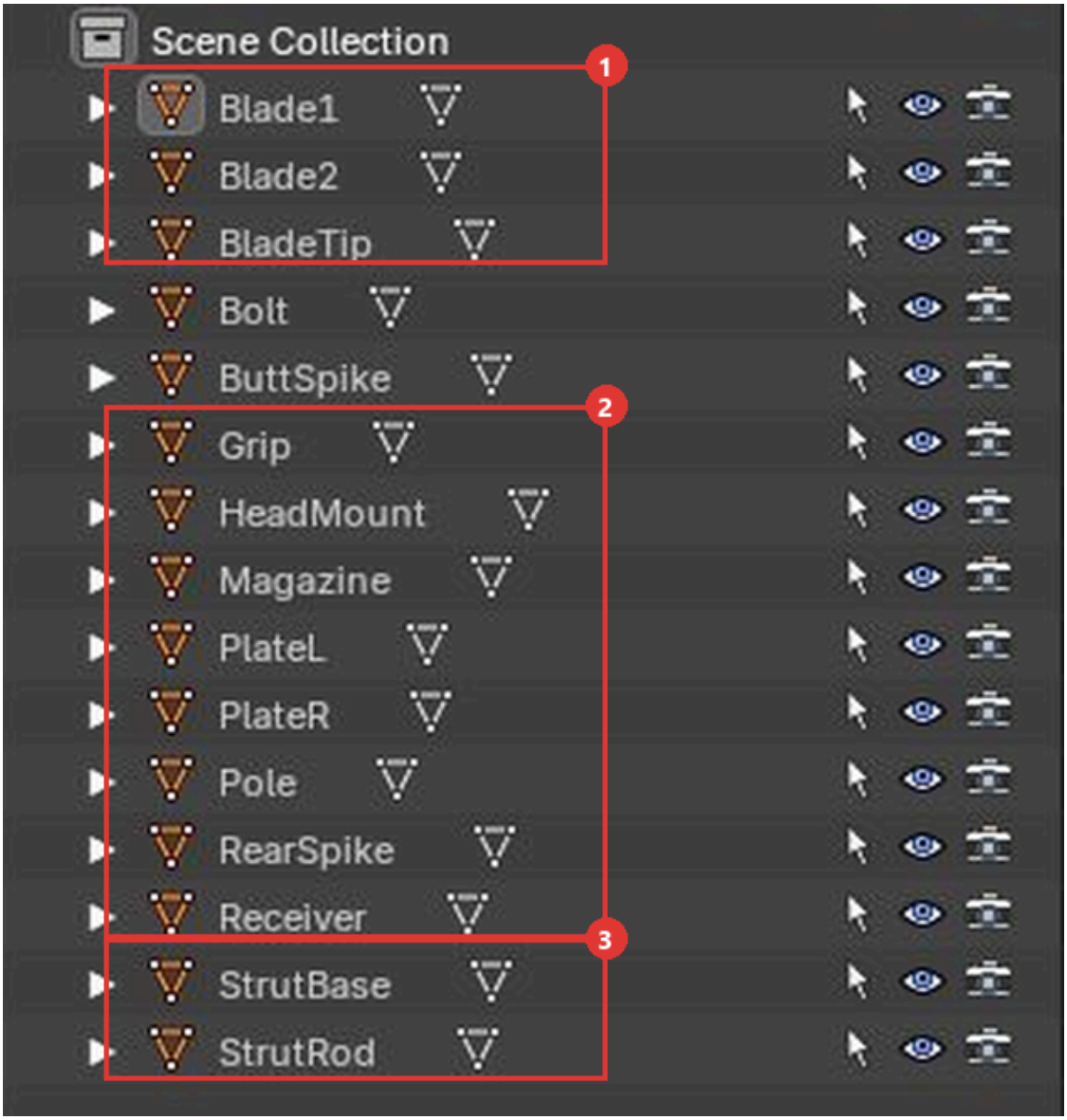
1: Click Material Preview. 2: Press Numpad 3 until the view says Right Orthographic; press Home to frame the weapon.

What you should be seeing: The scythe in red and black, about 2 m tall, with nothing selected.

Step 2 — Meet the parts

The Outliner lists the 15 objects. Each one is a piece that moves as one block:

Object	What it is	Rides on
`Receiver`	The rifle body with the scope, trigger and sight	the rig
`Grip`	The grip under the receiver	`Receiver`
`Magazine`	The cylinder magazine	`Receiver`
`Bolt`	The bolt	`Receiver`
`Pole`	The lower shaft and its collar	`Receiver`
`ButtSpike`	The spike at the bottom	`Pole`
`HeadMount`	The head block the blade hangs from	`Receiver`
`Blade1`	The first blade segment, nearest the head	`HeadMount`
`Blade2`	The second segment	`Blade1`
`BladeTip`	The tip	`Blade2`
`RearSpike`	The spike on the back of the head	`HeadMount`
`PlateL`, `PlateR`	The outer side plates of the head	`HeadMount`
`StrutBase`	The thin rod between the receiver and the head	`Receiver`
`StrutRod`	The notched sleeve around that rod	`HeadMount`

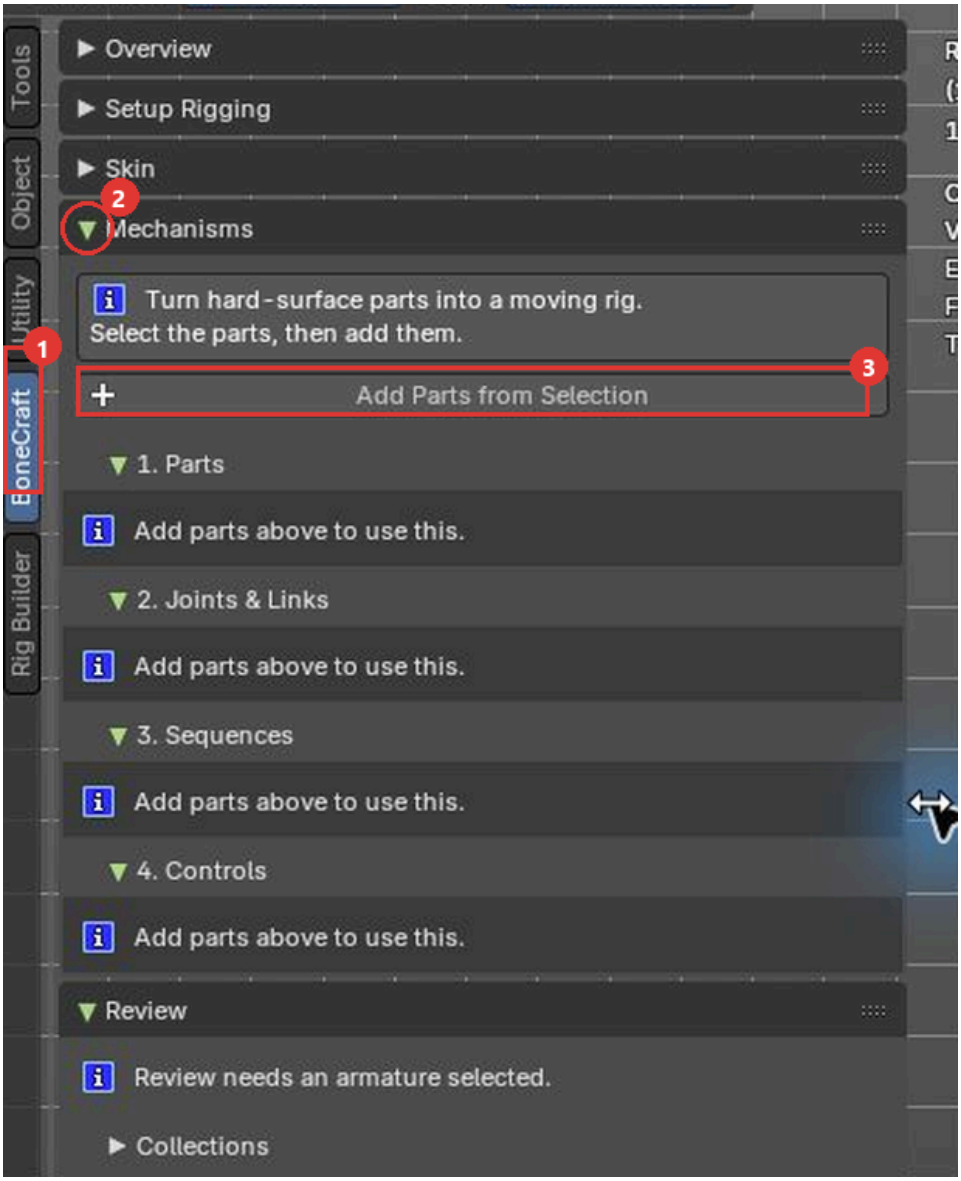


1: Identify the three blade segments. 2: Match the body, head and pole names to the table. 3: Keep StrutBase and StrutRod as separate objects.

For the original FBX, follow P1-P8 above. The supplied prepared file was made through those interface actions; the repository preparation script provides a reference grouping table.

Step 3 — Find Mechanisms

Open the BoneCraft tab on the left Tool Shelf (if hidden, press T over the viewport and choose Tools from the Region Toggle menu) and expand **Mechanisms**. Widen the shelf by dragging its right edge until the panel text has room. As an alternative, the same Mechanisms section is available in **Properties > Scene** when Scene Properties is set to Show in BoneCraft preferences. The screenshots use the Tool Shelf. With a mesh active and no rig yet, find **Add Parts from Selection** at the top of the section. Select all the parts in Step 4 before clicking it; the numbered subsections become usable after parts are added.

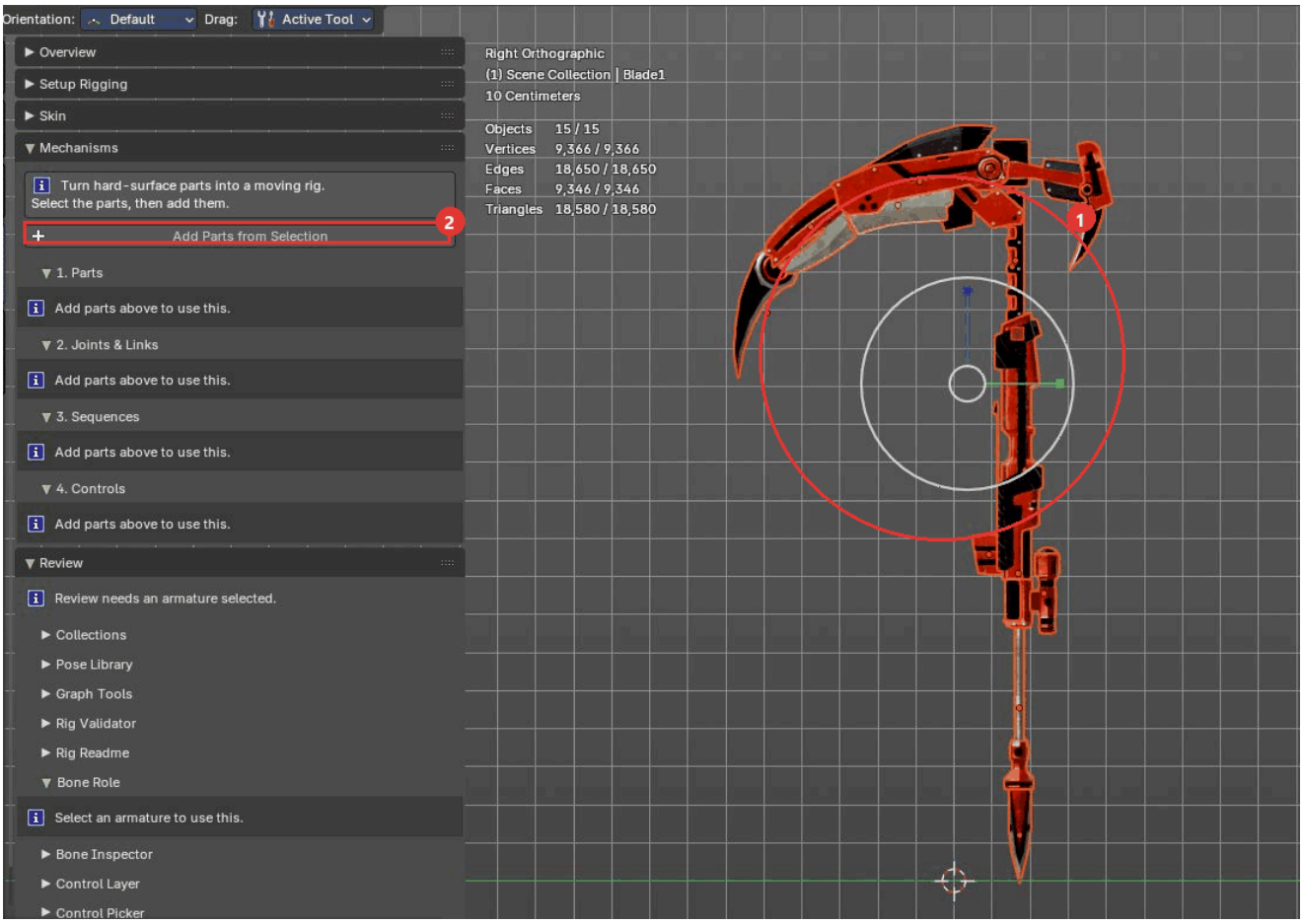


1: Click the BoneCraft tab. 2: Expand Mechanisms. 3: Find Add Parts from Selection; select the meshes before using it.

B. Parts

Step 4 — Add Parts from Selection

Hover the viewport and press A to select all 15 objects. Click **Add Parts from Selection**.

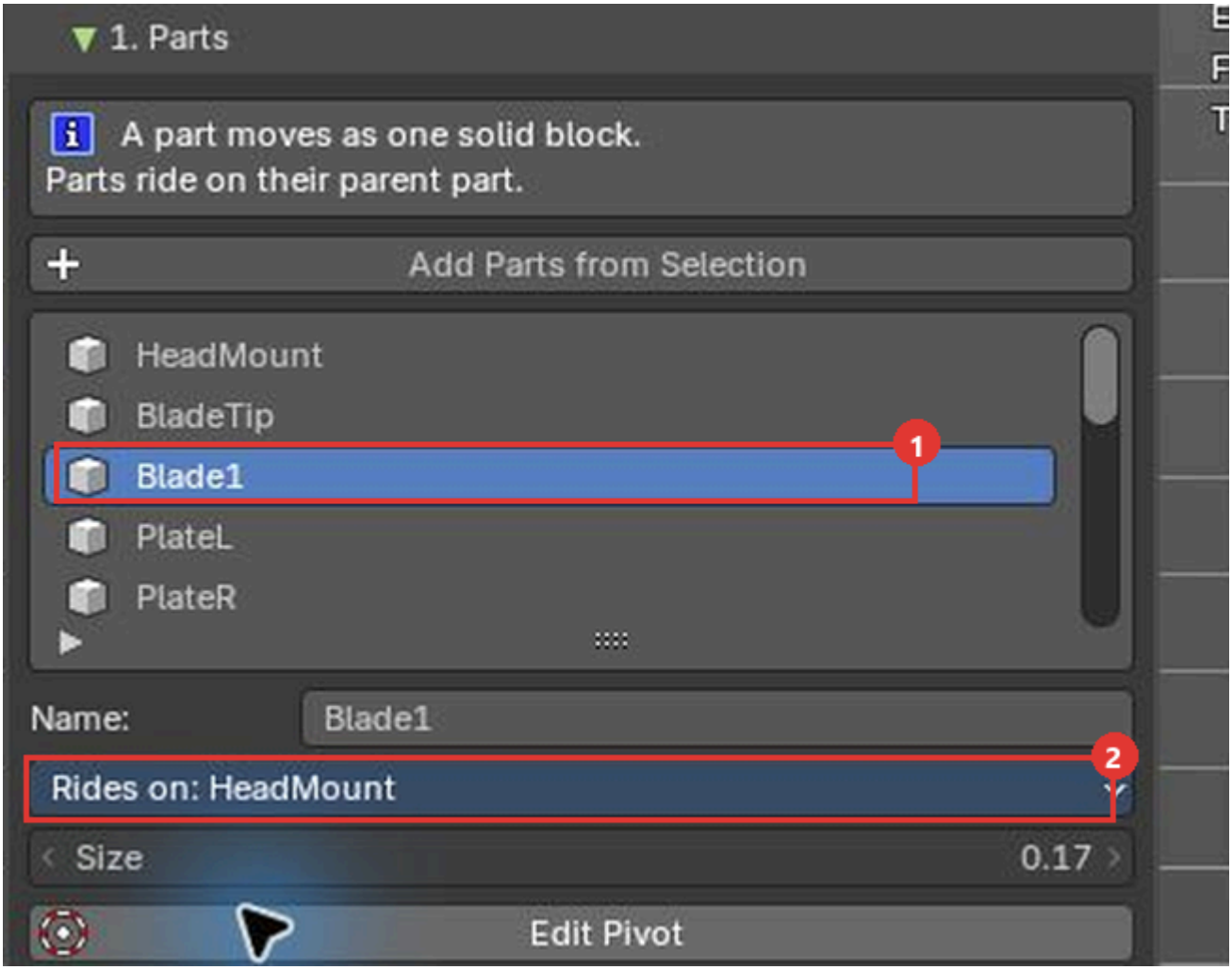


1: Hover the viewport and press A to select the 15 meshes. 2: Click Add Parts from Selection.

What you should be seeing: A new armature, "Mechanism Rig", appears in the Outliner with the parts under it. The mesh objects remain selected after the operation. **1. Parts** lists the 15 parts, and the BoneCraft tab now starts with the rig's Overview, which you can collapse.

Step 5 — Rides on

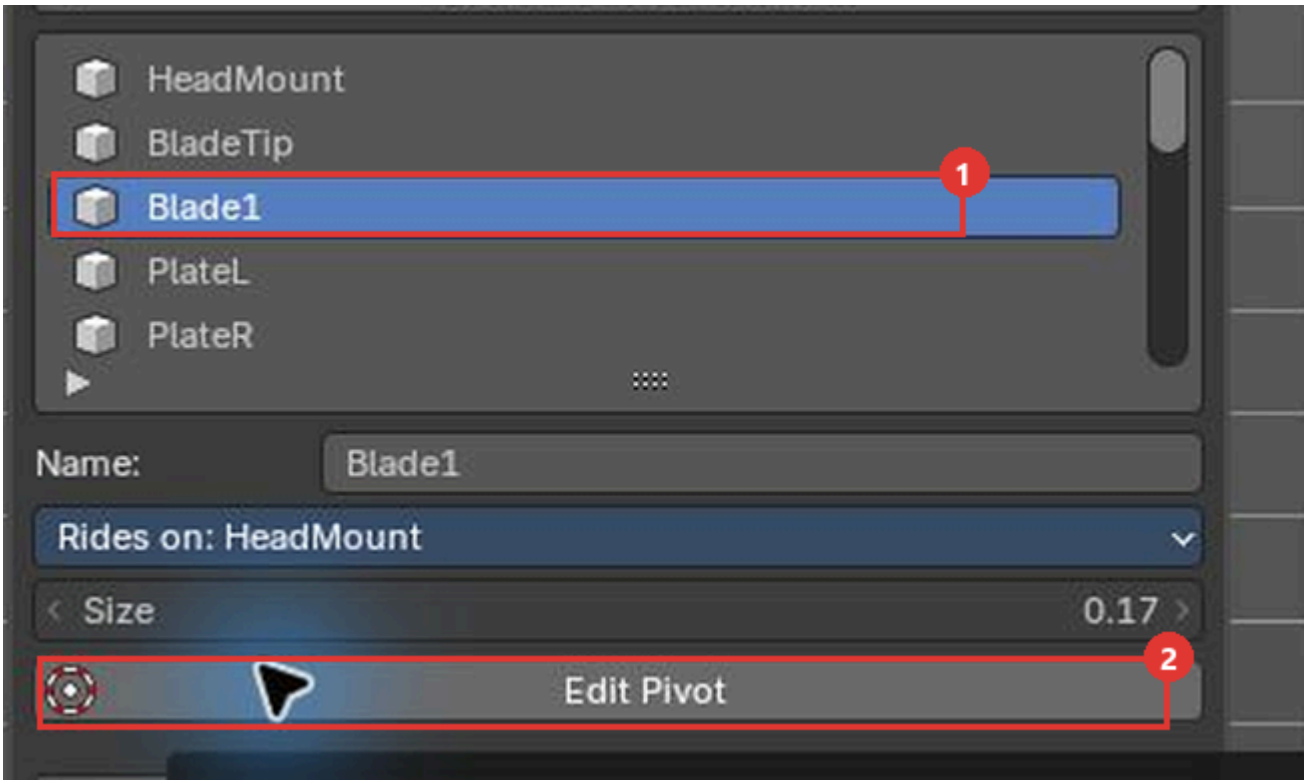
A part rides on the part it is attached to. Click `Blade1` in the **1. Parts** list, open the "Rides on" menu under the list and choose `HeadMount`. Do the same for every part in the table of step 2. `Receiver` keeps "the rig".



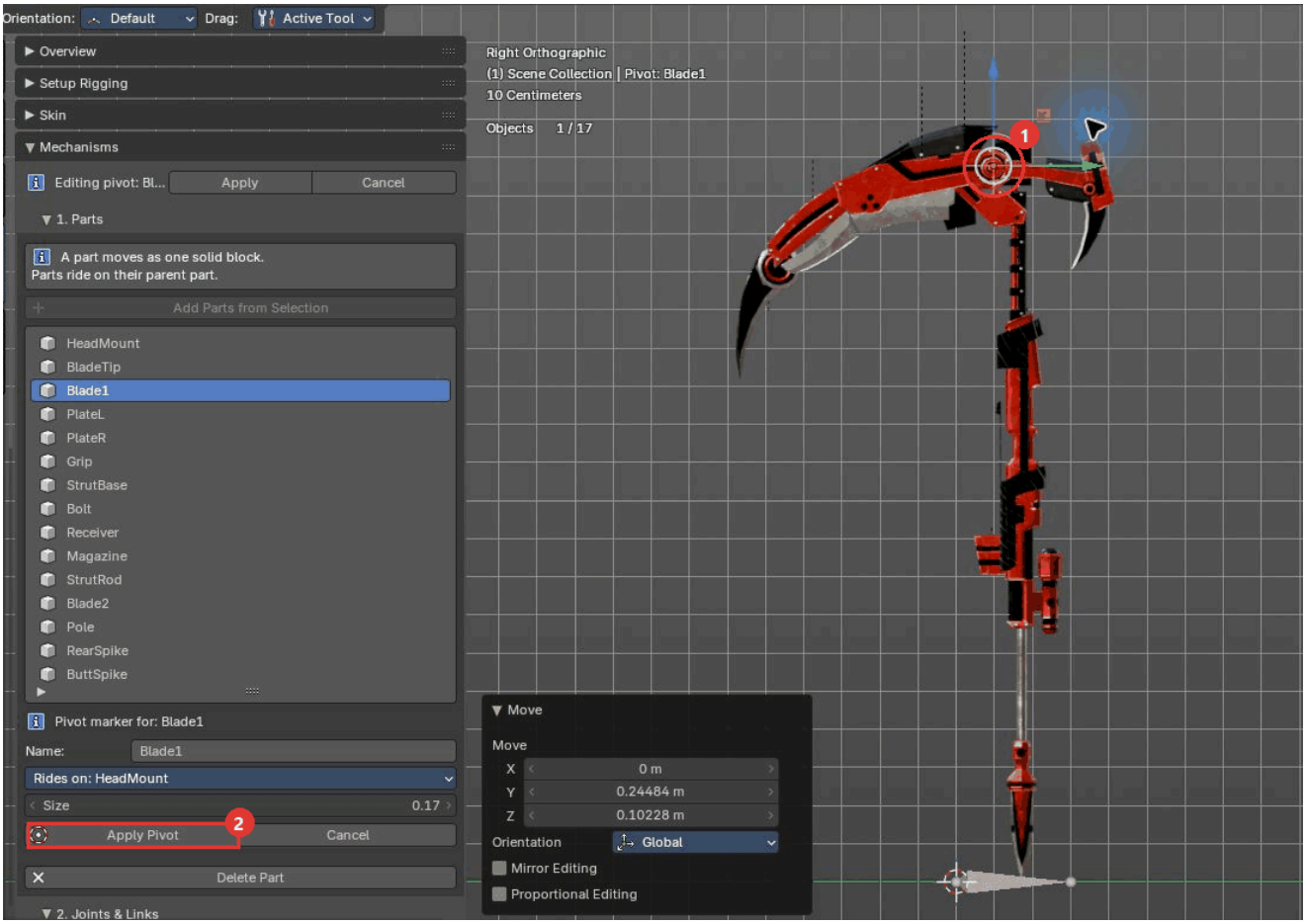
1: Select Blade1 in the Parts list. 2: Open Rides on and choose HeadMount. Repeat using the parent table.

Step 6 — Pivots

A hinge turns around its part's pivot. The blade segments, the head and the grip must turn at their joints, not at their centres. With `Blade1` selected in the list, click **Edit Pivot**, select the marker, press Ctrl+W for Move in Bforartists, click where the segment meets the head, and click **Apply Pivot**. Work in the orthographic side view so the sideways pivot position stays unchanged. Repeat for `Blade2` (where it meets `Blade1`), `BladeTip` (where it meets `Blade2`), `HeadMount` (its lowest point, where the strut meets it), `Grip` (its top edge), `RearSpike` (where it meets the head), `PlateL` and `PlateR` (the centre of the round bolt where Blade1 meets the head), `StrutBase` (its bottom) and `StrutRod` (its top). The screenshot's marker position is a guide; a few millimetres make no visible difference.



1: Select Blade1 in Parts. 2: Click Edit Pivot; then move and apply the marker as shown next.

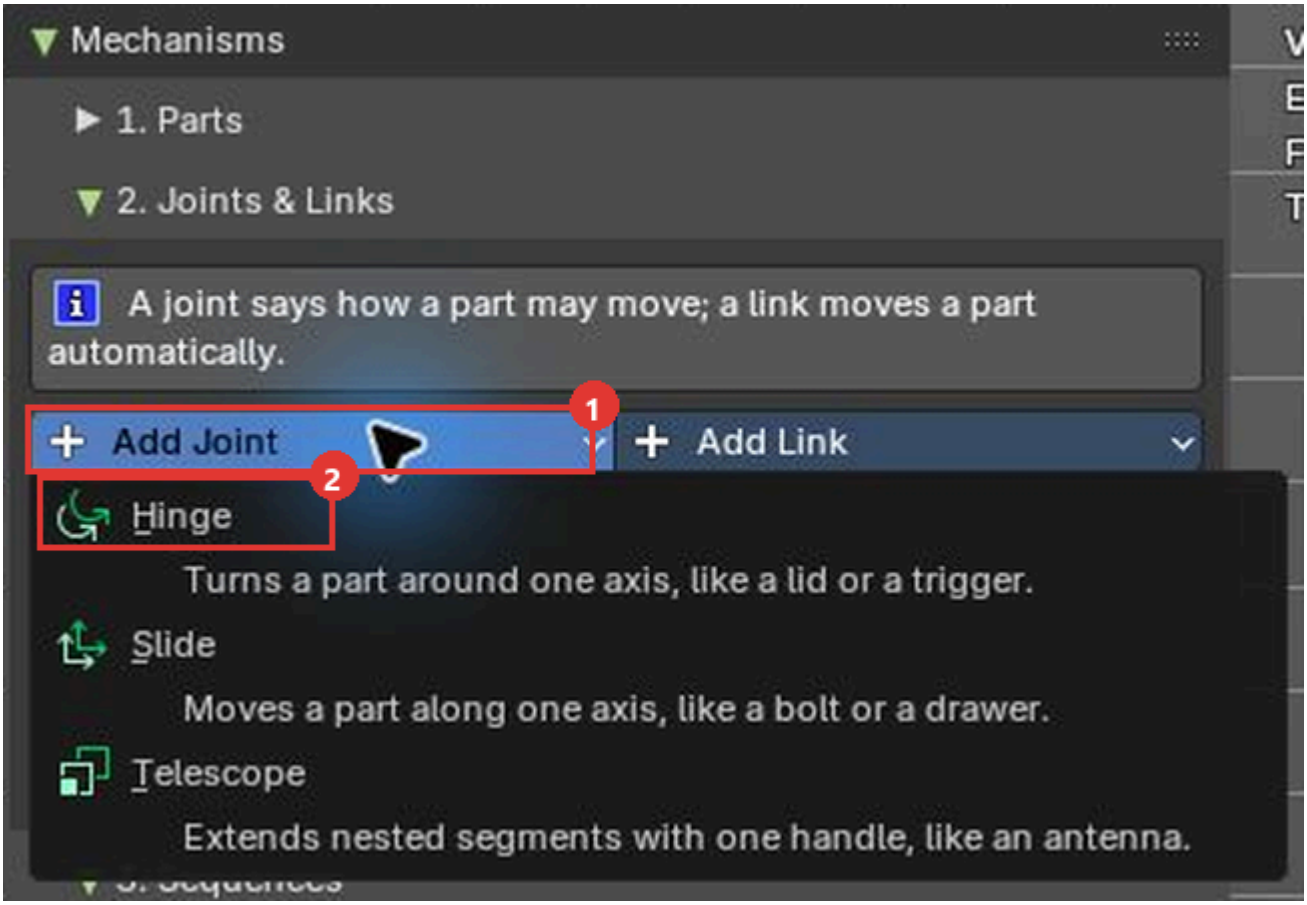


1: Move the pivot marker to the blade joint with Ctrl+W. 2: Click Apply Pivot to keep that location.

C. The head

Step 7 — Add Joint → Hinge

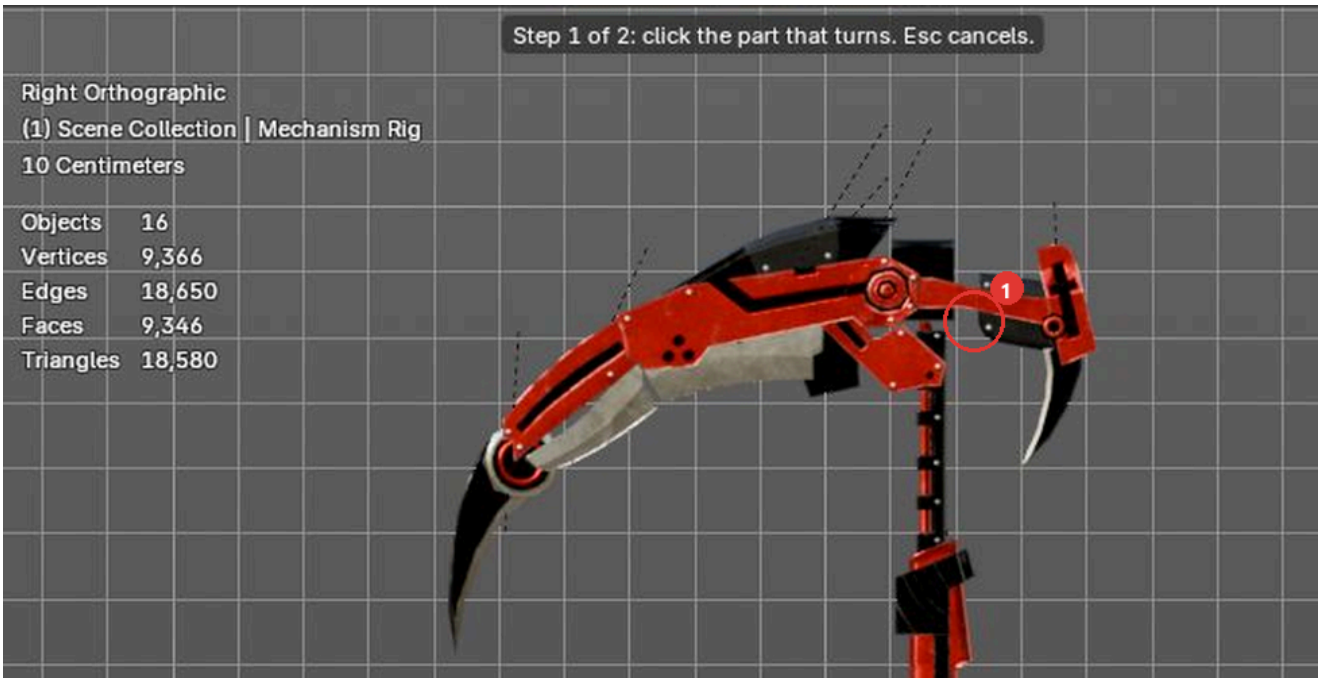
Open **Add Joint** → **Hinge**. A guided build starts: the header and the status bar tell you what to do at each step.



1: Open Add Joint. 2: Click Hinge. The guided build now waits for a part.

Step 8 — Click the head

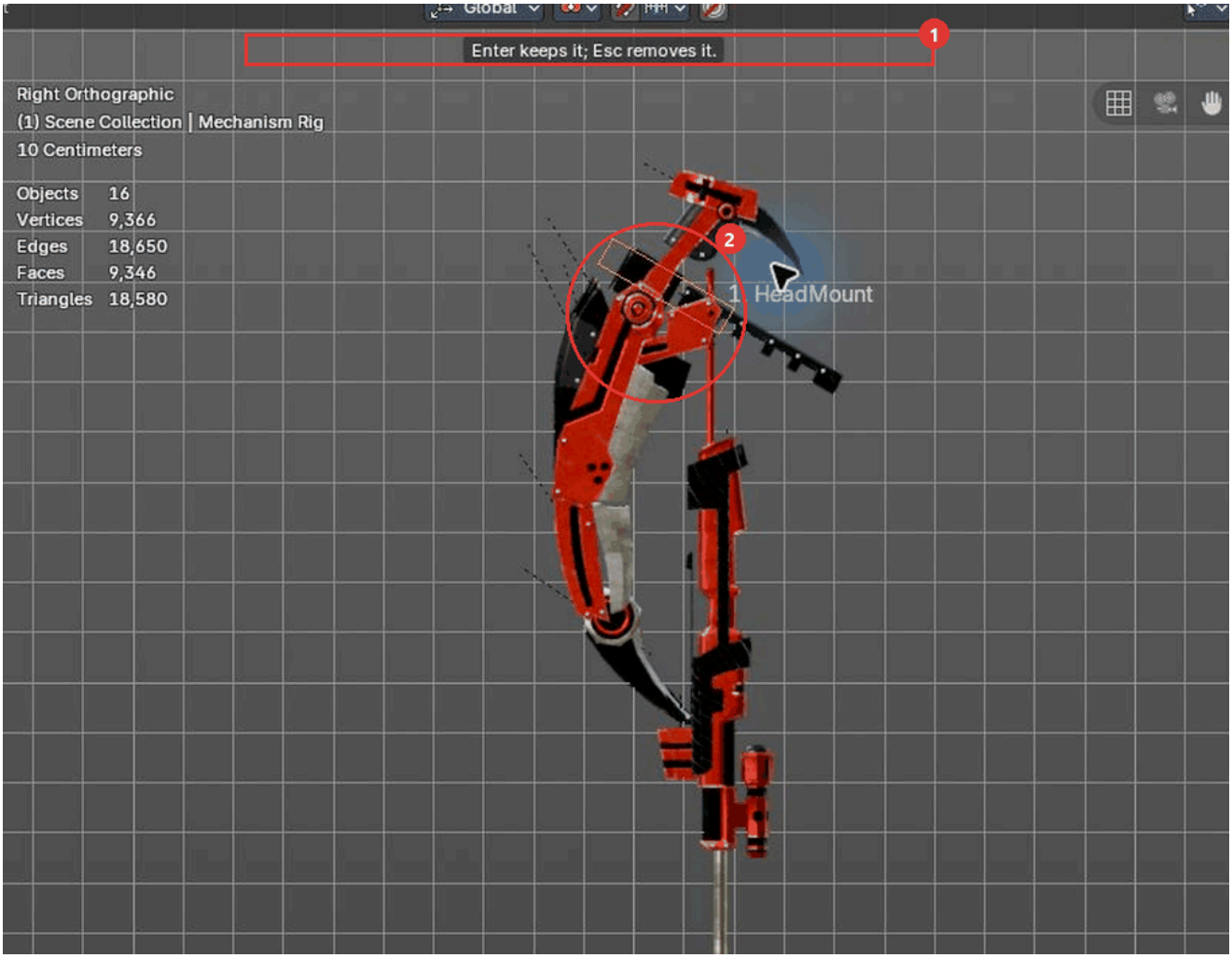
The header says "click the part that turns". In the Right view the outer plate sits in front of the head, so click the head's rear block, behind the plates (the circle). If the header names `PlateR`, press Esc, restart **Add Joint > Hinge**, and click further back. If you need to hide an obstruction, do it before starting the guide. The guide consumes ordinary keyboard shortcuts while active.



1: Click the rear black block of HeadMount, behind the red outer plate. Check the guide names HeadMount.

Step 9 — Press X, then Enter

The header now asks for the axis. Press X: the head turns around the sideways axis, and the guide previews the motion in a loop. Press Enter to keep it.

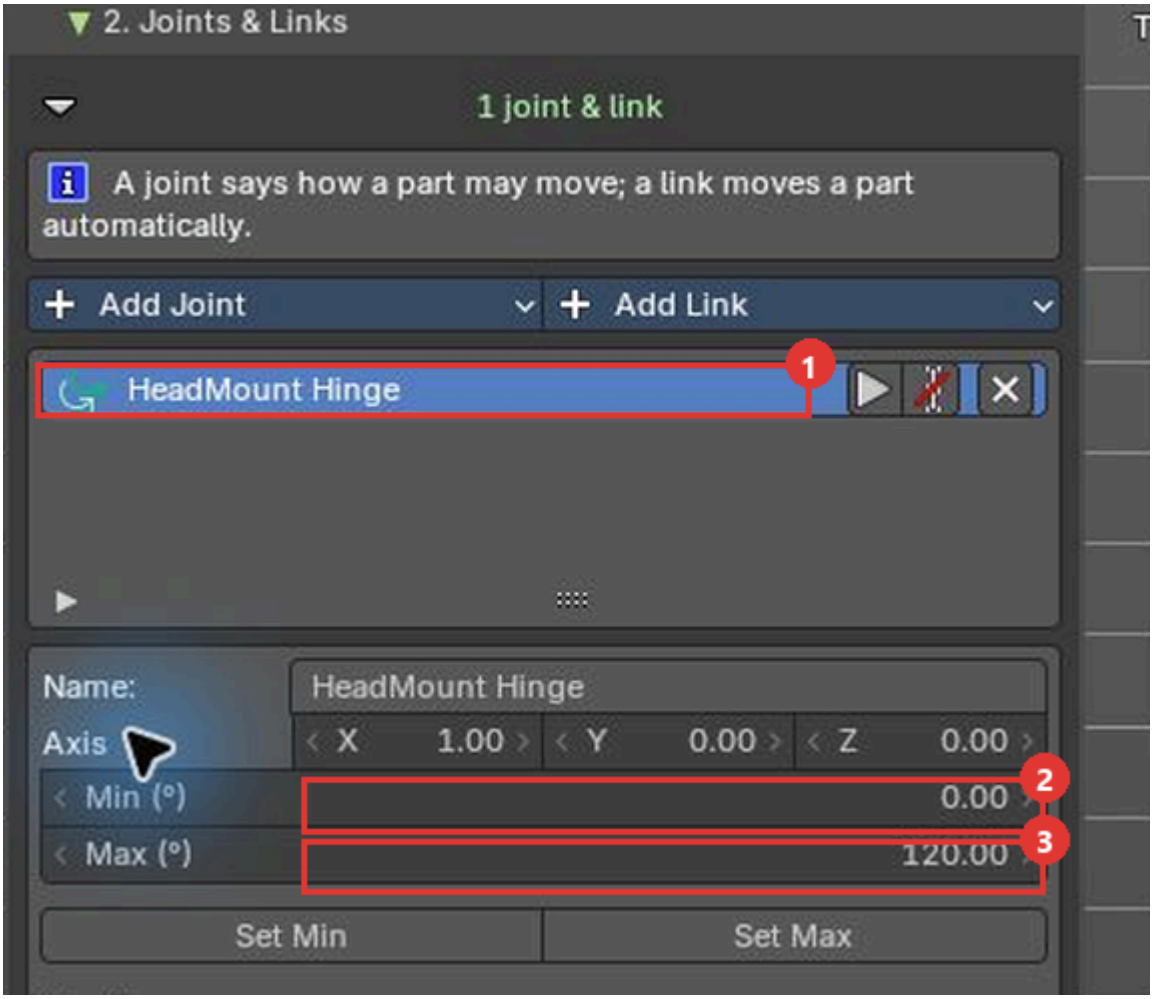


1: Follow the axis prompt: press X, then Enter. 2: Watch the head rotate around its pivot before confirming.

What you should be seeing: `HeadMount Hinge` in the **2. Joints & Links** list, and its slider in **4. Controls**. Use these sliders in Object Mode. To use the viewport bone handles, select the armature and switch to Pose Mode.

Step 10 — Limits

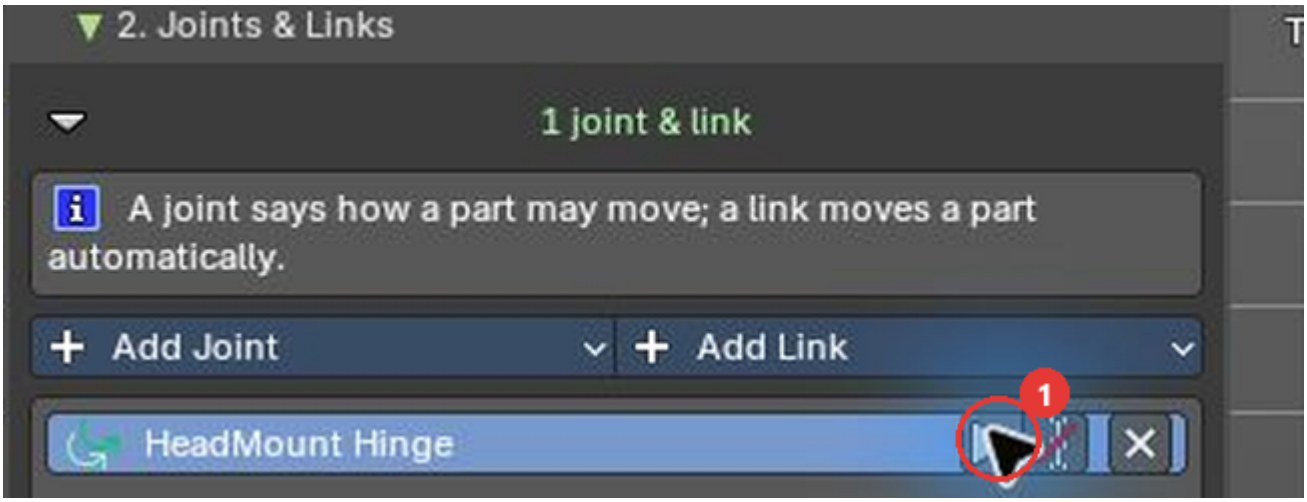
With `HeadMount Hinge` selected in the list, set **Min** to 0 and **Max** to 120. Positive values turn the head down and back. Keep the 120-degree travel limit; the Rifle sequence will use a 90-degree target together with the blade hinges.



1: Select HeadMount Hinge. 2: Set Min to 0. 3: Set Max to 120; double-click the number, type, and press Enter.

Step 11 — Test

Click the row's play button. The head sweeps to 120° and back, and the pose is restored.

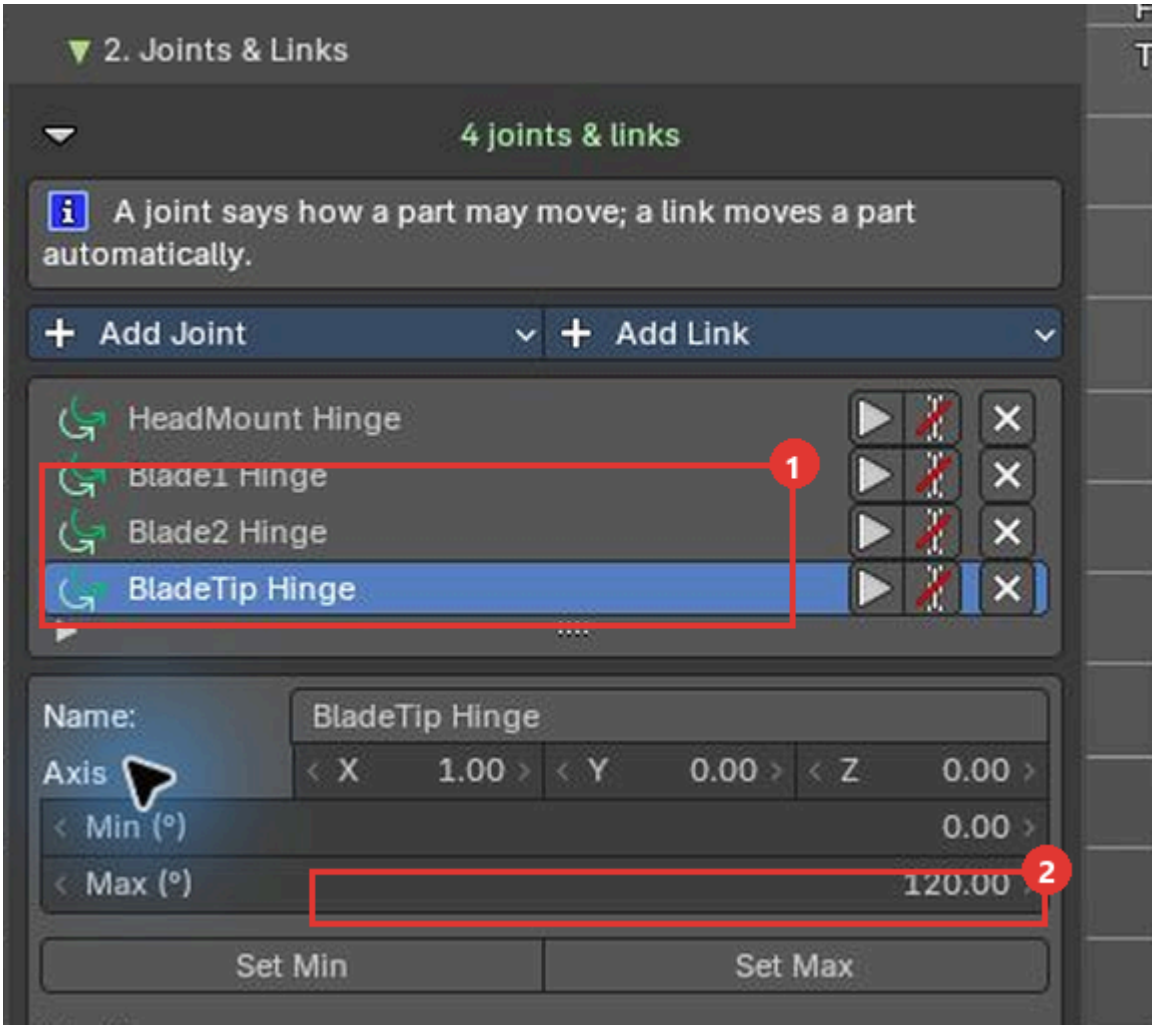


1: Click the small play triangle on the HeadMount Hinge row. Watch the sweep and return.

D. The blade chain

Step 12 — Three more hinges

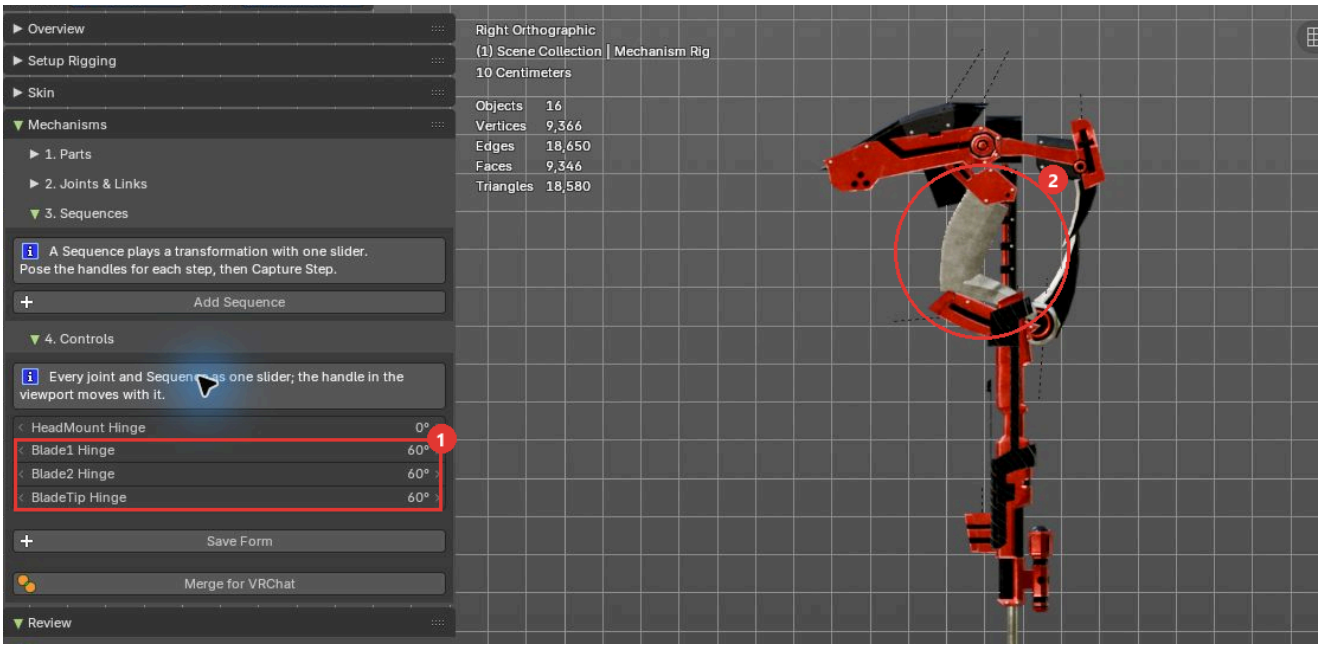
Repeat steps 7 to 10 for `Blade1`, `Blade2` and `BladeTip`: **Add Joint** → **Hinge**, click the segment, X, Enter, then **Max** 120. Each segment rides on the one before it, so the three hinges form a chain.



1: Create and select each blade hinge: Blade1, Blade2, BladeTip. 2: Set Max to 120 for each; use X as its axis.

Step 13 — Try the chain

Collapse **2. Joints & Links** to make room and expand **4. Controls**. In Object Mode, set `Blade1 Hinge`, `Blade2 Hinge` and `BladeTip Hinge` to 60 degrees, one at a time. Double-click a numeric field, press Ctrl+A, type the value and press Enter. The later segments move with the earlier ones and the chain curls. Set all three values back to 0 before continuing.

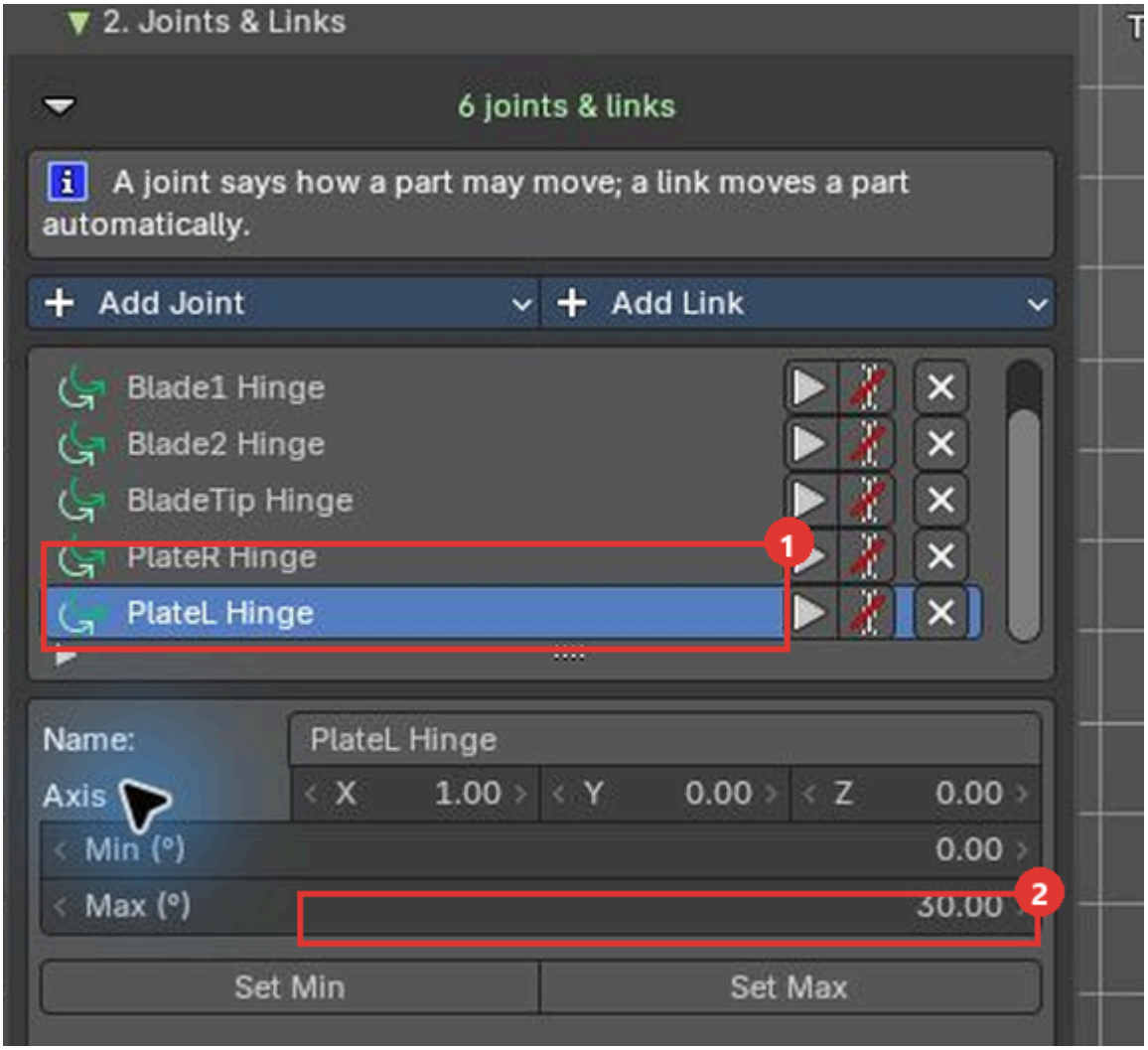


1: Enter 60 in each of the three blade sliders. 2: Check the chain curls together. Return all three sliders to 0.

E. Plates and strut

Step 14 — Hinges on the plates

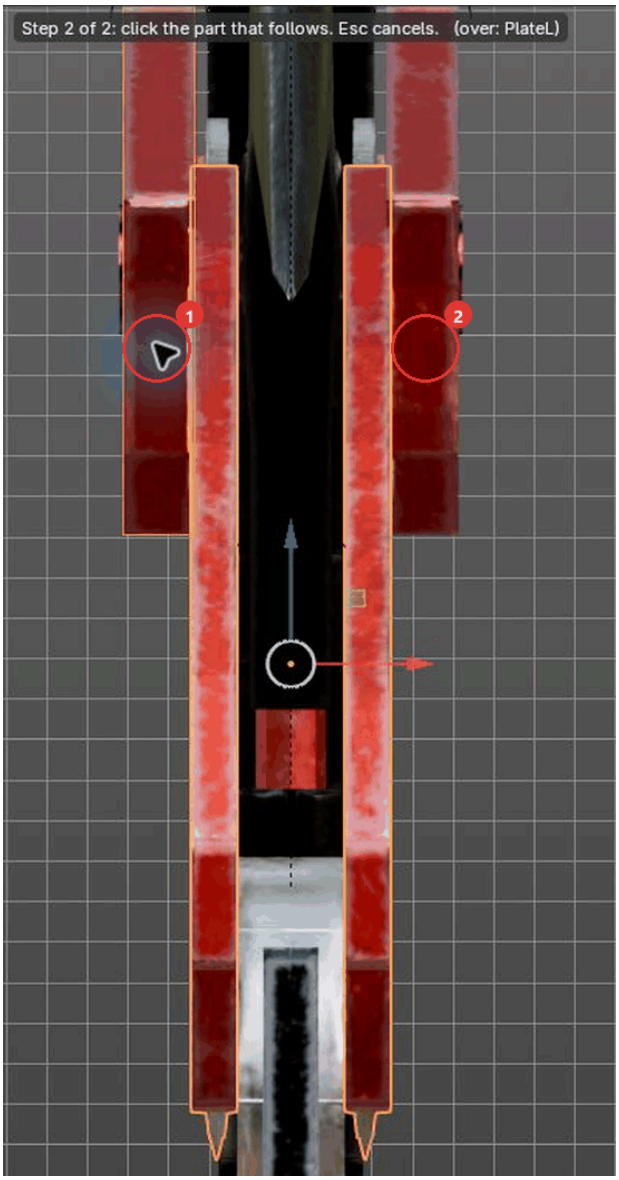
Add a Hinge on `PlateR` the same way, **Max** 30. `PlateL` is hidden behind the head in the Right view: press Ctrl+Numpad 3 for the Left view, add its Hinge, and come back with Numpad 3.



1: Make both PlateR and PlateL hinges; use the opposite side view to reach PlateL. 2: Set each Max to 30.

Step 15 — Add Link → Follow

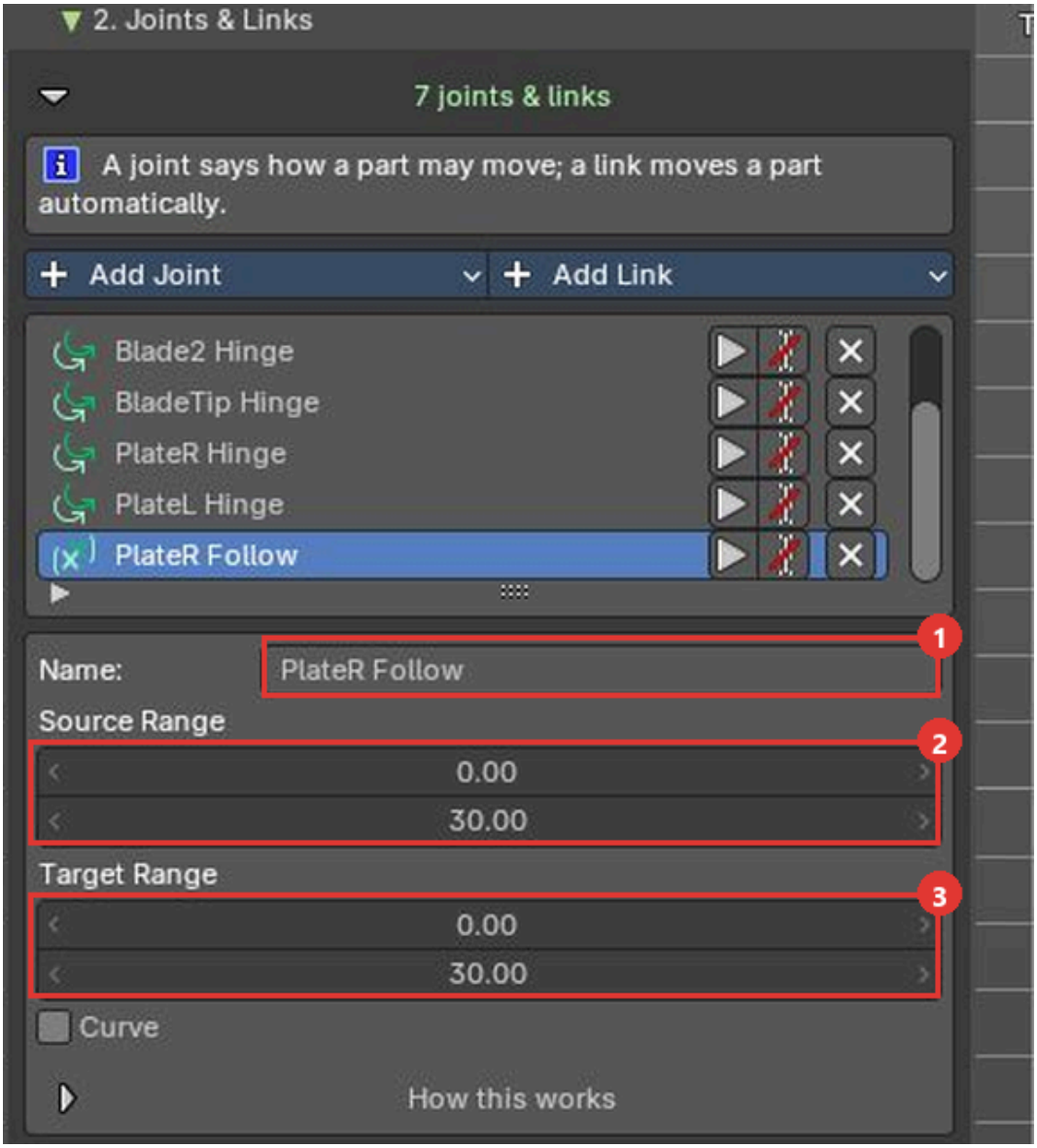
One handle should move both plates. Before starting the guide, press Numpad 1 for Front view, select `Blade2` and use F3 > Frame Selected to bring the head close. The two outer plates appear at the left and right edges. Open **Add Link > Follow**, click `PlateL` at the left edge (the leading part), then `PlateR` at the right edge (the following part), and press Enter.



1: Click PlateL first, the leading part. 2: Click PlateR second, the following part. Press Enter.

Step 16 — Ranges

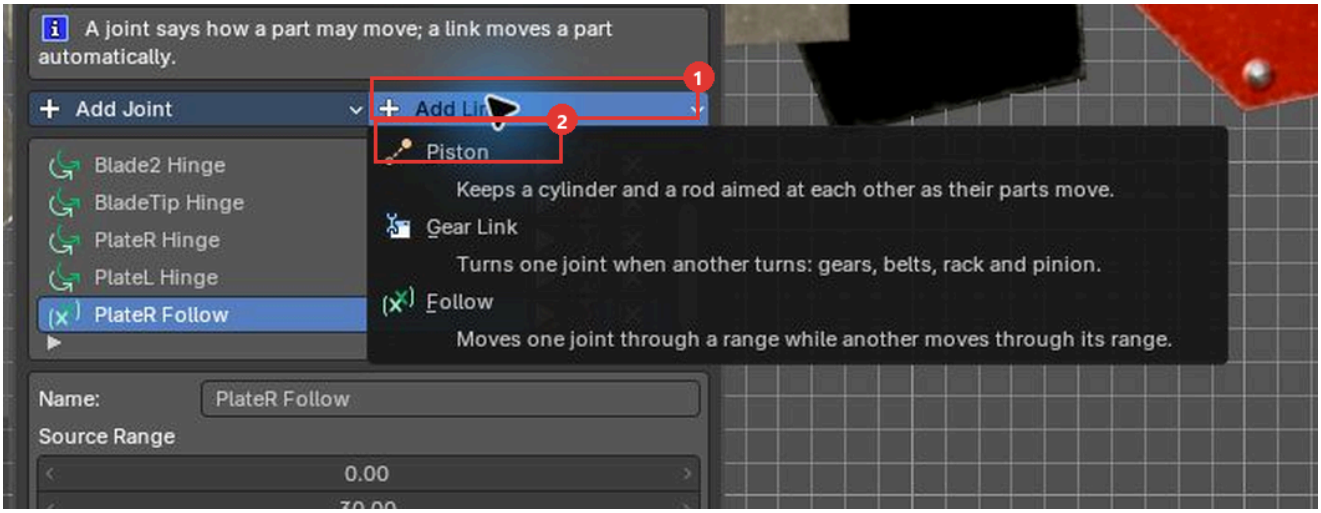
With the new Follow selected, verify the automatically populated **Source Range** 0…30 and **Target Range** 0…30. Rename it `PlateR Follow` in **Name**.



1: Name the link PlateR Follow. 2: Check Source Range is 0 to 30. 3: Check Target Range is 0 to 30.

Step 17 — Add Link → Piston

The strut must keep pointing between the receiver and the head while the head folds. Before starting the guide, press Numpad 3 for Right view, select `StrutRod` in the Outliner and use F3 > Frame Selected. Open **Add Link > Piston**. Set this close view first; navigation shortcuts are consumed while the guide is active.

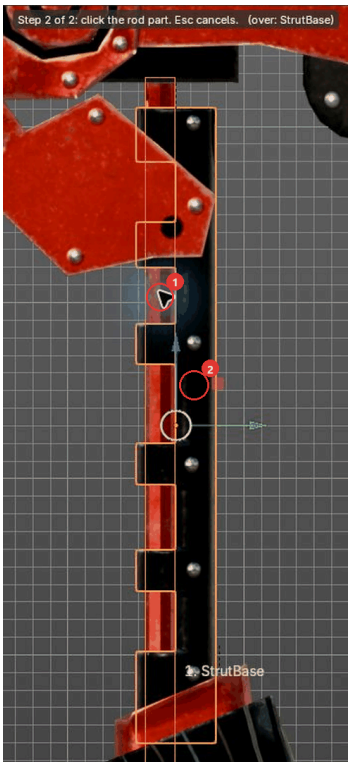


1: Open Add Link after framing StrutRod. 2: Choose Piston; the guide asks for the cylinder first.

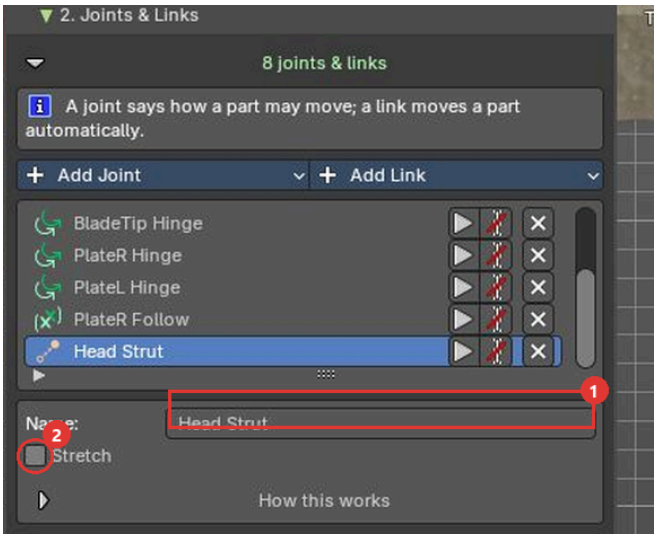
Step 18 — The cylinder and the rod

Click the cylinder part first: `StrutBase`, the thin rod, which shows in the sleeve's notches. Then click the rod part: `StrutRod`, the sleeve. Use the close view to click the exposed red rod first, then the black sleeve. The guide's cylinder/rod roles describe the link ends; this model's object names describe its source pieces. Press Enter to accept, rename the link `Head Strut`, and leave **Stretch** off.

What you should be seeing: Change `HeadMount Hinge` in Controls: the strut follows, each part aimed at the other. Restore the value to zero.



1: Click the exposed red StrutBase first. 2: Click the black StrutRod sleeve second. Press Enter.

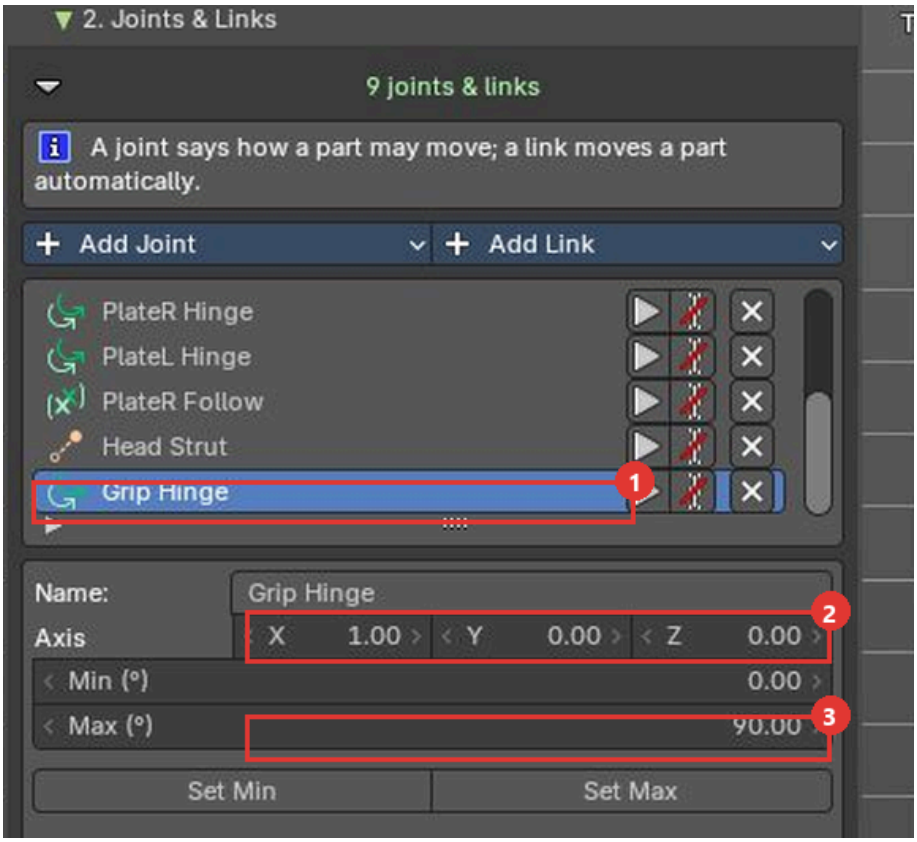


1: Rename the link Head Strut. 2: Leave Stretch unchecked; then test HeadMount Hinge and restore 0.

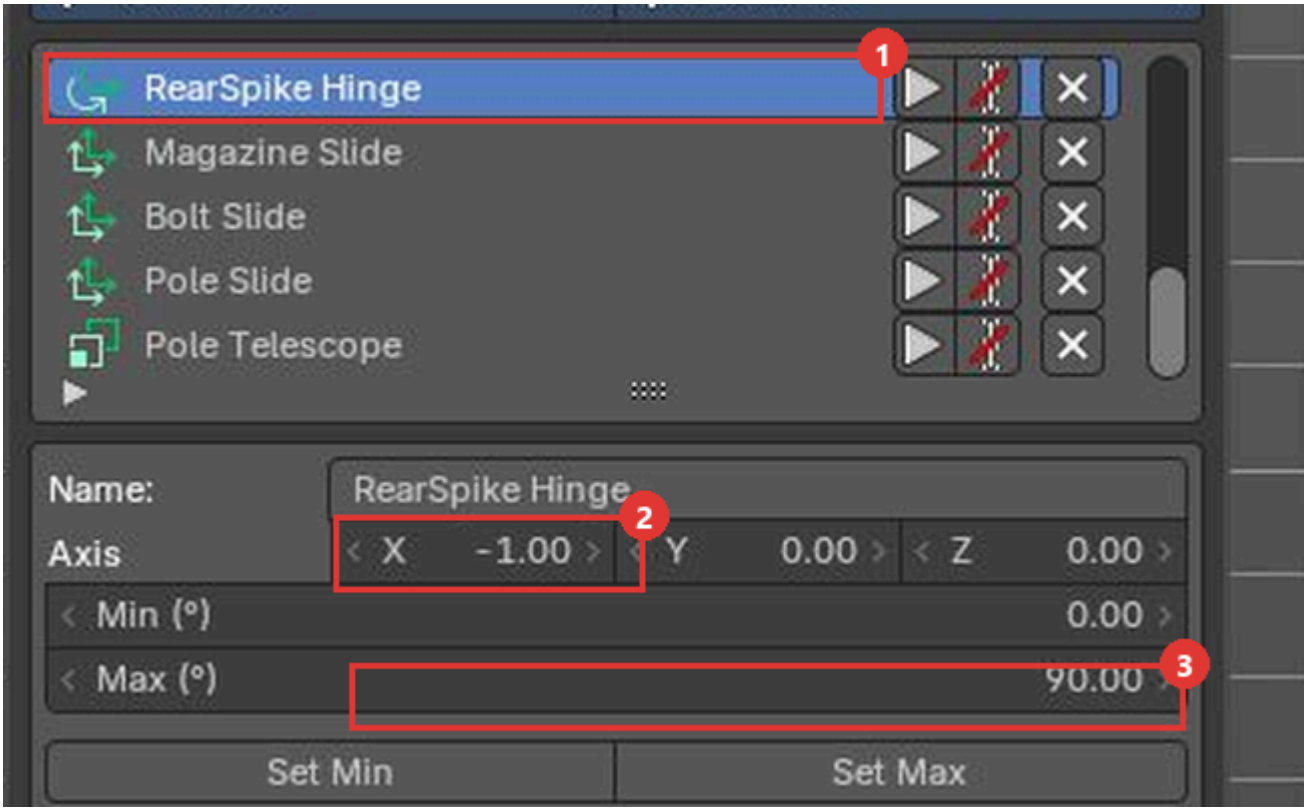
F. The rifle

Step 19 — Grip and rear-spike hinges

Add Joint → **Hinge** on `Grip`, X, Enter, **Min** 0, **Max** 90. Repeat on `RearSpike`, X, Enter, **Min** 0, **Max** 90. Then select `RearSpike Hinge` in the list and change its `Axis X to -1`, leaving Y and Z at 0. This reverses its fold direction so the rear spike turns back toward the folded head in Compact; keep Grip on positive X. Confirm that both `Grip Hinge` and `RearSpike Hinge` appear in the joint list. These two controls fold the grip and rear spike for the compact form in Step 25.



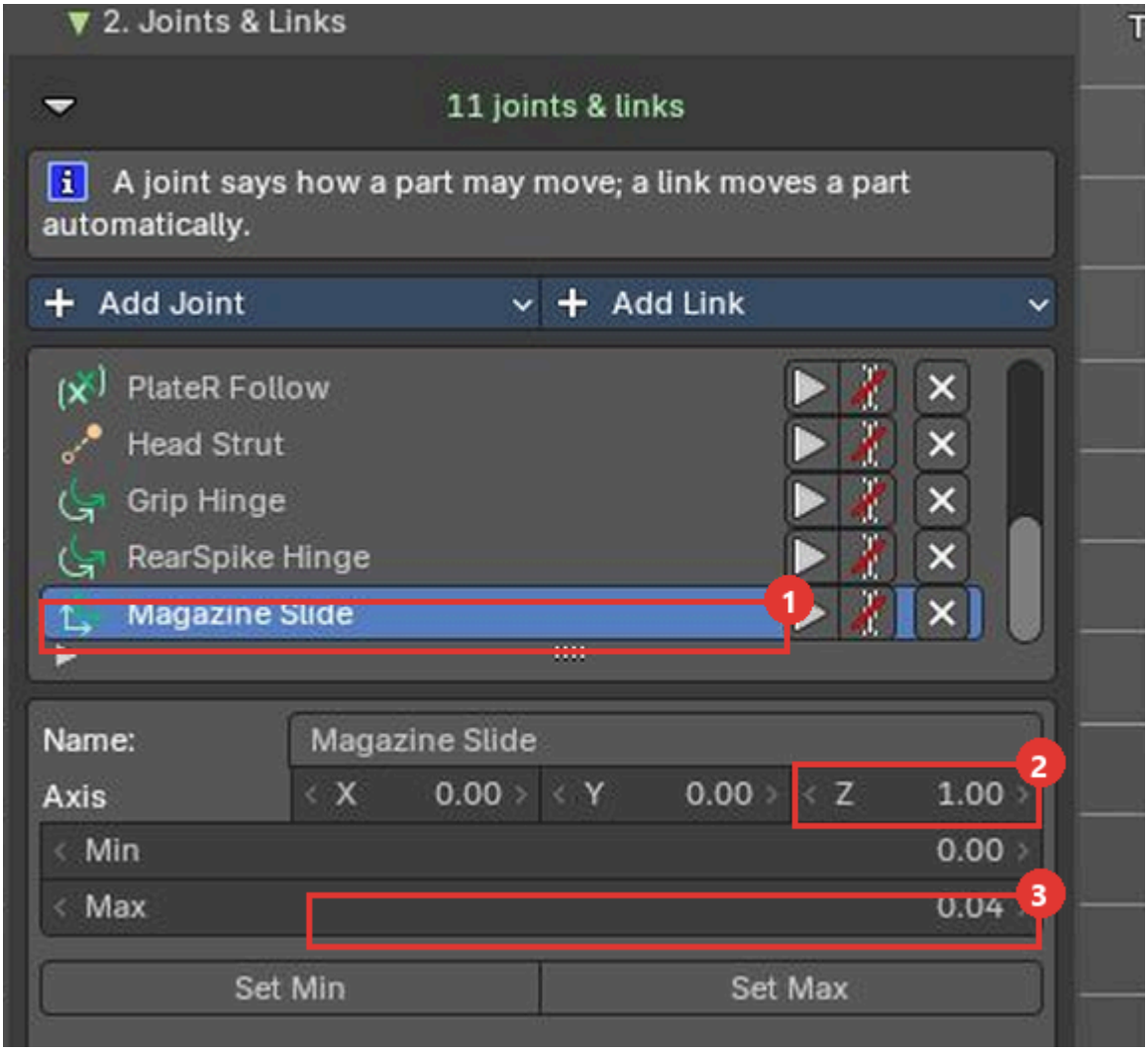
1: Select Grip Hinge. 2: Keep Axis X at 1, Y and Z at 0. 3: Set Max to 90; keep Min at 0.



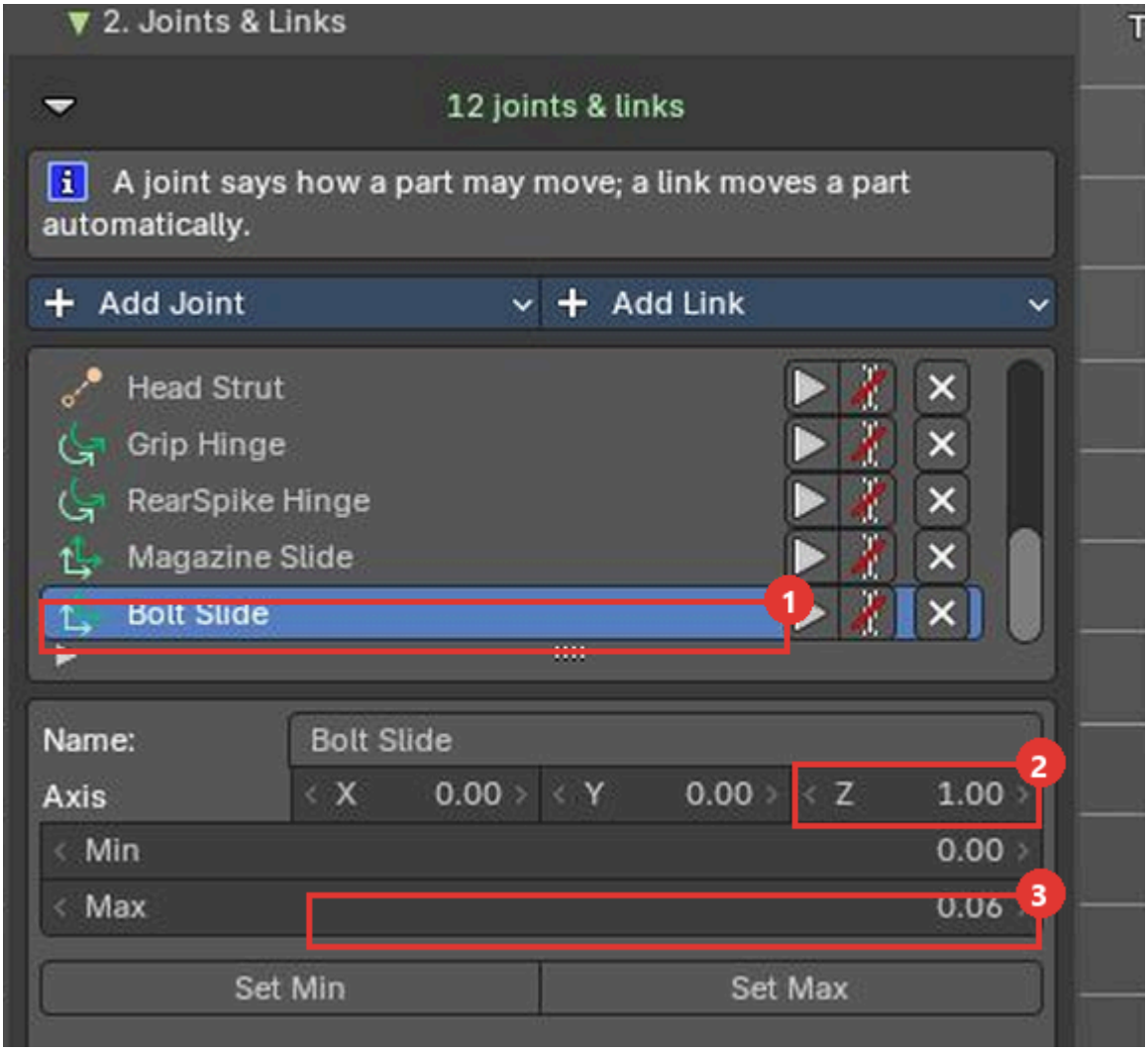
1: Select RearSpike Hinge. 2: Set Axis X to -1. 3: Set Max to 90; keep Min at 0.

Step 20 — Magazine and Bolt Slides

Add Joint → **Slide** on `Magazine`, press Z, Enter, **Max** 0.04. Then a Slide on `Bolt`, Z, Enter, **Max** 0.06. Slides measure in scene units: 0.06 is 6 cm.



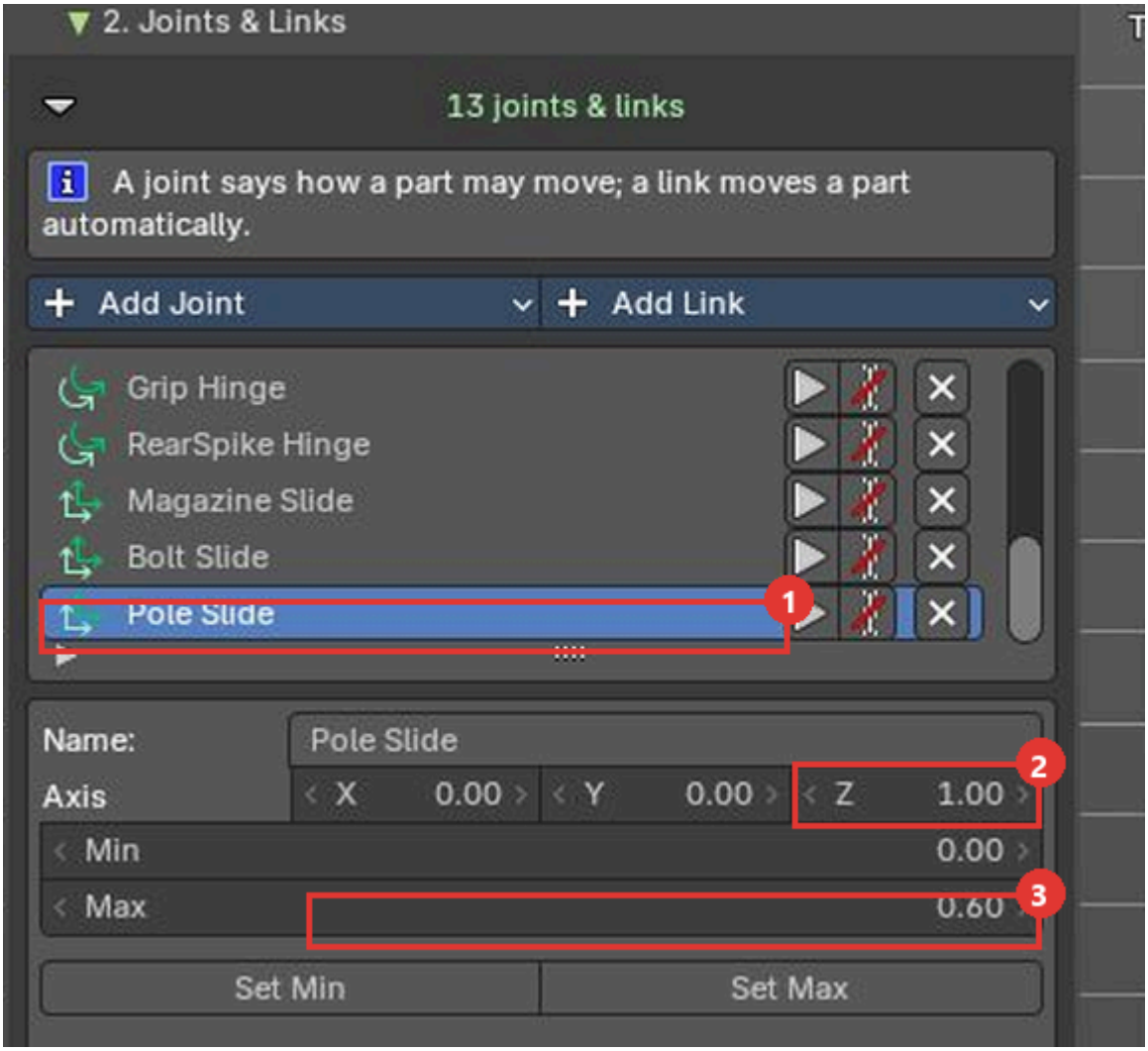
1: Select Magazine Slide after Add Joint > Slide. 2: Use the Z axis (Z = 1, X = Y = 0). 3: Set Max to 0.04; keep Min at 0.



1: Select Bolt Slide after Add Joint > Slide. 2: Use the Z axis (Z = 1, X = Y = 0). 3: Set Max to 0.06; keep Min at 0.

Step 21 — Pole Slide

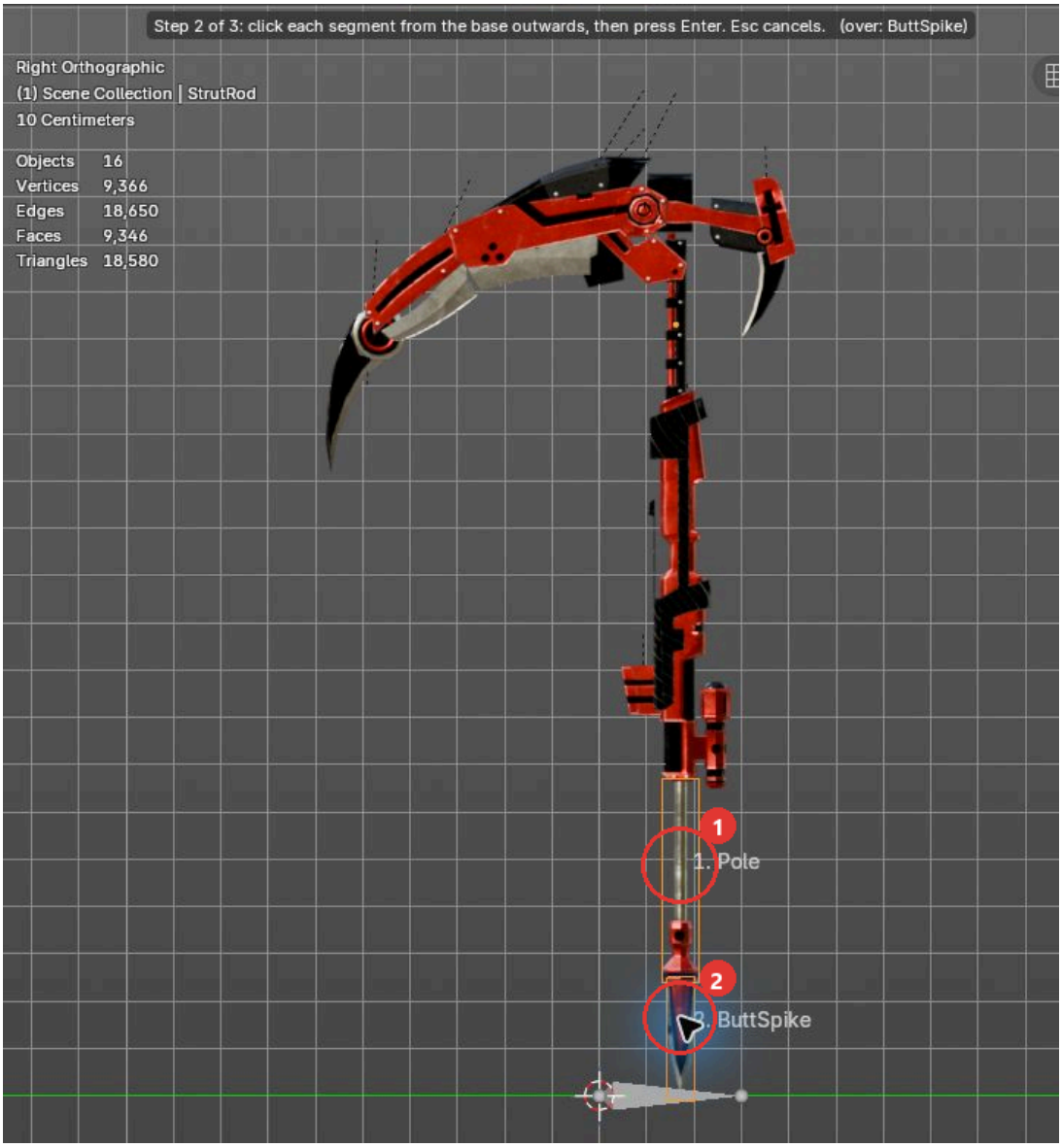
A Slide on `Pole`, Z, Enter, **Max** 0.6. Positive values move the pole up into the receiver. Set `Pole Slide` to 0.6 in **4. Controls** to see the shaft retract, then restore it to zero.



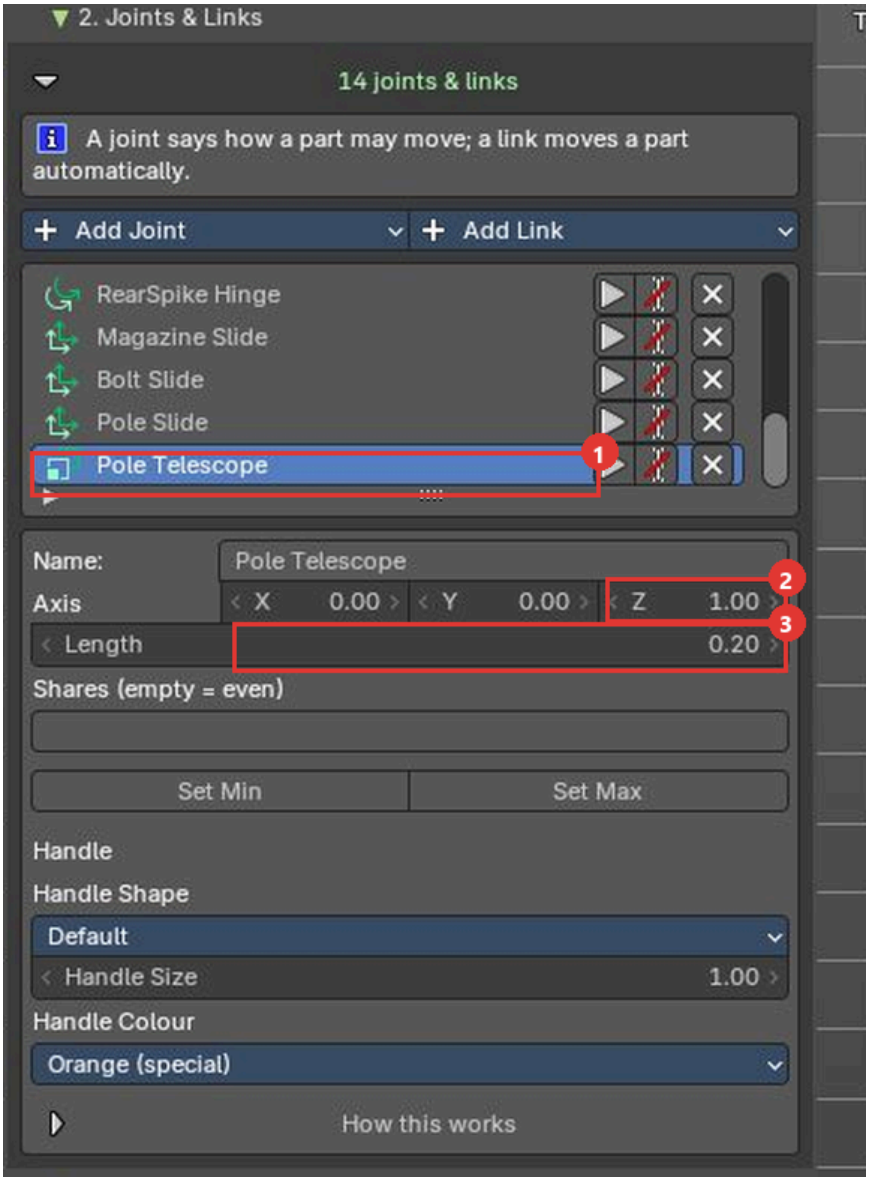
1: Select Pole Slide after Add Joint > Slide. 2: Use the Z axis (Z = 1, X = Y = 0). 3: Set Max to 0.6; keep Min at 0.

Step 22 — Pole Telescope

Add Joint → **Telescope**. Click `Pole` (the base), then `ButtSpike` (the segment), press Enter, press Z, press Enter. Set **Length** 0.2. Its separate control moves the spike up to 20 cm along positive Z, into the pole in this model. Test it and restore zero.



1: Click Pole as the base. 2: Click ButtSpike as its segment. Press Enter, then Z, then Enter.

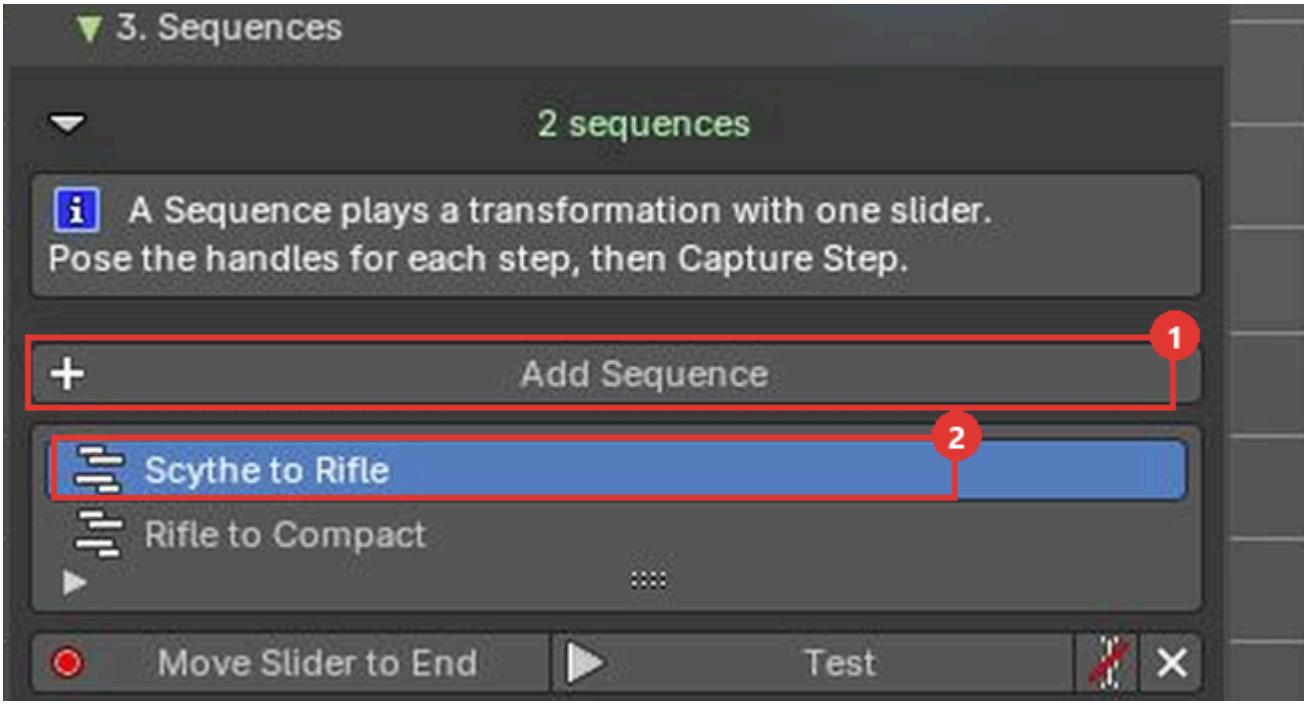


1: Select Pole Telescope. 2: Check Z is its axis. 3: Enter Length 0.2; test the control and restore zero.

G. Sequences

Step 23 — Add Sequence

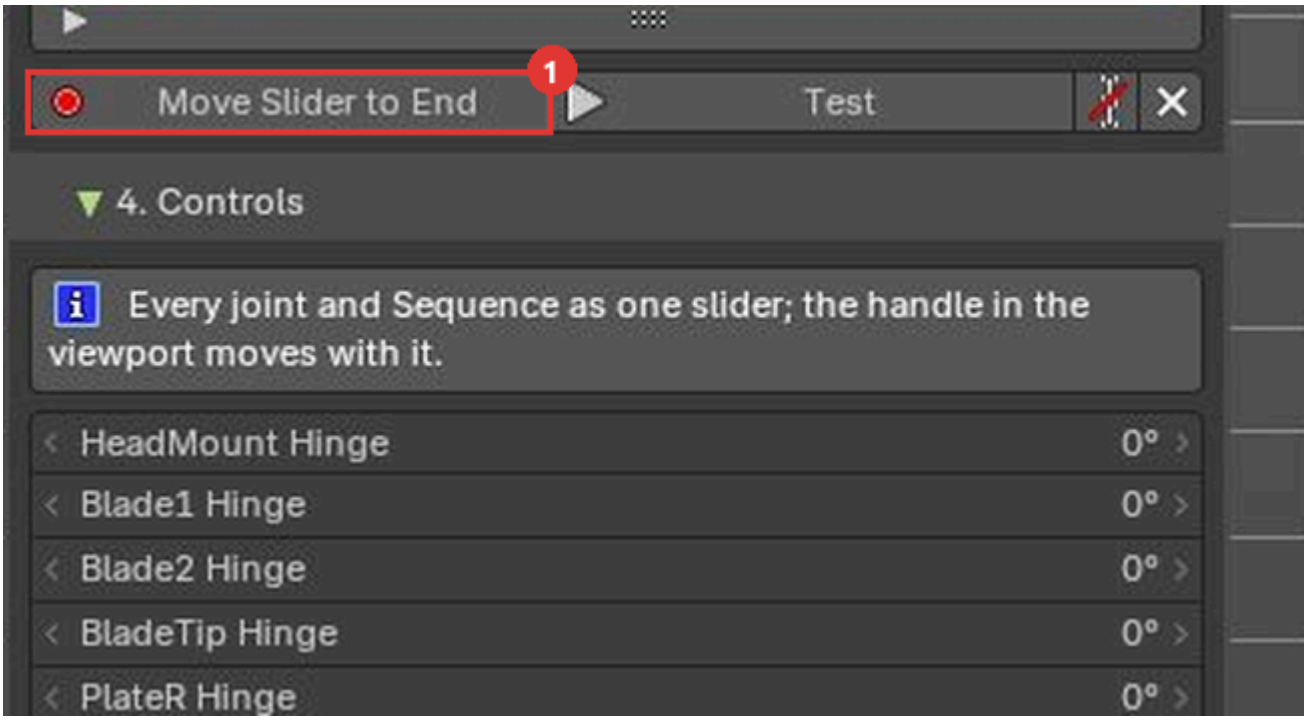
In **3. Sequences**, click **Add Sequence**, and double-click its name in the list to rename it `Scythe to Rifle`. Press Enter.



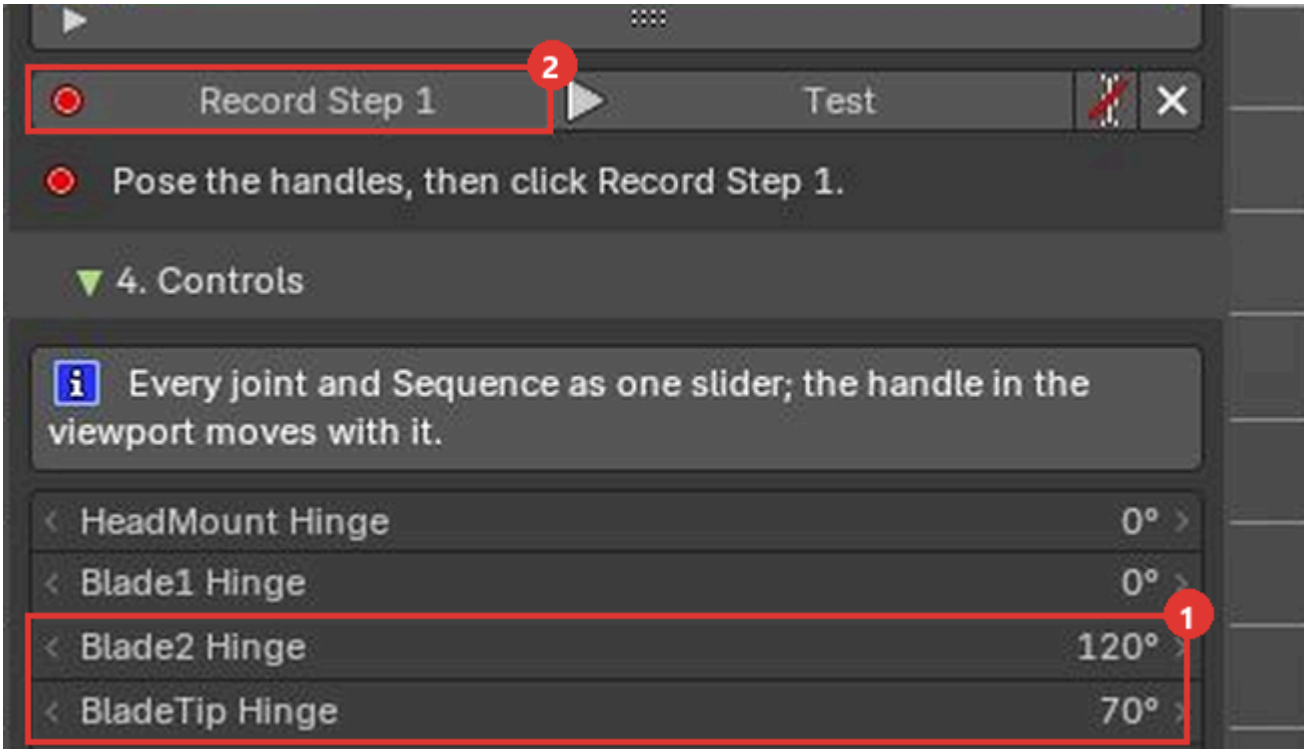
1: Click Add Sequence. 2: Double-click the new name and type Scythe to Rifle. Press Enter.

Step 24 — Record a motion group

Click **Move Slider to End**: the sequence's slider jumps to `100%` and the button becomes Record Step 1. Now pose the first motion group with the sliders in **4. Controls**: `BladeTip Hinge` 70 and `Blade2 Hinge` 120. Click Record Step 1. This stores two target entries, one for each moved handle, and resets those two joint sliders to zero.



1: Click Move Slider to End once. The button changes to Record Step 1.



1: Enter Blade2 = 120 and BladeTip = 70. 2: Click Record Step 1; both targets are stored and the joint sliders reset.

Remaining motion groups. Recording remains armed after each capture. Pose the next group and click the displayed Record Step button; do not arm it again:

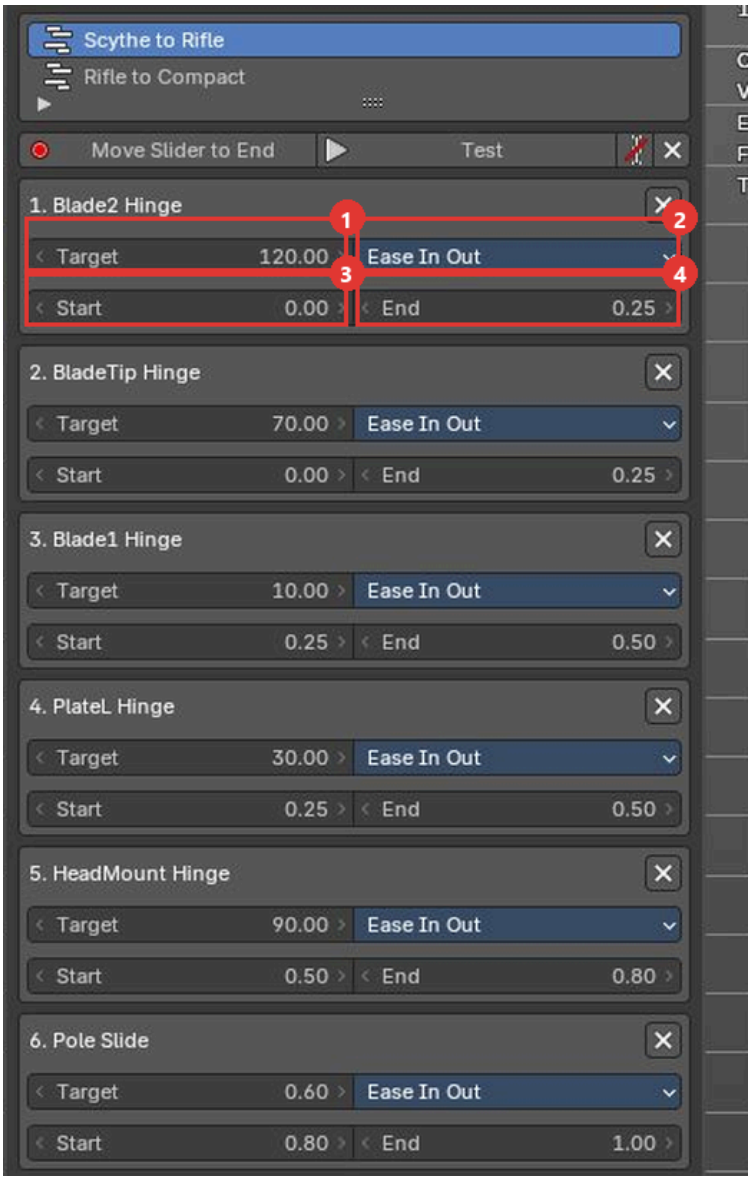
Motion group	Pose these handles	Record button after the preceding groups
2	`Blade1 Hinge` 10 and `PlateL Hinge` 30	Record Step 3
3	`HeadMount Hinge` 90	Record Step 5
4	`Pole Slide` 0.6	Record Step 6

The button number is the number of stored target entries plus one. Recording two handles advances it by two; it does not count recording clicks. The four motion groups above therefore produce six target entries. If you change which handles move, use the number actually displayed.

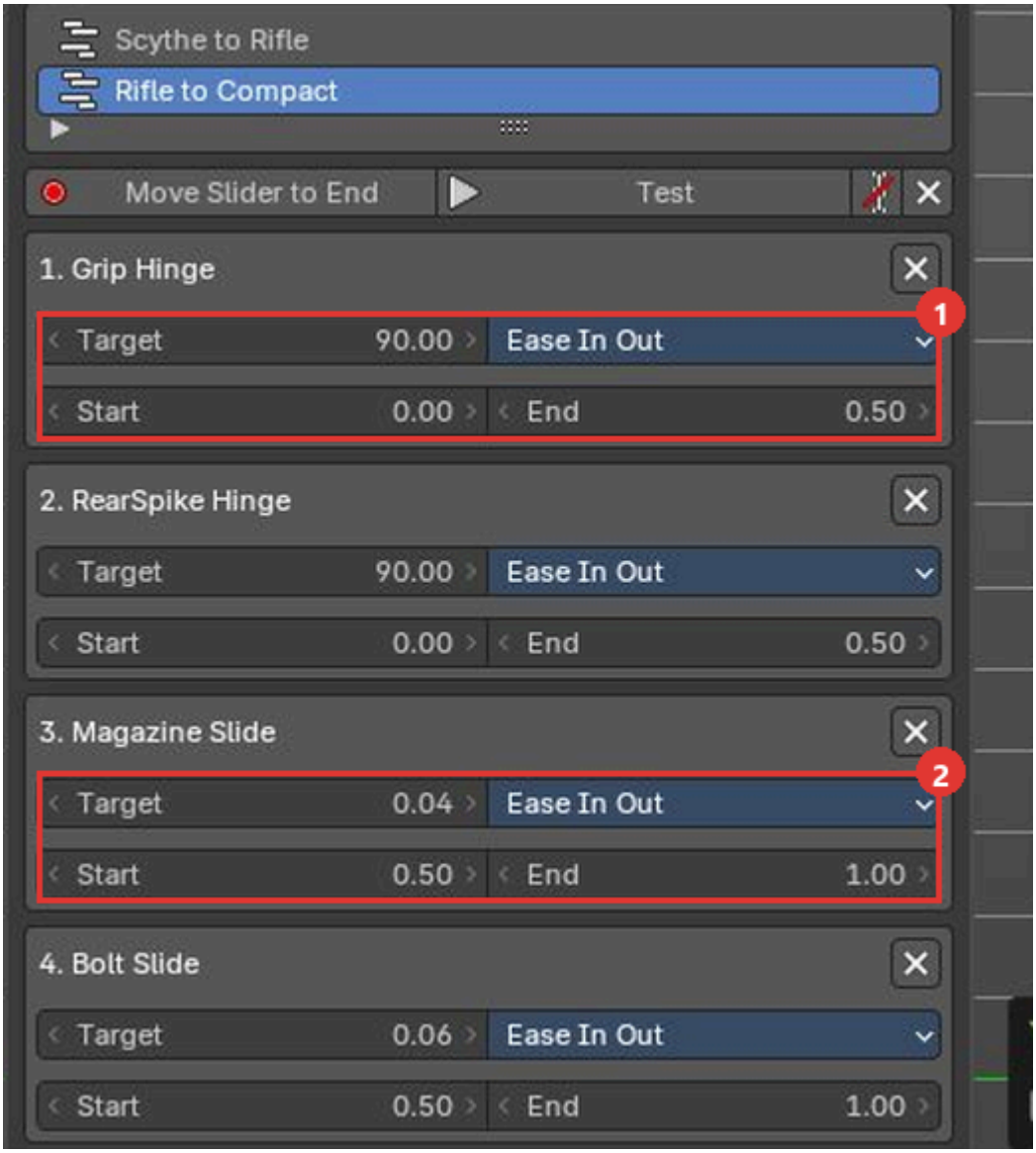
Step 25 — The steps

Each recorded target has its own entry under the sequence, showing its **Target**, its ease and its **Start** and **End** on the slider. Give both entries in each motion group the same Start and End: the blade folds first (0…0.25), the plates and the first segment next (0.25…0.5), the head third (0.5…0.8) and the pole last (0.8…1). Before its slice a step stays at its starting value; after its slice it holds its end value. The Start/End fields use 0-1, while the Controls sliders show 0-100%.

Keep each entry on `Ease In Out`. Then add a second sequence, `Rifle to Compact`: first record `Grip Hinge` 90 and `RearSpike Hinge` 90 (two entries, both 0…0.5); then record `Magazine Slide` 0.04 and `Bolt Slide` 0.06 (two entries, both 0.5…1). Its recording button changes from Record Step 1 to Record Step 3 for the second group. After each sequence is complete, click **Cancel** in the capture banner. This ends recording, retains the stored entries, and returns that sequence slider to zero. An overlap warning while armed describes the proposed next recording interval; do not add an extra empty group.



1: Target is the recorded joint value. 2: Choose Ease In Out. 3: Enter Start. 4: Enter End. Repeat these fields for all six entries using the shown values.

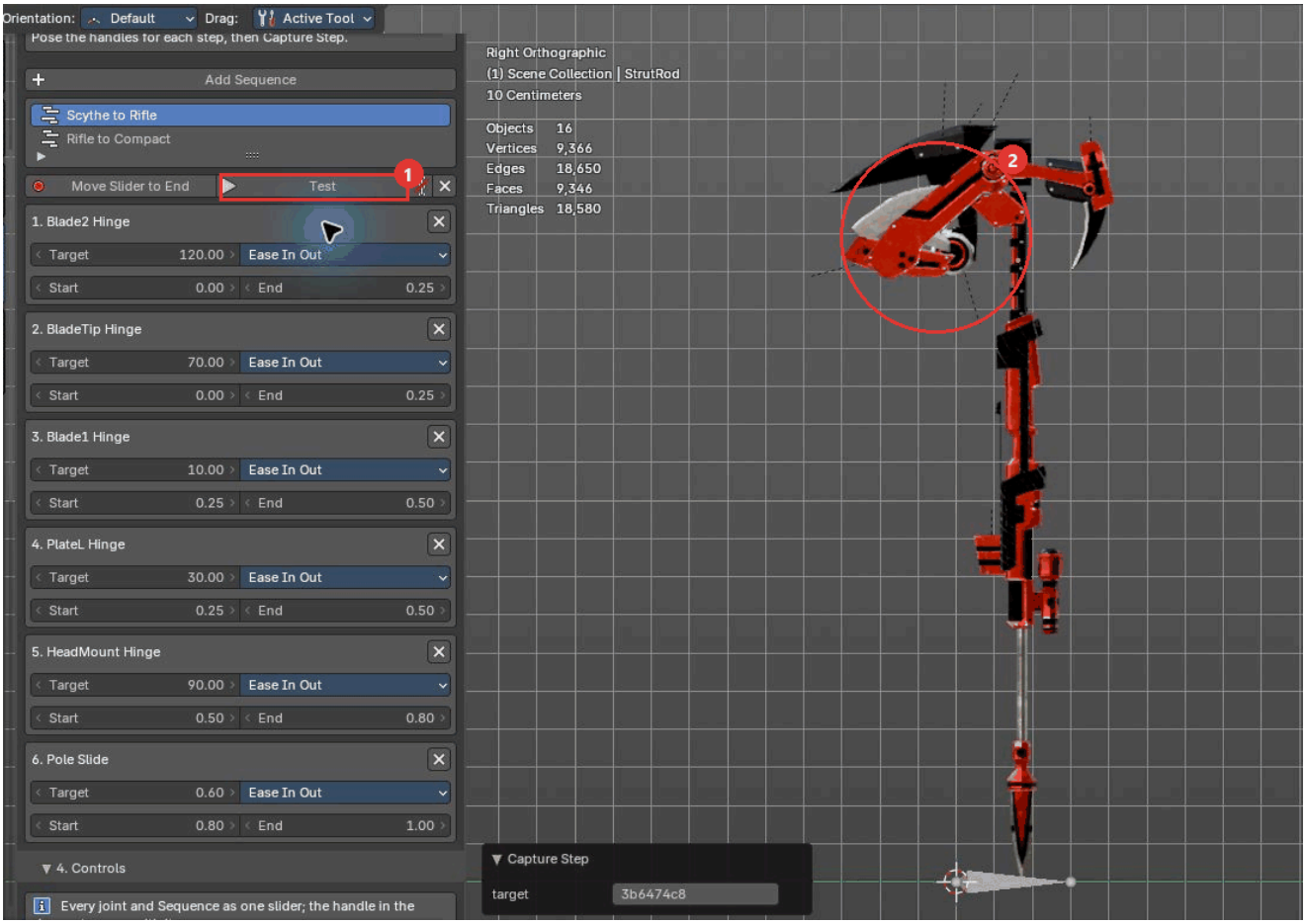


1: Grip and RearSpike use Target 90, Start 0, End 0.5. 2: Magazine and Bolt use 0.04 / 0.06, Start 0.5, End 1. Keep Ease In Out.

1: After finishing a sequence, click Cancel in this capture banner to end recording. This keeps the entries and returns the sequence to zero.

Step 26 — Test the sequence

Click **Test** next to the step-recording button. The slider sweeps from 0% to 100% to 0%, then restores its value from before the test. Test `Scythe to Rifle` from Scythe; test `Rifle to Compact` with the first sequence at 100%. Return both to 0% afterward.



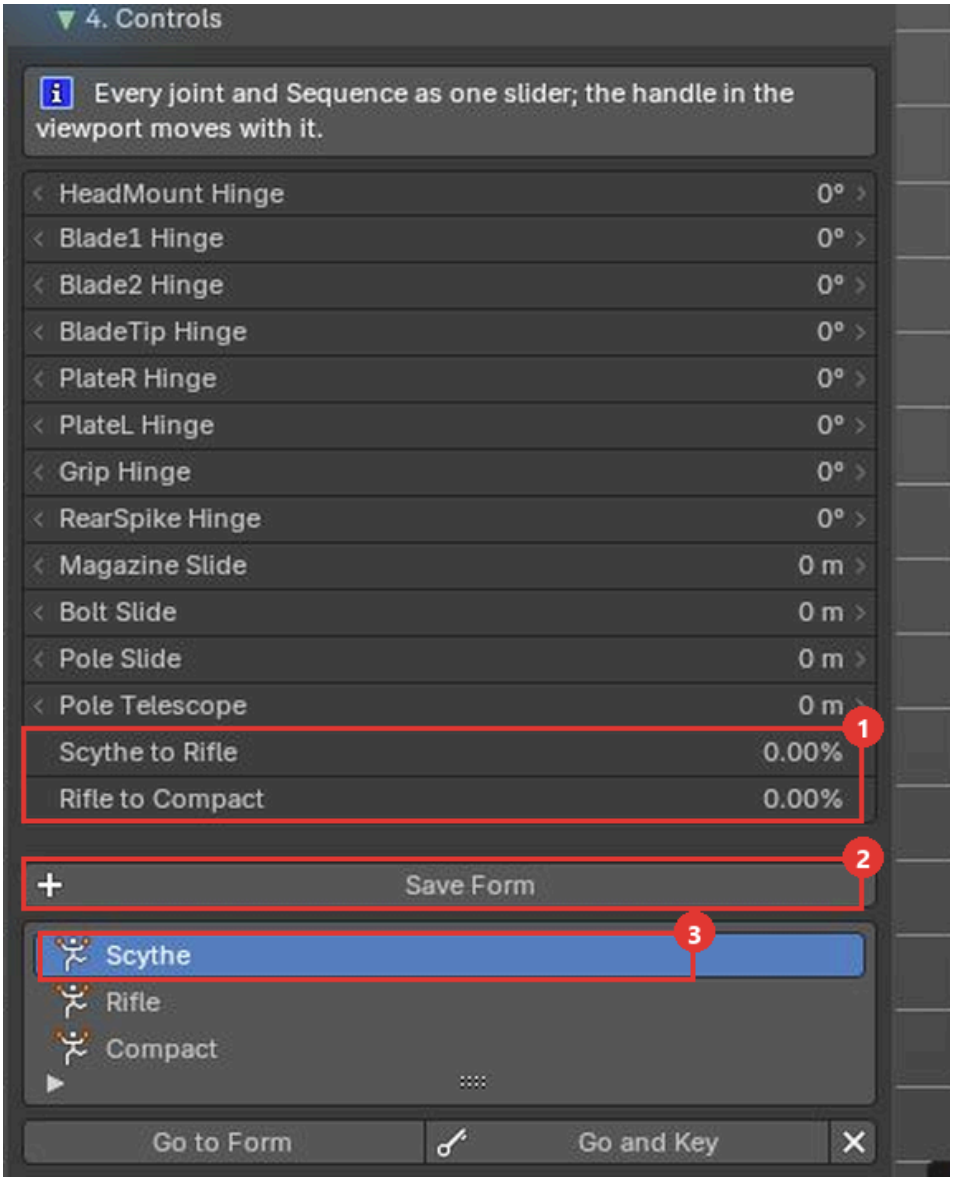
1: Click Test beside the recording button. 2: Watch the fold and return. Test Compact with Scythe to Rifle at 100%.

What you should be seeing: The blade curls, the plates close, the head folds down along the receiver, and the pole slides up. Then everything comes back.

H. Forms

Step 27 — Save Form

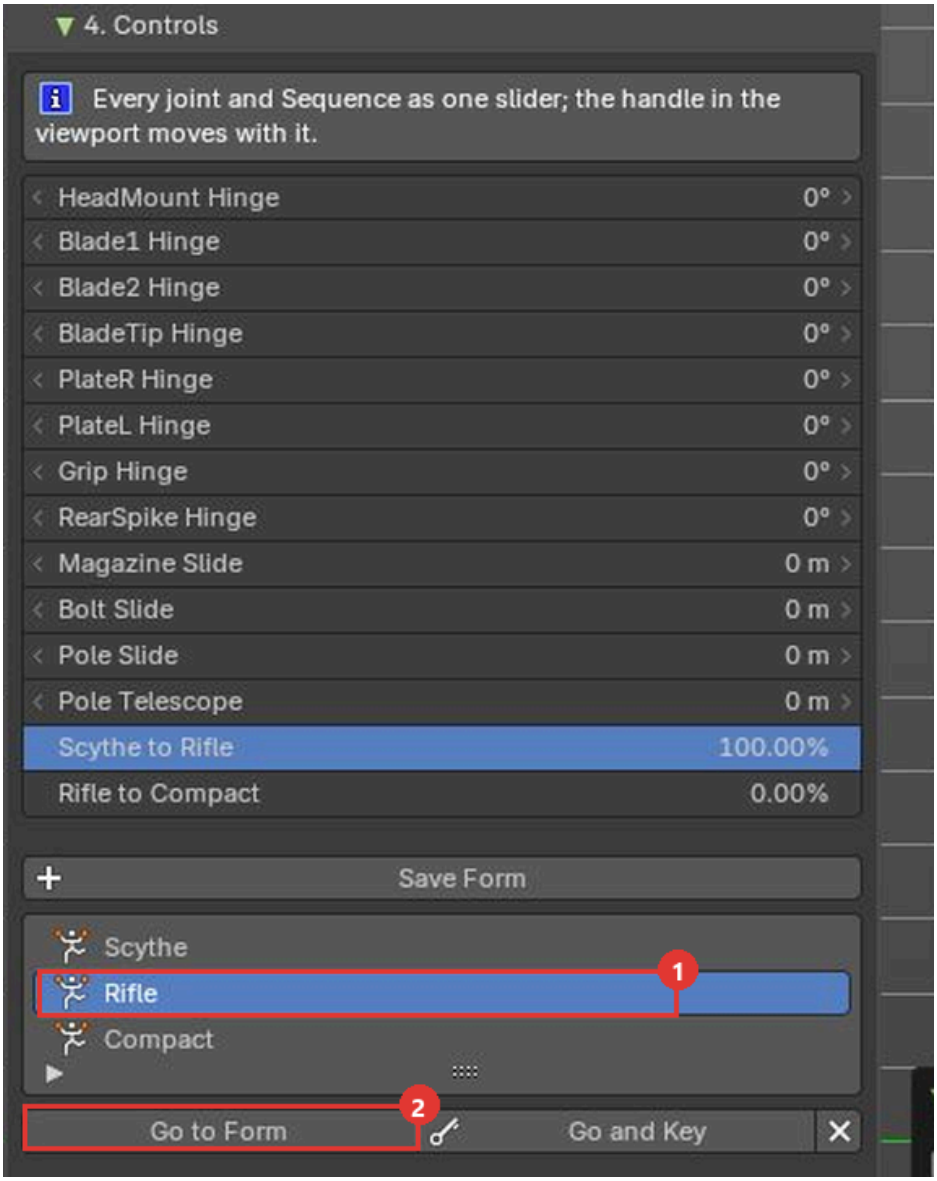
In **4. Controls**, with every slider at 0, click **Save Form**, then double-click the new list entry to name it `Scythe`. Set `Scythe to Rifle` to `100%` and save/name `Rifle`. Set `Rifle to Compact` to `100%` as well and save/name `Compact`. Leave all individual joint sliders at zero: the sequences provide their motion. Setting the selected sequence to its end can arm recording again; saving a Form does not record another step. Choosing `Scythe` and **Go to Form** returns both sequences to zero and clears the armed state. Going to Rifle or Compact can arm recording again because the selected sequence is at 100%; do not click Record Step unless you intend to add another target.



1: Start with both sequences and every joint at 0. 2: Click Save Form. 3: Double-click the new list entry and name it Scythe; repeat for Rifle and Compact.

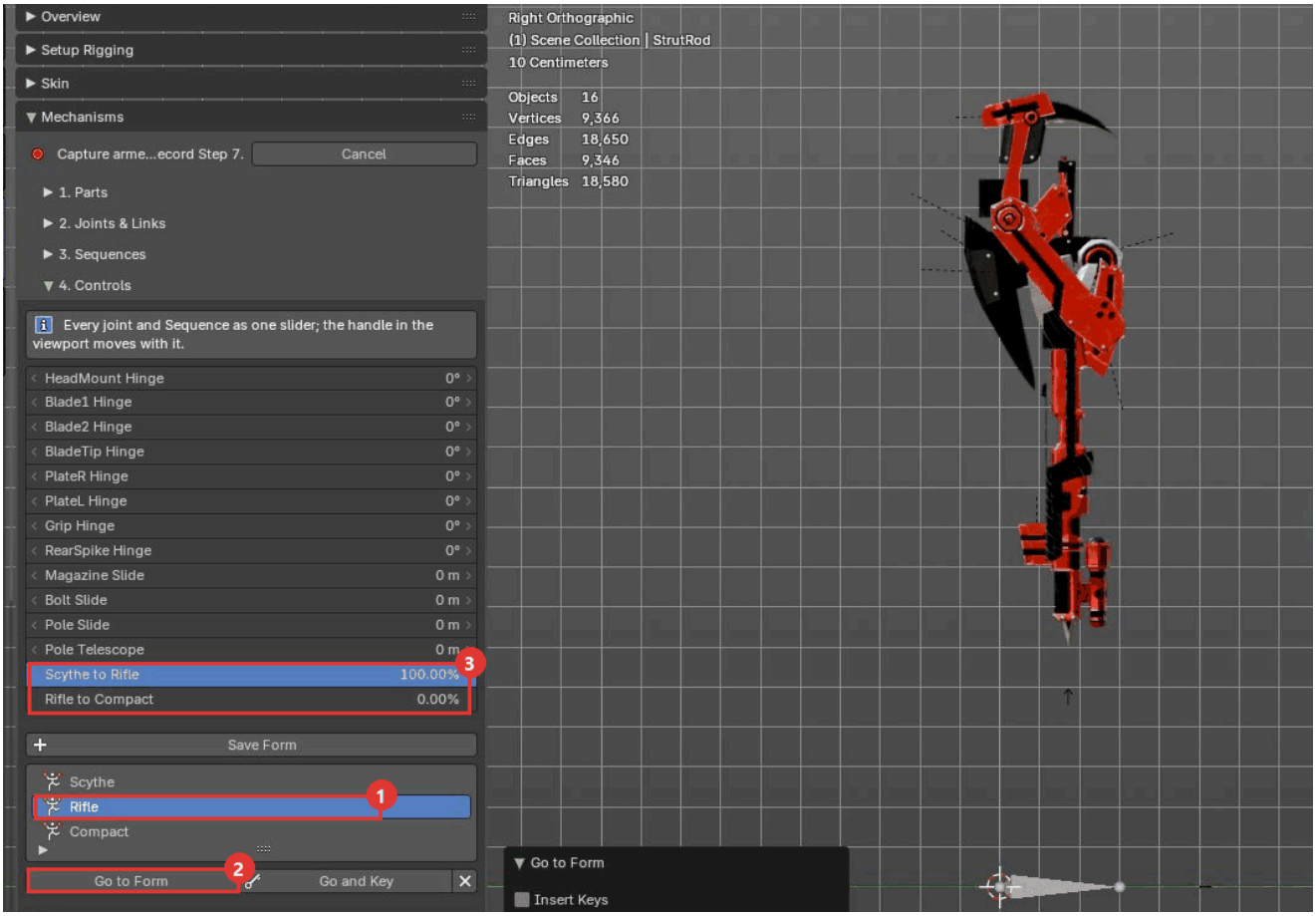
Step 28 — Go to Form

Click a form in the list and click **Go to Form**. **Go and Key** does the same and keys every control on the current frame, which is how you animate the transformation.



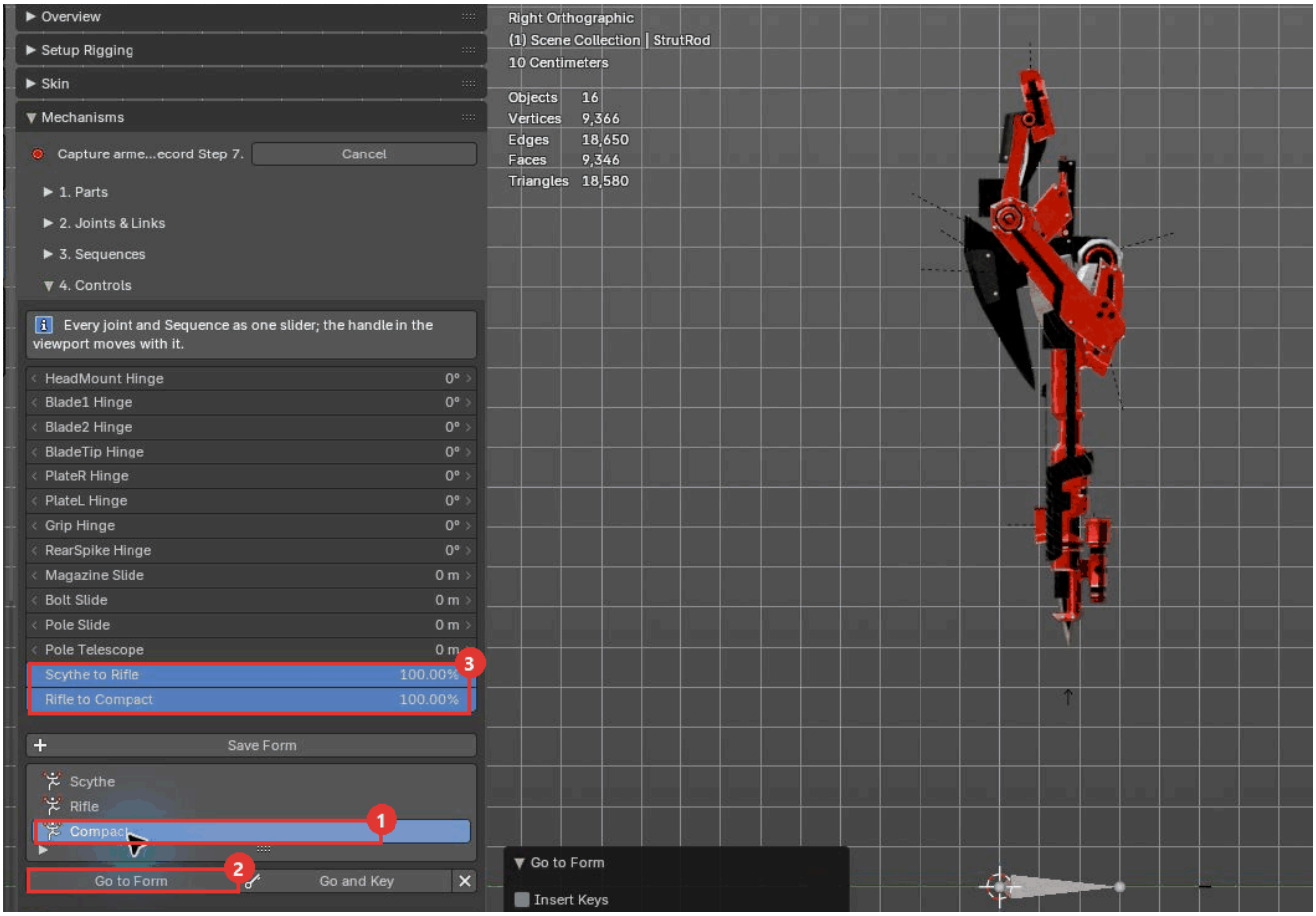
1: Select the saved form. 2: Click Go to Form. Go and Key also inserts animation keys.

Step 29 — The rifle



1: Select Rifle. 2: Click Go to Form. 3: Check the sequence percentages against the pictured pose.

Step 30 — Compact



1: Select Compact. 2: Click Go to Form. 3: Check the sequence percentages against the pictured pose.

Go back to `Scythe` and save the file.

What you built

Record	Bones	What it does
15 parts	`MCH-<part>` and `DEF-<part>`	The mesh of each part is parented to its `DEF-` bone
8 Hinges, 3 Slides, 1 Telescope	`CTRL-<part>`	The handles; their values are the sliders in **4. Controls**
1 Piston	`MCH-<part>-aim` and `MCH-<part>-target`	Keep the strut's two parts aimed at each other
1 Follow	none	`PlateR` turns with `PlateL`
2 Sequences	`CTRL-SEQ-<name>`	One slider each
3 Forms	none	Saved slider values

The refreshed example passed the Rig Validator with no reported issues. It also passed reload checks with Auto Run Python Scripts off, including every individual control and all three Forms.

Try next

- Tick **Curve** on `PlateR Follow` and reshape the mapping in the Graph Editor.
- Save a fourth form, `Carry`, with the blade folded but the head up.
- Key the forms over frames with **Go and Key**, then export the `DEF-` bones (see VRChat Tools).

Gear Link

Crescent Rose has no geared pair, so this rig has no Gear Link. A Gear Link turns one Hinge when another turns, by a tooth ratio: a crank and a wheel, or a rack that turns rotation into a slide. See Gear Link.

Credits

Crescent Rose model by amandangm, https://sketchfab.com/3d-models/crescent-rose-dcb6bfbb3b824c01a182fa131fe35824, CC BY 4.0; split into parts, scaled and rigged for this tutorial. Crescent Rose is a design of Rooster Teeth Productions; this is a fan-made tutorial asset.

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